Intraperitoneal and intravenous paclitaxel plus S-1 versus intravenous paclitaxel plus S-1 in gastric cancer patients with peritoneal metastasis: Results from the multicenter, randomized, phase 3 DRAGON-01 trial.

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Background: The optimal chemotherapeutic regimen for gastric cancer with peritoneal metastasis remains undefined. We evaluated the efficacy and safety of intraperitoneal and intravenous paclitaxel plus S-1 (NIPS Group) versus paclitaxel plus S-1 (PS Group) in gastric cancer patients with peritoneal metastasis. **Methods:** In this multicenter, randomized, controlled, phase 3 trial, patients were recruited from 9 cancer centers in China. Eligible patients were aged 18~75 years with an Eastern Cooperative Oncology Group performance status of 0~1, histologically confirmed gastric adenocarcinoma, peritoneal metastases from gastric cancer requiring definitive diagnosis by laparoscopy, without gastric outflow tract obstruction and intestinal obstruction, no prior treatment with chemotherapy, radiation therapy, targeted therapy or immunotherapy. Eligible patients were randomly assigned (2:1) to receive either intravenous paclitaxel at 50 mg/m^2 and intraperitoneal paclitaxel at 20 mg/m^2 on days 1 and 8, along with oral S-1 at a dose of 80 mg/m^2 on days 1~14, or intravenous paclitaxel at 70 mg/m^2 on days 1 and 8, along with oral S-1 at 80 mg/m^2 on days 1~14. The primary endpoint was overall survival in the intention-to-treat population. Safety was assessed in all participants. A sample size of 238 patients was calculated to provide 80% power with a one-sided alpha of 0.1, accounting for a 10% dropout rate. Survival was analyzed using Kaplan-Meier curves and compared with the log-rank test. Cox regression was used for multivariate analysis. A P-value < 0.05 was considered statistically significant. **Results:** From May 10, 2017, to March 9, 2022, 246 patients were screened and 222 were included in the modified intention-to-treat population, of whom 148 patients were assigned to the NIPS Group and 74 to the PS group. As of data cutoff (March 9, 2024), the median survival time was 19.4 months (95% CI, 17.1~22.9), in the NIPS group and 13.9 months (95% CI, 10.3~16.1) in the PS group (HR = 0.66; 95% CI 0.49 - 0.88; $P = 0.005$). The 1-year and 2-year overall survival rates were 69.6% and 37.2% in the NIPS group, compared to 54.1% and 20.3% in the PS group. The most common grade 3~4 adverse events were leukopenia (21.7% in the NIPS group, and 24.7% in the PS group) and neutropenia (19.9% in the NIPS group, and 23.4% in the PS group). No treatment-related deaths were reported. **Conclusions:** Intraperitoneal and intravenous paclitaxel plus S-1 significantly improved the overall survival compared to intravenous paclitaxel plus S-1 in gastric cancer patients with peritoneal metastasis, with manageable toxicity. Clinical trial information: ChiCTR-IIR-16009802. Research Sponsor: Shanghai Jiao Tong University School of Medicine Multicenter Clinical Research Project; DLY201602.

Outcomes of patients with peritoneal-limited metastatic gastric cancer (PLGC) undergoing bidirectional (BD) chemotherapy cytoreductive surgery (CRS) with heated intraperitoneal chemotherapy (HIPEC).

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Background: The prognosis of PLGC patients (pts) is poor with standard systemic therapy. PLGC pts who undergo CRS after BD chemotherapy (chemo) demonstrate prolonged survival. We retrospectively analysed the outcomes of pts who underwent BD chemo followed by CRS and HIPEC in a single institution, stratifying by peritoneal carcinomatosis index (PCI) and chemo regimen. **Methods:** PLGC pts who received BD chemo between January 2016 to September 2023 were enrolled and consented for data collection. BD chemo was either 3 weekly paclitaxel (PTX) (50mg/m2), S-1 (80mg/m2) and IP PTX (20mg/m2) or capecitabine (2g/m2), oxaliplatin (100mg/m2) (CAPOX) and IP PTX (40mg/m2). HIPEC utilised cisplatin 50mg/m2 at 40 for 60 minutes. Trastuzumab and nivolumab were given selectively. Pts without radiological progression after 24 weeks of BD chemo, with PCI7 and potential for complete resection underwent CRS including D2 gastrectomy and HIPEC. Post-resection, BD chemo was stopped. Survival was analysed by Kaplan Meier curves, adverse events (AEs) by CTCAE v5.0 and response by independent radiological review. **Results:** 34 pts were enrolled with median follow-up of 26.9 months (m). Median (range) age was 51 (17 – 79) and median PCI 12 (0-29). 52.9% were female, 8.8% had Cy1 only disease and 38% had prior gastrectomy. 3 out of 6 pts with evaluable disease had partial response. Median progression-free survival and overall survival (mOS) was 9.1m and 16.5m respectively. 7 pts (20.6%) underwent CRS with median (range) PCI of 1 (0-15.0). mOS in CRS vs non-CRS pts was 23.8m vs 14.6m (p=0.4); PCI <7 vs PCI ≥ 7 was 23.8m vs 14.6m (p=0.63); and CAPOX vs S1/PTX systemic chemotherapy was 23.9m vs 14.6m (p=0.29) respectively. Peritoneal cytology negativity increased from 25.8% to 86.4% after BD chemo. Peritoneal lesions biopsied post-BD chemo showed no tumour cells in 64% of cases. There was no treatment-related mortality. 1 pt had IP port removal due to complications. The commonest grade 3 AE was neutropenia (14.7%). Biomarker analysis is currently being performed. **Conclusions:** BD chemo CRS HIPEC was well tolerated with longer survival in patients undergoing CRS. Evaluation of tumour using PCI post-BD chemo may be sub-optimal. Research Sponsor: None.

LBA329 **Oral Abstract Session**

Preliminary results from the multicenter, randomized phase III trial (SCIENCE): Comparing chemotherapy plus sintilimab and chemoradiotherapy plus sintilimab versus chemoradiotherapy for neoadjuvant treatment in resectable locally advanced esophageal squamous cell carcinoma.

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The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

LBA331 **Oral Abstract Session**

EORTC-1203 GITC "INNOVATION": Integration of trastuzumab (T), with or without pertuzumab (P), into perioperative chemotherapy of HER-2 positive stomach cancer: Overall survival results.

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The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

Final analysis of the randomized phase 2 part of the ASPEN-06 study: A phase 2/3 study of evorpacept (ALX148), a CD47 myeloid checkpoint inhibitor, in patients with HER2-overexpressing gastric/gastroesophageal cancer (GC).

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Background: Evorpacept (Evo) is a high affinity, CD47-blocker with an inactive Fc region designed to safely magnify anticancer antibody dependent cellular phagocytosis. Evorpacept is being evaluated across cancers, in combination with anticancer antibodies and checkpoint inhibitors. **Methods:** The ASPEN-06 randomized phase 2 trial evaluates Evo in combination with standard trastuzumab (T), ramucirumab (R) and paclitaxel (P) for the treatment of patients with HER2+ GC. Patients with 2nd or 3rd line HER2+ advanced or metastatic GC that has progressed on or after prior anti-HER2 therapy were randomized to Evo (30 mg/kg Q2W) plus TRP or TRP. HER2 status was determined in the most recent GC tissue sample. Primary study objectives were to compare confirmed ORR of Evo-TRP to an assumed ORR of RP (=30%) with one-sided alpha error of 0.025 and to identify a clinically meaningful contribution of Evo to TRP in ORR (delta>10%). **Results:** Among the entire randomized population (N=127) the ORR was 40.3% (Evo-TRP) and 26.6% (TRP), respectively. The difference between Evo-TRP's ORR compared to historical RP's ORR was not statistically significant (p=0.095), but Evo-TRP's ORR demonstrated a meaningful delta of 13.7% over TRP's ORR (p=0.028 in an exploratory analysis). Median DOR for Evo-TRP and TRP was 15.7 and 7.6 months, respectively. In a prespecified population with HER2+ disease in fresh tumor tissues after prior anti-HER2 treatment (N=48), Evo-TRP's ORR (54.8%) compared favorably to historical RP's ORR (p=0.030) with a delta of 31.7% over TRP's ORR of 23.1% (p=0.004 in an exploratory analysis). Evo-TRP was well tolerated with a safety profile consistent with prior experience. Updated results including PFS will be presented at the meeting. **Conclusions:** The addition of Evo to TRP showed promising activities in HER2+ GC. The magnitude of response was greatest in tumors identified as HER2+ using fresh biopsies emphasizing the importance of biopsies post anti HER2 therapy. Consistent with Evo's mechanism of enhancing antibody dependent phagocytosis, these data support its ongoing investigation as an adjunct to anti-HER2 gastric cancer therapy. Clinical trial information: NCT05002127. Research Sponsor: ALX Oncology.

Efficacy and safety of albumin-bound docetaxel vs. docetaxel in patients with previously treated advanced gastric or gastroesophageal junction adenocarcinoma: A multicenter, randomized, phase II study.

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Background: Docetaxel has demonstrated promising activity in gastric cancer, both as monotherapy and in combination with other agents. Albumin-bound docetaxel (HB1801) is a new kind of taxane and has many advantages compared with docetaxel. Previous phase I study has demonstrated that HB1801 showed preliminary efficacy in patients with gastric cancer. **Methods:** In this phase II study, eligible patients were aged 18-75 years with histologically confirmed gastric or gastroesophageal junction (G/GEJ) adenocarcinoma, who progressed on at least first line of combined chemotherapy of platinum and fluorouracil. Patients were randomized (1:1) to receive HB1801 (100 mg/m^2) or docetaxel (Taxotere, 75 mg/m^2) administered every 3 weeks by intravenous infusion. Stratified factors included previous treatment with immunotherapy (yes or no) and ECOG PS (0 or 1). Primary endpoint was progression free survival (PFS). **Results:** As of June 25, 2024, 128 patients were randomized to the HB1801 group (n=65) or the docetaxel group (n=63). The two groups were comparable on baseline characteristics. Overall, the median age was 59.0 (range 27-75) years, 106 (82.8%) patients had ECOG PS of 1. 76 (59.4%) patients had received previous treatment with immunotherapy. 76 (59.4%) patients had ≥ 2 metastatic sites. Median PFS was 4.0 months (95%CI 2.8-4.2) with HB1801 versus 2.7 months (95%CI 1.8-3.0) with docetaxel, HR = 0.81(95%CI 0.53, 1.21). Objective response rate and disease control rate were 21.5% (1CR and 13 PRs, 95%CI 12.3-33.5) and 56.9% (20 SDs, 95%CI 44.0-69.2) in the HB1801 group, compared with 14.3% (1 CR and 8 PRs, 95%CI 6.8-25.4) and 42.9% (17 SDs, 95%CI 30.5-56.0) in the docetaxel group. Median overall survival was 11.3 months with HB1801 versus 7.8 months with docetaxel (HR = 0.59 [95%CI 0.35, 1.01], p=0.025). Treatment-related adverse events (TRAEs) occurred in 93.8% of patients receiving HB1801, and 93.7% of patients receiving docetaxel. Alopecia (46.2% vs. 39.7%), fatigue (43.1% vs 23.8%), anemia (40.0% vs 39.7%), hypoalbuminemia (29.2% vs 12.7%), decreased appetite (21.5% vs 12.7%), peripheral edema (16.9% vs 12.7%) and nausea (15.4% vs 19.0%) were the most common TRAEs in the HB1801 group and the docetaxel group. 20 patients (30.8%) in the HB1801 group and 19 patients (30.2%) in the docetaxel group experienced ≥grade 3 TRAEs, with the most common being anemia and fatigue. 3 patients in the docetaxel group experienced treatment emergent adverse events (TEAEs) leading to death, which one was considered to be treatment-related. No patients in the HB1801 group had TEAEs leading to death. No new safety signal was observed in HB1801 group. **Conclusions:** Compared with Taxotere, HB1801 results in 41% reduction in risk of death with a manageable safety profile in patients with previously treated advanced G/GEJ cancer. Clinical trial information: NCT05705635. Research Sponsor: CSPC ZhongQi Pharmaceutical Technology Co., Ltd.

Conversion effects of PD-1 inhibitor camrelizumab (Cam) combined with Nab-POF regimen in patients (pts) with initially unresectable locally advanced or limited metastatic gastric or gastroesophageal junction adenocarcinoma: FDZL-001 trial.

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Background: For initially unresectable locally advanced or metastatic gastric or gastroesophageal junction adenocarcinoma, the prognosis is poor and chemotherapy is the main treatment option. The AIO-FLOT3 Trial suggested that conversion therapy might improve outcome in pts with limited metastatic gastric cancer (GC). We conducted this prospective, single-arm, phase II trial to improve conversion efficacy of PD-1 inhibitor Cam combined with Nab-POF regimen (nab-paclitaxel , oxaliplatin, fluorouracil) in pts with initially unresectable locally advanced or limited metastatic gastric or gastroesophageal junction adenocarcinoma. **Methods:** Eligible pts received Cam 200mg, nab-PTX 125 mg/m^2, oxaliplatin 85 mg/m^2 or with trastuzumab if Her2 positive, each was an IV followed by fluorouracil 2400 mg/m^2 as a 48-h continuous IV on day 1, every 2 weeks. Patients were assessed for tumor response and surgical feasibility every 3 cycles, and if confirmed feasible for surgical resection by both surgical and medical experts after 6 cycles, they would undergo surgery within 3-6 weeks. The primary endpoint was R0 resection rate. **Results:** In total, 53 pts were enrolled up to 01/2024. 52 pts were evaluable. 50% are younger than 65 years of age. Male was 73.1%. The metastatic sites included liver 46.2% (24), retroperitoneal lymph nodes 40.4% (21), Krukenberg 5.8% (3), and locally advanced unresectable tumors 21.2% (11). 4 pts (7.7%) were dMMR/MSI-H and 10 (19.2%) were Her2 positive. ORR was 88.5% (46/52) and DCR was 98.1% (51/52). 45(86.5%) pts were assessed as resectable. 3 pts (6.8%) refused surgery and 3 (6.8%) were cCR under observation. The R0 resection rate was 75.0% (39/52). CR rate was 23.1% (cCR, 5.8%; pCR,17.3%). The mPFS was not reached (95% CI = 17.0-NE), and the mOS was not reached (95% CI = 25.4-NE), either. The 3-year PFS and OS rate were 56.9% (95% CI = 45.7%-86.3%) and 62.8% (95% CI = 41.6%-78.0%), respectively. The 3 cCR pts have survived utill the last follow-up date of 07/2025 without tumors for 17, 29, and 34 months, respectively. Pts tolerated treatment well, although all pts experienced treatment-related adverse events (AE). Grade 3/4 AEs were 42.3% (22/52) and neutrophil decrease was the most common at 36.5%. 10 patients (19.2%) experienced immune related AEs (irAEs), and 2 of them experienced grade 3 irAEs. **Conclusions:** The PD-1 inhibitor Cam combined with the Nab-POF regimen safely induced a high conversion rate with very high R0 resection and high 3-year PFS and OS rates, preliminarily demonstrating promising effects, providing a new conversion drug treatment option for pts with initially unresectable locally advanced or limited metastatic advanced GC. These findings warrant further prospective, randomized controlled investigations. Clinical trial information: NCT04510064. Research Sponsor: None.

Effect of SHR-1701 on chemotherapy (chemo)-induced myelosuppression: Data from a phase 3 study in HER2-negative gastric/gastroesophageal junction adenocarcinoma (G/GEJA).

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Background: Blocking TGF-β signaling has the potential to facilitate the recovery from chemo-induced myelosuppression. SHR-1701 is a bifunctional agent targeting PD-L1 and TGF-β. **Methods:** In the multicenter, randomized, double-blind phase 3 study (NCT04950322), SHR-1701 plus CAPOX demonstrated a statistically significant and clinically meaningful benefit in OS compared with placebo plus CAPOX in patients (pts) with previously untreated, unresectable locally advanced or metastatic HER2-negative G/GEJA (ESMO 2024, LBA60). Here, we reported the effect of SHR-1701 on chemo-associated myelosuppression. **Results:** In total, 364 and 366 pts in the SHR-1701 and placebo arm received assigned treatment. As of May 20, 2024, with a KM estimated median follow-up of 13.6 mo, all study medications showed similar exposure duration between the two arms (Table). Occurrence of treatment-related adverse events was generally comparable (any grade, 97.8% in SHR-1701 arm vs 98.4% in placebo arm; grade ≥3, 62.6% vs 59.0%). SHR-1701 reduced the incidence of any grade decreased platelet count, decreased neutrophil count, and decreased white blood cell count by 8.7%, 10.1%, and 11.2%, and for grade ≥3, by 9.1%, 4.3%, and 1.6%, respectively (Table). Compared with the placebo arm, less pts in the SHR-1701 arm used platelet growth factors (30.2% vs 42.9%) and white blood cell growth factors (32.7% vs 39.6%). **Conclusions:** SHR-1701 showed the capacity to suppress chemo-associated myelosuppression. Clinical trial information: NCT04950322. Research Sponsor: Jiangsu Hengrui Pharmaceuticals.

Safety.

	SHR-1701 plus CAPOX (N=364)		Placebo plus CAPOX (N=366)		Treatmentdifference (SHR-1701 vs placebo)	
Treatment exposure (day), median (IQR)						
SHR-1701/placebo	137.0 (65.5-251.5)		127.5 (62.0-190.0)		NA	
Capecitabine	122.0 (83.0-138.0)		122.0 (73.5-135.0)		NA	
Oxaliplatin	107.0 (66.0-123.0)		108.5 (64.0-121.0)		NA	
Treatment-related hematological toxicities, n (%) or %	**Any grade**	**Grade ≥3**	**Any grade**	**Grade ≥3**	**Any grade**	**Grade ≥3**
Platelet count decreased	217 (59.6%)	69 (19.0%)	250 (68.3%)	103 (28.1%)	-8.7%	-9.1%
Neutrophil count decreased	163 (44.8%)	44 (12.1%)	201 (54.9%)	60 (16.4%)	-10.1%	-4.3%
White blood cell count decreased	147 (40.4%)	13 (3.6%)	189 (51.6%)	19 (5.2%)	-11.2%	-1.6%

Effects of neighborhood deprivation index on survival in gastric cancers.

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Background: Previous studies have shown that living in disadvantaged neighborhoods resulted in poor outcomes in several malignancies. Neighborhood Deprivation Index (NDI) identifies key variables from 13 measures, including the following dimensions of socioeconomic (SES) status: wealth and income, education, occupation, and housing conditions. In this study, we aimed to investigate the effect of NDI among patients with gastric cancer (GC). **Methods:** We conducted a retrospective analysis using the Surveillance, Epidemiology, and End Results (SEER) database from 1996-2015. NDI was divided into two groups: less disadvantaged areas (NDI <60) and highly disadvantaged areas (NDI >60), respectively. Demographic and clinical characteristics were summarized by NDI groups. All associations were compared using Kruskal-Wallis and Chi-Square tests, respectively. Kaplan Meier curves were summarized by the NDI group for overall survival (OS) and disease-specific survival (DSS). All associations were compared using a log-rank test. Cox multivariate regression was performed to measure the association between NDI and OS/DSS. All analyses were conducted in RStudio v4.2.3 at a significance level of ≤ 0.05. **Results:** A total of 19,205 GC (11,137 patients with NDI<60 and 8,068 NDI>60) patients were analyzed. Disadvantaged areas (NDI>60) had a higher proportion of Black and Hispanic, single, uninsured, rural patients with later disease stage as compared to NDI <60. Median OS was 10.0 months for patients with NDI <60 versus 9.0 months for those with NDI >60 (p=0.0007). Median DSS was 12.0 months for NDI <60 and 10.0 months for NDI >60 (p=0.010). Multivariate analysis revealed that factors associated with worse OS included NDI >60 [HR 1.044, p=0.009], age (>60) [HR 1.28, p<0.001], male sex [HR 1.082, p<0.001], single marital status [HR 1.21, p<0.001], uninsured status [HR 1.13], p <0.001], advanced disease stage (III/IV) [HR 2.05, p<0.001], and higher grade [HR 1.32, p<0.001]. On the other hand, non-Hispanic Black [HR 0.98] and Hispanic [HR 0.94] race and urban location [HR 0.91, p<0.001] were associated with improved OS. Factors associated with inferior DSS included NDI >60 [HR 1.037, p=0.038], age (p<0.001), male sex [HR 1.05, p=0.005], single marital status [HR 1.15, p<0.001], advanced disease stage (III/IV) [HR 2.375, p<0.001], and advanced grade (III/IV) [HR 1.40, p<0.001). In contrast, Hispanic [HR 0.94], non-Hispanic Black [0.96], and urban location [HR 0.90, p<0.001] were associated with improved DSS. **Conclusions:** Patients from areas with higher NDI had worse OS and DSS for GC. Tailored healthcare policies that aim to improve healthcare delivery to areas with high deprivation may help improve GC outcomes. Research Sponsor: National Cancer Institute; P30CA016056.

Impact of race on end-of-life care among patients with esophageal cancer.

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Background: Esophageal cancer is an aggressive malignancy with rising incidence globally. Patients with advanced esophageal cancer suffer from a heavy burden of a wide array of symptoms including dysphagia and cachexia/anorexia, necessitating the need for focused symptom management and palliative care. Studies focused on other cancer types have shown that Black patients are more likely to receive aggressive care at end-of-life (EOL) and suffer from a decline in quality of life. We sought to investigate racial differences in EOL care among patients with advanced esophageal cancer in the US. **Methods:** The National Inpatient Sample (NIS) was queried to identify all hospitalizations with esophageal cancer utilizing ICD-10 codes C15x from 2016 to 2020. Hospitalizations of White and Black patients with documented inpatient mortality events were then extracted. Demographic and clinical data were analyzed using chi squared tests, independent sample t-tests, and binary logistic regression (adjusted for age, gender, and Charlson comorbidity index or CCI). Adjusted Odds ratios (aOR) are presented with 95% Confidence intervals (CI). **Results:** A total of 15,130 patients with esophageal cancer were included, of which 13,345 (88.2%) were White and 1785 (11.8%) were Black. Hospitalizations with White patients recorded higher age, more males, and higher comorbidities per CCI. Hospitalizations with Black patients had longer lengths of stay on average, higher total hospital charges, and significantly less likely to have a Do Not Resuscitate (DNR) code status (aOR 0.81, 0.74-0.9 95% CI). Rates of inpatient palliative care consultation at EOL were significantly lower among Black patients (aOR=0.53, 0.48-0.58 95% CI). Black patients were more likely to receive aggressive intervention at EOL demonstrated by rates of blood transfusion (aOR 1.9, 1.6-2.1 95% CI), mechanical ventilation (aOR 1.7, 1.5-1.8 95% CI), and vasopressor usage (aOR 1.4, 1.2-1.6 95% CI). **Conclusions:** We present a large retrospective analysis demonstrating significant racial disparities in EOL care amongst hospitalized patients with esophageal cancer. Black patients had longer lengths of stay, lower rates of palliative care consultation and DNR code status, and higher rates of aggressive interventions at EOL. These findings may have implications for improving health care decision making for EOL care amongst minority patients with esophageal cancer. Research Sponsor: None.

	White N=13345	Black N=1785	p-value
Age (in years)	68.24±10.67	65.12±10.36	**<.001**
CCI	9.4±3.3	8.6±3.9	**<.001**
Mean length of stay (days)	7.5±11.3	10.4±22.8	**<.001**
Total Cost (In thousands of dollars)	108.6±231.0	141.9±388.5	**<.001**
Palliative	60.9%	45.1%	**<.001**
Adjusted OR		**.53 (.48 - .58)**	
DNR	63.8%	58.0%	**<.001**
Adjusted OR		**.81 (.74 - .90)**	
Blood transfusion	13.0%	23.0%	**<.001**
Adjusted OR		**1.9 (1.6-2.1)**	
Mechanical ventilation	31.4%	45.9%	**<.001**
Adjusted OR		**1.7 (1.5-1.8)**	
Vasopressor	7.8%	10.6%	**<.001**
Adjusted OR		**1.4 (1.2-1.6)**	

The influence of depression on esophageal cancer outcomes: Using the National Inpatient Sample (NIS) 2021.

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Background: Esophageal cancer (EC) remains a challenging malignancy with high morbidity and mortality rates. Depression is prevalent among cancer patients and has been shown to affect various clinical outcomes. This study aims to evaluate the impact of depression on mortality, length of stay (LOS), and hospitalization charges in patients with esophageal cancer using data from the NIS. **Methods:** We conducted a retrospective, population-based study using the NIS data to analyze patients diagnosed with esophageal cancer. Depression was the primary exposure variable using the appropriate ICD-10 code. We applied survey-weighted descriptive statistics, logistic regression, and linear regression models to assess outcomes, adjusting for demographic and clinical factors, including age, sex, race, comorbidities (Charlson Comorbidity Index), hospital teaching status, and regional characteristics. **Results:** A total of 40,855 patients with esophageal cancer were included, with 11.4% (n = 4,855) having a concurrent diagnosis of depression. The unadjusted mortality rate for the cohort was 9.18%, with depressed patients showing a lower mortality rate (6.45% vs. 9.53%, $p = 0.0018$). After adjusting for covariates, depression was associated with significantly lower in-hospital mortality (adjusted OR 0.63, 95% CI 0.48–0.83, $p = 0.001$). While the unadjusted mean length of stay (LOS) was slightly higher in depressed patients (7.66 vs. 7.25 days, $p = 0.132$), adjusted models showed no significant difference ($\beta = 0.42$ days, $p = 0.132$). Depression was associated with lower unadjusted total hospitalization charges ($92,321 vs. $104,156, $p = 0.023$), and in adjusted analyses, depressed patients incurred significantly lower costs (β = -$10,327, 95% CI -$19,233 to -$1,421, $p = 0.023$). Significant predictors of increased hospitalization charges included teaching hospital status (β = $34,029, $p < 0.0001$) and large hospital size (β = $38,481, $p < 0.0001$). **Conclusions:** Depression, while common among patients with esophageal cancer, was unexpectedly associated with lower in-hospital mortality, despite a longer length of stay and higher comorbidity burden. This suggests that depressed patients may receive more comprehensive or prolonged care, which could contribute to their reduced mortality risk. Additionally, despite extended hospital stays, these patients incurred significantly lower hospitalization costs, indicating possible differences in resource utilization or treatment intensity. These findings highlight the complex relationship between mental health and cancer outcomes, emphasizing the need for further research to explore the mechanisms behind these associations and to better tailor care for esophageal cancer patients with depression. Research Sponsor: None.

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Incidence trends, demographic disparities, and survival outcomes of gastrointestinal malignancies in young adults: A focus on stomach and pancreatic cancers.

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Background: The incidence of gastrointestinal (GI) cancers has been rising among younger populations. While prior research has highlighted the increasing incidence of colorectal cancer (CRC) in adolescents and young adults (AYA), defined as individuals aged 15-39, there is less focus on non-CRC GI malignancies, such as stomach and pancreatic cancers, within this age group. This study aims to further elucidate the incidence, mortality, and demographic disparities in the AYA population for stomach and pancreatic cancers compared to older populations (>50 years old). **Methods:** Data was extracted from the SEER database spanning from 2000 to 2020. Demographic, clinical, and secondary cancer characteristics were summarized across the sample, stratified by cancer type. Overall survival (OS) and disease-specific survival (DSS) were analyzed using the Kaplan-Meier method, with comparisons performed via the log-rank test. Univariate and multivariate analyses were conducted to identify predictors of survival outcomes. **Results:** We identified 19,983 AYA patients diagnosed with non-CRC GI malignancies (including esophageal, stomach, small intestine, anal, hepatobiliary, and pancreatic cancers) between 2000 and 2020. Among these patients, the majority were male (58%), White (47%), and resided in metropolitan areas (90%). The median age at diagnosis was 38 years. Pancreatic cancer was 20% more common in the AYA group compared to those over 50 years old (12%), while stomach cancer was also more prevalent in the AYA group (27% vs. 17%). Survival outcomes were significantly better in the AYA group, with a median OS of 24 months compared to 9 months in the older population, and a median DSS of 52 months versus 13 months ($p<0.001$). Predictors of worse OS and DSS included male sex, African American race, income below $45,000/year, lack of surgical intervention, and delayed initiation of treatment. **Conclusions:** The incidence of stomach and pancreatic cancers is notably higher in the AYA population compared to older adults, with AYA patients demonstrating better overall and disease-specific survival outcomes. However, certain demographic factors, such as male sex, African American race, and lower income, are associated with worse survival, underscoring the need for targeted interventions to address these disparities. Early diagnosis and timely treatment are critical to improving outcomes in this younger population. Research Sponsor: None.

Predicting mortality in upper gastrointestinal tract cancer using long short-term memory neural networks.

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Background: Upper gastrointestinal (GI) tract cancers represent a significant global health burden, characterized by high morbidity and mortality rates, with an increasing incidence among aging populations and varying outcomes based on gender and ethnicity. Early prediction of mortality rates in upper gastrointestinal cancer can significantly enhance treatment strategies and patient outcomes. This study describes current trends in upper gastrointestinal cancer and proposes a predictive model, developed using Long Short-Term Memory (LSTM) neural networks to project future trends using demographic data. **Methods:** We extracted mortality data from the Center for Disease Control and Prevention (CDC) Wide-Ranging Online Data for Epidemiological Research (WONDER) Database spanning 1999, normalized it and encoded categorical variables such as age, sex, ethnicity, and race numerically. Using Studio R to analyze mortality trends, we developed an LSTM model known for its ability to capture and utilize long term data patterns. The model, trained with a sequence length of 20, predicted future data points based on the previous 20. We then assessed its accuracy and generalization with a test set. Ultimately, our model is able to utilize demographic variables to forecast annual mortality and survival rates. **Results:** The mortality rates for upper GI tract cancers are notably higher among elderly individuals, particularly those aged 80 and above, with males and certain ethnic groups, such as Hispanics. The accuracy of the trained model was evaluated by calculating the Mean Squared Error (MSE), a measure of the difference between the predicted and actual value, with zero indicating no difference between the two. Our LSTM model achieved a training MSE of 0.00085 and a validation MSE of 0.0016, indicating good reliability and accuracy. **Conclusions:** Our analysis highlighted clear mortality trends and demonstrated the model's effectiveness in improving mortality predictions in upper gastrointestinal cancer. However, limitations due to the dataset's lack of clinical variables (such as comorbidities, cancer stage, and prior treatments) and its reliance on a single dataset may affect external validity. Future research is needed to determine if the model can be applied for clinical prognosis or strategic planning. We ultimately created and validated an LSTM neural network model with the CDC WONDER dataset, showing promising results. Future development will aim to include clinical characteristics to create a more personalized predictive tool. Research Sponsor: None.

Safety and efficacy of first-line nivolumab plus chemotherapy for HER2-negative advanced gastric cancer in real-world patients: Updated analysis.

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Background: Based on the results of CheckMate-649 and ATTRACTION-4, nivolumab plus chemotherapy is the new standard of care as first-line treatment for patients with HER2-negative advanced gastric cancer (AGC) regardless of PD-L1 combined positive score (CPS) status in Japan. This study aimed to evaluate the safety and efficacy of this regimen in real-world patients with HER2-negative AGC with a longer follow-up. **Methods:** In this single-institutional retrospective study, patients with HER2-negative AGC who were treated with first-line nivolumab plus chemotherapy between September 2021 and April 2024 were consecutively enrolled and evaluated. Early tumor shrinkage (ETS) was defined as ≥20% tumor reduction at 8 ± 2 weeks. **Results:** Of 151 patients, median age was 66 years (range, 27–83); 58% were male; 81% had diffuse-type histology; the peritoneal metastasis was seen in 60%; PD-L1 CPS of <1, 1≤/<5, 5≤ was 30%/41%/29%; deficient mismatch repair and/or high microsatellite instability was 7%. At a median follow-up of 14.2 months, the median overall survival (OS) and progression-free survival (PFS) were 21.7 months (95% confidence interval [CI] 14.4–26.2) and 7.7 months (6.4–9.6), respectively. The median OS according to CPS subgroups (<1, 1≤/<5, 5≤) were 21.7 / not reached (NR) /25.0 months, respectively (log-rank P = 0.55). In patients with measurable lesions at baseline according to RECIST ver 1.1, the objective response rate and disease control rate were 62% and 87%, respectively, with a complete response rate of 10%. In the exploratory analysis by ETS, both median PFS and OS were significantly longer in the ETS group compared to the non-ETS group (PFS, NR vs. 6.2 months; hazard ratio [HR] 0.35; 95% CI 0.17–0.68; P = 0.001) (OS, 25.0 vs. 11.7 months; HR 0.44; 95% CI 0.20–0.93; P = 0.03). The immune-related adverse events (irAEs) of any grade were observed in 26% (grade 3–4 irAEs 12%), whereas no treatment-related death was documented. **Conclusions:** The updated analysis suggests that first-line nivolumab plus chemotherapy provides benefit in real-world patients with HER2-negative AGC regardless of PD-L1 status, and suggests that the magnitude of ETS may predict a significant impact on the duration of survival. Research Sponsor: None.

Real world data and biomarker analysis of the efficacy and safety of PD-1 inhibitor combined with chemotherapy in first-line treatment of advanced gastric cancer.

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Background: With the publication of several clinical research results, the efficacy of ICI combined with chemotherapy has gradually been recognized in HER-2 negative advanced GC field. We analyzed the efficacy of 1st-line PD-1 inhibitor combined with chemotherapy in advanced GC patients using real-world data and conducted exploratory analysis. We aim to identify factors that can predict treatment efficacy, and provide a basis for clinical practice. **Methods:** We prospectively collected data from GC patients between November 2020 and April 2024. The clinical staging was stage IV. All patients were HER2 negative and pMMR. All patients received SOX regimen chemotherapy combined with PD-1 inhibitor. Blood samples were collected for exploratory analysis. **Results:** A total of 190 GC patients with histological diagnosis of adenocarcinoma were included from November 1, 2020 to April 1, 2024. As of June 1, 2024, the median follow-up time was 11.75 (95% CI 10.65-13.00) months. Among them, 174 had measurable lesions. 94 patients achieved PR, and 55 patients achieved SD. The ORR and DCR were 61.9% (95% CI 48.8-73.4%) and 96.8% (95% CI 89.0-99.6%), respectively. The survival analysis results indicated that the mPFS and mOS were 9 (95% CI 7.9-14.2) and 27 (95% CI 20- not reached) months. We conducted risk factor analysis on patients' age, gender, CPS, HER2 expression, EGFR expression and EBER status. Age, gender, CPS, EBER status and EGFR expression are not risk factors affecting survival. Specifically, the expression level of HER2 was significantly correlated with therapeutic efficacy. The results indicated that the HER2 expression level was significantly correlated with ORR. In terms of survival analysis, the PFS of treated patients with HER2 '2+' was significantly longer than that of treated patients with HER2 '1+', and there was a trend of prolonged OS, but the statistical difference was not reached. We collected baseline blood samples from 62 patients and performed ctDNA testing. In terms of PFS, the patients with SNV in SMARCA4 and RASA1 was significantly shorter than that of wild-type patients with SMARCA4 ($p = 0.02$) and RASA1 ($p = 0.043$). The patients with CNV in EXT1 was significantly shorter than those with wild-type in EXT1 ($p = 0.00046$). In terms of OS, patients with SNV in BRCA2, TP53, and STK11 had significantly shorter OS than those with wild-type BRCA2 ($p = 0.033$), TP53 ($p = 0.048$), and STK11 ($p = 0.0083$). The patients with CNV in MYC and EXT1 had shorter OS than those with wild-type MYC ($p < 0.0001$) and EXT1 ($p < 0.0001$) patients. **Conclusions:** Real-world data showed that 1st-line treatment with PD-1 inhibitor combined with SOX had shown good efficacy in advanced GC patients. The HER2 expression level in HER2 negative GC may become a direction for further exploration. The EXT1 mutation can serve as a potential predictor, and further investigation is warranted. Clinical trial information: NCT06034964. Research Sponsor: CAMS Innovation Fund for Medical Sciences (CIFMS); 2022-I2M-C&T-B-066.

Retreatment with immune checkpoint inhibitors (ICIs) for patients with advanced gastric cancer: A retrospective, real-world study.

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Background: Immunotherapy combined with chemotherapy has become the standard first-line treatment for advanced gastric cancer and is recommended by multiple guidelines and consensuses. After the first line immunotherapy, there is still controversy over whether immune inhibitors can continue to be used across lines after progression. **Methods:** This is a single-center, retrospective, observational, real-world study. We collected clinical information of patients with advanced gastric cancer hospitalized at the First Affiliated Hospital of Zhengzhou University from January 2018 to July 2024. All patients received combination therapy with ICIs inhibitors for at least 2 cycles in the first-line and at least 2 cycles of ICI inhibitor-based therapy in the second-line after first-line progression. The efficacy and safety of treatment were evaluated, including overall survival (OS), progression-free survival (PFS), objective response rate (ORR), disease control rate (DCR), the progression pattern of retreatment with immune checkpoint inhibitor, grade 3-5 treatment-related adverse events (TRAEs), and immune-related adverse events (irAEs). Cox regression model was used to investigate the influence of multiple factors on survival. **Results:** We included 217 patients with a median age of 59.0 (18-84) years, of which 70.0% were male. The median OS and PFS were 9.8 (95% CI 8.8-10.8) months and 4.6 (95% CI 4.0-5.3) months. The ORR and DCR were 22.7% (95% CI 17.1-29.0) and 56.2% (95% CI 49.0-63.1). For the second-line treatment regimens, there were 87 (40.1%) patients and 130 (59.9%) patients received immunotherapy combined with chemotherapy (I+C) and immunotherapy combined with chemotherapy and anti-angiogenic drugs(I+C+A) respectively. The median OS was 9.3 (95% CI 8.5-10.1) months and 11.5 (95% CI 9.3-13.7) months, p=0.032. The median PFS was 4.2 (95% CI 3.1-5.3) months and 5.2 (95% CI 4.3-6.1) months, p=0.082. Cox regression model analysis showed that ECOG 0-1, PFS≥6 months in the first-line treatment, second-line regimen (I+C+A), and second-line medication > 3 cycles were independent prognostic factors for OS in the second-line treatment. The incidence of grade ≥3 TRAEs was 20.7%, and the incidence of irAEs was 6.5%. No treatment-related deaths occurred. **Conclusions:** This real-world study results showed that retreatment with immune checkpoint inhibitors had promising clinical benefit for patients with advanced gastric cancer and manageable safety. Research Sponsor: None.

Efficacy of different groups.

	n	mOS (m)	mPFS (m)	ORR (%)	DCR (%)
ITT	217	9.8	4.6	22.7	56.2
ECOG					
0-1	180	9.8	4.7	19.5	56.8
2	37	8.6	4.0	38.2	52.9
1-Line PFS					
< 3m	61	5.5	2.2	1.7	17.2
≥3m	156	11.7	6.0	31.0	71.7
< 6m	136	7.4	3.2	2.4	38.7
≥6m	81	15.1	8.0	54.4	83.5
Second-line treatment					
I+C	87	9.3	4.2	16.9	54.2
I+C+A	130	11.5	5.2	26.7	57.5
Second-line treatment cycle					
≤3	97	7.4	2.9	4.8	31.0
> 3	120	13.1	6.2	35.3	73.9

Nimotuzumab combined with chemotherapy plus immunotherapy as first-line treatment for advanced esophageal squamous cell carcinoma.

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Background: Esophageal squamous cell carcinoma (ESCC) is the primary type of esophageal cancer with poor prognosis in China. Chemotherapy plus immunotherapy as first-line standard treatment has limited benefit in advanced ESCC patients. Nimotuzumab, a humanized anti-EGFR antibody, exhibited favorable efficacy in advanced ESCC with good safety profile. We explored the efficacy and safety of nimotuzumab as first-line therapy for patients with advanced ESCC. **Methods:** In this retrospective study, patients were aged ≥ 18 years with histologically diagnosed advanced ESCC, ECOG of 0-3, given nimotuzumab (400 mg, Q3W) combined with chemotherapy (paclitaxel: 300-400mg and platinum: 400-500mg or tegafur: 60mg/bid, Q3W) plus immunotherapy (PD-(L)1: 200-240 mg, Q3W). They assigned into two groups: metastasis group and locally advanced group. The primary endpoint was overall survival (OS). Secondary endpoints were progression-free survival (PFS), objective response rate (ORR) and safety. **Results:** As of July 2024, 58 patients were enrolled (metastasis group: n=20, locally advanced group: n=38). The median age was 59.5 years (range: 44-80). The median follow-up was 32.59 months (95% CI: 23.0, 36.7). The ORR was 74.1% (95% CI: 0.6, 0.8). Survival analysis showed that the 1-year OS was 79.16% (95% CI: 64.5, 88.3) and 2-year OS was 57.38% (95% CI: 41.3, 70.5), 1-year PFS and 2-year PFS was 64.41% (95% CI: 49.4, 76.0) and 32.80% (95% CI: 19.7, 46.5), respectively. The median OS (mOS) was 31.01months, the median PFS (mPFS) was 17.84 months. Nimotuzumab plus chemotherapy and immunotherapy displayed a favorable survival outcomes both in metastasis group and locally advanced group with OS (1-year OS: 77.76% and 80.37%; 2-year OS: 62.21% and 55.91%) and PFS (1-year PFS: 53.46% and 69.99%; 2-year PFS: 25.06% and 36.66%), the mOS was not reached and 31.01 months, mPFS was 18.27 months and 16.69 months, respectively. There were 74.1% patients occurred grade 1-2 adverse events (AEs), and 13.8% occurred grade 3-5 AEs. The most common AEs included nausea (60.3%), myelosuppression (51.7%), leukopenia (41.4%), thrombocytopenia (29.3%), and neutropenia (24.1%). No serious treatment-related AEs or death recorded. **Conclusions:** This study demonstrated a favorable survival profile and acceptable toxicity in first-line treatment of advanced ESCC patients with nimotuzumab regimen. Further prospective study needed to certify in order to benefit more patients. Research Sponsor: None.

Real-world treatment patterns and outcomes for patients with neoadjuvant treatment between 2019-2022 for gastric or gastroesophageal junction cancer (GC/GEJC) in the US community setting.

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Background: Treatment options for patients with early-stage GC/GEJC are still limited and the prognosis remains poor, despite recent therapy approvals. With the latest adoption of FLOT use in the US, it is important to understand patients' characteristics, treatment patterns and outcomes in the US community oncology setting. **Methods:** This retrospective observational cohort study used electronic health records data from 278 adult patients with stage II-IVA GC/GEJC adenocarcinoma initiating neoadjuvant treatment (index) within The US Oncology Network between 1/1/2019 and 12/31/2022, based on Food and Drug Administration approval of FLOT in 2019. Patients were followed through 9/30/2023, last patient record or death, whichever occurred first. Descriptive analyses were conducted to evaluate patient and treatment characteristics. Kaplan-Meier was used to assess overall survival (OS) and event-free survival (EFS) from index. Adjusted Cox proportional hazard models were used to analyze the association of surgery as a time-dependent variable with OS and EFS. **Results:** Median (interquartile range [IQR]) age at index was 67.5 (58.2, 74.2) years; most patients were male (68.0%), and half were Caucasian (51.1%). Stage III was the most common initial stage (50.0%), followed by IIB (29.1%), and II (not otherwise specified)/IIA (19.8%). Most patients had an ECOG (0-1) 65.5%. Median (IQR) follow-up from index was 15.6 (8.6, 27.8) months. The most common neoadjuvant regimens were FLOT (51.1%), doublet (28.4%), and CROSS (18.0%). Median OS (months) (95% confidence interval [CI]) for patients with surgery was not reached (NR) (47.7, NR) and those without was 16.1 (8.6, 35.1). Patients with surgery had a longer median EFS (95% CI) (months) than those without 38.3 (25.0, NR) vs 5.6 (4.5, 6.5) (log rank p-value <0.0001). Adjusted hazard ratios (95% CI) for OS and EFS indicated that patients who received surgery were less likely to experience an event than those without, 0.252 (0.139, 0.456) and 0.268 (0.169, 0.426), respectively ($p<0.0001$). **Conclusions:** Study findings showed a high proportion of patients receiving FLOT since its approval in US, with a directional OS benefit similar to the FLOT4 trial. Also, receipt of surgery was a strong predictor for OS and EFS. While the results are promising, most patients who did not receive surgery progressed within 6 months and died a little over a year from index. However, given the small portion of patients who did not receive surgery, further research is needed to understand if there is an unmet need for this group. Research Sponsor: None.

Variable	Doublet	CROSS	FLOT
N	79	50	142
Surgery Rate (%)	60.8%	62.0%	71.8%
GC Rate (%)	79.8%	<20%	85.9%
OS, median (95% CI), months	26.0 (20.9, NR)	37.7 (27.8, NR)	52.1 (47.7, NR)
EFS, median (95% CI), months	10.8 (8.4, 23.2)	20.9 (11.3, 26.3)	37.7 (18.0, NR)

Prognostic analysis of pathological complete response after neoadjuvant therapy for locally advanced gastric cancer: A national, multicenter, retrospective study.

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Background: Several research findings indicate that some patients with locally advanced gastric cancer (LAGC) who undergo neoadjuvant therapy followed by curative surgery can achieve pathological complete response (pCR). Whether pCR is equivalent to cure, serves as a good prognostic indicator for long-term survival, and whether patients achieving pCR require subsequent adjuvant therapy are still unknown. Therefore, we conducted a nationwide multicenter retrospective study to investigate the long-term efficacy of patients with LAGC who achieved pCR after neoadjuvant therapy. **Methods:** Clinical and pathological data were collected from 351 patients with locally advanced gastric cancer who achieved pCR after neoadjuvant therapy between January 2018 and October 2023 at 14 medical centers in China. These patients were matched 1:1 with non-pCR patients based on age, gender, clinical T stage, and N stage. This study evaluated general clinical data, neoadjuvant treatment regimens, cycles, surgical outcomes, and follow-up information including postoperative adjuvant therapy, recurrence, metastasis, and survival status up to November 2023. **Results:** Negative Serum tumor markers, non-signet ring cell carcinoma and neoadjuvant treatment regimens were closely associated with pCR. Patients achieving pCR were more likely to reach ypN0 status after neoadjuvant therapy compared to non-pCR patients. Survival analysis showed significantly higher overall survival (OS) and disease-free survival (DFS) in pCR patients compared to matched non-pCR patients. Moreover, OS and DFS were significantly higher in ypN0 patients within the pCR group compared to non-pCR patients. However, there was no significant difference in OS and DFS between ypN+ patients in the pCR group and non-pCR patients. Additionally, postoperative adjuvant chemotherapy did not significantly impact OS and DFS in the pCR group. **Conclusions:** pCR is an independent prognostic factor for OS and DFS. However, survival benefits from pCR are limited to patients achieving ypT0N0 status, while ypT0N+ patients do not benefit from pCR. Furthermore, adjuvant chemotherapy may not improve the prognosis of patients achieving pCR after neoadjuvant therapy. Research Sponsor: None.

Immune checkpoint inhibitors (ICIs) in deficient mismatch repair (dMMR)/ microsatellite instability (MSI) non-colorectal cancers: Real world Indian data and the use of alternative dosing.

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Background: Immune checkpoint inhibitors, predominantly pembrolizumab and dostarlimab, have been approved in deficient mismatch repair (dMMR)/ microsatellite instability (MSI) non-colorectal cancers. There is also an emerging data to suggest significant clinical activity for low-dose ICIs (LD-ICIs) in these cancers. **Methods:** A multi-institutional retrospective analysis in Indian patients with dMMR/MSI-H advanced/metastatic non-colorectal carcinomas treated with ICIs (irrespective of dose or schedule) between 2017 and 2024 was conducted. The primary objective of the study was to evaluate 12-month Overall survival (OS) for the whole cohort as well as LD-ICI and standard dose ICI (SD-ICI) cohorts, while other secondary objectives included assessment of overall response rates (ORR) (assessed by RECIST v1.1 as per local radiological reporting), 12-month progression free survival (PFS) and incidence of immune related adverse events (IRAEs). **Results:** A total of 51 patients were available for analysis. The most common tumour types were gastrointestinal (64%, commonly gastric cancer, oesophageal cancer and cholangiocarcinoma) and endometrial cancers (23%). Eighty-four percent of patients received ICI monotherapy, while the remaining 16% received varying combinations of ICIs with chemotherapy and tyrosine kinase inhibitors. Nivolumab was used in 24 patients (47%) while pembrolizumab was used in 17 patients (33%). With a median follow-up of 14 months, 12-month OS was 72% and 12-month PFS was 62% in the entire cohort. The corresponding 12-months OS and 12-month PFS for those receiving SD-ICIs (n=29) was 69% and 62%, while it was 72% and 64%, respectively for the LD-ICIs cohort (n=22). The ORR was 61%, with 10 patients (20%) achieving complete response (CR) and 21 (41%) achieving partial response (PR). The most common IRAEs (all grades) were fatigue (62%), thyroid disturbances (37%), and pruritus (23%). **Conclusions:** The current study from India highlights the importance of treating non-colorectal dMMR/MSI-H cancers with ICIs and shows outcomes commensurate with published trial data. Within the confines of a small retrospective study with a short follow-up, low dose ICIs should be explored in these patients in scenarios where standard dosing schedules are not feasible due to logistic reasons. Key Words – Low dose immunotherapy; non-colorectal dMMR/MSI-H cancers; Overall survival. Research Sponsor: None.

Real-world biomarker testing and first-line treatment patterns among locally advanced, unresectable or metastatic gastric and gastroesophageal junction adenocarcinoma patients in the US.

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Background: Advanced-stage gastric cancer and gastroesophageal junction adenocarcinoma (advG/GEJC) are leading causes of cancer-related morbidity and mortality. Chemotherapy is the traditional first-line (1L) therapy for patients in the advanced setting, with targeted and immunotherapy considered based on biomarker expression. This study aimed to describe real-world HER2 and PD-L1 testing practices and 1L treatment patterns among advG/GEJC patients. **Methods:** A retrospective cohort study was conducted among adult patients diagnosed with advG/GEJC between 01 Jan 2017 and 30 Jun 2023 usingthe US-based Flatiron Health electronic health record-derived de-identified database. HER2 and PD-L1 testing practices were evaluated prior to and up to 1-month post-1L treatment initiation or 6 months post-advG/GEJC diagnosis date in untreated patients. 1L treatment regimens were described overall and stratified by HER2 and PD-L1 expression. **Results:** Among 4,256 advG/GEJC patients, the average age was 67 years with the majority male (68%), non-Hispanic white (44%), and with metastatic disease (64%). Nearly three quarters (72%) of patients included received a HER2 test and 55% had a HER2 test with a valid result within the timeframe mentioned above. More than a quarter of patients (27%) received a PD-L1 test and 25% had a PD-L1 test with a valid result. PD-L1 testing before the approval of nivolumab in 1L (i.e. before 2021) showed testing at 19% (490/2,629; 17% with a valid result) and from 2021 onward at 40% (643/1,627; 37% with a valid result). Across the full cohort, half of the patients received chemotherapy alone (52%) and one quarter received no treatment (25%). Similar treatment patterns were observed in the subset of 1,117 patients who did not receive a HER2 or PD-L1 test; 54% received chemotherapy alone and 36% no treatment. Of the 340 patients who were tested for HER2 and were HER2 positive, half (52%) received trastuzumab combination therapy, 23% received chemotherapy alone, and 18% had no treatment. Of the 301 patients who received a PD-L1 test with a combined positive score (CPS) $\geq$5 from 2021 onward, 41% received immunotherapy, 34% received chemotherapy alone and 18% had no treatment. Prior to 2021, most patients with a PD-L1 test and CPS $>$5 used chemotherapy alone (57%; 132/233), 13% (31/233) received PD-L1 targeted therapies and 19% (43/233) received no treatment. **Conclusions:** This study describes a potential subset of US advG/GEJC patients who do not receive a HER2 or PD-L1 test and/or do not receive eligible targeted or immunotherapy as part of routine clinical practice. As precision medicine options continue to evolve, these data suggest an opportunity to further understand the drivers for biomarker testing as well as the factors impacting the use of biomarker-informed treatment options. Research Sponsor: Amgen.

The association of antibiotic use and survival among patients receiving first line immunotherapy-based treatment for advanced gastroesophageal cancers.

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Background: The advent of immune checkpoint inhibition (ICI) has improved the survival for patients with gastroesophageal cancer (GEC). It is vital to understand the relationship between the tumor microenvironment and the ability of the immune system to prevent and eliminate cancer cells. Antibiotics (abx), frequently used in patients on cancer-directed therapy, can alter gut microbiota leading to dysbiosis and altered immune responses. Some research shows its use is associated with decreased ICI efficacy in RCC, NSCLC, and melanoma. The goal of our study is to investigate the association of abx timing and survival with 1L ICI-based treatments for advanced GEC. **Methods:** The retrospective, multicenter analysis used the Flatiron Health deidentified electronic health record-derived database of real world (rw) patients with advanced esophageal, GEJ, and gastric cancers (EC/GEJC/GC) who received ICI +/- chemotherapy (chemo) as 1L therapy. Abx exposure was defined by antibiotic use from 42 days before to 42 days after starting ICI. The cox proportion hazards models were used to compare rw overall survival (OS) and progression-free survival (PFS) across exposure groups. **Results:** A total of 990 patients were included for the main group analysis with comparison of survival and abx usage and timing: 876 did not receive abx, 58 received abx within 42 days after starting ICI, and 56 received abx within 42 days prior to starting ICI. The median age was 66 with 44.4% EC, 33.5% GC, and 22% GEJC. The majority of patients had adenocarcinoma (87.9%). The unadjusted OS HR for abx (within 42 days) before ICI was 0.99 (0.68-1.46, p=0.971) and (within 42 days) after ICI was 1.56 (1.11-2.19, p=0.011), suggesting worse OS with abx use after ICI initiation. HR trended towards worse PFS for those receiving abx after 1L ICI [1.36 (0.97-1.90 p= 0.073)]. In subgroup analysis, the negative effect of starting abx after 1L ICI over abx before ICI was more prominent in those receiving ICI monotherapy compared to those receiving chemo/ICI. **Conclusions:** Our study identifies a poorer survival association of abx administration after 1L ICI-based therapy for advanced GEC, primarily driven by ICI monotherapy. Research Sponsor: None.

Main group comparison between abx usage and timing: OS	Abx Use/Timing	n (%)	HR
	No	876 (88.5)	
	Yes, after 1L IO	58 (5.9)	1.56 (1.11-2.19, p=0.011)
	Yes, before 1L IO	56 (5.7)	0.99 (0.68-1.46, p=0.971)
Subgroup comparison between abx usage and timing by treatment type			
Chemo + IO: OS	Yes, after 1L IO	44 (6.9)	1.32 (0.88-1.98,p=0.177)
	Yes, before 1L IO	38 (6.0)	1.01 (0.62-1.65, p=0.964)
IO monotherapy: OS	Yes, after 1L IO	12 (4.9)	2.64 (1.38-5.06, p=0.003)
	Yes, before 1L IO	12 (4.9)	1.09 (0.53-2.23, p=0.810)

Prognostic and biological divergence of upper gastrointestinal adenocarcinomas.

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Background: Esophageal (EAC), gastroesophageal junction (GEJ) or gastric (GAC) adenocarcinoma share similar etiologies and treatment approaches. However, it is not clear if they possess similar or different prognostic and biological characteristics. We aimed to examine overall survival (OS) in associations of the common genomic alterations with these three anatomically closely linked adenocarcinomas. **Methods:** Eligible patients had recurrent or de novo metastatic GAC, GEJAC or EAC whose tumors had undergone next generation sequencing (NGS) performed from November 2017 to December 2023. We used Cox regression modeling to examine the association between GAC, GEJAC, EAC and OS, adjusting for demographics, performance status, Charlson comorbidity index, receipt of chemotherapy, and HER2 overexpression or amplification (HER2 positive), *p53* (mutp53), *KRAS* (mutKRAS), *CDKN2A*, *PIK3CA* co-mutations and *MYC* amplification. **Results:** Of 875 total eligible patients, 173 had EAC, 276 had GEJAC and 426 had GAC. GEJAC had substantially better OS than EAC (HR = 0.68, [95% CI, 0.54-0.86]), and modestly better OS than GAC (HR = 0.85, [95% CI, 0.67-1.09]). HER2 positivity was associated with substantially better OS among EAC (HR = 0.62; [95% CI, 0.40-0.96]) and GEJAC (HR = 0.59; [95% CI, 0.38-0.87]) but not among GAC (HR = 0.98; [95% CI, 0.78-1.23]) patients. In addition, p53 gain-of-function versus non-gain-of-function mutations were associated with substantially worse OS among GAC (HR = 1.41; [95% CI, 0.98-2.01]) but not among EAC or GEJAC. MutKRAS was associated with substantially worse OS among EAC (HR = 1.81; [95% CI, 1.10-2.99]) but not among GEJAC or GAC. Surprisingly, *MYC* amplification was associated with dramatically better OS among EAC (HR = 0.19; [95% CI, 0.08-0.42]) but substantially worse OS among GEJAC (HR = 1.98, 95% CI, 1.14-3.40]) and GAC (HR = 1.75; [95% CI, 1.10-2.80]). **Conclusions:** GEJAC, EAC and GAC possess substantially different OS that appears differentially associated with HER2 status, mutp53, mutKRAS and *MYC* amplification. These results suggest distinct prognostic and biological characteristics of these three anatomically closely linked adenocarcinomas that could have important implications in clinical practice and on further investigations on their biological and topological mechanisms. Research Sponsor: None.

Results of a multicenter retrospective cohort study evaluating the safety and efficacy of FTD/TPI plus ramucirumab in advanced gastric cancer (HGCSG2302).

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Background: In advanced gastric cancer, trifluridine/tipiracil (FTD/TPI) is one of the standard treatments for later-line chemotherapy. Ramucirumab (RAM), an anti-VEGFR-2 monoclonal antibody, has shown promising efficacy in combination with FTD/TPI in phase II studies conducted in Japan, suggesting that this combination could improve treatment outcomes. However, there remains a paucity of real-world clinical data comparing the efficacy and safety of FTD/TPI monotherapy and FTD/TPI plus RAM combination therapy in general clinical practice. **Methods:** This retrospective analysis to evaluate safety and efficacy included patients with gastric cancer who were received FTD/TPI plus RAM combination therapy or FTD/TPI monotherapy between August 2019 and March 2023 at 21 institutions. **Results:** 164 patients were analyzed, 38 patients in the RAM combination group and 126 in the FTD/TPI monotherapy group. The distribution of treatment lines (third/fourth/fifth) was 10/13/15 in the RAM group and 26/54/46 in the monotherapy group, respectively. RAM had been previously administered to 34 patients (89.5%) in the RAM group. Objective response rate and disease control rate were 11.1% / 55.6% (OR 2.0, 95% C.I. 0.84-4.77, p=0.126) in the RAM group and 5.5% /38.5% (OR 2.0, 95% C.I. 0.84-4.77, p=0.126) in the monotherapy group, respectively. Median progression-free survival (PFS) was 3.5 months in the RAM group and 2.1 months in the monotherapy group (HR 0.69, 95% C.I. 0.49-1.02, p=0.06), respectively. Median overall survival (OS) was 8.4 months in the RAM group and 5.8 months in the monotherapy group (HR 0.82, 95% C.I. 0.54-1.25, p=0.350), respectively. Grade 3 neutropenia was reported in 48.6%/52.4% (p=0.707), and there were no significant differences in other adverse event profiles between the two groups. Additionally, in the RAM group, a comparison between patients receiving standard dosing (2 weeks on/2 weeks off; n=7) and those on biweekly dosing (1 week on/1 week off; n=24) showed no notable differences in efficacy or safety. **Conclusions:** FTD/TPI plus RAM combination therapy showed a trend toward prolonged PFS compared to FTD/TPI monotherapy, but no significant difference in OS. Research Sponsor: None.

Impact of prior nivolumab use on the efficacy of second-line taxane-based chemotherapy in advanced gastric or esophagogastric junction adenocarcinoma.

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Background: Taxane-based chemotherapy remains the standard second-line treatment after first-line treatment including anti-PD-1 antibody therapy for patients with advanced gastric or esophagogastric junction adenocarcinoma (GA/EGA). The CheckMate 649 trial suggested that prior use of nivolumab in the first-line setting was associated with longer progression-free survival (PFS) with subsequent treatment. However, there have been few reports on the difference in efficacy of taxane-based chemotherapy as second-line treatment with or without prior nivolumab use in the real-world setting. **Methods:** We retrospectively reviewed the medical records of patients with advanced GA/EGA who received solvent-based paclitaxel or nanoparticle albumin–bound (nab)-paclitaxel chemotherapy regimens as second-line treatment after first-line platinum-based chemotherapy between January 2018 and August 2024. We divided the patients into two groups: those who received nivolumab with the first-line chemotherapy (Nivo group) and those who did not (non-Nivo group). PFS and overall survival (OS) were estimated using the Kaplan–Meier method and compared between the two groups using the log-rank test. Hazard ratios (HRs) and 95% confidence intervals (95%CIs) were estimated by univariate and multivariate Cox regression analysis. **Results:** A total of 115 patients were included: 27 patients in the Nivo group and 88 in the non-Nivo group. Patient backgrounds did not differ significantly in terms of age (median 67/67 years), sex (male 59%/57%), performance status (PS; ≥2 11%/15%), site of primary tumor (EGA 11%/9%), number of metastatic sites (≥3 33%/18%), and ramucirumab use (yes 82%/82%). For the taxane regimens, nab-paclitaxel-based treatment was significantly more common in the non-Nivo group (11.1% /55.7%, $p<0.001$). There was a tendency toward longer PFS in the Nivo group compared with the non-Nivo group (median 4.2 months [95%CI 2.8-6.0] vs. 2.3 months [95%CI 2.0-3.0], HR 0.65 [95%CI 0.40-1.03]; p=0.067.) The objective response rate was numerically higher in the Nivo group compared with the non-Nivo group (29.4% vs. 17.6%, p=0.315). OS was not significantly different between the Nivo and non-Nivo groups (median 9.6 months [95%CI 4.9-NA] vs. 6.6 months [95%CI 5.5-10.8], HR 0.92 [95%CI 0.52-1.64]; p=0.785). In the non-Nivo group, 64 patients (73%) received subsequent chemotherapy, and 63 patients (72%) received nivolumab. Multivariate analysis identified prior use of nivolumab was a favorable prognostic factor for PFS (HR 0.55 [95%CI 0.31-0.97]; p=0.039). However, prior use of nivolumab did not show a significant benefit for OS (HR 1.09 [95% CI: 0.56-2.13]; p = 0.802). **Conclusions:** Prior use of nivolumab in first-line platinum-based chemotherapy had a positive impact on PFS with taxane-based second-line chemotherapy in advanced GA/EGA. Research Sponsor: None.

Real-world biomarker testing patterns, first line treatment and drivers of treatment choice in patients with advanced/metastatic gastric/gastroesophageal junction adenocarcinoma in the United States and Europe.

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Background: For first line (1L) treatment (tx) in HER2- gastric/gastroesophageal junction (G/GEJ) adenocarcinoma (AC), clinical guidelines recommended platinum-fluoropyrimidine (PF) chemotherapy (CT) in patients (pts) with a PD-L1 combined positive score (CPS) <5 and PF CT in combination with immunotherapy (IO) for CPS ≥5. There is little contemporary evidence into IO tx uptake in the real world. This real world study examined biomarker testing, 1L tx characteristics, and reasons for 1L tx choice in pts with HER2- G/GEJ AC to identify unmet needs and opportunities for optimal clinical practice. **Methods:** Data were derived from the Adelphi G/GEJ Cancer Disease Specific Programme: a systematic approach involving multi-national, multi-faceted data sources including cross-sectional surveys capturing data from physicians and their consulting pts. Physicians reported pt demographics, clinical characteristics, tx received and reasons for tx decisions, via chart review. Data were collected across US and Europe (EU: France, Germany, Spain, the UK) between Oct 2022 to Apr 2023. All pts were alive at data collection and analyses were descriptive. **Results:** Overall, 243 treating physicians (US n=60/EU n=183) provided data on 642 HER2- pts (US n=83/EU n=559) receiving 1L tx at data collection. For pts from US/EU, median age was 66/67 years, 84%/79% had an ECOG of 0-1 and 52%/75% had metastatic disease at data collection. In the US/EU, 87%/80% of pts were tested for PD-L1 status respectively, of which 63%/69% had a known CPS. For US/EU, 67%/45% of pts (n=30/n=139) had a CPS ≥5. Differences in biomarker testing rates were observed in US v EU pts respectively for MSI/dMMR (66% v 58%), NTRK-gene fusion (31% v 4%) and FGFR2 (27% v 3%). Of US/EU physicians, 70%/85% (n=60/n=183), reported the main reason for conducting biomarker testing was 'to inform tx decisions'. Cost related concern (50%/39%) and time constraints (23%/15%) were the main perceived barriers to testing. Among pts with PD-L1 CPS <5, 27% of US pts and 2% of EU pts received IO + CT; in pts with CPS ≥5, 63% of US pts and 76% of EU pts received IO + CT. Whilst progression free survival/overall survival benefit were the most common reasons selected by physicians for 1L tx choice across the US/EU (75%/74% and 60%/61%), differences across regions were reported for tx choice based on maintaining and improving pt's QoL (4%/28%) and pt's performance status (2%/24%). **Conclusions:** PD-L1 testing was not performed by 19% of physicians and approximately one third of pts with CPS ≥5 across the US/EU were not receiving IO + CT in 1L. As informing 1L tx choice was a key driver for biomarker testing, there may be scope for improvement in 1L tx by increasing testing rates. Further studies are needed to address barriers to biomarker testing and uptake of IO in pts who can benefit from 1L IO tx. Research Sponsor: Gilead Sciences.

Q-TWiST analysis of pembrolizumab plus trastuzumab and chemotherapy for patients with metastatic HER2-positive gastric or gastroesophageal junction (G/GEJ) adenocarcinoma in the KEYNOTE-811 trial.

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Background: In the phase 3 KEYNOTE-811 trial (NCT03615326), first-line pembrolizumab plus trastuzumab and chemotherapy (pembrolizumab group) statistically significantly improved PFS and OS versus placebo plus trastuzumab and chemotherapy (placebo group), with manageable safety, in patients with metastatic HER2-positive G/GEJ adenocarcinoma, while maintaining HRQoL, specifically in patients with PD-L1 combined positive score (CPS) ≥1 tumors. The objective of this analysis was to assess quality-adjusted time without symptoms of disease progression or toxicity (Q-TWiST) for patients treated in the KEYNOTE-811 trial. **Methods:** Q-TWiST is a measure of survival weighted by health states utility values. The analysis categorized survival time into 3 health states: time with grade ≥3 adverse events (toxicity [TOX]) and time without symptoms or toxicities (TWiST)—both before disease progression or death—and relapse (REL), defined as time from disease progression to death. Q-TWiST was calculated as the restricted mean survival time spent in each state, with a weighted health state utility for that state. Average utility weights were derived based on pooled EQ-5D-5L scores using the US mapping algorithm and standardized base case utility weights. Relative gains in Q-TWiST of ≥10% and ≥15% were "clinically important" and "clearly clinically important," respectively, as reported in the literature. Treatment difference 95% CIs were generated using the non-parametric bootstrapping method. Data are reported for patients with PD-L1 CPS ≥1 tumors. The data cutoff date was March 20, 2024. **Results:** At the maximum follow-up of 63 months, patients in the pembrolizumab group had a 2.36-month (95% CI, −0.24 to 4.56) longer restricted mean time in TOX (6.98 vs 4.63 months), a 3.40-month (95% CI, −0.44 to 7.45) longer restricted mean time in TWiST (12.03 vs 8.62 months), and a 1.45-month (95% CI, −5.84 to 3.09) shorter restricted mean time in REL (8.87 vs 10.32 months) versus the placebo group. For the restricted mean Q-TWIST at month 63, based on the US mapping algorithm, the difference between the pembrolizumab and placebo groups favored the pembrolizumab group by 3.90 months (95% CI, 1.30 to 6.72), representing a 16.57% relative Q-TWiST gain. Based on the standardized base case utility weights, the difference also favored the pembrolizumab group by 3.86 months (95% CI, 1.47 to 6.67), representing a 16.38% relative Q-TWiST gain. **Conclusions:** Pembrolizumab plus trastuzumab and chemotherapy produced a clearly clinically important improvement in quality-adjusted survival time based on Q-TWiST analyses compared with placebo plus trastuzumab and chemotherapy in patients with previously untreated metastatic HER2-positive G/GEJ adenocarcinoma with PD-L1 CPS ≥1. Clinical trial information: NCT03615326. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

Focusing on clinical trial ineligibility: Nivolumab plus chemotherapy for patients with advanced gastric cancer.

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Background: Nivolumab demonstrates promising efficacy when combined with chemotherapy as first-line treatment in patients with advanced gastric cancer (AGC). However, a large proportion of patients are perceived as ineligible for clinical trials in clinical practice. The aim of this study is to analyze the treatment outcomes and safety among patients deemed ineligible in clinical trials for AGC. **Methods:** A retrospective examination was conducted involving 187 patients who received treatment for HER2-negative unresectable or recurrent AGC between December 2014 and October 2023 at six institutions. Eleven patients were excluded due to massive ascites or poor performance status (PS). Ineligible patients were defined as meeting any of the seven criteria: age 75 years or over, PS 2, bone marrow dysfunction, hepatic dysfunction, renal dysfunction, serious complications, and extended bone metastasis. Of the 104 patients meeting these criteria, individuals were divided into the ICI group (nivolumab plus chemotherapy) and the non-ICI group (chemotherapy only). A comparative assessment of survival outcomes and tumor responses between the two groups was performed. Additionally, the incidence of immune-related adverse events (irAE) in the ICI group was assessed. **Results:** The ICI group comprised 57 patients, while the non-ICI group comprised 47 patients. The median follow-up for survival analysis was 11.6 months. In the ICI group, response rate (RR), median progression-free survival (mPFS), and median overall survival (mOS) were 48.7% among patients with measurable lesions, 7.4 months (95%CI, 5.3-10.0), and 15.1 months (95%CI, 13.1-21.0). In the non-ICI group, RR, mPFS, and mOS were 33.3%, 6.4 months (95%CI, 4.1-7.8), and 11.8 months (95%CI, 10.1-14.6). The OS in the ICI group was significantly better than those in the non-ICI group (log-rank p = 0.0436). Furthermore, irAEs were observed in eight patients (14.0%), including hypothyroidism (n=3), arthritis (n=2), hypopituitarism (n=2), and dermatitis (n=1). No patients discontinued treatment due to intolerable irAEs. **Conclusions:** The combination of nivolumab and chemotherapy showed a favorable outcome even in AGC patients with ineligible group. Research Sponsor: None.

Prognostic impact of postoperative complications after neoadjuvant therapy followed by esophagectomy for esophageal cancer: An exploratory analysis of phase III trial JCOG1109.

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Background: We previously reported that postoperative complications after esophagectomy were significantly associated with poor long-term prognosis in the prospective randomized trial JCOG9907 (Ann Surg, 2017). JCOG1109 evaluated the efficacy and safety of neoadjuvant docetaxel plus cisplatin plus 5-FU (DCF), radiation with cisplatin plus 5-fluorouracil (CF-RT) compared with cisplatin plus 5-fluorouracil (CF) for locally advanced esophageal squamous cell carcinoma, and demonstrated the superiority of DCF compared with CF in improving overall survival (OS). In this exploratory analysis, we investigated the association between postoperative complications and prognosis in JCOG1109, where more intensive preoperative therapies and thoracoscopic esophagectomy (TE) were introduced. **Methods:** Patients with potentially resectable advanced thoracic esophageal cancer were randomly assigned to CF, DCF, or CF-RT therapies and followed by open esophagectomy (OE) or TE with regional lymphadenectomy in JCOG1109. The impacts of postoperative complications (≥ Grade 2) on long-term prognosis including OS and progression-free survival (PFS) were investigated in each of the three preoperative therapies. **Results:** Between December 2012 and July 2018, 601 patients were randomized (CF/DCF/CF-RT; 199/202/200). Of 589 eligible patients, 541 patients underwent esophagectomy (183/181/177). Any postoperative complications, pneumonia, anastomotic leakage, recurrent laryngeal nerve paralysis, or infectious complications had no significant impact on OS in each of the three arms. When divided into OE or TE, the impact of any postoperative complications on OS was attenuated by introducing TE in each of three arms; from neoadjuvant CF followed by OE group [hazard ratio (HR) 1.557, 95% confidence interval (CI) 0.881-2.752] to TE group [HR 0.802, 95% CI 0.424-1.515], from neoadjuvant DCF followed by OE group [HR 1.151, 95% CI 0.575-2.306] to TE group [HR 0.703, 95% CI 0.3363-1.471], and from neoadjuvant CF-RT followed by OE group [HR 1.548, 95% CI 0.840-2.852] to TE group [HR 1.186, 95% CI 0.627-2.245]. **Conclusions:** JCOG1109 showed no association between postoperative complications and long-term prognosis. The prognostic impact of postoperative complications might be attenuated by the introduction of minimally invasive esophagectomy (open to thoracoscopy) even after more intensified preoperative therapy (CF to DCF or CF to CF-RT). Clinical trial information: CRB3180009. Research Sponsor: None.

Prognosis of signet ring cell (SRC) gastric carcinoma (GC) compared with other histological subtypes.

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Background: Signet ring cell (SRC) gastric cancer (GC) has its unique clinicopathological features with poor tissue differentiation, high invasiveness, diffuse growth pattern and poor prognosis. Survival rates and clinicopathological characteristics of patients with SRC GC regardless of the stage of the disease has not been well studied. **Methods:** The retrospective analysis was undertaken of 993 patients with diagnosed gastric cancer. These patients received surgical resection in period from January 2013 to December 2018 in N.N. Blokhin National Medical Research Center of Oncology. We compared clinical and pathological features as well as prognostic factors between these groups. **Results:** For early SRC carcinoma, the median survival rates weren't reached; 3- and 5-year survival rates were 100% and 89.0 %; for non-SRC carcinoma 91.4 and 85.3 % (HR=0.73; 95 % CI 0.22–2.42, p = 0.6) respectively. For locally advanced/metastatic SRC carcinoma, the median survival rates, 3- and 5-year survival rates were 38 months, 53.0 and 38.4 %; for non-SRC carcinoma 51.1 months, 59.2 and 48.0 % (HR=1.2; 95 % CI 0.91–1.54, p = 0.2). At univariate analysis SRC morphology was found as independent risk factor for overall survival (HR=1.19; CI 95% 0.94 – 1.50; p<0.0001). Multivariate analysis did not reveal SRC morphology to be significant and independent risk factor for overall survival (HR=1.00; CI 95% 0.72 – 1.38; p=0.98). **Conclusions:** Overall survival rates in patients with early (mucosal and submucosal tumor regardless of lymph nodes status) SRC gastric cancer did not show any significant differences in comparison with other histological types of GC. Thus, the prognosis of early SRC GC is equivalent or better than that of other histological subtypes of GC. Overall survival rates of locally advanced/metastatic SRC gastric cancer are worse than non-SRC cancer, but the differences are not statistically significant.
Research Sponsor: None.

Postoperative exercise in gastric cancer patients undergoing gastrectomy: A randomized controlled trial.

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Background: Faster physical recovery after gastrectomy may lead to better physical and psychological health in patients with gastric cancer. Post-surgical exercise may prevent physical dysfunction and improve recovery, but its effects on physical function are not fully explored. **Methods:** In this trial, 52 gastric cancer patients (Stage 1-3) were randomly assigned to early postoperative exercise rehabilitation or usual care (1:1 ratio). Interventions Inpatient Supervised Exercise sessions were conducted for 15 minutes once a day on Postoperative Day (POD) 1 and 2 in the ward. Home-based Exercise involved 30-minute remote exercise sessions at home once a week following discharge, with additional recommendations for daily 30-minute sessions of low to moderate-intensity aerobic exercises and bodyweight exercises. Moreover, the exercise group was provided with an exercise log and instructional videos to record the duration of bodyweight exercises and aerobic exercises. Measurements were taken three times (baseline, POD3, and the first outpatient visit 1 month after surgery). The primary outcome was a Short Physical Performance Battery (SPPB), with secondary outcomes including grip strength, 30-second sit-to-stand (STS), body composition, C-reactive protein (CRP), and quality of life (QOL). **Results:** A total of 46 (88%) participants (mean [SD] age, 60.2 [12.24] years; 26 [57%] male) completed the trial. Compliance with intervention in-patient exercise was 96% and remote exercise was 79%. The significant difference in SPPB scores of 11.4±1.3 in the exercise group compared with 10.7±1.43 usual care group (mean difference, -1.09 score; 95%CI, -1.7 to -0.4; p=<.001). Moreover, there were significant differences between groups in the STS exercise group 14.3±2.8 usual care group 13.6±4.5 (mean difference, 1.86; 95%CI, -3.3 to -0.4; p=.028), and CRP in POD3 exercise group 66.9±40.73 usual care group 103.7±54.25 (mean difference, -22.54 score; 95%CI, -46.5 to 1.4; p=.016). However, there was no significant difference in grip strength, body composition, and QOL. **Conclusions:** Postoperative exercise presents as a beneficial strategy to enhance physical recovery in patients undergoing gastrectomy, improving key physical function markers and reducing inflammation. Clinical trial information: NCT06260293. Research Sponsor: None.

Impact of combined resection of pancreas on long-term survival in patients with gastric cancer.

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Background: In patients with advanced gastric cancer invading adjacent organs, extended multivisceral resection is required to achieve R0 resection. However, the survival benefit of combined resection of pancreas remains controversial. We investigated the safety and efficacy of combined pancreatic resection in gastric cancer. **Methods:** This study included 64 patients who underwent gastrectomy with pancreatectomy (pancreaticoduodenectomy (PD) or distal pancreatectomy with splenectomy (PS)) for primary gastric cancer with invasion to the pancreas by primary tumor or lymph node metastasis and invasion to the duodenum. Patients who underwent R2 resection were excluded. Short- and long-term outcomes were assessed. **Results:** PD was performed in 18 patients (28.1%) and PS in 46 patients (71.9%). Macroscopic invasion to the pancreas by primary tumor was observed in 47 patients (73.4%), by lymph node in 13 patients (20.3%), and the invasion to the duodenum by primary lesion was in 4 patients (6.3%). Pathologically, pancreatic invasion was confirmed in 27 out of 47 patients (57.4%). Postoperative intra-abdominal infectious complications (PIIC) of Clavien-Dindo (CD) Grade III or higher were observed in 33 patients (51.6%). The most common postoperative complication was pancreatic fistula, which was observed in 31 patients (48.4%). There was no mortality within 30 days after surgery. Univariate analysis revealed that invasion the duodenum was highly associated with PIIC ($p=0.038$). Twelve patients (18%) resulted in R1 resection due to positive peritoneal cytology in 11 patients and positive surgical margin in one patient. The 5-year overall survival (OS) rate was 42.9%, which was 53.6% in R0 resection and 0% in R1 resection. Multivariate analysis revealed R1 resection was an independent prognostic factor for OS (HR:5.315, $p<0.001$). The 5-year recurrence-free survival (RFS) rate was 44.5%. The 5-year RFS rate was significantly worse in patients with PIIC (28.7%) than in patients without it (58.7%). Multivariate analysis revealed that PIIC was an independent prognostic factor for RFS (HR:2.345, $p=0.036$). **Conclusions:** Pancreatic resection for gastric cancer is considered safe and effective when R0 resection is possible. However, surgeon should pay particular attention to avoid the postoperative complications. Research Sponsor: None.

Association of early tumor shrinkage with survival in patients with HER2-positive advanced gastric cancer treated with trastuzumab deruxtecan.

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Background: Trastuzumab deruxtecan (T-DXd), an antibody-drug conjugate composed of an anti-human epidermal growth factor receptor 2 (HER2) antibody and a cytotoxic topoisomerase I inhibitor, has been approved as third or later-line treatment for unresectable HER2-positive advanced gastric cancer (AGC) based on the results of phase II DESTINY-Gastric 01 trial. However, there are few studies on real-world clinical outcomes and clinicopathological factors associated with efficacy of T-DXd in patients with HER2-positive AGC. **Methods:** This single-center retrospective study enrolled patients with AGC treated with T-DXd as third- or later-line chemotherapy between March 2018 and December 2023. Early tumor shrinkage (ETS) was defined as the relative change in the sum of longest diameters of RECIST target lesions at 8 $\pm$ 2 weeks compared with baseline. Prognostic outcomes were assessed using the log-rank test and Cox proportional hazards regression model. **Results:** A total of 57 patients were enrolled. The median age was 66 years (range, 31–82); 68% were male; 79% had HER2 3+; 23% had pulmonary metastasis; 67% were treated with T-DXd in the third-line setting. At a median follow-up of 27.6 months, the median progression-free survival (PFS) was 4.5 months (95% confidence interval [CI], 3.8–5.5), and the median overall survival (OS) was 7.3 months (95% CI, 4.9–9.4). Among 40 patients with target lesions, the overall response rate was 35.0% and the disease control rate was 90.0%. Patients with ETS had notably longer PFS (median PFS (months) 8.6 vs 4.1, $P < 0.01$) and OS (median OS (months) 9.1 vs 6.1, $P < 0.01$) than those without ETS. In multivariate analysis, ETS was associated with significantly longer PFS (hazard ratio [HR], 0.23; 95% CI, 0.08–0.63; $P = 0.004$) and OS (HR, 0.27; 95% CI, 0.12–0.65; $P = 0.004$), whereas pulmonary metastasis was associated with significantly shorter OS (HR 2.46, 95% CI, 1.07–5.65; $P = 0.03$). **Conclusions:** This study showed that achieving an ETS was significantly associated with a favorable prognosis in patients with HER2-positive AGC treated with T-DXd.
Research Sponsor: None.

Preliminary results of the ALTER-E009 study: Efficacy and safety of anlotinib combined with radiotherapy in patients with locally advanced or metastatic esophageal squamous cell carcinoma (ESCC)—A multicenter, multi-cohort retrospective exploratory study.

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Background: Therapeutic options for metastatic esophageal squamous cell carcinoma (ESCC) primarily involve immunotherapy combined with chemotherapy. For locally advanced ESCC, concurrent chemoradiotherapy (CRT) or radiotherapy is the standard, though outcomes remain suboptimal. Targeted therapies, particularly anti-angiogenic agents, hold promise to enhance CRT efficacy. Anlotinib, a multi-target anti-angiogenic agent inhibiting VEGFR, PDGFR, FGFR, and c-KIT, has shown potential to enhance radiotherapy by alleviating tumor hypoxia. This multicenter, retrospective study evaluates the efficacy and safety of anlotinib combined with radiotherapy in locally advanced or metastatic ESCC. **Methods:** This study screened patients (pts) with locally advanced or metastatic ESCC treated with anlotinib (8-12 mg, p.o., qd, d1-14, q3w) and radiotherapy from June 2018 to June 2024. Two cohorts were included: Cohort 1 (metastatic ESCC) and Cohort 2 (locally advanced ESCC). Cohort 2 was divided into an observational group (OG, anlotinib + radiotherapy ± chemotherapy) and a control group (CG, radiotherapy ± chemotherapy). Radiotherapy doses for primary lesions were 45-60 Gy (1.8-2.2 Gy per fraction), while metastatic lesions received either standard fractionation (≥45 Gy) or stereotactic body radiotherapy (SBRT, ≥3 Gy per fraction, total dose ≥30 Gy). Chemotherapy regimens were not restricted. Cohort 1 aimed to enroll 50 pts, while Cohort 2 included 100 in the OG and 200 in the CG. The primary endpoint was overall survival (OS), with secondary endpoints including objective response rate (ORR), disease control rate (DCR), progression-free survival (PFS), duration of response (DoR), and safety. **Results:** By September 10, 2024, 128 pts had been enrolled: 34 in Cohort 1 and 94 in the OG of Cohort 2. In Cohort 1, all 34 pts (100%) were stage IVB, with a median OS of 24.0 months (95% CI: 12.44-35.60) and a median PFS of 9.2 months (95% CI: 7.60-10.80). The 24-month OS and PFS rates were 54.0% (95% CI: 27.20-74.73) and 30.8% (95% CI: 10.21-54.53), respectively. ORR was 52.9% (95% CI: 35.1-70.2), with 1 complete response (CR) and 17 partial responses (PR), DCR was 88.2% (95% CI: 72.5-96.7). In Cohort 2, 50 pts (53.2%) were stage III, and 28 (29.8%) were stage IVA. The median OS is not yet mature, but the median PFS was 20.1 months. The 36-month OS and PFS rates were 61.9% (95% CI: 46.12-74.34) and 48.4% (95% CI: 29.58-64.87). ORR was 46.8% (95% CI: 36.4-57.4), with 6 CRs and 38 PRs, DCR was 95.7% (95% CI: 89.5-98.8). Safety data collection is ongoing. **Conclusions:** Anlotinib combined with radiotherapy exhibited promising efficacy in locally advanced or metastatic ESCC, though further data are needed to confirm these preliminary findings. Research Sponsor: None.

362 **Poster Session**

Gastric cancer patient insights: Patient-reported outcomes and real-world perspectives from the gastric cancer patient community.

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Background: According to the National Cancer Institute Surveillance, Epidemiology, and End Results Program, an estimated 26,890 patients in the United States will be diagnosed with gastric (stomach) cancer in 2024. Estimated deaths from the disease in 2024 will reach 10,880, with a five-year relative survival rate of 36%. Understanding the need for increased gastric cancer education and awareness, the GI Cancers Alliance created a 12-month series of engagement opportunities to better understand and listen to the gastric cancer patient voice and perspectives. **Methods:** Our 12-month patient-reported outcomes (PRO) research (September 2023 – August 2024) included an online survey, in person and virtual focus groups, and personalized follow-up interviews. 651 gastric cancer patients and survivors participated in these engagement opportunities, with 89% of participants asking for follow-up personalized interviews. One of our program goals was to reach and interact with patients who often experience barriers in receiving quality cancer care. Our PRO research provided an opportunity to amplify the patient voice and lived experience by engaging patients in medically underserved communities, rural and urban populations, diverse ethnic communities, and patients receiving care in community-based and federally funded oncology centers, and charity (indigent) care. **Results:** Overarching results from our study suggest that while advances have been made, health equity, racial disparities and barriers in cancer care remain a challenge. 651 patient participants (43% male, 54% female, 3% non-binary) identified their race/ethnicity as: 24% Black/African American, 22% Hispanic, 21% Multiracial, Non-Hispanic/White 19%, Asian/Pacific Islander 7%, Native American 2%, and 5% Other. 69% of patients reported that their clinician did not suggest biomarker testing, nor discuss the role of biomarkers to help make informed treatment decisions. 68% of patients were concerned that they lacked knowledge and options to make informed decisions regarding their treatment. 64% of patients reported they suffered from mental health concerns ranging from severe depression, feelings of hopelessness and anxiety; 63% of patients struggled to afford nutritious food to meet their changing dietary needs, especially post-surgery. **Conclusions:** Our PRO research underscores the need for increased education and outreach for gastric cancer patient communities experiencing barriers to high quality cancer prevention, early detection, and treatment. By partnering with our 100+ advocacy partner organizations, we will continue to amplify the gastric cancer patient voice by establishing new programming, initiatives, and enhanced outreach addressing the challenges and barriers to quality cancer care for all patients. Research Sponsor: GI Cancers Alliance; Amgen; Astellas; Astra-Zeneca.

363 **Poster Session**

Real-world validation of the Khorana risk score in gastric cancer patients receiving immune checkpoint inhibitors.

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Background: Thrombosis is a common cancer-related complication, particularly in gastric cancer. Studies show that immune checkpoint inhibitors (ICIs) increase the risk of venous thromboembolism (VTE) and arterial thrombosis. The Khorana risk score is widely used to assess thrombosis risk before starting chemotherapy. This study aimed to validate the Khorana score in gastric cancer patients receiving ICIs using a large global database. **Methods:** We used the TriNetX Global Collaborative Network, which contains de-identified data from over 120 million patients. We included patients diagnosed with gastric cancer who received ICIs between January 2014 and July 2023. The Khorana risk score was calculated based on lab values and BMI within two weeks before ICI initiation. Patients were divided into two cohorts: elevated risk (score 3-6) and intermediate risk (score 2). Cox-proportional hazard analyses were conducted over a one-year follow-up. The primary outcome was VTE (deep vein thrombosis [DVT] and pulmonary embolism [PE]). Secondary outcomes were arterial thrombosis, including myocardial infarction (AMI) and ischemic stroke, and all-cause mortality. **Results:** The study included 551 patients, with 362 in the intermediate risk and 189 in the elevated risk cohorts. The mean age was 64 in both cohorts, with the elevated group consisting of more males (76% vs. 63%) and a higher proportion of Black patients (11% vs. 6%). Pembrolizumab (49%) and nivolumab (47%) were the most used ICIs. Patients with elevated risk had a significantly higher risk of VTE (HR 1.74, 95% CI 1.18-2.58), DVT (HR 1.82, 95% CI 1.07-3.10), and PE (HR 1.74, 95% CI 1.07-2.84). However, no significant associations were found with arterial thrombosis (HR 1.28, 95% CI 0.74-2.20), including cerebral infarction (HR 1.05, 95% CI 0.51-2.16), or AMI (HR 1.46, 95% CI 0.70-3.05). The elevated risk cohort also had significantly higher mortality (HR 1.44, 95% CI 1.11-1.87). **Conclusions:** An elevated Khorana risk score is associated with a higher risk of VTE, but not arterial thrombosis, in gastric cancer patients treated with ICIs. Further studies are needed to determine how the Khorana score can be applied to predict ICI-related thrombosis and guide prophylactic anticoagulation. Research Sponsor: None.

	Khorana risk score 3-6		Khorana risk score 2			
Outcomes	**At risk patients**	**Cases**	**At risk patients**	**Cases**	**Hazard ratio (95% CI)**	**P-value (Log-rank)**
Venous thromboembolism	189	45	362	58	1.74 (1.18-2.58)	0.005
Deep vein thrombosis	189	25	362	30	1.82 (1.07-3.10)	0.025
Pulmonary embolism	189	29	362	36	1.74 (1.07-2.84)	0.025
Arterial thrombosis	189	21	362	34	1.28 (0.74-2.20)	0.375
Cerebral infarction	189	11	362	22	1.05 (0.51-2.16)	0.901
Acute myocardial infarction	189	12	362	17	1.46 (0.70-3.05)	0.317
All-cause mortality	189	93	362	145	1.44 (1.11-1.87)	0.006

Correlation between PD-L1 CPS and clinical outcomes in patients treated with first-line nivolumab plus chemotherapy for advanced gastric or gastroesophageal junction cancer: A multi-center retrospective study in Japan.

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Background: The CheckMate 649 trial showed the superiority of first-line nivolumab plus chemotherapy (Nivo-CT) compared with chemotherapy alone for advanced gastric and gastroesophageal junction (GEJ) cancer with a programmed cell death ligand 1 (PD-L1) combined positive score (CPS) >5, while survival benefit was poor in PD-L1 CPS <1 and <5. However, there were few studies on correlation between PD-L1 CPS and efficacy of first-line Nivo-CT for advanced gastric or GEJ cancer in real world data. **Methods:** This multi-center retrospective study included patients with histologically confirmed advanced gastric or GEJ cancer who were started on first-line Nivo-CT between December 2021 and December 2023. PD-L1 CPS was assessed before the start of Nivo-CT by classified into three groups at each center: <1, 1~5 and >5, using the Dako PD-L1 immunohistochemistry 28-8 pharmDx assay. The efficacy was evaluated by overall survival (OS), progression-free survival (PFS) and objective response rate (ORR). **Results:** A total of 271 patients were included (median age, 70 [range 18–85] years; male/female, 64/36%; performance status (PS) 0/1/2, 47/49/4%; primary tumor site gastric/GEJ, 90/10%; status unresectable/recurrent, 81/19%; histology intestinal/diffuse, 30/70%). PD-L1 CPS status were evaluated 234 (86%) patients and PD-L1 CPS <1, 1~5 and >5 was found in 35 (15%), 76 (33%) and 123 (53%) patients, respectively. Median follow-up time was 11.3 (range 1-30) months. The median PFS was 9.7 (95%CI, 5.9-16.3), 8.8 (95%CI, 7.2-13.0) and 6.3 (95%CI, 5.3-9.0) months in the patient with PD-L1 CPS <1, 1~5 and >5, respectively. The median OS was 18.7 (95%CI, 17.4-NA), 23.4 (95%CI, 17.7-NA) and 18.3 (95%CI, 13.3-22.5) months, and the ORR were 30.8, 66.7 and 61.3% ($p = 0.075$) in the patient with PD-L1 CPS <1, 1~5 and >5, respectively. Liver metastases were found in 43 (35.0%) of the patients with PD-L1 CPS >5, significantly more than in the patients with PD-L1 CPS <5 ($p < 0.001$). Multivariate analysis indicated that PD-L1 CPS status was not associated with prognosis (PD-L1 CPS ≥5 vs. <1, HR 1.18 [95%CI, 0.75-1.86], $p = 0.48$). In the overall population, grade ≥3 adverse events were observed in 102 (37.6%) patients and grade ≥3 immune-related adverse events were observed in 26 (9.6%) patients. **Conclusions:** There were no significant differences in prognosis and response between PD-L1 CPS in first-line Nivo-CT for advanced gastric or GEJ cancer in real world data.
Research Sponsor: None.

Impact of tislelizumab + chemotherapy versus placebo + chemotherapy on patient-reported symptoms and disease progression by programmed death-ligand 1 expression in gastroesophageal adenocarcinoma: A post hoc analysis of the RATIONALE-305 trial.

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Background: The relationship between patient-reported outcome (PRO)-based symptom scores, recurrent PRO-based symptomatic deterioration events (RDEs), and terminal events such as progression-free survival (PFS) are rarely examined in the oncology therapeutic domain. Thus, we applied a 3-component joint model (JM) framework aiming to illuminate more clinically interpretable associations between PRO-based treatment effects, RDEs, and PFS among subgroups of patients with programmed death-ligand 1 (PD-L1) expression of ≥1%, ≥5%, and ≥10%. **Methods:** The final analytic sample included 378 patients in the tislelizumab + chemotherapy arm (T+C) vs 401 in the placebo + chemotherapy arm (P+C) for the PD-L1 ≥1% subgroup, 238 in the T+C arm vs 237 in the P+C for the PD-L1 ≥5% subgroup, and 118 in the T+C arm vs 125 in the P+C arm for the PD-L1 ≥10% subgroup. Symptom domain scores from the EORTC QLQ-C30 and QLQ-STO22 symptom were modeled. Change from baseline (CFBL) in each domain was analyzed every cycle up to cycle 6, then every other cycle thereafter (up to cycle 25). The joint model was applied to all PD-L1 subgroups and comprised three components: 1) linear mixed model (LMM) predicting CFBL symptom scores, 2) Cox proportional hazard (CPH) model for disease progression (PFS as terminal event), and 3) frailty (random effects for RDEs) CPH model for time to PRO-based RDEs. Osoba's 10-point threshold was used to define RDEs. **Results:** In the LMM, significant treatment efficacy for the T+C arm compared with the P+C arm was observed for the GHS/QoL (P=0.0080) and dietary restriction scores (P=0.0475) in the PD-L1 ≥5% subgroup. In the PD-L1 ≥1% subgroup, compared with the P+C arm, the T+C arm was associated with less worsening in pain/discomfort (P=0.0376), upper gastrointestinal symptoms (P=0.0088), and dietary restrictions (P=0.0352). In the CPH model, the average hazard ratio indicated that compared with the P+C arm, the T+C arm was associated with a reduction in risk of disease progression across all PRO domains and PD-L1 subgroups. Lastly, the PRO-based RDE frailty predictions were strongly associated with PFS risk in the PD-L1 ≥1% and ≥5% subgroups for GHS/QoL, and in the PD-L1 ≥1% subgroup for physical functioning, fatigue, dysphagia-odynophagia, pain/discomfort, and dietary restrictions, as well as for pain/discomfort and dietary restrictions in the PD-L1 ≥5% subgroup. **Conclusions:** Through the 3-component JM, PRO-based effects were detected and demonstrated strong associations between PRO-based RDEs, and investigator-assessed PFS among varying PD-L1 expression levels. This framework may be a promising alternative for analyzing and interpreting PRO data in the context of PFS. Research Sponsor: BeiGene, Ltd.

Impact of tislelizumab + chemotherapy versus placebo + chemotherapy on patient-reported symptoms and overall survival (OS) by programmed death-ligand 1 (PD-L1) expression in advanced or metastatic esophageal squamous cell carcinoma (ESCC): A post hoc analysis of the RATIONALE-306 trial.

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Background: While improved survival has been previously demonstrated, the impact of immunotherapy on HRQoL in ESCC has not been well examined. Traditional PRO-based analyses in oncology trials, such as time to deterioration (TTD) and mixed models for repeated measures (MMRMs), are limited by discounting recurrent PRO events. Thus, we applied a 3-component joint model (JM) framework to define more clinically interpretable associations between patient-reported symptoms, treatment effects, and OS among subgroups of patients with ESCC from RATIONALE-306, which met its primary endpoint, with PD-L1 expression of ≥1%, ≥5%, and ≥10%. **Methods:** The final analytic sample included 226 patients in the tislelizumab + chemotherapy arm (T+C), 242 in the placebo + chemotherapy arm (P+C) for PD-L1 ≥1%, 113 in the T+C arm and 103 in the P+C arm for PD-L1 ≥5%, and 168 in the T+C arm and 178 in the P+C arm for PD-L1 ≥10%.From EORTC QLQ-C30 and OES18, 7 keysymptom domains were modeled (GHS, physical functioning, fatigue, dysphagia, pain, reflux, dietary restrictions). PRO data were collected at baseline and at every treatment cycle (up to 6 cycles), then every other cycle, and at safety follow-up, and change from baseline (CFBL) was analyzed. The joint model comprised three components: 1) linear mixed model predicting CFBL symptom scores; 2) Cox proportional hazard model (CPH) for time to OS; and 3) frailty (random effects for recurrent deterioration events [RDEs]) CPH model for time to PRO-based RDEs. Osoba (1998) 10-point threshold was used to define RDEs. **Results:** Adjusted completion rates were >90% in the ITT population for both arms. Significant T+C treatment effects wereobserved for physical functioning in PD-L1 ≥5% (P=0.0476), and pain in PD-L1 ≥1% (P=0.0028) and PD-L1 ≥5% (P=0.0149) subgroups, but not in PD-L1 ≥10%. For other 5 symptoms (ie, GHS, fatigue, reflux, dysphagia, dietary restrictions), there were no statistically significant differences between treatment arms. However, T+C was associated with significant reductions in the risk of death across all 7 key symptoms and PD-L1 subgroups. As one example, with respect to interaction between pain and OS, T+C was associated with a 22% (HR, 0.78 [95% CI, 0.652-0.931]), 33% (HR, 0.67 [95% CI, 0.515-0.860]), and 47% (HR, 0.50 [95% CI, 0.344-0.720]) reduction in the risk of death in PD-L1 ≥1%, ≥5%, and ≥10%, respectively compared with P+C. **Conclusions:** In this analysis, the addition of tislelizumab to chemotherapy was associated with significantly less deterioration in multiple PRO symptoms including physical functioning and pain, alongside a lower risk of death, after adjusting for recurring deterioration events using a frailty model across multiple PD-L1 expression subgroups. Research Sponsor: BeiGene, Ltd.

Anlotinib combined with penpulimab and nab-paclitaxel as first-line treatment for advanced esophageal squamous cell carcinoma (ESCC): A single-arm, open-label phase II clinical trial.

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Background: Recently, PD-1 blockades combined with dual chemotherapy regimens in first-line setting exhibited encouraging efficacy for patients with ESCC. However, the safety profile of conventional dual chemotherapy remained unsatisfactory. Therefore, PD-1 blockades combined with anti-angiogenic tyrosine kinase inhibitors (TKIs) and single chemotherapy regimen might offer a promising strategy. Anlotinib, a novel multitarget TKI primarily targeting VEGFR1-3, demonstrated promising therapeutic activity as first-line combination therapy or second-line monotherapy for ESCC patients in China clinically. Therefore, this study was designed to explore the efficacy and safety of anlotinib combined with penpulimab (PD-1 blockade) and nab-paclitaxel as first-line therapy in advanced ESCC. **Methods:** Patients with previously untreated metastatic or locally advanced ESCC were recruited and treated with anlotinib (12mg, po, d1~14, q3w) and penpulimab (200mg, iv, d1, q3w) plus nab-paclitaxel (220mg/m^2, iv, d1, q3w) until disease progression or unacceptable toxicity. The tumor response was assessed according to RECIST 1.1 using CT scans every two cycles. Adverse events were recorded by severity in accordance with the NCI CTC AE Version 5.0. The predefined sample size was 30. Primary endpoint was PFS and secondary endpoints included safety, ORR, DCR and OS. **Results:** From Jul 2022 to Sep 2024, a total of 30 patients were enrolled, with 29 patients who had received first tumor response were included in this analysis. The best overall response indicated that there were 1 CR (3.4%), 23 PR (79.3%), 2 SD (6.9%) and 3 NE (10.3%). Therefore, the preliminary ORR was 82.8% (95%CI: 64.2%-94.2%), DCR was 89.7% (95%CI: 72.6%-97.8%). The primary median PFS of the 29 patients was 10.84 months (95%CI: 7.57-14.11). Additionally, safety profile exhibited that the regimen was tolerable. The most common treatment-emergent adverse events among the 29 patients with the incidence >20% were anemia (62%), leukopenia (31%), peripheral neurotoxicity (28%) and rash (21%). Common grade ≥3 treatment-emergent adverse events were leukopenia (7%), glutamic-pyruvic transaminase was elevated (3%), glutamic oxalacetic transaminase increased (3%), hypertension (3%), diarrhea (3%) and rash (3%). **Conclusions:** The combination of anlotinib plus penpulimab and nab-paclitaxel as first-line therapy for advanced ESCC demonstrated promising efficacy and manageable safety profile. And the conclusions needed to be confirmed in subsequent trials. Clinical trial information: ChiCTR2400089133. Research Sponsor: None.

Safety and efficacy of perioperative chemotherapy combined with PD-1 inhibitor versus chemotherapy with D2 gastrectomy plus PAND in gastric cancer with PALD metastasis: A single-center retrospective cohort study.

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Background: Gastric cancer with para-aortic lymph node (PALD) metastasis is associated with a poor prognosis. When the metastasis is limited to the PALD, preoperative systemic therapy combined with D2 gastrectomy plus para-aortic lymph node dissection (PAND) may enhance survival. The benefit of perioperative chemotherapy combined with immune checkpoint inhibitors for these patients is unknown. **Methods:** A total of 133 patients with gastric cancer and limited No.16a2/b1 lymph node metastasis underwent perioperative systemic therapy combined with D2 gastrectomy plus PAND. Patients receiving chemotherapy combined with a PD-1 inhibitor were assigned to Group A (n=62), while those receiving chemotherapy alone were assigned to Group B (n=71). Pathological response, safety, efficacy, and survival outcomes were compared between the two groups. **Results:** Adverse events occurred at comparable rates between the two groups. No treatment-related deaths occurred in either group. In Group A, all patients (100%) achieved R0 resection, compared to 70 patients (98.6%) in Group B. Group A showed significantly better pathological outcomes than Group B, with higher rates of pathological complete response (pCR) (37.1% vs. 4.2%, $P < 0.001$), and major pathological response (MPR) (58.1% vs. 22.5%, $P < 0.001$). Group A showed a higher 2-year overall survival (OS) rate compared to Group B (90.3% vs. 62.9%, $P = 0.011$). The median overall survival (mOS) in Group B was 35.0 months, while follow-up data for Group A are currently insufficient to assess mOS. **Conclusions:** The addition of a PD-1 inhibitor to chemotherapy significantly improves the pathological response rate and prognosis in gastric cancer patients with PALD. Notably, the combination did not increase adverse events. These findings suggest that combining PD-1 inhibitor with chemotherapy could be a promising strategy for improving outcomes in this patient population, though further prospective studies are necessary to confirm long-term survival benefits. Research Sponsor: None.

Postoperative outcomes and pathological response.

Variable	Group A (n=62)	%	Group A (n=71)	%	P Value
Vascular tumor embolus					0.005
Negative	43	69.4%	32	45.1%	
Positive	19	30.6%	39	54.9%	
Nerve infiltration					0.004
Negative	44	71.0%	33	46.5%	
Positive	18	29.0%	38	53.5%	
No.16a2/b1					0.057
Negative	53	85.5%	51	71.8%	
Positive	9	14.5%	20	28.2%	
Tumor resection					0.970
Distal	26	41.9%	30	42.3%	
Total	36	58.1%	41	57.7%	
Serum CEA (U/mL)					0.805
Normal	56	90.3%	65	91.5%	
>5	6	9.7%	6	8.5%	
Serum CA125 (U/mL)					0.280
Normal	30	48.4%	41	57.7%	
>35	32	51.6%	30	42.3%	
Serum CA199 (U/mL)					0.783
Normal	56	90.3%	65	91.5%	
>37	6	9.7%	6	8.5%	
pT-stage					<0.001
≤T2	37	59.7%	19	26.8%	
> T2	25	40.3%	52	73.2%	
pN-stage					0.001
N0/1	42	67.7%	27	38.0%	
N2/3	20	32.3%	44	62.0%	
TRG					< 0.001
0	23	37.1%	3	4.2%	
1	13	21.0%	13	18.3%	
2	22	35.5%	47	66.2%	
3	4	6.5%	8	11.3%	
pCR	23	37.1%	3	4.2%	< 0.001
MPR	36	58.1%	16	22.5%	< 0.001

Epidemiological characteristics of gastric and pancreatic cancers in Latin America: GASPAR (LACOG 0222) study.

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Background: Gastric (GC) and pancreatic cancer (PC) have poor prognosis representing major public health challenges, particularly in Latin America. Real-world data are limited. GASPAR (LACOG 0222) study addresses this gap by creating a multicenter database to analyze epidemiological, clinical, and pathological data, treatments, and outcomes for patients (pts) with GC and PC. **Methods:** This retrospective and prospective observational cohort study enrolled pts from 7 research sites in 3 LATAM countries with metastatic/locally advanced not amenable to curative intent GC (Cohort A) and PC (Cohort B) diagnosed from January 2019. Data were collected from medical records. (NCT05924789). **Results:** 201 pts were included: 122 with GC (Cohort A) and 79 with PC (Cohort B). 150 (75%) from Brazil, 29 (14%) from Argentina, and 22 (11%) from Colombia. Median age at diagnosis was 69 years (IQR 57-78), 70 (35%) had history of tobacco use, and 153 (76%) were covered by private healthcare (Table). In Cohort A (GC), HER2 status was assessed in 105 (86%) pts, with 12 (11%) HER2-positive; PD-L1 was tested in 87 (71%): 55 (63%) had ≥1% expression and 28% had ≥10%; all 88 (72%) patients who had MMR tested had MMR proficient. First-line treatment was given to 107 (88%) pts: 78 (73%) chemotherapy (CT) alone (FOLFOX in 43 [40%]; FLOT in 16 [15%]); 23 (21%) CT + immunotherapy; and 7 (7%) CT + anti-HER2. Only 59 (48%) pts received second-line treatment and 31 (25%) third-line. With median follow-up (mFUP) of 29 months (CI95% 18.2-50.4), the median progression-free survival (mPFS) at first-line was 5.6 months (CI95% 4.2-6.3) and the median overall survival (mOS) was 11.2 months (CI95% 9.2-14.1). In Cohort B (PC), 67 (85%) pts received first-line treatment: 41 (61%) FOLFIRINOX and 14 (21%) gemcitabine-nab-paclitaxel. 27 (34%) pts received second-line treatment and 10 (13%) third-line. With mFUP of 21 months (CI95% 15.6-NR), the mPFS at first-line was 5.4 months (CI95%3.9-6.4) and the mOS was 10.2 months (CI95% 8.6-13.1). **Conclusions:** The GASPAR study provides valuable real-world data on GC and PC in Latin America. Notably, the mPFS in both cohorts was slightly lower than historical controls for first-line treatments, while the mOS was similar, suggesting earlier disease progression despite comparable overall survival. Improved therapeutic strategies and healthcare access are needed to enhance outcomes for these populations. Clinical trial information: NCT05924789. Research Sponsor: Funding: Astellas Pharma Inc. Sponsor: Latin American Cooperative Oncology Group (LACOG).

Clinical characteristics.

Characteristic	GC (Cohort A) N=122	PC (Cohort B) N=79
Median age (IQR), yr	66 (51–78)	71.0 (61–79)
Male – n. (%)	58 (56)	40 (51)
Race – n. (%)		
White	67 (55)	71 (90)
American Indian or Alaska Native	15 (12)	6 (8)
Black	7 (6)	0
Other	7 (6)	0
Not reported	26 (21)	2 (2)
Disease status at study entry		
Locally advanced disease	4 (3)	14 (18)
Metastatic disease	118 (97)	65 (82)
ECOG status in first-line advanced disease – n. (%)		
0-1	75 (70)	45 (67)
≥ 2	13 (12)	3 (4.5)
Not reported	19 (17.8)	19 (28.4)

Patterns of immune checkpoint inhibitor utilization and PD-L1 testing in advanced gastroesophageal cancers using real-world data.

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Background: Globally, gastroesophageal cancer (GEC) is a leading cause of morbidity and mortality, with poor 5-year overall survival (OS). The addition of immune checkpoint inhibitors (ICIs) such as nivolumab and pembrolizumab to chemotherapy has resulted in significant improvement in OS of those patients (pts) with metastatic GEC (mGEC), most notably in those with tumors exhibiting high PD-L1 CPS scores. However, the pattern of ICI utilization in associated PD-L1 testing since the adoption of this treatment (tx) has yet to be elucidated. We evaluated the patterns of ICI utilization and PD-L1/CPS testing in the first-line tx of mGEC pre and post FDA approval of ICI use in this setting. **Methods:** Pts ≥ 18 years of age, with recurrent/metastatic squamous or adenocarcinoma GEC diagnosed after 2011 were included from the de-identified nationwide Flatiron Health, electronic health record-derived database. We presented the proportion of pts receiving ICI therapy, stratified by line of tx, as well as the proportion of patients who received PD-L1 testing at 6-month intervals. Pts demographics and clinical characteristics, including age, gender, race, ECOG score, documented insurance, and stage at diagnosis (dx) were described and compared using T or Wilcoxon rank-sum tests for continuous variables, and chi-square or Fisher's exact tests for categorical data. We compared tx rates and PDL-1 testing across four time points: time 1: Jan-June 2020, prior to ASCO data presentation; time 2: July-Dec 2020, ASCO data presentation; time 3: Jan- June 2021, FDA label change; time 4: July 2021 to 6 m after FDA label change using Chi-square test. **Results:** A total of 9,573 pts were included with 1,593 (16.6%) receiving ICI-based tx. Among these, 62.3% were White, 7% Black. Descriptive statistics indicated that pts with ICI- tx tend to be younger (median age 65y vs. 66.7; ($p < 0.001$). Within the total cohort, 74% pt were dx before FDA approval of ICI tx, while 26% pts were dx after FDA approval. PD-L1 testing was conducted in 4,065 (42.5%) pts. A total of 1,268 (13%) pts received first-line ICI-based tx. Of these, 154 pts (12.1%) were treated before FDA approval, while 1,114 pts (87.9%) received tx following FDA approval. The rate of first-line ICI tx significantly increased at each subsequent time point Time 2: 11.2% ($p < 0.0001$), Time 3: 32.9% ($p < 0.0001$), Time 4: 37.4% ($p < 0.0001$). PD-L1 testing rates also increased over time, particularly after FDA label change. Testing rates at Time 1 was 68%. Comparatively, testing rates at subsequent time points were: Time 2 (67.2%; p 0.88), Time 3 (71.5%; p 0.324) and Time 4 (77.4%; p=0.004). **Conclusions:** The approval of ICIs for first-line tx of mGEC led to a significant increase in both ICI utilization and PD-L1/CPS testing rates. Despite these advancements, ongoing disparities in tx access and testing highlight the need for further research and efforts to ensure equitable care. Research Sponsor: None.

Comparison of long-term outcomes for neoadjuvant chemoradiation (NA-CRT) followed by surgery versus (V) definitive chemoradiation (D-CRT) in localized gastroesophageal cancer (GEC): A single-center analysis.

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Background: The therapeutic approach in locally advanced GEC is evolving. NA-CRT followed by surgery has been recommended based on the *CROSS* trial. The *SANO* trial reports active surveillance may be an alternative in clinically complete response after D-CRT in both squamous cell carcinoma (SCC) and adenocarcinoma (AC). SCC and AC in GEC are different diseases with differences in tumor location, etiology, and biology. Despite the evolving landscape, there still may be a role for both *CROSS* and *SANO* approaches. In our large observational cohort study, we sought to compare the long-term outcomes in localized GEC across all histological subtypes treated with NA-CRT+Surgery V D-CR+Surveillance. **Methods:** A retrospective analysis comparing NA-CRT + surgery V D-CRT in localized GEC between 2007-2023 treated at the Princess Margaret Cancer Centre was completed. Baseline characteristics including age, gender, race, performance status and Charlson Comorbidity index (CCI) were reviewed. The primary and secondary endpoints were overall survival (OS) and disease-free survival (DFS), respectively and to compare these outcomes in both histologies. Cox proportional hazards analysis and Kaplan-Meier methods were employed. Outcomes were adjusted for baseline characteristics. **Results:** There were 529 patients (pts) included, 321 (61%) received NA-CRT+Surgery and 208 (39%) D-CR+Surveillance. Median age was 64yrs, 75% were male and 66% had N1+ disease. The proportion of pts with SCC treated with NA-CRT+Surgery and D-CR+Surveillance was 25% and 63%, respectively. There was a statistically significant difference in median OS and DFS between pts who received NA-CRT+surgery and D-CRT+surveillance (Table). Multivariate regression analysis shows a significant association between OS and age (adjusted HR 1.02, p=0.005), male sex (HR 1.55, p=0.003) but no significant association between OS and CCI (HR 1.1 p=0.36). In our subgroup analyses, there was no statistically significant difference in OS (p=0.33) and DFS (p=0.18) between SCC and AC. However, this was significant in D-CRT (OS $p<0.0001$, DFS $p<0.001$) favoring SCC. **Conclusions:** In contrast to the *SANO* trial, we found that NA-CRT+surgery was associated with a significantly better OS and DFS compared to D-CRT when subtypes were combined. However, in the D-CRT subgroup, SCC was associated with improved OS and DFS compared to AC. Thus, we continue to recommend current approaches in SCC. Awaited updates from *SANO* will inform whether active surveillance may also be appropriate in pts with AC subtype. Research Sponsor: None.

	Median (95% CI) NA-CRT Surgery	Median (95% CI) D-CRT	Log-rank test p-value
OS (mo)	28.2 (15.1, 57.6)	15.6 (7.7, 33.7)	P=<0.0001
DFS (mo)	17.9 (9.6, 40.8)	9.4 (5.0, 20.7)	P=<0.0001
pathCR (%)	14	-	
R0 resection (%)	89	-	

Efficacy and safety of intraperitoneal paclitaxel based regimens in patients with gastric cancer and peritoneal metastasis: A systematic review and meta-analysis.

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Background: Peritoneal metastasis (PM) occurs in about 5% - 20% of patients with gastric cancer (GC) at the time of diagnosis, and prognosis remains poor with limited treatment options. There are conflicting results on the efficacy of intra-peritoneal (IP) paclitaxel in patients with gastric cancer with PM. We performed a meta-analysis to examine the current evidence on the effectiveness of IP paclitaxel-based regimens in this patient population. **Methods:** A comprehensive search was conducted across Pubmed, EMBASE, Cochrane library, and Clinicaltrials.gov, resulting in the initial identification of 253 articles. Following screening, 35 articles, comprising only clinical trials (CT) and Randomized Controlled Trials (RCTs), were selected for detailed analysis. Screening and data extraction were conducted by two reviewers concurrently, with quality assessment performed using RoB (Risk of Bias) 2.0 tool and NIH quality assessment questionnaire. Proportional meta-analysis was done to estimate the pooled efficacy of IP paclitaxel based on the Median Survival Time (MST) and 1-year Overall Survival (OS), and tumor response was assessed using RECIST criteria. Subgroup analysis was conducted for patients who underwent conversion gastrectomy. Safety was assessed through the analysis of adverse events. **Results:** A total of 855 patients were included in the evaluation of MST, revealing an overall pooled MST of 20.21 months for GC patients with PM treated with IP paclitaxel-based regimens. Additionally, 1-year OS data was available for 785 patients, revealing an overall cumulative 1-year OS of 71% (95% CI: 66-76). Furthermore, among the 332 patients who underwent conversion gastrectomy, the pooled overall MST was 27 months. For 168 participants who underwent conversion gastrectomy after receiving therapy with IP PAC-based regimens, the 1-year OS was 84% (95% CI: 77-89). The most common adverse events (AE) observed were alopecia (75%) and anemia (60%). Overall, most studies exhibited low RoB, with only a few demonstrating poor quality. **Conclusions:** IP paclitaxel-based regimens demonstrate both safety and encouraging efficacy in the treatment of GC patients with PM. Subsequent conversion gastrectomy in eligible patients can further improve the survival outcomes.
Research Sponsor: None.

The incidence and risk of cardiovascular events among gastroesophageal cancer patients receiving immune-checkpoint inhibitors.

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Background: Immune checkpoint inhibitors (ICIs) have been associated with an increased risk of cardiovascular events. However, these data were mostly derived from patients with lung cancer or melanoma. To date, there is limited data describing the incidence and risk of cardiovascular adverse events associated with the use of ICIs among patients with gastroesophageal cancers. **Methods:** We performed a propensity score-matched cohort study using the TriNetX Analytics Network database, which comprises de-identified data from over 120 participating institutions and 101 million individuals. We defined the ICI cohort as patients who received any of the ICIs and chemotherapy. We defined the control cohort as patients who received only chemotherapy but not an ICI. We matched 2 cohorts based on predetermined variables including age, race, gastroesophageal cancer-directed therapy, cardiovascular agents, underlying comorbidities, and surgical history. The outcomes were myocarditis, pericarditis, myocardial infarction, heart failure, cerebral ischemia/infarction, atrial fibrillation, conduction disorders, and venous thromboembolism (VTE). All outcomes were captured within 1-year of the start of ICI therapy and identified using the International Classification of Diseases (ICD)-10 codes. **Results:** We identified 2005 patients who received ICI and chemotherapy (ICI cohort) and 14048 patients who received chemotherapy only (control cohort). We matched 952 patients in 2 cohorts. The mean age was 62.8 ± 12.7 and 62.8 ± 12.3 for the ICI and chemotherapy cohorts, respectively. We observed a significant increase in cardiovascular events in the ICI cohort compared to the chemotherapy cohort. In particular, the risk of myocarditis (4 cases in the ICI cohort and no cases in the chemotherapy cohort), pericarditis (Hazard ratio (HR), 2.53 [95% CI: 1.52-4.19]), conduction disorders (HR, 1.64 [95% CI: 1.04-2.59]), and VTE (HR, 1.56 [95% CI: 1.17-2.10]) were higher in the ICI cohort than the chemotherapy cohort. **Conclusions:** The use of ICI was associated with an increased risk of cardiovascular events, particularly myocarditis, pericarditis, conduction disorders, and VTE among patients with gastroesophageal cancer. Further studies are needed to identify strategies to stratify patients at high risk of developing ICI-associated cardiovascular events. Research Sponsor: None.

	ICI + Chemotherapy (ICI cohort)		Chemotherapy only (Control cohort)		
Outcomes	**Cases**	**At risk patients**	**Cases**	**At risk patients**	**Hazard ratio (95% CI)**
Myocarditis	4	952	0	952	-
Pericarditis	52	952	21	952	2.53 (1.52-4.19)
Myocardial infarction	18	952	12	952	1.49 (0.72-3.11)
Cerebral Ischemia/Infarction	22	952	17	952	1.30 (0.69-2.44)
Atrial fibrillation	56	952	48	952	1.16 (0.79-1.71)
Conduction disorder	49	952	30	952	1.64 (1.04-2.59)
Venous thromboembolism	112	952	73	952	1.56 (1.16-2.10)
Heart Failure	118	952	114	952	1.04 (0.80-1.34)

Envafolimab in combination with lenvatinib and albumin paclitaxel for previously treated advanced GEJ adenocarcinoma: Latest analysis of a phase II, two-cohort study.

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Background: The combination of first-line chemotherapy and PD-1 has emerged as a standard treatment, enhancing the efficacy in advanced gastric/gastroesophageal junction (GEJ) cancer. The efficacy of the combination of second-line chemotherapy and PD-1/PD-L1 remains unclear. This study aims to observe the efficacy of Envafolimab (PD-L1 inhibitor) plus tyrosine kinase inhibitors (TKIs) and second-line chemotherapy in the treatment of patients (pts) with advanced GEJ adenocarcinoma. **Methods:** Eligible pts with HER2-negative, microsatellite stable (MSS), advanced GEJ adenocarcinoma were enrolled in this study. Pts who never received PD-1 or PD-L1 inhibitors before would assign into Group A. Pts in Group B have failed first-line PD-1 antibody combined chemotherapy, with optimal efficacy as or above stable disease (SD). Pts in both groups received Envafolimab (200mg, hypodermic injection, days 1, 15, Q4W) combined with Lenvatinib (8mg/12mg, po, once a day) and albumin paclitaxel (100mg/m2, IV, days 1, 8 and 15, Q4W, up to 6 cycles) until disease progression, unacceptable toxicity, or pts refusal to continue treatment. The primary endpoint was objective response rate(ORR) per iRECISTv1.1, and the secondary endpoints included disease control rate (DCR), progression free survival (PFS), overall survival (OS) and safety. **Results:** From Oct 2022 to Apr 2024, 32 pts were enrolled. Both groups enrolled 16 pts equally, and 15 pts of each group were evaluable for efficacy analysis. The average ages were 60.07 ± 11.73 y and 50.13 ± 12.18 y in group A and B respectively. In group A, 7 pts had been tested for PD-L1 status, of which 2 (13.3%) was positive and 5 (33.3%) were negative. PD-L1 status was negative in 5(33.3%) pts, positive in 9 (60.0%), and untested in 1(6.7%) in group B. As of Sep 14, 2024, The ORR was 60.0% (9/15) in group A and 46.7% (7/15) in group B. The DCR was 100.0% (15/15) in both groups. After a median follow-up of 17.0 mo (IQR: 8.0-18.8) in group A, the mPFS was 8.2 mo (95% CI = 6.1-10.4), and the mOS was 14.8 mo(95% CI = 7.4-22.2). With median follow-up of 9.0 mo (IQR: 6.1-16.2) in group B, the mPFS was 5.9 mo (95% CI= 3.8-8.1) and the mOS was 11.5 mo (95% CI = 3.1-19.9). Overall incidences of adverse events (AEs) of any grade was 100% (30/30), and 80.0% (24/30) pts had ≥ grade 3 AEs during treatment. Most treatment-related adverse events (TRAEs) of grade ≥3 included decreased neutrophil count (73.3%), decreased leukocyte count (63.33%), Anemia (10%), febrile neutropenia (6.67%). **Conclusions:** In pts of advanced gastric cancer who have previously received PD-1 inhibitors or not, this second-line combination therapy demonstrates promising preliminary effects. Pts who have not previously received ICIs might benefit more from this combination therapy. Clinical trial information: NCT06030934. Research Sponsor: None.

Q-TWiST analysis of pembrolizumab or placebo plus chemotherapy for patients with advanced HER2-negative gastric or gastroesophageal junction (G/GEJ) cancer in KEYNOTE-859.

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Background: In the phase 3 KEYNOTE-859 study (NCT03675737; N = 1579), patients (pts) with locally advanced or metastatic HER2-negative G/GEJ cancer treated with first-line pembrolizumab (pembro) plus chemotherapy (chemo) had a significant and clinically meaningful improvement in OS with manageable toxicity vs placebo (pbo) plus chemo while maintaining HRQoL in the intention-to-treat and prespecified PD-L1 combined positive score (CPS) populations. We report data from an analysis of quality-adjusted time without symptoms of disease progression or toxicity (Q-TWiST). **Methods:** Q-TWiST is a measure of survival weighted by health states utility values. The analysis categorized survival time into 3 health states: time with grade ≥3 AEs (toxicity [TOX]) and time without symptoms or toxicities (TWiST)—both before disease progression or death—and relapse (REL), defined as time from disease progression to death. Q-TWiST was calculated as the restricted mean survival time (RMST) spent in each state, with a weighted health state utility for that state. Average utility weights were derived based on pooled EQ-5D-5L scores using the US mapping algorithm and standardized base case utility weights. Relative gains in Q-TWiST of ≥10% and ≥15% were "clinically important" and "clearly clinically important," respectively, as reported in the literature. Treatment difference 95% CIs were generated using the nonparametric bootstrapping method. Data are reported for all randomly assigned pts and pts with PD-L1 CPS ≥1 and CPS ≥10 tumors. The data cutoff date was August 22, 2023. **Results:** At the maximum follow-up of 56 mo for pembro plus chemo vs pbo plus chemo, RMST was 5.16 vs 2.86 mo for TOX, 7.77 vs 5.87 mo for TWiST, and 6.85 vs 7.12 mo for REL. Between-treatment differences are shown in the table. The difference in restricted mean Q-TWiST and relative gain favored pembro plus chemo. **Conclusions:** A clearly clinically important Q-TWiST gain was observed for pembro plus chemo vs pbo plus chemo in all pts with advanced HER2-negative G/GEJ cancer and pts with PD-L1 CPS ≥1 and CPS ≥10 tumors. Clinical trial information: NCT03675737. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

	All pts N = 1579	PD-L1 CPS ≥1 n = 1235	PD-L1 CPS ≥10 n = 553
RMST in TOX, mo (95% CI)	2.30 (1.28 to 3.44)	2.65 (1.46 to 3.99)	4.28 (2.08 to 6.41)
RMST in TWiST, mo (95% CI)	1.90 (−0.02 to 3.79)	2.28 (0.07 to 4.49)	3.72 (0.43 to 7.19)
RMST in REL, mo (95% CI)	−0.28 (−2.43 to 2.01)	−0.22 (−2.88 to 2.39)	−0.45 (−4.52 to 3.71)
Restricted mean Q-TWiST, mo (95% CI)[a]	3.31 (2.10 to 4.59)	3.98 (2.60 to 5.43)	6.45 (4.26 to 8.74)
Relative Q-TWiST gain, % (95% CI)[a]	20.90 (12.49 to 30.56)	25.34 (16.04 to 36.26)	38.05 (23.21 to 56.59)
Restricted mean Q-TWiST, mo (95% CI)[b]	2.91 (1.58 to 4.14)	3.50 (2.02 to 4.88)	5.63 (3.31 to 7.99)
Relative Q-TWiST gain, % (95% CI)[b]	18.38 (9.78 to 27.46)	22.27 (12.65 to 32.52)	33.19 (18.76 to 49.97)

[a]Based on US mapping algorithm.
[b]Based on standardized base case utility weights.

FOLFIRI-ramucirumab (FOLFIRI-Ram) versus ramucirumab-paclitaxel (Ram-Pac) in the second line (2L) for patients with advanced upper gastrointestinal (UGI) cancer: A real-world propensity-score matched analysis of survival.

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Background: The approach to 2L treatment of advanced UGI (gastric, esophageal, and gastroesophageal junction) cancers is guided primarily by the results of the RAINBOW trial, a global randomized phase III trial that demonstrated the benefit of Ram-Pac over paclitaxel. However, the onset of peripheral neuropathy limits its use in the palliative-intent setting. The RAMIRIS trial demonstrated the feasibility of 2L FOLFIRI-Ram compared to Ram-Pac, however it failed to meet the prespecified endpoint of ≥ 65% 6-month overall survival (OS). Despite this, FOLFIRI-Ram has emerged as an alternative 2L regimen and now endorsed in major society guidelines. Yet, comparative outcomes between FOLFIRI-Ram and Ram-Pac therapy in the real world are lacking. **Methods:** This study used the nationwide Flatiron Health electronic health record-derived deidentified database. During the study period (January 2011-June 2024), longitudinal patient-level data were derived from approximately 280 cancer clinics. Patients were eligible for inclusion if they had advanced unresectable or metastatic UGI cancer and received 2L treatment with either FOLFIRI-Ram or Ram-Pac. Demographics and lab values at the time of 2L initiation were extracted. Given anticipated imbalances in sample size, patients were matched 1:6 (FOLFIRI-Ram:Ram-Pac) for inclusion into the analytic dataset using a greedy match based on a logit model to predict propensity scores from key clinical and laboratory characteristics. The primary endpoint was OS from the start of 2L therapy, analyzed using the Kaplan-Meier method, log-rank test, and Cox proportional hazards model. A hybrid approach was used to construct a multivariate Cox model. **Results:** A total of 15,908 patients were included in the database, of whom 5189 received a 2L treatment. Of these, 631 patients received 2L Ram-Pac and 40 received 2L FOLFIRI-Ram. After matching, the final sample size was 240 patients who received Ram-Pac and 40 who received FOLFIRI-Ram. Baseline clinical and laboratory characteristics were well-balanced. The median OS from initiation of 2L FOLFIRI-Ram was 9.7 months (95% CI 6.9-12.3) and 7.6 months with 2L Ram-Pac (95% CI 6.3-9.2 months). The hazard ratio (HR) for death with FOLFIRI-Ram was 0.77 (95% CI 0.52-1.15, log-rank P=0.20) in reference to Ram-Pac. The adjusted HR death for FOLFIRI-Ram after adjustment for baseline albumin and white blood cell count was 0.87 (95% CI 0.58-1.30, p=0.49) in the final multivariate model. **Conclusions:** This real-world propensity-score matched analysis of patients with advanced UGI cancers receiving FOLFIRI-Ram or Ram-Pac supports the ongoing use of FOLFIRI-Ram in the 2L as an alternative to Ram-Pac. These data are consistent with results from the RAMIRIS study, however further prospective study is needed. Research Sponsor: Flatiron Health.

Association between adjuvant component of perioperative chemotherapy and survival outcomes in resected gastric cancer (GC).

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Background: Perioperative chemotherapy is the standard of care for resectable GC but many patients do not proceed to or complete adjuvant chemotherapy (AC). It remains unclear whether AC following neoadjuvant chemotherapy (NAC) improves survival. **Methods:** Patients with GC who received NAC and underwent curative-intent gastrectomy from 2006-2022 were identified from an institutional database. Patients with mismatch repair-deficient/microsatellite instability-high tumors, GEJ tumors, M1 disease, neoadjuvant radiotherapy, total neoadjuvant intent treatment, incomplete NAC (<66% of planned cycles), and incomplete data were excluded. A lymph node ratio (LNR) risk cutoff was determined using maximally selected rank statistics. Kaplan-Meier analysis with log-rank tests were used to compare overall survival (OS) and disease-free survival (DFS). Cox proportional hazards regression models were used to identify independent predictors of OS and DFS. All analyses were performed in R statistical software. **Results:** Of 340 patients included, 264 (78%) received AC. Most common reasons for incomplete AC were postoperative complications (20%), poor performance status (16%), minimal pathologic response to NAC (14%), patient choice (14%), and NAC toxicity (9%). Median follow-up was 6.4 years. In the overall cohort, there was no significant difference in OS (median 10 vs. 8.3 years; p=0.14) or DFS (7.0 vs. 7.5 years; p=0.12) between AC and non-AC groups. In node-negative (ypN-) patients (n=170), there was no association between receipt of AC and OS (12 vs. 10 years; p=0.6) or DFS (12 vs. 10 years; p=0.8). On univariable analysis, lack of AC receipt was not associated with shorter OS (p=0.3) or DFS (p=0.4). In the node-positive (ypN+) cohort (n=170), AC receipt was associated with significantly longer DFS (4.8 vs. 2.7 years; p=0.031) but not OS (5.5 vs. 2.8 years; p=0.071). However, on multivariate analysis, AC only showed a non-significant trend towards longer DFS (HR 0.77 [95% CI 0.49-1.19]; p=0.20). High-risk patients based on LNR (LNR>0.0684; n=112) who received AC (n=87; 78%) had longer OS (p=0.033) and DFS (p=0.014). No such differences were observed in the low risk LNR subgroup (LNR<0.0684; n=227). **Conclusions:** Following NAC, AC was not associated with improved survival in patients with ypN- GC. The association between AC and survival in ypN+ GC remains unclear but LNR may be used to identify patients with ypN+ disease who may benefit from AC. These observations require prospective validation but may have implications for clinical care and future trial design. Research Sponsor: None.

379 **Poster Session**

Incidence of mediastinal lymph node positivity in patients undergoing robotic transthoracic esophagectomy.

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Background: Esophageal cancer is increasing in incidence worldwide. Robotic approaches have quickly become mainstream due to the improved visualization and possible increase in nodal harvest. There remains significant debate on the utility of mediastinal lymph node dissection in esophageal cancer patients. We sought to analyze the incidence of mediastinal lymph node positivity and to evaluate its impact on survival in patients with esophageal cancer undergoing robotic assisted trans-thoracic esophagectomy (RATE). **Methods:** Utilizing a prospectively maintained esophageal database, we identified patients who were diagnosed with EC and underwent RATE between 2009-2024. We then identified patients with positive lymph nodes and stratified by location (celiac or mediastinal). Baseline univariate comparisons were made for continuous variables using both the Mann-Whitney U and Kruskal Wallis tests. Pearson's Chi-square test was used to compare categorical variables. Survival analysis was performed via Kaplain-Meier method. **Results:** We identified 201 patients who underwent RATE with a median age of 70 (41-91). There were 160 (79.6%) males and 41 (20.4%) females. The mean BMI was 26.5 ± 5.5 and mean ASA score was 3.1 ± 0.6. Adenocarcinoma was the predominate histology at 162(80.6%). Locally advanced T3/T4 tumors were identified in 53 (26.4%) patients and 63(31.3%) had clinical node + disease. Neoadjuvant therapy was utilized in 159(79.1%) patients. The median lymph nodes resected 18 (14 – 22), and length of hospitalization 8 (7 – 11) days. The 30 and 90-day mortality were 2.2% and 2.6% respectively. R0 resections was performed in 200 (99.5%) of patients and the pathologic complete response rates were 42.1%. The incidence of mediastinal lymph node positivewas 6.3% vs 76.3% abdominal. There were 17.6% of patients that had both mediastinal and abdominal lymph nodes positive. The median survival was 23 months for abdominal LN+, 19.5 months for mediastinal LN+ p=0.23. Multivariate analysis revealed that age, neoadjuvant radiation, nodal positivity were predictors of survival. **Conclusions:** The incidence of mediastinal positive lymph nodes are much higher than previously reported. Perhaps the improved visualization and manipulation of tissue with robotic approaches have enhanced our ability to resect these nodes. Mediastinal lymph node positivity is a poor prognostic factor for survival in patients undergoing robotic esophagectomy. Given these findings, we would continue to support aggressive nodal resection in patients diagnosed with esophageal cancer. Research Sponsor: None.

The prevalence of Lynch syndrome and associated clinicopathologic features in patients with mismatch repair deficient gastric cancer.

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Background: Approximately 2-4% of patients with primary colorectal cancer are estimated to be associated with Lynch syndrome. However, the prevalence of Lynch syndrome and associated clinicopathologic features in patients with mismatch repair-deficient (dMMR) gastric cancer (GC) is not well understood. In this study, we investigated the frequency of dMMR GCs in unselected GCs and the prevalence of Lynch syndrome, as well as associated clinicopathologic features and molecular alterations in dMMR GCs. **Methods:** A total of 847 cases of unselected GCs from 2010 to 2012 were collected at Jeju National University Hospital. Immunohistochemistry for mismatch repair proteins (MMR), including MLH1, MSH2, MSH6, and PMS2, and MSI analysis were performed on representative tumor sections of surgically resected specimens. Follow-up genetic counseling and genetic and/or epigenetic alterations of the MMR genes were also investigated. **Results:** Loss of one or more MMR proteins and microsatellite instability-high were observed in 72 patients; 67 patients showed loss of MLH1/PMS2, 2 patients showed loss of PMS2 alone, and 3 patients with MSH2/MSH6. Genetic counseling demonstrated that ten patients had a family history of LS-associated cancers and nine patients had a history of metasynchronous cancers. Of the 72 patients, 65 revealed hypermethylation of the promoter region of MLH1. The subsequent genetic germline tests demonstrated 2 patients with MLH1 mutation, 3 with MSH2 mutation, and 2 with PMS2 mutation. **Conclusions:** dMMR tumors were identified in 72 (8.5%) of 847 unselected primary GCs. This study estimated that 0.8% of patients with unselected GC and 9.7% of patients with dMMR GC fulfilled the revised Bethesda criteria and are associated with Lynch syndrome. The detection of dMMR GCs and necessary genetic counseling/tests based on revised Bethesda criteria would help find Lynch syndrome in patients with GC. Research Sponsor: None.

Clinical and immunophenotypical features of patients with dMMR GC.

Age	
≤ 50 years	5 (6.9)
> 50 years	67 (93.1)
Other malignancy history except GAC	
No	63 (87.5)
Yes	9 (12.5)
History of metachronous cancers	
Esophageal cancer	1 (1.4)
Maxillary sinus cancer	1 (1.4)
Papillary thyroid cancer	1 (1.4)
Colorectal cancer	1 (1.4)
Pancreatic cancer	1 (1.4)
Breast cancer	1 (1.4)
Breast cancer and malignant GIST	1 (1.4)
Multiple HCCs	2 (2.8)
History of Family with LS related cancers	
No	62 (86.1)
Yes	10 (13.9)
Fulfillment of the rBG	
No	63 (87.5)
Yes	9 (12.5)
MMR proteins	
Loss of MLH1 and PMS2	67 (93.1)
Loss of PMS2	2 (2.8)
Loss of MSH2 and MSH6	3 (4.2)
Germline mutation test	
MLH1 mutations	2 (2.8)
MSH2 mutations	3 (4.2)
PMS2 mutations	2 (2.8)

Effect of adjuvant radiotherapy after endoscopic resection of early esophagus cancer to prevent new intraluminal mucosal recurrence.

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Background: Endoscopic resection (ER) is effective for treating T1a early esophageal squamous cell carcinoma (ESCC). Occasional distant metastasis and recurrence are inevitable invasion in depth of muscularis mucosa(MM). Additional surgical resection is an effective means for cure and prevent recurrence, but it imposes a heavy physical burden. Chemoradiation therapy (CRT) is a comparable alternative to surgery. Not all patients will want those additional treatment. Another problem is the occurrence of new esophagus intraluminal mucosal recurrence. After endoscopic treatment, there is an increased risk of scarring and deformity. But there are few reports of the effectiveness of additional RT in suppressing new lesions. This study was designed to verify the efficacy of not only curability but also suppressive effect the occurrence of new lesions additional treatment of T1a(MM) ESCC. **Methods:** This study was a retrospective study. We reviewed 71 consecutive patients who underwent ER followed by clinical stage I (T1a: MM) ESCC. We compared patients in the radiotherapy group or non-radiotherapy group after ER. Patients of radiotherapy group received radiation within 3 months after ER. All patients underwent regular follow-up. Recurrence, recurrence-free survival, cancer-specific survival, overall survival, and complications were evaluated. **Results:** There were 34 cases of radiotherapy group and 37 cases of non-radiotherapy group. 3 patients (8.9 %) in the radiotherapy group experienced intraluminal mucosal recurrence, and 2 patients (6.6 %) experienced local recurrence. On the other hand, in the non-radiotherapy group, 9 patients (24.3 %) experienced intraluminal mucosal recurrence, and 7 patients (18.9 %) experienced local recurrence (intraluminal mucosal recurrence: $p< 0.05$, local recurrence $p=0.14$). The recurrent group received chemotherapy. The 3-year cumulative recurrence-free survival was 97 % in the radiotherapy group and 83.7 % in the non-radiotherapy group ($p< 0.05$). No severe radiation toxicities were recorded. **Conclusions:** The radiotherapy after ESD reduced the risk of lymph nodes recurrence not only local lymph node recurrence but also new intraluminal mucosal recurrence, and safe and effective therapeutic strategy. Research Sponsor: None.

382 **Poster Session**

Optical coherence tomography and elastography for the visualization of architecture and stiffness differences in gastric cancer and dysplasia.

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Background: Individuals at high risk for gastric cancer may be recommended to undergo risk-reducing gastrectomy, which leads to life-long morbidity. Surveillance with endoscopy can be an appropriate and effective option for avoiding unnecessary surgery. Gastric cancer is commonly characterized by changes in tissue architecture and increased tissue stiffness, but visualization of these alterations is challenging for white light endoscopy methods. We utilized optical coherence tomography (OCT), a non-destructive imaging modality that uses reflected near-infrared light to generate cross-sectional images of tissue, to visualize microstructure and relative tissue stiffness. **Methods:** We utilized two modes of OCT for visualization of *ex vivo* tissue samples collected from patients undergoing gastrectomy for gastric cancer, endoscopic mucosal resection, or endoscopic biopsy. Images were collected with a benchtop OCT system (Thorlabs Telesto TEL221). Structural OCT was used to visualize changes in tissue architecture including thickened mucosa and irregular put structure of the mucosa. Optical coherence elastography (OCE) was used to extract relative measures of tissue stiffness using an optical palpation method. A phantom of known mechanical properties was placed on the tissue and an axial force was applied. The resulting deformation (strain) of the phantom provided relative measurements of tissue stiffness. Classification of normal and cancerous tissue was confirmed by histology. **Results:** Structural OCT imaged architectural features of tissue with high resolution (<10 um) and depth of imaging up to 1.5 mm. Strain maps generated with OCE data showed that normal gastric tissue had relatively consistent stiffness, whereas stomach cancer and dysplasia exhibited increased tissue stiffness compared to normal. **Conclusions:** Structural OCT and OCE captured changes in tissue architecture and increased stiffness in patient samples of gastric cancer and dysplasia. These tissue features are classic characteristics of cancer, making this technique a useful screening tool for several cancer types. Future work aims to implement endoscopic OCT and OCE to detect cancer at early stages *in vivo*. Research Sponsor: University of Arizona International Research Grant.

383 **Poster Session**

Young onset gastrointestinal cancer: A needs analysis.

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Background: The global incidence of Young Onset (YO) cancers diagnosed in adults under 50 years, is increasing for unknown reasons. YO Gastrointestinal (GI) cancers, which account for approximately 27% of newly diagnosed cancer cases and 37% of cancer deaths present distinct challenges in areas including sexual health, finance, career and family. We assessed the holistic needs of these patients through a mixed methods approach. **Methods:** A single institution needs analysis study was undertaken in YO GI patients diagnosed 2014-2024 through an anonymous survey and Focus Group Discussions (FGD). The surveys asked about their experiences of sexual health and function, psychosocial concerns, financial concerns, career developments and when in their cancer journey would they prefer to speak to a health care professional about these. FGD's used nominal group technique to deepen the analysis. **Results:** 88 participants responded via survey and 10 participated in a FGD. The average age at diagnosis was 43 years. 80% of participants reported they would have benefitted from a conversation about fertility, sexual health/function (82%) and financial supports (98%) at diagnosis. 93% of patients were out of work during cancer treatment with average length of time > 1 year. **Conclusions:** This needs analysis highlights the importance of specialised clinical pathways for young onset GI cancer patients focusing on these unique and complex needs. Financial supports, conversations regarding sexual health/function, fertility preservation and psychosocial support are critical areas requiring structured intervention. Research Sponsor: None.

2014-2024 young onset SJH.

	UGI (N=233)	Colorectal (N=275)
Clinical Stage at diagnosis	Oeso Stage 4 = 39% Gastric stage 4 = 26%	Stage 3= 46% Stage 4 = 20%
Treatment intent %	Radical =54% Palliative =38% Radical/Palliative = 8%	Radical =75% Palliative 25%
Mortality rate	66%	25%
GI Survey results (N=236)	Support offered Y/N	Preferred timing of support
Sexual health/ function	No=85% Yes =15%	82% At diagnosis
Financial support	No=95% Yes= 5%	98% At diagnosis
Fertility preservation	No = 65% Yes=35%	80% At diagnosis

Clinical utility of nurses' telephone follow-up program (Tsukiji Call) for resectable locally advanced esophageal squamous cell carcinoma patients received neo-adjuvant DCF therapy.

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Background: The standard neoadjuvant treatment in Japan for resectable locally advanced esophageal squamous cell carcinoma (ESCC) is docetaxel (DTX) + cisplatin (CDDP) + 5-fluorouracil (5-FU) (DCF) based on the results of the JCOG1109 study. Neoadjuvant DCF therapy is administrated triweekly; in the first week in a hospital, and the remaining two weeks at hom. After the administration of neoadjuvant DCF therapy, patients might suffer from febrile neutropenia or gastrointestinal symptoms such as nausea, vomiting, constipation, diarrhea, and stomatitis even during remaining two weeks at home. Therefore, nurses in our hospital teach self-care at home for these patients who received neoadjuvant DCF therapy. However, some patients experienced emergency admission due to worsening adverse events. To prevent these events, nurses in our hospital conducted a telephone follow-up program (Tsukiji Call) for patients at home. **Methods:** This retrospective analysis included locally advanced ESCC patients who received neoadjuvant DCF therapy in our hospital between Feb 2023 and Jul 2024. Three courses of DCF (DTX [day 1]: 70 mg/m2, CDDP [day 1]: 70 mg/m2, 5-FU [continuous infusion on days 1-5]: 750 mg/m2, every three weeks) were planned as neoadjuvant treatment. On days 8-14 of the first course of neoadjuvant DCF therapy, Tsukiji Call was conducted by nurses in our hospital to check fever and nausea, vomiting, constipation, diarrhea, and stomatitis based on the CTCAE ver. 5.0 for patients at home before the first outpatient clinic. Eligible patients were divided into the Tsukiji Call and non-Tsukiji Call groups. We used Fisher's exact test to evaluate the frequency of grade 2 (Gr.2) or more adverse events (AEs) between two groups at the first outpatient clinic. **Results:** This study identified 90 patients (Tsukiji Call: 20, non-Tsukiji Call: 70). The median age (range) and gender male/female, PS 0/1/2, cStage I/II/III/IVA/IVB were 66.5 years (45-76 years) and 15 (75%)/5(25%), 19(95%)/1(5%)/0, 1(5%)/4(20%)/11(55%)/1(5%)/3(15%) in Tsukiji Call group, and were 64 years (46-79 years) and 49(70%)/21(30%), 67(95%)/2(2%)/1(1%), 1(1%)/13(18%)/50(71%)/0/6(8%) in non-Tsukiji Call group, respectively. In the first outpatient clinic, the frequency of Gr.2 or more AEs on fever, nausea, vomiting, constipation, diarrhea, stomatitis and any events between in Tsukiji Call and non-Tsukiji Call groups were 0/0(p = 1), 2(10%)/1(1%) (p = 0.123), 0/0 (p = 1), 0/0 (p = 1), 0/4(5%) (p = 0.572), and 0/10(14%) (p = 0.109), 2(10%)/15(21%) (p = 0.342), respectively. One patient in each group was emergently admitted within 7 days after discharge. **Conclusions:** Non-hematologic toxicities during DCF therapy may be reduced by Tsukiji Call by nurses in the hospital, which seemed to be in preventing the worsening of adverse events. Research Sponsor: None.

385 **Poster Session**

Precision medicine and multidisciplinary care in gastric and gastroesophageal junction (G/GEJ) cancers: Challenges and practice gaps in community cancer clinics.

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Background: The advent of biomarker-driven therapeutic strategies brings increasing precision to the treatment of G/GEJ cancer, placing biomarker testing at the forefront of patient management. The management of evolving logistics for testing and related decision-making can be challenging, especially in community clinics. As part of a multi-center quality improvement initiative, we evaluated current gaps in biomarker testing and practice patterns including multidisciplinary care for G/GEJ cancers in community oncology centers. **Methods:** Between 5/2024 and 8/2024, oncology team members (N = 60) from 5 US-based community oncology practices completed surveys assessing attitudes, knowledge, competence, practice patterns, challenges, and gaps regarding biomarker testing and multidisciplinary care in G/GEJ cancers. **Results:** Despite practice guidelines, many patients with G/GEJ cancer did not receive testing for recommended biomarkers. While most providers report that biomarkers HER2 (67%) and PD-L1 (62%) are routinely reported in pathology reports, less than half said testing for MSI/MMR, TMB, and BRAF p.V600E were reported, and less than 25% reported analysis of *NTRK1/2/3*, *RET*, CLDN.18.2, and *FGFR2*. Biomarkers most frequently requested if not included in the original pathology work-up were PD-L1 (48%), TMB (32%), and BRAF p.V600E (32%). Top challenges reported by providers in integrating biomarker testing into practice were *prioritizing testing for limited tissue* (38%), *determining whether to begin treatment prior to receipt of molecular testing results* (35%), and *optimizing specimen collection for molecular testing* (27%). Only 23% of providers reported feeling very or extremely comfortable (4/5 on 5-point Likert scale) with interpreting and applying biomarker testing results to treatment decision-making. Additional challenges included turnaround time for biomarker testing, uncertainty interpreting biomarker results, and inconsistent or infrequent multidisciplinary tumor boards. Indeed, some providers (12%) reported not participating in multidisciplinary tumor boards at all, while others reported participating weekly (16%), once or twice a month (21%), every other month (12%), or less frequently (23%). Providers believed *improving test ordering process* (47%), *decreasing testing turnaround time* (42%), and *standardizing panels for biomarker testing* (40%) would be most impactful in implementation of biomarker testing into clinical workflows. **Conclusions:** Although most oncology teams reported use of some form of biomarker testing in treatment of G/GEJ cancer, testing was inconsistent and not as comprehensive as currently recommended. These data will be leveraged to develop multidisciplinary action plans with participating clinics to improve biomarker testing for treatment selection in G/GEJ cancers. Research Sponsor: Merck Sharp & Dohme LLC.

Development and validation of machine learning model for mortality after curative gastrectomy for gastric cancer.

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Background: The risk for postoperative complications can significantly impact perioperative planning. The aim of this study is to develop and validate a model based on various machine learning techniques to predict postoperative mortality (PM) after curative radical gastrectomy for gastric cancer. **Methods:** Data from a nationwide survey led by the Korean Gastric Cancer Association was used for the development set, which included 12,722 patients who underwent gastrectomy in 2019 across 68 institutions in South Korea. The validation set comprised 4,887 patients from 8 institutions between 2009 and 2017. PM was defined as all-cause death within 30 days after surgery or in-hospital death beyond 30 days. Five machine learning models (CatBoost, Gradient Boosting, Light GBM, Random Forest, XGBoost) were evaluated using perioperative clinical characteristics. Model performance was assessed using the area under the receiver operation characteristic curve (AUROC) with 5-fold cross-validation. **Results:** The PM rates were 0.83% and 1.04% in the development and validation set, respectively. In the development set, the Random Forest model demonstrated the highest discrimination capacity with an AUROC of 0.772, followed by Gradient Boosting at 0.767, CatBoost at 0.742, XGBoost at 0.732, and LightGBM at 0.717. However, in the validation set, CaBoost showed the highest AUROC at 0.747, while the AUROC for Random Forest decreased to 0.688. **Conclusions:** The CatBoost model demonstrated robust and consistent performance in predicting PM risk across both development and validation datasets. To further enhance the performance of the model and ensure stable clinical application, future research should focus on developing methods for quantitative measurement of perioperative factors such as surgical techniques. Research Sponsor: Korean Gastric Cancer Association; KGCA2022IC5.

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Outcome of radiation related esophageal stricture from endoscopic dilation.

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Background: Esophageal strictures present a significant challenge, particularly in cases of radiation-induced strictures. Esophageal dilation is a commonly used intervention aimed at improving swallowing function and nutritional status. However, radiation-induced strictures are amongst the most refractory to dilation. Understanding the success and limitations of this procedure is crucial for optimizing patient outcomes in the cancer population. **Methods:** This was a single-center retrospective chart review of 736 patients who received radiation as well as an EGD between 2014 and 2024. Patients without a radiation-induced stricture that was dilated were excluded. All continuous data are summarized as medians with interquartile ranges (IQR). Independent t-test was used for comparison among different groups. **Results:** 192 were eligible for this study. Our population was mostly male (58%) with a median age of 66 (IQR 67-74). The leading cancer diagnosis were esophageal (37%), head and neck (30.2%), and lung (17.1%). Radiation targets were head and neck (50.5%), lung (45.3%), both (1.6%), or other (1.6%). Patients were receiving active chemotherapy in 9.9% of cases. Mortality by the end of the study period was 51%. The median radiation dose was 5040 cGy (IQR 5040-6600) in a median of 28 fractions (IQR 26-33). Common indications for dilation were dysphagia (91.1%), weight loss (21.9%), and tube feed dependency (17.2%). Strictures were most often dilated 2 to 4 times (39.6%) to a median dilation size of 14 mm (IQR 12-15). Of note, 18.2% of patients needed 8 or more dilations. Dilation was associated with weight gain in 46.4% of patients, PEG tube removal in 3.1%, both forms of improvement in 9.9%, and no change in 40.6%. Complications included perforation (3.6%), aspiration pneumonia (3.6%), chest pain (0.5%), and hospitalization (0.5%). No significant differences were found in maximum dilation size, number of dilations, initial luminal diameter of the stricture, and initial length of the stricture between those who had clinical improvement as compared to those who did not. **Conclusions:** Our study reported 59.4% clinical improvement from esophageal dilation for radiation-induced strictures with nearly 20% of patients needing 8 or more dilations. The stricture dilation parameters did not appear to affect the success rate. This is the largest known single study of esophageal dilation for radiation-induced strictures. Research Sponsor: None.

The impact of nutritional status on recurrence-free survival (RFS) in patients with locally advanced oesophagogastric adenocarcinoma (LA-OGA) treated with FLOT therapy.

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Background: Malnutrition and weight loss were reported as poor prognostic factors in operable OGA. This study examines the relationship between oncological outcomes and nutritional status in LA-OGA patients under the current standard of care, perioperative FLOT therapy. **Methods:** From 2017 to 2023, 423 patients who had radical resection at Royal Marsden Hospital were screened. Inclusion criteria included LA-OGA (cT2≤ and Nany or Tany and N+), at least one cycle of neoadjuvant chemotherapy (NAC), and ECOG PS 0–2. Nutritional status was assessed by body weight and prognostic nutritional index (PNI), with significant weight loss defined as a 5% loss within six months before diagnosis or during NAC. PNI was calculated as serum albumin (g/L) + 0.005 × lymphocyte count /μL, with categories of severe (<45), mild to moderate (45–49.9), and normal (≥50). The primary endpoint was 3-year recurrence-free survival (3y-RFS). Restricted mean survival time (RMST) at 36 months was calculated to analyze survival time. Multivariate analyses were performed to identify prognostic factors for RFS. The pathological response was assessed using tumour regression grade (TRG) (Mandard criteria). **Results:** Of the 423 patients, 210 met the inclusion criteria. Primary tumour sites were the oesophagus (52.4%), the oesophagogastric junction (18.1%), and the stomach (29.5%). The median follow-up was 26.5 months. Weight loss at baseline and after NAC, as well as PNI, did not significantly impact RFS (p=0.13, p=0.91, p=0.69, p=0.45). However, a PNI decrease from baseline was associated with significantly shorter 3y-RFS (46% vs. 69%, p<0.001), and the RMST difference was 5.46 months (p<0.001). Multivariable analyses showed ypN positivity (p<0.001), R1 resection (p<0.001), and PNI decrease (p=0.03) as negative prognostic factors for RFS. Pathological response did not differ regardless of PNI change (28.2% vs. 30.4%, p=0.75). **Conclusions:** A decrease of PNI is a significant poor prognostic factor for RFS in LA-OGA patients undergoing FLOT, suggesting that maintaining nutritional status during NAC could enhance survival outcomes. Research Sponsor: None.

Assessment of nutritional status in elderly patients with gastroesophageal cancer: A post hoc analysis.

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Background: Elderly patients with gastroesophageal cancer (GEC) frequently suffer from malnutrition due to tumor location and treatment-related toxicity. Malnutrition is associated with increased morbidity and mortality in this population. This study investigates the relationship between body mass index (BMI), Mini Nutritional Assessment (MNA) scores, unintentional weight loss (UWL), and chemotherapy toxicity while identifying key factors influencing nutritional status. **Methods:** This post hoc analysis utilized data from a prospective, multicenter study of patients aged 65 and older with all stages of GEC undergoing chemotherapy. Nutritional assessments were conducted using validated tools, focusing on BMI, MNA, and UWL over three months post enrollment. We defined malnourished patients as those with abnormal BMI (<18.5 or >30), low MNA (0-11), and significant UWL (≥10%). We analyzed associations between nutritional status and geriatric assessment comorbidities, depression score, functional status, cognition, hospitalization, treatment toxicity, and social support using Fisher's exact tests. **Results:** Eighty-two patients were enrolled, with a median age of 73 years (range 65-91 years). The cohort was predominantly male (74%) and primarily presented with stage III-IV disease (82%), most receiving first-line therapy (79%). Primary sites included gastric (32%), esophageal (43%), and gastroesophageal junction (26%). Notably, 34% of patients had abnormal BMI, with 2 patients classified as underweight (BMI <18.5) and 19 as obese (BMI >30). Approximately 69% reported UWL, of 10% or greater. MNA scores indicated malnutrition in about 66% of patients (severe (0-7): 15 patients; moderate (8-11): 37 patients), while 31% scored within the normal range (12-14).Hospital admissions were significantly higher in patients with low MNA (33% vs. 3%; p=0.002) and UWL (29% vs. 8%; p=0.047). No significant correlations were observed between nutritional status and treatment type, treatment toxicity, or other geriatric assessment variables, including functional status and social support. **Conclusions:** MNA and UWL are effective indicators of nutritional status in elderly GEC patients, while BMI is not a reliable measure. Malnutrition significantly correlates with increased hospital admissions, highlighting the need for reliable tools to measure malnutrition to implement necessary interventions. Further research is required to explore the broader implications of malnutrition on treatment outcomes and quality of life in this vulnerable population. Research Sponsor: None.

An exploratory phase II trial of low-dose decitabine and sintilimab combined with chemotherapy in HER2-negative gastric or gastroesophageal junction adenocarcinoma.

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Background: Combining epigenetic modulator with PD-1 inhibitor may enhance treatment outcomes in cancer patients. This study aimed to evaluate the efficacy and safety of epigenetic drug decitabine, sintilimab (a PD-1 inhibitor), and chemotherapy in gastric cancer. **Methods:** In this single-arm, exploratory phase II trial, patients with HER2-negative, locally advanced or metastatic gastric or gastroesophageal junction (G/GEJ) adenocarcinoma were enrolled between August 12, 2021, and October 22, 2022. Treatment included low-dose decitabine (10 mg/m^2 for 5 days), sintilimab (a PD1 inhibitor, 200 mg every 21 days), and XELOX. The primary endpoint was objective response rate (ORR) and second endpoints included disease control rate (DCR), progression-free survival (PFS), overall survival (OS), and safety. The protocol was approved by the ethics committee of the Changzhi Medical College Affiliated Changzhi People's Hospital (No. 2021K002). All patients provided written informed consent form before study participation. The trial is registered with Chinese Clinical Trial, number ChiCTR2100043312. **Results:** Sixteen patients were initially enrolled, with a median follow-up of 14.8 months. Eleven patients exhibited a PD-L1 combined positive score (CPS) < 5. In the per-protocol sets, 11 patients remained eligible for efficacy analysis. In the intention-to-treat analysis of all 16 patients, 9 achieved partial response and 2 had stable disease, resulting in an ORR of 69% and a DCR of 56%. As of the data cutoff on November 4, 2023, the median PFS was 11.1 months (95% CI, 5.1-not reached) and the median OS was 13.2 months (95% CI, 5.1-not reached). Treatment-related adverse events were generally well-tolerated, with most being grade 1-2. Three patients experienced severe AEs, including one fatal case of grade 4 immune-related pneumonia. **Conclusions:** The combination of low-dose decitabine, sintilimab, and chemotherapy appears promising strategy for locally advanced or metastatic G/GEJ adenocarcinoma, especially in patients with negative HER2 and low PD-L1 expression. These preliminary findings support further investigations to confirm the benefits and explore the underlying mechanisms. Clinical trial information: ChiCTR2100043312. Research Sponsor: Shanxi Provincial Health and Wellness Committee Research Project; 2020129.

Objective response and disease response.

	Full analysis set (n=16)	Efficacy analysis set (n=11)
Complete response	0	0
Partial response	9	9
Stable disease	2	2
Progressive disease	0	0
Non-evaluable*	5	NA
Objective response	9 (56%)	9 (81.8%)
Disease control	11 (69%)	11 (100%)

NA=not applicable.
*Included patients without any post-baseline response assessment.

Is treatment delivery of perioperative FLOT therapy in locally advanced OGA associated with recurrence-free survival?

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Background: Although perioperative FLOT therapy is a standard for locally advanced oesophagogastric adenocarcinoma (LA-OGA), completing all cycles, especially in the adjuvant chemotherapy (ACT) phase, is often challenging. This study examined the survival impact of treatment delivery in LA-OGA patients receiving FLOT therapy. **Methods:** 423 patients who underwent radical resection at The Royal Marsden Hospital from 2017 to 2023 were screened. Major inclusion criteria included patients with LA-OGA (cT2≤ and Nany or Tany and N+), treated with FLOT (at least one cycle of neoadjuvant chemotherapy) and radical resection, with an ECOG performance status of 0–2. Patients were divided into therapy incomplete (Tx Incomp, less than eight cycles of FLOT) and therapy (Tx comp) groups. We performed univariate and multivariate analyses, as well as propensity score matching (PSM) analysis, to evaluate whether treatment delivery was associated with recurrence-free survival (RFS) and overall survival (OS). Kaplan-Meier curves and restricted mean survival times (RMST) up to 36 months were used to evaluate the survival time. The primary endpoint was the 3-year RFS and OS rate. **Results:** Of the screened patients, 210 met the inclusion criteria, with 79 (38%) in the Tx Incomp and 131 (62%) in the Tx comp group, and 50 (24%) patients did not receive ACT. The median follow-up time was 26.5 months. The 3-year RFS and OS rates in the Tx Incomp and Tx comp groups after PSM were 54% vs 59% (p=0.14) (RMST difference: 4.04 months, p=0.09) and 58% vs 78% (p=0.04) (RMST difference: 6.15 months, p<0.001), respectively. Multivariable analysis showed a significant impact of Tx comp on OS (HR 0.53, 95% CI: 0.32-0.88). In the subgroup of ypN-positive patients, a significant difference of RMST in OS rate was observed between those who started ACT and those who did not start ACT (RMST difference: 7.89 months, p=0.01), whereas no significant difference was found in the ypN-negative group (RMST difference: 0.65 months, p=0.75). **Conclusions:** This study suggests that completing FLOT therapy is associated with better OS. The impact of ACT might differ according to ypN stage. Research Sponsor: None.

First-line (1L) nivolumab (NIVO) plus chemotherapy (chemo) vs chemo in patients (pts) with advanced gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma (GC/GEJC/EAC): 5-year (y) follow-up of Chinese pts from CheckMate 649.

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Background: NIVO + chemo demonstrated clinically meaningful survival benefit and an acceptable safety profile vs chemo in previously untreated Chinese pts with advanced GC/GEJC/EAC from CheckMate 649, consistent with the overall study population. 1L NIVO + chemo is currently approved for pts with advanced non-HER2+ GC/GEJC/EAC in China and other countries. We report 5-y results of NIVO + chemo vs chemo in Chinese pts from CheckMate 649. **Methods:** Adults with previously untreated, unresectable advanced or metastatic, non-HER2+ GC/GEJC/EAC were enrolled regardless of programmed death ligand 1 (PD-L1) expression. Randomized pts received NIVO + chemo, NIVO + ipilimumab, or chemo. Dual primary endpoints for NIVO + chemo vs chemo were overall survival (OS) and progression-free survival (PFS) by blinded independent central review (BICR) in pts with PD-L1 combined positive score (CPS) ≥ 5. **Results:** 208 Chinese pts were randomized to NIVO + chemo or chemo. At 61-month (mo) minimum follow-up, NIVO + chemo continued to demonstrate OS and PFS benefit vs chemo in pts with PD-L1 CPS ≥ 5 and all randomized pts (Table). 5-y OS rate was 24% with NIVO + chemo vs 8% with chemo in pts with PD-L1 CPS ≥ 5 and 20% vs 7% in all randomized pts. Objective response rate (ORR) was higher and responses were more durable with NIVO + chemo vs chemo in pts with PD-L1 CPS ≥ 5 and all randomized pts (Table). No new safety signals were identified with longer follow-up. **Conclusions:** NIVO + chemo continued to demonstrate clinically meaningful long-term survival benefit, more durable responses, and acceptable safety vs chemo in Chinese pts after 5 y of follow-up, consistent with earlier reports and with the overall study population of pts with advanced non-HER2+ GC/GEJC/EAC. These results further support NIVO + chemo as a standard 1L treatment option for Chinese pts. Clinical trial information: NCT02872116. Research Sponsor: Bristol Myers Squibb.

	PD-L1 CPS ≥ 5		All randomized	
Efficacy	**NIVO + chemo (n = 75)**	**Chemo (n = 81)**	**NIVO + chemo (n = 99)**	**Chemo (n = 109)**
mOS (95% CI), mo	15.5 (11.9–21.1)	9.6 (8.0–12.1)	14.3 (11.5–16.5)	10.3 (8.1–12.1)
HR (95% CI)	0.57 (0.40–0.82)		0.63 (0.46–0.85)	
mPFS[a] (95% CI), mo	8.5 (6.0–14.0)	4.3 (4.1–6.5)	8.3 (6.2–12.4)	5.6 (4.2–6.8)
HR (95% CI)	0.51 (0.34–0.76)		0.57 (0.41–0.80)	
ORR[a,b] (95% CI), %	68 (56–79)	48 (36–60)	66 (55–76)	45 (35–56)
mDOR[a,c] (95% CI), mo	12.5 (7.2–23.4)	6.9 (3.9–8.5)	12.5 (7.2–17.7)	5.6 (4.4–8.3)

[a]Per BICR.
[b]In pts with measurable target lesions at baseline.
[c]In responders. DOR, duration of response; m, median.

A randomized phase II trial of systemic chemotherapy with or without trastuzumab followed by surgery in HER2 positive advanced gastric or esophagogastric junction adenocarcinoma with extensive lymph node metastasis: 3-year follow-up data of the Japan Clinical Oncology Group study JCOG1301C (Trigger study).

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Background: We previously reported better objective and pathological response with the addition of trastuzumab to preoperative chemotherapy in the randomized phase II trial comparing the efficacy of systemic chemotherapy with or without trastuzumab followed by surgery in HER2 positive advanced gastric or esophagogastric junction adenocarcinoma with extensive lymph node metastasis. Here, we report 3-year follow-up data. **Methods:** Eligible patients with HER2-positive gastric cancer having extensive lymph node metastasis were randomized to receive preoperative chemotherapy with S-1/cisplatin (SP: S-1 80–120 mg/body on days 1–14, and cisplatin 60 mg/m2 on day 1, q21 days, 3-4 courses) in arm A or SP plus trastuzumab (first course 8 mg/kg followed by 6 mg/kg, day 1 on each course) in arm B. After gastrectomy, adjuvant chemotherapy with S-1 was added for 1 year in both arms. The primary endpoint was overall survival (OS), and the sample size was planned to be 130 patients in total, expecting a 10% increase in the 3-year OS (70% vs 80%) with a 1-sided alpha of 0.2, a power of 0.75. **Results:** The study was terminated in March 2021 due to slow patient accrual. In total, 46 patients were randomized to either arm A (22 patients) or arm B (24 patients). Patient characteristics were well balanced between the arms. Planned preoperative chemotherapy was completed in 20 patients (90.9%) in arm A and in 23 patients (95.8%) in arm B. Objective response rate tended to be higher in arm B (66.7% [16/24]) than in arm A (36.4% [8/22], p=0.08), and R0 resection was achieved in 20 patients (90.9%) in arm A and 22 patients (91.7%) in arm B. The adjuvant chemotherapy was completed in 54.6% (12/22) in arm A and in 54.2% (13/24) in arm B. With the median observation period of 4.1 years, the 3-year OS was 68.2% in arm A and was 78.9% in arm B with a hazard ratio of 0.698 (95% confidence interval (CI), 0.221-2.209). The 3-year progression-free survival of arm A and arm B was 39.3% and 62.3%, respectively, with a hazard ration of 0.562 (95% CI, 0.229-1.377). **Conclusions:** Preoperative chemotherapy with SP plus trastuzumab was feasible, and tended to improve the objective response rate, OS and progression-free survival. An integrated analysis with other clinical trials evaluating perioperative trastuzumab is being planned. Clinical trial information: s031180006.
Research Sponsor: None.

Alternative dosing regimens of tislelizumab proposed using modeling and simulation.

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Background: Tislelizumab (TIS; BGB-A317) is approved for the treatment of multiple solid tumors, administered at 200 mg every 3 weeks (Q3W), and has demonstrated a flat exposure–response relationship across a wide range of doses. We evaluated alternative dosing regimens of TIS at 150 mg every 2 weeks (Q2W), 300 mg every 4 weeks (Q4W), and 400 mg every 6 weeks (Q6W) using a model-based approach with the aim of alleviating patient burden by providing longer dosing intervals and/or treatment flexibility compatible with background chemotherapy to meet the needs of patients and healthcare practitioners. **Methods:** A previously developed population pharmacokinetic (PK) model was used for simulating PK exposure of the alternative regimens, which were selected by exposure-matching to the reference dose of 200 mg Q3W. PK-based criteria (peak concentration [C_{max}] within 25% and trough concentration [C_{trough}] within 20% of the reference dose) were also used. Alternative dosing regimen exposures in the first least common time interval and at steady state were compared with the reference. Deviations from PK-based criteria were bridged using appropriate safety and efficacy references and exposure–response analyses using data from four phase 1, 2, and 3 clinical trials of TIS in patients with solid tumors, including gastric cancer and esophageal squamous cell carcinoma. **Results:** Simulations at steady state shown here demonstrate that the TIS alternative dosing regimens of 150 mg Q2W, 300 mg Q4W, and 400 mg Q6W produce comparable exposures to the 200 mg Q3W reference regimen. Although the simulated C_{max} at 300 mg Q4W and 400 mg Q6W were higher than with the 200 mg Q3W reference dose, these were below the C_{max} of the 5 mg/kg Q3W safety reference (Table). And while the C_{trough} for the 400 mg Q6W dosing regimen was lower than with the 200 mg Q3W reference dose, it was 10.7% higher compared with the 2 mg/kg efficacy reference dose; therefore, it was within the concentration range where a flat exposure–efficacy relationship of TIS has previously been established. **Conclusions:** TIS alternative dosing regimens of 150 mg Q2W, 300 mg Q4W, and 400 mg Q6W are expected to result in similar safety and efficacy profiles as the 200 mg Q3W reference dosing regimen and may be used interchangeably for indications where 200 mg Q3W is approved. Clinical trial information: NCT04716634, NCT05014828, NCT04379635, NCT02407990, NCT04068519, NCT03430843 and NCT03358875. Research Sponsor: BeiGene Ltd.

Steady-state PK exposure metrics for alternative TIS doses vs reference doses.

Parameter	150 mg Q2W vs 200 mg Q3W	300 mg Q4W vs 200 mg Q3W	300 mg Q4W vs 5 mg/kg Q3W[a]	400 mg Q6W vs 200 mg Q3W	400 mg Q6W vs 5 mg/kg Q3W[a]	400 mg Q6W vs 2 mg/kg Q3W[b]
C_{max}	−5.8 (Yes)	31.4 (No)	−18.7 (Yes)	52.2 (No)	−5.8 (Yes)	–
$C_{average}$	12.3 (Yes)	12.6 (Yes)	–	0.4 (Yes)	–	–
C_{trough}	27.2 (Yes)	0.1 (Yes)	–	−28.4 (No)	–	10.7 (Yes)

Data are presented as difference (% = [$GM_{test} - GM_{reference}/GM_{reference}$] × 100) vs the 200 mg Q3W reference, [a]safety reference, or [b]efficacy reference (meets regulatory criteria, Yes/No).
$C_{average}$, average concentration; GM, geometric mean.

Does chemotherapy regimen matter for first-line immunochemotherapy in low programmed cell death ligand 1 (PD-L1)-expressing esophageal squamous cell carcinoma (ESCC)? A systemic review and meta-analysis.

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Background: Anti-programmed death 1 (PD-1) therapy plus chemotherapy has become standard first-line therapy for patients with high PD-L1-expressing advanced ESCC. The benefit for patients with low PD-L1-expressing ESCC remains debatable. **Methods:** Eligible studies were phase III trials investigating anti-PD-1/PD-L1 therapy plus chemotherapy (immunochemotherapy) vs chemotherapy as first-line therapy of advanced ESCC. Study-level pooled analyses of hazard ratios (HR) for progression-free survival (PFS) and overall survival (OS) with corresponding 95% confidence intervals (CI), were used to compare the different groups in random-effects model. PD-L1 expression levels with tumor proportion score (TPS) $\geq$ 1% vs $<$ 1%, tumor area positivity (TAP) $\geq$ 10% vs $<$ 10%, and combined positive score (CPS) $\geq$ 10 vs $<$ 10 were collectively categorized as high vs low PD-L1 expression. The impact of two different chemotherapy regimens, TP (paclitaxel plus platinum) and PF (platinum pus fluoropyrimidine), stratified by PD-L1 expression levels, on the survival benefit of immunochemotherapy vs chemotherapy was investigated. **Results:** Eight randomized controlled trials including 4733 subjects were enrolled. Immunochemotherapy vs chemotherapy was associated with significantly improved PFS and OS irrespective of PD-L1 expression levels or chemotherapy regimens (Table). The PFS and OS benefits were more pronounced in high PD-L1 group than low PD-L1 group. In patients with high PD-L1 expression, the PFS and OS benefits were not significantly different between two chemotherapy regimen subgroups. However, in patients with low PD-L1 expression, the PFS benefit was more significant in TP subgroup than PF subgroup, and OS benefit trended to be in favor of TP subgroup. **Conclusions:** The addition of anti-PD-1/PD-L1 therapy to TP chemotherapy may associate with more significant survival benefit than to PF chemotherapy in advanced ESCC with low PD-L1 expression. Research Sponsor: None.

	PFS (immunochemotherapy vs chemotherapy)		OS (immunochemotherapy vs chemotherapy)	
Population	**HR (95% CI)**	**P value (subgroup differences)**	**HR (95% CI)**	**P value (subgroup differences)**
All patients	0.63 (0.57-0.69)		0.69 (0.64-0.74)	
High PD-L1 group	0.54 (0.49-0.60)	$P < 0.01$	0.60 (0.53-0.67)	$P < 0.01$
Low PD-L1 group	0.71 (0.62-0.81)		0.78 (0.70-0.87)	
TP group	0.56 (0.50-0.62)	$P < 0.01$	0.66 (0.59-0.74)	P = 0.32
PF group	0.68 (0.62-0.75)		0.72 (0.65-0.79)	
Among high PD-L1 patients				
TP subgroup	0.56 (0.45-0.69)	P = 0.75	0.60 (0.46-0.78)	P = 0.90
PF subgroup	0.53 (0.45-0.62)		0.59 (0.50-0.69)	
Among low PD-L1 patients				
TP subgroup	0.59 (0.48-0.74)	P = 0.01	0.72 (0.55-0.93)	P = 0.32
PF subgroup	0.82 (0.72-0.94)		0.84 (0.72-0.97)	

Perioperative chemotherapy with either S-1 or 5-fluorouracil plus oxaliplatin and docetaxel for locally advanced gastric or gastroesophageal junction adenocarcinoma: A prospective trial and matched comparison.

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Background: Perioperative triplet chemotherapy improves the survival and surgical outcome of patients with locally advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma in addition to curative surgery. 5-fluorouracil (5-FU), docetaxel, oxaliplatin and leukovorin (FLOT) has been suggested as the treatment of choice. S-1, an oral fluoropyrimidine chemotherapeutic, also shows compelling efficacy as one of the triplet backbone and is exempt from a prolonged infusion time. However, a comparison over efficacy and tolerability between S-1 or 5-FU-containing triplet has not been investigated. **Methods:** The study contains two parts. The first part was a prospective single-arm multi-centered phase II trial with eight cycles of perioperative leucovorin, oxaliplatin, docetaxel and S-1 (LOTS) in patients with ≥T3 any N or ≥T2N(+) non-metastatic gastric/GEJ adenocarcinoma. The primary endpoint was the treatment response assessed by pathological tumor regression (TRG). After confirming the result passing the statistical threshold, we compared them with an identical and independent patient population who received perioperative FLOT under a prospective observation. A propensity score matching (PPSM) was used to balance the heterogeneity between the two groups. The efficacy, survival and tolerability were compared in a post-hoc manner. **Results:** From Feb 2017 to Jan 2024 with a comparable median follow-up time (LOTS, 18.7 ms; FLOT, 18.9 ms), a total of 45 and 154 patients were enrolled and received either LOTS or FLOT treatment. After a one-to-two PPSM procedure, the baseline characteristics of patients were comparable (LOTS, n=45; FLOT, n=90). Patients had similar curative resection rates (LOTS, 91%; FLOT, 86%; p=.359) and above nearly complete pathological regression rates (TRG 0-1: LOTS, 23%; FLOT, 27%; p=.669). The 24-ms recurrence-free (RFS) and overall survival (OS) rates were also comparable (LOTS vs. FLOT: 66 vs. 59% in RFS, p=.463; 73 vs. 68% in OS, p=.491). The toxicities were tolerable, except that more diarrhea and nausea events were reported in the LOTS group, whereas neutropenia, thrombocytopenia and mucositis were significantly prominent in the FLOT group. **Conclusions:** Perioperative LOTS shows reasonable therapeutic efficacy and survival as compared with FLOT in patients with locally advanced gastric/GEJ cancer. While the toxicity profile is distinctive from FLOT, it could be a potential alternative regimen with the convenience to save the prolonged infusion time. Clinical trial information: NCT04999332. Research Sponsor: Clinical Trial Facilitation Program, Ministry of Health and Welfare, Taipei, Taiwan; Year 2023; TTY Biopharm, Taipei, Taiwan.

Pathological outcome.

	Prior to LOTS treatment (n=45)		After LOTS treatment (n=43)		Prior to FLOT treatment (n=90)		After FLOT treatment (n=86)		p in LOTS vs. FLOT
CAP TRG, n (%)									0.669
0	-	-	1	(2.3)	-	-	3	(3.5)	
1	-	-	9	(20.9)	-	-	20	(23.3)	
2	-	-	14	(32.6)	-	-	23	(26.7)	
3	-	-	19	(44.2)	-	-	40	(46.5)	

Efficacy of PD-1 inhibitor plus chemotherapy versus chemotherapy in advanced esophageal carcinoma: A meta-analysis of phase III clinical trials.

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Background: Programmed death cell death protein-1 (PD-1) inhibitors combined with chemotherapy have demonstrated survival benefits in patients with advanced esophageal squamous cell carcinoma (ESCC) and adenocarcinoma (EAC). Our pooled analysis aims to analyze the efficacy of PD-1 inhibitor plus chemotherapy in advanced esophageal carcinoma (EC) in phase III randomized clinical trials (RCTs). **Methods:** We collected data from eligible phase III RCTs searched through PubMed, EMBASE, ClinicalTrials.gov, and meeting abstracts till July 5, 2024. Initial screening revealed 792 articles. We excluded duplicates, review articles and irrelevant studies, and we included eight RCTs that reported objective response rate (ORR), treatment-related adverse events (TRAEs), overall survival (OS) and progression-free survival (PFS). Odds ratio (OR) for ORR and TRAEs, and hazard ratio (HR) for OS and PFS were computed along with a 95% confidence interval (CI) and p-value for pooled analysis, using a random effect model in RevMan v.5.4. We performed a subgroup analysis based on a combined positive score (CPS) for programmed cell death ligand-1 (PDL-1) expression and tumor histology. **Results:** We included eight phase III RCTs (Jupiter-06, Checkmate 648, Keynote 590, ESCORT-1st, Orient-15, Checkmate 649, ESCORT-NEO, and ASTRUM-007), including 4131 patients with 2137 in group A (PD-1 inhibitor + chemotherapy) and 1994 in group B (placebo + chemotherapy, or chemotherapy). PD-1 inhibitors against EC included nivolumab, pembrolizumab, camrelizumab, sintilimab, toripalimab, and serplulimab. A total of 9.9% of patients were diagnosed with EAC, and 90.1% were ESCC. A pooled analysis showed significant achievement in ORR in group A compared to group B with an OR of 2.19 (CI: 1.77-2.69, $p < 0.05$, $I^2 = 60\%$). In terms of OS benefit, group A led to a 29% reduction in death risk compared to group B (HR: 0.71, CI: 0.66-0.76, $p < 0.05$, $I^2 = 0$). Group A also showed a significantly reduced risk of disease progression compared to group B (HR: 0.62, CI: 0.54-0.70, $p < 0.05$, $I^2 = 60\%$). Safety analysis revealed more TRAEs (OR: 1.93, CI: 1.43-2.60, $p < 0.01$, $I^2 = 11\%$) and grade ≥3 AEs (OR: 1.34, CI: 1.08-1.67, $p < 0.05$, $I^2 = 54\%$) in group A compared with group B. Sub-group analysis based on CPS revealed that patients with CPS ≥10 showed more OS (pooled HR: 0.59 vs 0.75) ($p = 0.04$) and PFS (pooled HR: (0.59 vs 0.64) ($p = 0.18$) benefits as compared to patients with CPS <10. Survival analysis between ESCC and EAC also exhibited statistically non-significant OS benefits ($p = 0.28$). **Conclusions:** PD-1 inhibitor plus chemotherapy as a first-line therapy in advanced EC, mainly in patients with CPS ≥10, showed improved antitumor efficacy in terms of superior ORR, OS, and PFS, with a manageable toxicity profile providing a benchmark for future studies. Research Sponsor: None.

Nivolumab (NIVO) + chemotherapy (chemo) vs chemo as first-line (1L) treatment for advanced gastric cancer/gastroesophageal junction cancer/esophageal adenocarcinoma (GC/GEJC/EAC): 5-year (y) follow-up results from CheckMate 649.

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Background: At 4-y follow-up, 1L NIVO + chemo continued to demonstrate clinically meaningful overall survival (OS) and progression-free survival (PFS) benefit vs chemo with acceptable safety in patients (pts) with advanced non-HER2+ GC/GEJC/EAC from CheckMate 649. We report efficacy and safety results of NIVO + chemo vs chemo at 5-y follow-up. **Methods:** Adults with previously untreated, unresectable, advanced or metastatic, non-HER2+ GC/GEJC/EAC were enrolled, regardless of programmed death ligand 1 (PD-L1) expression. Pts were randomized to NIVO (360 mg Q3W or 240 mg Q2W) + chemo (XELOX Q3W or FOLFOX Q2W), NIVO + ipilimumab, or chemo. Primary endpoints for NIVO + chemo vs chemo were OS and PFS by blinded independent central review (BICR) in pts with PD-L1 combined positive score (CPS) ≥ 5. **Results:** Pts were randomized to NIVO + chemo (n = 789) or chemo (n = 792). NIVO + chemo continued to show OS and PFS benefit vs chemo in pts with PD-L1 CPS ≥ 5, pts with PD-L1 CPS ≥ 1, and all randomized pts at 60-month (mo) minimum follow-up (Table). OS rates at 60-mo were higher with NIVO + chemo vs chemo in pts with PD-L1 CPS ≥ 5, pts with PD-L1 CPS ≥ 1, and all randomized pts (Table), and OS benefit with NIVO + chemo continued to be observed in most prespecified subgroups. Objective response rates (ORRs) were higher and responses were more durable with NIVO + chemo vs chemo in pts with PD-L1 CPS ≥ 5, pts with PD-L1 CPS ≥ 1, and all randomized pts (Table). No new safety signals were identified. **Conclusions:** These results represent the first report of 5-y follow-up for anti–PD-1 + chemo combination therapy in GC/GEJC/EAC to our knowledge. NIVO + chemo continued to provide sustained long-term survival vs chemo with an acceptable safety profile after 5 y of follow-up. These data continue to support the use of NIVO + chemo as a standard 1L treatment for advanced GC/GEJC/EAC. Clinical trial information: NCT02872116. Research Sponsor: Bristol Myers Squibb.

	PD-L1 CPS ≥ 5		PD-L1 CPS ≥ 1		All randomized	
Efficacy	**NIVO + chemo (n = 473)**	**Chemo (n = 482)**	**NIVO + chemo (n = 641)**	**Chemo (n = 656)**	**NIVO + chemo (n = 789)**	**Chemo (n = 792)**
mOS (95% CI), mo	14.4 (13.1–16.2)	11.1 (10.1–12.1)	13.8 (12.4–14.8)	11.4 (10.7–12.3)	13.7 (12.4–14.5)	11.6 (10.9–12.5)
HR (95% CI)	0.71 (0.61–0.81)		0.76 (0.67–0.85)		0.79 (0.71–0.88)	
60-mo OS rate (95% CI), %	16 (12–19)	6 (4–9)	13 (11–16)	5 (4–7)	12 (10–14)	6 (4–8)
mPFS[a] (95% CI), mo	8.3 (7.0–9.4)	6.1 (5.6–6.9)	7.5 (7.0–8.5)	6.9 (6.2–7.1)	7.8 (7.1–8.6)	6.9 (6.7–7.2)
HR (95% CI)	0.71 (0.61–0.82)		0.77 (0.68–0.87)		0.79 (0.71–0.89)	
ORR[a,b] (95% CI), %	60 (55–65)	45 (40–50)	60 (55–64)	46 (42–51)	58 (54–62)	46 (42–50)
mDOR[a,c] (95% CI), mo	9.6 (8.3–12.4)	7.0 (5.7–8.0)	8.6 (7.9–10.5)	6.9 (5.8–7.6)	8.5 (7.7–9.9)	6.9 (5.9–7.6)

[a]Per BICR.
[b]In pts with measurable target lesions at baseline.
[c]In all measurable responders.
DOR, duration of response; m, median.

Real-world safety of trastuzumab deruxtecan with gastric cancer: All-patient post-marketing surveillance study in Japan.

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Background: Interstitial lung disease and pneumonitis (ILD/p) is an important identified risk in the Japanese risk management plan of trastuzumab deruxtecan (T-DXd). In Japan, an all-patient post-marketing surveillance (PMS) study was conducted to investigate the incidence of ILD/p and factors associated with its development among patients with gastric cancer treated with T-DXd in the real-world setting. **Methods:** This is an observational, multicenter, PMS study (jRCT2001200001) with an observation period of 12 months that enrolled all patients who initiated T-DXd treatment for HER2-positive unresectable advanced or recurrent gastric cancer between Sep 2020 and Dec 2021. All suspected ILD/p events reported by physicians were adjudicated by an independent ILD adjudication committee; events adjudicated as drug-related ILD/p were summarized. The factors associated with the development of adjudicated drug-related ILD/p were investigated using a Cox proportional hazards model. Effectiveness endpoints included best overall response rate according to the RECIST 1.1 and overall survival, which was estimated using the Kaplan–Meier method. **Results:** The safety analysis set included 1070 patients (77.4% male, 89.9% ECOG performance status ≤1) with median age of 70 years (range: 23-100). Median initial dose and treatment duration of T-DXd were 6.4 mg/kg (range: 3.1-6.4) and 3.9 months (range: 0.7-12.0), respectively.The incidence of any grade, grade ≥3, and grade 5 adjudicated drug-related ILD/p were 9.6% (n=103), 2.8% (n=30) and 1.2% (n=13), respectively. Within 24 weeks from onset, 73.8% of adjudicated drug-related ILD/p had resolved, were resolving, or resolved with sequelae.A multivariate analysis identified that medical history and/or comorbidity of ILD/p [hazard ratio (HR) = 3.372, 95% confidence interval (95% CI): 1.431, 7.840], radiation pneumonitis (HR = 8.523, 95% CI: 2.006, 36.215), COPD or emphysema (HR = 1.962, 95% CI: 1.032, 3.731) and age ≥75 years (HR = 1.648, 95% CI: 1.058, 2.567) were factors of interest associated with the development of ILD/p in patients with gastric cancer. The overall response rate at 12 months post-treatment initiation of T-DXd was 34.2%. Overall survival after 364 days post-treatment initiation of T-DXd was 43.8% (median survival time: 303.0 days). **Conclusions:** The results suggest that the risk of ILD/p in this PMS study does not differ from that in the Destiny Gastric-01 study, and no new safety concerns were identified. Medical history and/or comorbidity of ILD/p, radiation pneumonitis, COPD or emphysema and age ≥75 years were identified as factors of interest associated with the development of ILD/p in patients with gastric cancer. Further investigation into the identified factors of interest may offer insights into the development of ILD/p among T-DXd treated gastric cancer patients in Japan. Clinical trial information: jRCT2001200001. Research Sponsor: Daiichi Sankyo Co., Ltd.

Prognostic factors for gastric cancer patients with positive peritoneal cytology who underwent upfront surgery.

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Background: Positive peritoneal cytology (CY1) is classified as Stage IV disease, the standard treatment for which is systemic chemotherapy. However, CY1 is often diagnosed after surgery in patients for whom staging laparoscopy is not indicated. In addition, the treatment strategy for patients with CY1 diagnosed during surgery is controversial. The Japanese guidelines recommend upfront surgery followed by adjuvant chemotherapy for patients diagnosed with CY1 during surgery. To identify patients with CY1 who are most likely to benefit from upfront surgery, we investigated prognostic factors in patients with CY1 who underwent upfront surgery. **Methods:** A total of 169 gastric cancer patients diagnosed as P0CY1 during or after surgery who underwent R1 resection other than positive resection margins were included. The diagnosis of CY1 was obtained by staging laparoscopy or during laparotomy in 69 patients and after surgery in 87 patients. Fourteen patients were initially diagnosed as CY0 by staging laparoscopy but were rediagnosed as CY1 after surgery. The clinicopathological factors of overall survival (OS) and progression-free survival (PFS) were investigated. Prognostic factors were identified using Cox regression models. **Results:** The median patient age was 72 years, and there were 108 males. The histological type was undifferentiated type in all of 102 patients. The macroscopic types were 0 in 10) patients 1 in 3, 2 in 27 3 in 84 and 4 in 45). cT grades and cN grades were cT1 in 3, cT2 in 7, cT3 in 12, cT4 in 147, cN0 in 65cN1 in 30), cN2 in 46 and cN3 in 28. The median OS and PFS were 25.1 months and 14.6 months, with 5-year OS rate and PFS rate of 30% and 23%, respectively. A multivariate analysis for OS identified macroscopic type 0-2, Charlson Comorbidity Index<3, and adjuvant chemotherapy as independent prognostic factors. Similarly, macroscopic type 0-2, tumor size <80 mm, and adjuvant chemotherapy were identified as independent prognostic factors for PFS. In a subgroup analysis of patients who received adjuvant chemotherapy, macroscopic type 0-2 was the only independent prognostic factor for PFS. The 5-year PFS rate in patients with macroscopic type 0-2 was 42%. **Conclusions:** Patients with macroscopic type 0-2 who are suitable for adjuvant chemotherapy may have benefit from upfront surgery, even if peritoneal cytology is positive. Research Sponsor: None.

Safety and efficacy of conversion therapy for metastatic esophageal cancer: Exploratory analysis of JCOG1314.

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Background: Patients with esophageal cancer (EC) with distant metastasis were treated with systemic chemotherapy. Recent advances in multimodal treatments have made conversion therapy (CT) a viable option for patients with metastatic EC. JCOG1314 was a randomized phase III trial to confirm the survival benefit of docetaxel plus cisplatin plus 5-fluorouracil (DCF) versus cisplatin plus 5-fluorouracil (CF) as initial treatment for unresectable or recurrent esophageal cancer. The purpose of this study was to evaluate the safety and survival impact of CT in patients enrolled in JCOG1314. **Methods:** CT was defined as surgery or chemoradiotherapy (CRT) aiming at cure after initial treatment for tumors that were initially unresectable due to distant metastasis. Clinicopathologic factors, surgical outcomes, toxicities during CRT, and survival were compared between CT and non-CT groups. CTCAE v4.0-JCOG was used to evaluate postoperative complications and adverse events during CRT. **Results:** Between September 2014 and April 2021, 240 patients were randomized to CF or DCF. Excluding patients with recurrent disease, 154 patients with initially unresectable esophageal cancer due to distant metastasis were selected for this study. Among them, 21 patients received CT which included conversion surgery (n=5) and conversion CRT (n=16). There was no significant difference in the number of metastatic organs, depth of invasion of the primary tumor, lymph node metastasis, and chemotherapy regiment between CT and non-CT groups. The most common M1 factor in CT group was the thoracic lymph node, followed by abdominal lymph node and cervical node. In conversion surgery group, there was no incidence of Grade 3 or higher pneumonia/leakage. Regarding the Grade 3 or higher non-hematological toxicities during conversion CRT, the incidence of appetite loss/ oral mucositis/pneumonia was 2 (13%)/2 (13%)/1 (6%), respectively. The 3-year overall survival (OS) of CT and non-CT groups was 52.4% and 14.3%, respectively. On multivariable analysis, patients with CT showed significantly better OS compared with the non-CT group (HR 0.36, 95% CI 0.19-0.67, $p<0.01$). In patients who underwent CT, there was no significant difference in OS between conversion surgery and CRT groups. **Conclusions:** This study demonstrated that CT was safely performed with favorable prognosis. A prospective study is warranted to investigate whether CT would improve survival in metastatic esophageal cancer.
Research Sponsor: None.

Short- and long-term outcomes of conversion surgery for initially unresectable advanced gastric cancer: A Japanese multicenter retrospective cohort study.

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Background: The efficacy of conversion surgery in patients with unresectable advanced gastric cancer at the initial diagnosis who converted to be resectable due to remarkable response to chemotherapy is unclear. To evaluate short- and long-term outcomes of conversion surgery for initially unresectable cStage IVB/pStage IV gastric cancer as the basic data for the JCOG2301 trial, which is a randomized controlled trial comparing conversion surgery with continuation of chemotherapy, a multicenter retrospective cohort study was conducted in the Stomach Cancer Study Group of the Japan Clinical Oncology Group. **Methods:** Inclusion criteria were as follows; (1)gastric cancer with at least one of following distant metastasis at the initial diagnosis; (i) peritoneal metastasis (PER) diagnosed by imaging examination or P1b or P1c disease diagnosed by laparoscopy or laparotomy, (ii) multiple liver metastasis (HEP) more than three lesions, and (iii) distant lymph node metastasis (LYM) beyond paraaortic lymph node station 16a2/b1, (2) conversion to be resectable after systemic chemotherapy, (3) surgery aiming at an R0 resection of the primary and remaining lesions between April 2010 and September 2017. Clinical data were retrospectively obtained from electronic patient records of 58 institutions. **Results:** Atotal of 241 patients were included to the study. Initial unresectable factors were PER, HEP, and LYM in 132(55%), 37(15%), and 63(26%) patients, respectively, and 9(4%) patients had two or more factors. Conversion to be resectable obtained during the first and second/later line chemotherapy was in 209 (87%) patients and 32 (13%), respectively. 182(76%), 29(12%), and 30(12%) patients underwent R0, R1, and R2 resection, respectively. The incidence of postoperative complication of Clavien-Dindo Grade II or higher was 24% and no surgery-related death was observed. The median overall survival (mOS) from surgery was 45.4 (95%CI 30.9-61.4) months in R0 resection, 16.6 (11.5-18.5) months in R1, and 20.9 (11.3-35.6) months in R2. In case of R0 resection, the mOS from surgery was 34.4 (95%CI 26.5-59.2) months in PER, 55.9 (21.8-96.7) months in HEP, 54.6 (30.9-98.3) months in LYM, and not reached (19.7-NA) in two or more factors. **Conclusions:** Conversion surgery appears to be safe and a promising treatment strategy for patients who have initially unresectable advanced gastric cancer and presented remarkable response to systemic chemotherapy. Research Sponsor: None.

Factors and timing of decreased adherence to S-1 in postoperative docetaxel + S-1 therapy for gastric cancer.

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Background: The combination of docetaxel + S-1 (DS) therapy in postoperative adjuvant treatment for gastric cancer has proven to be effective and has become a standard therapy. Clarification of factors involved in and the timing of S-1 non-adherence is important to minimize non-adherence. **Methods:** Among 92 patients who started DS treatment as post-operative adjuvant chemotherapy for gastric cancer between November 1, 2015 and April 30, 2021 at the Cancer Institute Hospital of the Japanese Foundation for Cancer Research, 90 were eligible for this study. The S-1 adherence rate per cycle was defined as the number of times a patient took S-1 divided by 28. In this study, adherence to S-1 was assessed through pill counts and by asking patients about reasons for non-adherence in interviews at a pharmaceutical outpatient clinic. Elicited reasons for non-adherence were categorized and analyzed. **Results:** Percentage of remaining S-1 was 3.9% in the first cycle, 7.4% in the second cycle, and 2.0% in the seventh cycle. Median relative dose intensity of S-1 was 80.9%. As reasons for non-adherence, patients most commonly cited nausea/vomiting (18.9%), missed dose (17.6%), and diarrhea (9.9%). From the first to the seventh course, the highest percentage of remaining S-1 was in the second course. **Conclusions:** Nausea/vomiting, missed dose, and diarrhea were identified as the most common factors for decreased adherence to S-1 in postoperative DS therapy for gastric cancer. Between the first and seventh courses, S-1 adherence decreased the most during the second course, when docetaxel was started. Using this information, we intend to develop methods to improve S-1 adherence in the future. Research Sponsor: Japan Society for the Promotion of Science KAKENHI.

The role of immunotherapy for real-world patients with HER2-negative advanced gastric cancer between 2011-2023.

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Background: Although the appearance of immunotherapy has apparently brought benefits for all patients (pts) with advanced gastric cancer (AGC), the magnitude of the benefit in real-world pts with HER2-negative AGC has not been fully investigated. **Methods:** We conducted a retrospective study on pts with HER2-negative AGC who received first-line platinum-based chemotherapy between 2011 and 2023. We evaluated the clinical outcomes across different periods of immunotherapy approval in Japan: Group A (pre-immunotherapy approval): 2011-2017, Group B (approved for third-line treatment or later): 2018-2021, and Group C (approved for first-line treatment): 2022-2023. **Results:** A total of 949 pts were enrolled (A: n = 477; B: n = 344; C: n = 128). Patient's characteristics were comparable except for the proportion of those with age ≥75 years (P = 0.002), prior gastrectomy (P = 0.03), and liver metastases ($P < 0.001$). The median overall survival (OS) was 16.2, 15.2, and 21.3 months in group A, B, and C, respectively, with no significant difference between groups (log-rank P = 0.50). Pts who had received first-line immunotherapy plus chemotherapy (n = 173) showed significantly improved OS compared with pts without having received any immunotherapy-containing treatment at 2011-2017 (n = 382) (hazard ratio [HR], 0.78; 95% confidence interval [CI], 0.61-0.99; P = 0.04). In the multivariate analysis, the use of immunotherapy at first-line treatment was not significantly associated with worse OS, whereas the use of immunotherapy at any lines was prognostic (HR, 0.54; 95% CI, 0.41-0.71; $P < 0.0001$). The proportion of pts who had received any second-line treatment was comparable among groups: 76%, 80%, and 71%, respectively. **Conclusions:** Our study suggests that the immunotherapy might partially have an impact on the survival improvement of real-world pts with HER2-negative AGC, which calls for the need for appropriate treatment strategies, including other agents. Research Sponsor: None.

Investigation of synergistic effects in trials of combination therapies with immune-checkpoint inhibitors in advanced gastric or gastroesophageal junction cancer.

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Background: Combinations of immune checkpoint inhibitors (ICI) and chemotherapy (CT) have been approved for gastric cancer. However, there is a hypothesis that this combination may blunt antitumor immune responses because most chemotherapeutic agents also target lymphocytes. The primary objective was to investigate that the ICI and CT does not interfere each other's therapeutic effects in advanced gastric cancer (GC) or gastroesophageal junction cancer (GEJC) patients. **Methods:** The reconstructed individual patient data was electronically extracted from the Kaplan-Meier curve of phase III randomized controlled trials (RCTs). The observed PFS curve of each constituent monotherapies was used to estimate simulated PFS curves expected under a model of independent drug action. If the observed curve demonstrated significantly better PFS than simulated curve, the combination of ICI and CT may have a synergistic effect, implying a superior outcome compared to simply adding the component monotherapy. **Results:** The study included 2,538 unresectable advanced, recurrent, or metastatic GC or GEJC patients from three RCTs comparing pembrolizumab (KEYNOTE-061, KEYNOTE-062 and KEYNOTE-859). In patients with programmed cell death ligand 1 (PD-L1) combined positive score (CPS) of 1 or greater, the 1-year and median PFS of the observed and simulated curves were 28.0% vs. 27.9%, and 6.89 months vs. 6.88 months, respectively. One sample log-rank test showed no significant differences between the observed and simulated curves ($p = 0.107$). In the subgroups with PD-L1 CPS ≥ 10 or <1, the 1-year PFS of the observed and simulated curves was 34.7% vs 32.8%, and 28.5% vs 26.8%. **Conclusions:** The observed PFS of ICT involving pembrolizumab was comparable to the simulated PFS estimated from the data for each monotherapy regardless of the magnitude of PD-L1 CPS. Although it was not clear whether potential synergies existed for ICT, these findings at least suggest that the benefits of ICI and CT are not interfering each other, thereby providing theoretical support for the efficacy of ICT in patients with advanced GC or GEJC. Research Sponsor: None.

PD-1 inhibitor (sintilimab) and fruquintinib plus SOX as conversion therapy for initially unresectable gastric/gastroesophageal junction adenocarcinoma (GC/GEJC): Updated results from a single-arm, phase 2 clinical trial.

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Background: The evidence of conversion therapy for unresectable GC/GEJC is limited. Previously, preliminary results of the ongoing open-label, phase 2 study (NCT05177068) have showed promising R0 surgical conversion rate (62.1%), R0 resection rate (100%) and acceptable tolerability in unresectable GC/GEJC with sintilimab and fruquintinib plus SOX (ASCO 2024). Here we present updated results with longer follow-up duration, including more enrolled patients. **Methods:** Eligible patients were administered fruquintinib (4mg/d, po, qd, d1-14), sintilimab (200mg, iv, d1), oxaliplatin (130 mg/m^2, iv, d1) and S-1 (40-60mg based on BSA, po, bid, d1-14) every 3 weeks for 3 or 6 cycles. One more cycle of sintilimab plus SOX was given before resection. Patients were assessed for tumor response and surgical feasibility by radiologic imaging & MDT every 3 cycles. Primary endpoint was R0 resection rate. Secondary endpoints included pathological response, ORR, PFS, OS, and safety. **Results:** As of July 30, 2024, 42 pts (37 males/5 females) with a median age of 64 years (range: 43–76), 71% ECOG PS1, 55% GEJC were enrolled. Nine (21.4%) of the 42 pts had distant metastasis and five (11.9%) pts had more than one unresectable factors. The most common unresectable factors were extensive or bulky lymph nodes 54.8% (23), liver metastasis 11.9% (5), peritoneal metastasis 4.8% (2), para-aortic lymph node metastasis 2.4% (1), and local progression 38.1% (16). Of 35 evaluable pts, ORR was 68.6% and DCR was 97.1%. Among the 29 pts who had completed surgical conversion, the R0 resection rate was 100% (29/29) and R0 surgical conversion rate was 82.9% (29/35). 10.3% (3/29) pts achieved pathological complete response (pCR), and 20.7% (6/29) pts reached tumor regression grade (TRG) 0-1. Additionally, the pathological response rate (pRR) according to JCGC (≥ Grade 1b) was 93.1% (27/29). Seven pts (16.7%) had grade 3 treatment-emergent adverse events, including 4 cases of anemia, 2 cases of neutrophil count decreased, and 1 case of each (lymphocyte count decreased, gastrointestinal hemorrhage, diarrhea, and immune-related pneumonitis). No severe surgery-related complication was observed. **Conclusions:** Fruquintinib combined with sintilimab and SOX yielded very high R0 conversion rate and R0 resection rate with a manageable safety profile in unresectable GC/GEJC, representing a potential and feasible conversion therapy regimen for this population. Survival data will continue to be followed up. Clinical trial information: NCT05177068. Research Sponsor: None.

A phase II study of fruquintinib plus sintilimab as a second-line therapy for advanced gastric and gastroesophageal junction adenocarcinoma (GC/GEJC): Updated results.

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Background: Preliminary results of the open-label, single-arm phase II study (NCT05625737) had demonstrated the combination of fruquintinib and sintilimab as a second-line therapy had a promising anti-tumor activity and an acceptable safety profile in pts with advanced GC/GEJC (ASCO-GI 2024. Poster 332). Here, we update the results with a focus on subgroup analyses. **Methods:** Patients (pts) aged 18-75 years who were HER2-negative and had failed first-line standard treatment were enrolled. Eligible pts received 4mg of fruquintinib orally once daily on days 1-14 and 200 mg of sintilimab intravenously on day 1, with treatment repeated every 3 weeks. An optimal Simon two-stage design was employed. The primary endpoint was objective response rate (ORR). Secondary endpoints included disease control rate (DCR), progression-free survival (PFS), overall survival (OS), and safety. **Results:** At data cut-off (August 25, 2024), 29 pts (7 in the first stage; 22 in the second stage) with median age 51 (range 33–73) years old had been enrolled. 41.4% were male, 37.9% had ECOG PS 1, 69% had lymph node metastasis and 72.4% had previously received immunotherapy. Among the 24 efficacy evaluable pts, the ORR was 33.3% (95% CI 15.6–55.3) and DCR was 66.7% (95% CI 44.7–84.4). After a median follow-up of 13.14 months, the median PFS was 5.13 (95% CI: 2.73–7.03) months and OS result was pending maturity. Pts who had lymph node metastasis exhibited higher ORR compared to pts without lymph node metastasis (ORR-47.1 vs 0%). In particular, pts without prior immunotherapies were more likely to achieve higher response rate (37.5 vs 33.3%) and had an obvious trend for longer PFS than pts with prior immunotherapies (9.00 vs 5.13 months). Majority of treatment-emergent adverse events (TEAEs) were grade 1-2 and the most frequently reported ones (≥20%) were white blood cell decreased (33%), hypertriglyceridemia (25%), lymphocyte count decreased (21%), abdominal pain (21%) and decreased appetite (21%). Grade 3/4 TEAEs occurred in only one patient, consisting of increased blood bilirubin levels, aspartate transaminase elevation, and alanine transaminase elevation. **Conclusions:** Fruquintinib combined sintilimab was well tolerated, with encouraging antitumor activity as second-line treatment for advanced GC/GEJC, especially in pts without prior immunotherapies. Further studies in larger cohorts to validate these findings are warranted. Clinical trial information: NCT05625737. Research Sponsor: None.

Conversion from radical esophagectomy to definitive chemoradiotherapy after neoadjuvant chemotherapy for advanced esophageal squamous cell carcinoma: Treatment options based on chemotherapy response.

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Background: Radical esophagectomy after neoadjuvant chemotherapy (NAC) is the established strategy for resectable advanced esophageal cancer. However, some patients convert to definitive chemoradiotherapy (dCRT) after NAC due to reasons such as their wishes or disease progression, and their prognosis remains uncertain. We aimed to investigate the prognosis of patients who converted to dCRT. **Methods:** Patients who underwent NAC for resectable advanced esophageal squamous cell carcinoma between 2006 and 2020 were enrolled retrospectively. The prognostic impact of subsequent treatment after NAC, planned surgery or conversion to dCRT, was compared. **Results:** We analyzed 686 patients, 70 who converted to dCRT (dCRT group) and 616 who underwent surgery (Surg group). The dCRT group had a poorer prognosis than the Surg group, with more advanced tumors and poorer response to NAC. Therefore, further analysis was conducted by categorizing patients as Responders (complete or partial response) and Non-responders (stable or progressive disease) to NAC. Among Responders in the dCRT group, 76.7% achieved a complete response, and the 5-year esophageal preservation survival rate was 66.9%. The 5-year survival rates for Responders were 77.5% and 71.3% in the dCRT and Surg groups. Multivariable analysis showed that dCRT did not worsen prognosis (P = .706, hazard ratio = 1.13; 95% confidence interval, 0.59 to 2.16). Conversely, in Non-responders, dCRT had a significantly poorer prognosis, with 5-year survival rates of 22.3% and 45.1% in the dCRT and the Surg groups ($P < .001$). **Conclusions:** For patients responding to NAC, conversion to dCRT is considered a potential treatment option. The possibility of chemoselection has been suggested in the choice of treatment for esophageal squamous cell carcinoma. Research Sponsor: None.

Zimberelimab platform study: Safety and efficacy of zimberelimab in combination with futibatinib and chemotherapy in patients with first-line advanced or metastatic esophageal carcinoma.

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Background: Zimberelimab: AB122 (Zim) platform study is a Phase 1 a/b study with multiple cohorts designed to evaluate the safety and efficacy of Zim, an anti-PD-1 antibody. Futibatinib (Futi) is an FGFR inhibitor that covalently binds FGFR1–4; FGFR inhibitors have been reported to modulate the tumor immune microenvironment, including reducing MDSCs and inhibiting CAFs. These immunomodulatory effects are expected to enhance the efficacy of the anti-PD-1 antibodies. The safety and efficacy of the combination of Zim, Futi, and chemotherapy were evaluated in patients (pts) with first-line advanced or metastatic esophageal cancer (EC) in Japan. **Methods:** Key eligibility criteria included age ≥ 18 years, histologically confirmed advanced or metastatic EC with measurable lesions by RECIST v1.1 regardless of FGFR and PD-L1 status, previously untreated, adenocarcinoma or squamous cell carcinoma, and ECOG PS 0 or 1. Patients received Zim 360 mg on day 1, Futi 20 mg QD, fluorouracil 800 mg/m2 on days 1–5, and cisplatin 80 mg/m2 on day-1 of a three-week cycle. The primary endpoint was dose-limiting toxicity (DLT). The secondary endpoints included adverse events, antitumor activity, pharmacokinetics, pharmacodynamics, and pharmacogenomics. **Results:** As of May 10, 2024, 43 pts were enrolled. At the data cutoff date for this analysis (July 15, 2024), treatment was ongoing in 23 pts. Safety and efficacy were evaluated in 43 and 41 pts, respectively. The median age of the pts was 61 years (53–70). An ECOG PS score of 0 was observed in 31 pts. DLTs were evaluated in the first three pts, and none were reported. Common treatment-emergent adverse events (TEAEs)were hyperphosphatemia (81.4%), diarrhea (67.4%), nausea (62.8%), decreased appetite (62.8%), constipation (55.8%), anemia (53.5%), stomatitis (51.2%), neutrophil count decreased (37.2%), alanine aminotransferase increased (34.9%), and palmar-plantar erythrodysaesthesia syndrome (30.2%). Grade ≥ 3 TEAEs occurred in 31 pts (72.1%), and common events (≥ 30%) were anaemia (30.2%) and neutrophil count decreased (30.2%). The objective response rate was 70.7% (95% CI: 54.5–83.9), and the confirmed overall response rate was 58.5% (95% CI: 42.1–73.7). The disease control rate was 92.7% (95% CI: 80.1–98.5). Median duration of response was 5.6 months (95% CI: 4.1–5.6), and median progression-free survival was 4.9 months (95% CI: 3.4–6.7). **Conclusions:** The combination of Zim, Futi, and chemotherapy has a manageable safety profile without new safety signals and shows promising antitumor activity in pts with advanced or metastatic EC. As more than half of the enrolled pts are still undergoing treatment, these are preliminary data; follow-up is needed to evaluate the viability of this combination appropriately. Clinical trial information: NCT04999761. Research Sponsor: TAIHO PHARMACEUTICAL CO., LTD.

Neoadjuvant SHR-1701 (a bifunctional anti-PD-L1/TGF-βRII agent) combined with chemoradiotherapy for resectable locally advanced esophageal squamous cell carcinoma (ESCC): A phase II trial.

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Background: Neoadjuvant chemotherapy or chemoradiotherapy followed by surgery is the standard of care for resectable locally advanced ESCC. SHR-1701, a new bifunctional fusion protein composed of a monoclonal antibody against PD-L1 fused with the extracellular domain of TGF-β receptor II, may enhance antitumor activity in combination with neoadjuvant standard therapies in ESCC patients (pts). The aim of this phase II trial is to determine the safety and efficacy of neoadjuvant chemoradiotherapy plus SHR-1701 followed by esophagectomy in pts with locally advanced resectable ESCC. **Methods:** Pts with resectable thoracic ESCC, diagnosed as clinical stage of cT1b-cT2N+M0 or cT3-cT4aNxM0 per AJCC 8th were eligible. Preoperative therapy included SHR-1701 (30 mg/kg every 3 weeks for 2 cycles), albumin paclitaxel (50 mg/m^2, once a week for 5 weeks), carboplatin (AUC=2, once a week for 5 weeks) and radiotherapy (41.4Gy in 23 fractions). Following esophagectomy, pts received SHR-1701 up to 1 year. The primary endpoint was pathological complete response (pCR) rate in the per-protocol population. Secondary endpoints included R0 resection rate, major pathological response (MPR) rate, disease free survival (DFS) and safety. **Results:** Between Dec. 2021 and Feb. 2023, 43 pts were screened and 41 met the inclusion criteria, among whom the clinical stage I, II, III, and IVa at baseline were 1 (2.4%), 8 (19.5%), 31 (75.6%), and 1 (2.4%), respectively. All 41 pts received neoadjuvant SHR-1701 combined with chemoradiotherapy. As of Mar. 2024, 30 pts underwent surgery, and all achieved R0 resection. There was no in-hospital and postoperative 30-day mortality. 9 (30.0%) pts achieved pCR in both primary tumor and lymph nodes (ypT0N0), and 12 (40.0%) pts had complete pathological response of the primary tumor with residual disease in lymph nodes alone (ypT0N+). Treatment-related adverse events (TRAEs) occurred in all pts, and most of TRAEs were grade 1-2. Notable toxicity included pneumonitis (31.7%) and anastomotic leak (12.2%). **Conclusions:** The addition of SHR-1701 to neoadjuvant chemoradiotherapy in ESCC demonstrated promising efficacy with acceptable toxicity, and might be a promising approach for neoadjuvant treatment. Clinical trial information: ChiCTR2000041562. Research Sponsor: None.

Bifunctional anti-PD-L1/TGF-βRII agent SHR-1701 plus chemotherapy as first-line treatment for advanced esophageal squamous cell carcinoma (ESCC): Survival results of a phase II trial.

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Background: Recently, several clinical trials have demonstrated synergistic efficacy by combining immune checkpoint inhibitors (ICIs) with chemotherapy. However, only a subset of patients (pts) derived benefits. Combining ICIs with agents blocking immunosuppressive pathway may expand the clinical benefit of ICIs to more pts. Transforming growth factor β (TGF-β) participates in tumor immune escape. SHR-1701, a novel bifunctional fusion protein consisting of a monoclonal antibody targeting PD-L1 fused with the extracellular domain of TGF-β receptor II, was evaluated in a phase II trial (ChiCTR2000039909) to assess its efficacy and safety when combined with chemotherapy for pts with unresectable locally advanced, recurrent or metastatic ESCC in China. **Methods:** This trial enrolled treatment-naive pts with histologically or cytologically confirmed unresectable locally advanced, recurrent, or metastatic ESCC. Eligible pts received SHR-1701 (30mg/kg, iv, d1, q3w) in combination with up to six cycles of albumin-bound paclitaxel (125mg/m^2, iv, d1, d8, q3w) and cisplatin (75mg/m^2, iv, d1, q3w). For those without progressive disease, maintenance treatment was administered with SHR-1701 monotherapy until disease progression or unacceptable toxicity. The primary endpoint was objective response rate (ORR). Secondary endpoints included progression-free survival (PFS), overall survival (OS), disease control rate (DCR) and safety. **Results:** At the data cut-off, March 1, 2024, 24 pts had received at least one dose of the study treatment and were included in both the intention-to-treat (ITT) analysis and safety analysis sets. 3 and 12 pts achieved confirmed complete and partial responses, respectively. The confirmed ORR was 62.5% and DCR was 87.5%. With a median follow-up of 20.8 months (95% CI, 15.5-NR), the median duration of response was 9.1 months (95% CI, 6.9-NR). The median PFS was 10.8 months (95% CI, 7.8-NR) and the median OS was 26.1 months (95% CI, 8.6-NR). 12-month PFS and OS rates were 41.4% (95% CI, 22.5-76) and 64.8% (95% CI, 47.8-87.9), respectively. Grade 3-4 treatment-related adverse events occurred in 11 (45.8%) pts and no treatment-related death was observed. **Conclusions:** The results highlighted that SHR-1701 plus chemotherapy as first-line therapy maintained long-term, durable survival benefit in pts with advanced ESCC. Clinical trial information: ChiCTR2000039909. Research Sponsor: None.

Evaluation of event-free survival (EFS) as a surrogate for overall survival (OS) in patients with locally advanced esophageal carcinoma (LA EC) receiving definitive chemoradiotherapy (dCRT).

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Background: Drug development in esophageal cancer is rapidly evolving, and identifying surrogates for patient outcomes to accelerate clinical drug development remains an important unmet medical need. Although early clinical endpoints such as EFS, disease free survival (DFS) are proven to be valid surrogates for overall survival (OS) in early esophageal (Ajani JA *et al.* 2022), gastric (Oba K *et al.* 2013; Wainberg ZA *et al.* 2023), and gastroesophageal junction tumors (Ronellenfitsch U. *et al.* 2013) after disease resection, there are no such surrogacy data evaluating patients who underwent non-operative management. Specifically, there has been no prior systemic evaluation of the predictive value of EFS as a surrogate for OS among LA EC patients treated with dCRT. This study evaluated the trial-level correlation between EFS and OS in unresectable LA EC patients receiving dCRT. **Methods:** A systematic literature review was conducted to identify randomized controlled trials (RCTs) of dCRT interventions in unresectable LA EC (search cutoff date: April 2023). Studies reporting EFS (or similar outcomes such as progression free survival (PFS), defined as time from randomization to local, regional, or distant recurrence, or death) and OS were included. EFS and OS hazard ratios (HRs) of each treatment comparison were extracted. Correlation between the log(HR) of EFS and OS was assessed using weighted linear regression (WLR). The coefficient of determination (R^2) was used to evaluate the strength of the correlation. A good surrogate was defined as an $R^2 \geq 0.65$ (Ciani *et al.* 2017) or an $R^2 \geq 0.75$ (Lassere *et al.* 2008). Leave-one-out cross-validations and sensitivity analyses were conducted to assess the robustness of the WLR model and correlation analyses, respectively. **Results:** The final analysis included 14 between-treatment comparisons from 11 RCTs (total number of patients = 2,812). We found a strong statistical correlation between EFS and OS, with an estimated slope coefficient of 0.91 ($p < 0.001$) and R^2 of 0.80 (95% confidence interval [CI]: 0.38, 0.96). In the leave-one-out cross validation, 71.4% of observed OS HRs fell within the predicted 95% CIs. Sensitivity analyses showed R^2 of 0.75 among studies conducted in China and R^2 of 0.73 after including studies that did not report EFS definition. **Conclusions:** In a pooled analysis of 11 clinical trials corresponding to over 2800 patients, we observed a strong statistical correlation between EFS and OS outcomes. This analysis supports the utility of EFS as a valid surrogate of OS in patients with unresectable LA EC receiving dCRT.
Research Sponsor: Merck & Co., Inc.

Response characteristics of tislelizumab (TIS) plus chemotherapy (chemo) in first-line (1L) treatment of locally advanced or metastatic esophageal squamous cell carcinoma (ESCC): A post hoc analysis of RATIONALE-306.

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Background: The global phase 3 RATIONALE-306 study (NCT03783442) demonstrated superior OS, PFS, ORR and manageable safety profile of TIS + chemo as 1L treatment of ESCC, which was well maintained at 3 years. This post-hoc analysis aims to explore the response characteristics of responders in TIS + chemo arm from RATIONALE-306. **Methods:** Advanced or metastatic ESCC patients (pts) were randomly assigned (1:1) to receive 1L TIS + chemo or placebo + chemo. Assessments of tumor response were done every 6 weeks for the first 48 weeks, then every 9 weeks thereafter per RECIST v1.1. Time to response (TTR), depth of response (DpR) and time to maximum response (TTMR) were assessed in responders (pts who achieved CR or PR). TTR was defined as the time from randomization to the first occurrence of a CR or PR, DpR as the percentage of maximal tumor reduction from the baseline of target lesion sum of diameters, and TTMR as the time from randomization to the maximum tumor shrinkage. **Results:** As of 24 November 2023, 207 pts (68.3% of the ITT population) in the TIS + chemo arm achieved responses, with a median TTR of 6.1 weeks (range: 3.3 - 101.3). First response was detected in 64.7% of responders within 8 weeks, 22.2% between > 8 and ≤ 14 weeks, and 13.0% after 14 weeks; the median OS was 21.8, 23.1 and 29.0 mos, respectively in pts within the three TTR categories, and the median OS post first response was 21.3, 21.3 and 23.2 mos (Table). DpR > 30% to ≤ 50%, > 50% to ≤ 80%, and > 80% to ≤ 100% was observed in 29.8%, 50.2% and 20.0% of responders, respectively, and the median OS was 16.1, 23.7, and 34.5 mos. The median TTMR was 19 weeks (range: 3.3 - 201.0), with 32.7% of responders achieved maximum response within 14 weeks, 34.6% between > 14 and ≤ 28 weeks, and 32.7% after 28 weeks, respectively. Median OS of pts with the three TTMR categories was 14.3, 17.4, and 39.9 mos, respectively, and median OS post-maximum response was 12.6, 14.3, and 24.8 mos. **Conclusions:** In the TIS + chemo arm, 64.7% of responders achieved response within 8 weeks, while still 35.2% of responders achieved response at later tumor assessments, and the OS benefits were comparable between early responders and late responders. Responders with deeper tumor response and/or longer TTMR tended to have longer OS. Clinical trial information: NCT03783442. Research Sponsor: BeiGene, Ltd.

	Cutoff	Patients, n (%)	OS, median (95% CI), mos	OS from response*, median (95% CI), mos
TTR	> 0 to ≤ 8w	134 (64.7)	21.8 (16.3, 25.0)	21.3 (16.0, 24.5)
	> 8 to ≤ 14 w	46 (22.2)	23.1 (16.3, 26.0)	21.3 (14.4, 24.3)
	> 14w	27 (13.0)	29.0 (20.4, NA)	23.2 (15.9, NA)
DpR	> 30% to ≤ 50%	61 (29.8)	16.1 (13.2, 20.8)	\
	> 50% to ≤ 80%	103 (50.2)	23.7 (19.6, 25.6)	\
	> 80% to ≤ 100%	41 (20.0)	34.5 (23.7, NA)	\
TTMR	> 0 to ≤ 14w	67 (32.7)	14.3 (8.7, 17.6)	12.6 (8.1, 15.4)
	> 14 to ≤ 28w	71 (34.6)	17.4 (15.8, 24.1)	14.3 (11.8, 20.1)
	> 28w	67 (32.7)	39.9 (28.6, NA)	24.8 (17.5, 39.4)

*From first response for TTR, and from maximum response for TTMR.

Tislelizumab (TIS) + chemotherapy (chemo) vs placebo (PBO) + chemo as first-line (1L) treatment in gastric/gastroesophageal junction adenocarcinoma (GC/GEJC) patients with/without peritoneal or liver metastases: A post hoc analysis of RATIONALE-305 study.

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Background: Gastric cancer patients with peritoneal or liver metastases had poor prognosis, and the efficacy of immunotherapy in these patients remains unclear. The global phase 3 RATIONALE-305 study (NCT03777657) demonstrated that tislelizumab combined with chemotherapy could bring survival benefits to 1L treatment of GC/GEJC patients. Here, we assessed the efficacy of TIS + chemo vs PBO + chemo in patients with/without peritoneal or liver metastases in RATIONALE-305. **Methods:** Patients with systemic treatment-naïve GC/GEJC were randomly assigned (1:1) to receive either TIS + chemo or PBO + chemo. Regression analyses were conducted to explore the associations between peritoneal or liver metastases and OS. Relative treatment effect between tislelizumab and placebo was assessed in each subgroup. The Kaplan-Meier method was used to estimate the median OS, and hazard ratios (HRs) for OS were estimated using Cox proportional hazards models. **Results:** Among the 997 randomized patients (TIS + chemo, n=501; PBO + chemo, n=496), 434 (43.5%) had peritoneal metastases (220 in TIS arm; 214 in PBO arm) and 378 (37.9%) had liver metastases (190 in TIS arm; 188 in PBO arm) at baseline. Regression analyses showed peritoneal and liver metastases were significantly associated with shorter OS. Baseline characteristics were balanced between TIS and chemo arms within each subgroup, including PD-L1 expression levels. As of data cut-off on Feb 28, 2024, OS was longer in the TIS arm compared with the PBO arm in patients with peritoneal metastases (HR= 0.78, 95% CI 0.64-0.96) or without (HR = 0.79, 95% CI 0.65-0.95). OS improvement was also observed in patients with liver metastases (HR = 0.77, 95% CI 0.62-0.96) or without (HR = 0.80, 95% CI 0.67-0.95). **Conclusions:** This post hoc analysis demonstrated OS improvement with TIS + chemo vs PBO + chemo in GC/GEJC patients with/without peritoneal or liver metastases. To our knowledge, RATIONALE-305 is the first global pivotal study reporting survival benefits with TIS + chemo as 1L treatment for GC/GEJC, irrespective of peritoneal or liver metastases. Clinical trial information: NCT03777657. Research Sponsor: BeiGene, Ltd.

	With metastases		Without metastases	
	TIS arm	PBO arm	TIS arm	PBO arm
Peritoneum	n=220	n=214	n=281	n=282
Median OS, mo (95% CI)	12.3 (10.6-14.3)	11.8 (10.6-13.0)	17.3 (15.0-20.3)	14.0 (12.6-16.0)
HR (95% CI)	0.78 (0.64-0.96)		0.79 (0.65-0.95)	
Liver	n=190	n=188	n=311	n=308
Median OS, mo (95% CI)	13.9 (11.4-15.6)	12.9 (10.9-14.5)	16.0 (13.6-18.0)	12.9 (11.9-14.4)
HR (95% CI)	0.77 (0.62-0.96)		0.80 (0.67-0.95)	

Clinical outcomes and changes in body composition following endoscopic submucosal dissection versus total gastrectomy in patients with proximal early gastric cancer.

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Background: Sarcopenia is associated with poor surgical outcomes and prognosis in patients with gastric cancer. However, the relationship between changes in skeletal muscle following endoscopic submucosal dissection (ESD) versus total gastrectomy (TG) in patients with proximal early gastric cancer (EGC) remains unclear. This study aimed to investigate the clinical outcomes and skeletal muscle mass changes between ESD and TG in patients with proximal EGC. **Methods:** We retrospectively reviewed 329 patients who underwent either ESD (n=127) or TG (n=202) for cT1N0M0 gastric cancer in the upper third of the stomach between 2015 and 2019 at a tertiary care center. The skeletal muscle index (SMI), visceral/subcutaneous fat, and abdominal circumference were measured preoperatively, and at 1 and 3 years after surgery, using cross-sectional computed tomography. We compared these parameters between the two groups and assessed both short- and long-term outcomes, as well as prognostic factors. **Results:** No significant differences were observed between the two groups with respect to overall survival (P=0.266), recurrence-free survival (P=0.079), or disease-specific survival (P=0.24). The recurrence rate was four (3.1%) in the ESD group and two (0.9%) in the TG group. Of the four recurrences in the ESD group, two were cured by gastrectomy, and two by ESD. Two recurrences in the TG group died of cancer progression. In the ESD group, all deaths were unrelated to gastric cancer, compared to 83.3% in the TG group (P=0.376). Patients in the ESD group showed more preserved SMI (P<0.001), visceral/subcutaneous fat (P<0.001), and abdominal circumference (P<0.001) at 1 and 3 years after surgery, compared to the TG group. The incidence of sarcopenia was significantly lower in the ESD group at postoperative years 1 (22.2% vs. 34.7%, P=0.02) and years 3 (25.7% vs. 40.6%, P=0.001). Sarcopenia had a significant impact on survival, with 5-year OS rates being lower in patients with preoperative sarcopenia (85.3% vs. 95.6%, P=0.005). Multivariate analysis revealed that the prognostic nutritional index was significantly associated with OS (hazard ratio, 0.86; 95% confidence interval: 0.75–0.99, P=0.03). **Conclusions:** Although OS was comparable between the ESD and TG groups, ESD was associated with better preservation of skeletal muscle and visceral/subcutaneous fat. These results suggest that ESD should be considered for patients with proximal EGC who are at high risk for sarcopenia. Research Sponsor: None.

Outcomes of neoadjvuant chemotherapy versus chemoradiation for esophageal adenocarcinoma: A National Cancer Database analysis.

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Background: While neoadjuvant chemoradiation (NCRT) has been the acceptable standard of care for esophageal adenocarcinoma (EAC), there has been a paradigm shift from recent trials have shown that neoadjuvant chemotherapy (NAC) is either equivalent or superior to NCRT. We sought to examine the outcomes of NAC versus NCRT in esophageal cancer from the National Cancer Database (NCDB). **Methods:** Utilizing the NCDB, we identified esophagectomy patients with EAC who underwent multiagent NCT versus multiagent NCRT. Overall Survival (OS) was analyzed with the Kaplan-Meier method and multivariate analysis (MVA) was performed to identify predictors of OS. Propensity score matching (PSM) was used to correct for baseline differences between groups. **Results:** After PSM, we identified 1007 patients in each group. Groups were equally balanced in age, gender, T/N stage, grade, and facility volume. There was improved lymphadenectomy and higher use of adjuvant therapy in NCT patients. There were significant improvements in R0 resection and pathologic response associated with NCRT patients. There was a significant improvement in OS in NCT versus NCRT. The median and 5-year OS was 42.7 months and 42% in NCT patients versus 34.2 months and 35% in NCRT patients (p = 0.001). For pathologic complete responders, median and 5-year OS was 101 months and 64% in NCT patients versus 71.2 months and 53% in NCRT patients (p = 0.04). On MVA, improved mortality was associated with female gender, pathologic N0, >10 nodes removed, pathologic partial or complete response, R0, well/moderate grade, NCT, adjuvant chemotherapy, and higher facility volume. There was also no survival benefit to adjuvant chemoradiation. **Conclusions:** Despite higher R0 and pathologic response associated with NCRT, NCT was superior to NCRT for OS in EAC patients. NCRT needs to be re-examined in its role as a treatment recommendation for operable EAC patients. Research Sponsor: None.

Multivariate analysis for overall survival.

Variable	HR	95% CI	*P* value
Age	1.01	1.00 – 1.02	0.004
Sex			
Male	Reference		
Female	0.79	0.64 – 0.98	0.03
Charlson Deyo score			
0	Reference		
1	1.05	0.91 – 1.21	0.54
Pathologic N-stage			
N0	Reference		
N+	1.82	1.58 – 2.10	<0.001
LN removed			
<10	Reference		
>10	0.81	0.68 – 0.97	0.02
Response			
None	Reference		
Partial	0.81	0.69 – 0.94	0.008
Complete	0.61	0.50 – 0.75	<0.001
Margin status			
R0	Reference		
R1	1.24	1.01 – 1.52	0.04
Grade			
Well/moderate	Reference		
Poor	1.35	1.18 – 1.54	<0.001
Preop radiation			
Yes	Reference		
No	0.77	0.67 – 0.88	<0.001
Adjuvant therapy			
None	Reference		
Chemotherapy	0.77	0.65 – 0.92	0.004
Chemoradiation	0.83	0.64 – 1.08	0.17
Facility volume			
Low (≤10/year)	Reference		
Medium (11-19/year)	0.86	0.72 – 1.03	0.10
High (≥20/year)	0.78	0.67 – 0.90	<0.001

Comparison between neoadjuvant chemotherapy followed by surgery and definitive chemoradiotherapy for survival in patients with resectable esophageal squamous cell carcinoma (JCOG2305A).

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Background: Neoadjuvant triplet chemotherapy followed by surgery (Neo-S) has become the standard treatment for resectable esophageal squamous cell carcinoma (ESCC) based on the results of JCOG1109. However, definitive chemoradiotherapy (CRT) remains an option for organ preservation with curative intent. This analysis aimed to explore subgroups of patients undergoing CRT, including salvage treatment, to identify those with survival outcomes potentially equivalent to Neo-S. **Methods:** Pooled data from two clinical trials of patients with resectable (T1N1-3M0 and T2-3NanyM0) ESCC. JCOG0909 was a single-arm trial that evaluated upfront CRT consisting of 5-FU (1000 mg/m^2 on day1-4, 29-32) and cisplatin (80 mg/m^2 on day1, 29) combined with radiotherapy at a dose of 50.4 Gy, including salvage treatment. If there was no progression at the time of salvage treatment, it was not regarded as an event for progression-free survival (PFS) in JCOG0909. JCOG1109 compared neoadjuvant treatment with doublet chemotherapy, triplet chemotherapy, and doublet chemotherapy with radiotherapy followed by surgery. The patients were enrolled in JCOG0909 between 2010 and 2014 and in JCOG1109 between 2012 and 2018. Subgroup analyses of clinical data, including tumor location, cT, cN, and clinical stage, were conducted on overall survival (OS) and PFS using Cox proportional hazards regression models for patients with JCOG0909 and JCOG1109. When a subgroup with a multivariable hazard ratio (HR) of more than 0.91 was detected, CRT including salvage treatment was considered not to be inferior to Neo-S in that subgroup. **Results:** Ninety-three patients in the CRT group from JCOG0909 and 196 patients in the Neo-S group from the triplet chemotherapy arm of JCOG1109 were included in this analysis. Although there were no significant differences, PFS and OS tend to be better in the Neo-S group than in the CRT group (PFS, HR 0.72, 95% confidence interval [CI] 0.50-1.03; OS, HR 0.89, 95% CI 0.59-1.36). For patients with Ut/Mt or cT1-2 disease, OS in the Neo-S group had a HR of 0.91 or more compared with those in the CRT group (Table). **Conclusions:** Subgroups in which OS after CRT including salvage treatment was not inferior to that of Neo-S were identified. Research Sponsor: None.

Hazard ratios of Neo-S to CRT by subgroups in multivariable Cox regression.

	Progression-free survival		Overall survival	
Subgroups	**Hazard ratio**	**95% confidence interval**	**Hazard ratio**	**95% confidence interval**
Ut/Mt	0.85	0.56-1.28	1.13	0.70-1.83
Lt	0.45	0.21-0.97	0.44	0.19-1.00
cT1-2	0.88	0.46-1.68	1.45	0.65-3.24
cT3	0.66	0.42-1.02	0.72	0.44-1.16
cStage IB+II	0.79	0.45-1.36	1.06	0.56-2.00
cStage III	0.75	0.45-1.23	0.92	0.52-1.62
Overall	0.72	0.50-1.03	0.89	0.59-1.36

Ut upper thoracic esophagus, Mt middle thoracic esophagus, Lt lower thoracic esophagus.

Phase II study of neoadjuvant chemotherapy with fluorouracil, leucovorin, oxaliplatin and docetaxel for resectable esophageal squamous cell carcinoma.

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Background: Based on the JCOG1109 trial, neoadjuvant docetaxel, cisplatin, and fluorouracil (DCF) followed by surgery has become the standard of care for resectable locally advanced esophageal squamous cell carcinoma (ESCC). Although the combination of fluorouracil, leucovorin, oxaliplatin, and docetaxel (FLOT) demonstrated benefits for esophageal adenocarcinoma as a perioperative therapy, its safety and efficacy for locally advanced ESCC have not been evaluated. **Methods:** We conducted a multicenter phase II study of neoadjuvant FLOT therapy for ESCC. Patients with cT1N1-3M0-1 or cT2-3N0-3M0-1 (only supraclavicular lymph node (SCLN) metastasis is included as M1) based on the 8th edition of the UICC TNM staging system were eligible. Neoadjuvant chemotherapy consisted of oxaliplatin (85 mg/m^2), docetaxel (50 mg/m^2), and l-leucovorin (200 mg/m^2) on day 1, and continuous infusion of fluorouracil (2600 mg/m^2/day) for 24 hours. This regimen was repeated every 2 weeks with a maximum of four cycles. The prophylactic antibody and G-SCF were not used mandatory. After completion of neoadjuvant chemotherapy, esophagectomy with extended lymphadenectomy was performed. Adjuvant treatment was prohibited for all patients. The primary endpoint was the pathological response rate (pRR), defined as the Grade 2 (more than two-thirds of the tumor is necrotic or fibrotic) or 3 (no viable tumor cells), based on the Japanese Classification of Esophageal Cancer. The sample size was determined based on an expected pRR of 38%, aiming for the lower bound of the 95% confidence interval to exceed the predetermined threshold of 20%. The expected number of patients to be enrolled was 60, with enrollment to be stopped when 45 patients with negative SCLN were enrolled. **Results:** Fifty-four patients were enrolled between September 2020 and January 2024. Patients with cStage I/II/III/IVB were 3/16/27/8. Of 54 patients, 45 patients were M0 without SCLN metastasis. Excluding 1 patient who did not receive any treatment after enrollment, 53 patients were included in the full analysis set. During chemotherapy, the most common grade 3 or 4 toxicities were neutropenia (73.6%), and leukopenia (22.6%). Febrile neutropenia was observed in 1 patient (1.9%). Finally, 46 patients underwent surgery. No treatment-related deaths were observed and the incidence of operative morbidity was tolerable. The pRR was 43.4% (23/53) (95% CI 29.8-57.7, p=00002). This study met the primary endpoint. The radical resection rate was 83.0% (44/53). The pathological complete response rate was 13.2% (7/53). **Conclusions:** Neoadjuvant FLOT therapy showed a promising pathological response with acceptable toxicities. It was noteworthy that the incidence of febrile neutropenia was relatively lower with compared to neoadjuvant DCF therapy. This regimen might be a treatment option for locally advanced ESCC. Clinical trial information: jRCTs031200094. Research Sponsor: None.

Therapeutic value of para-aortic lymph node dissection in gastric cancer with extensive lymph node metastasis: Integrated analysis of three phase II trials (JCOG2212A).

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Background: Gastric cancer (GC) sometimes forms metastatic bulky nodes around celiac artery or its branches (Bulky N), or metastasizes to para-aortic node (PAN) without any other distant metastasis. These severe nodal metastases are seldom cured by surgery alone. Japan Clinical Oncology Group (JCOG) defined these tumors as an extensive lymph node metastasis (ELM), and has developed multidisciplinary treatment through 3 phase II trials (JCOG0001, 0405, and 1002). JCOG0405 and 1002 reported the efficacy of D2 gastrectomy and PAN dissection (PAND) after neoadjuvant chemotherapy for GC with ELM with favorable survival outcomes. However, it remains unclear whether PAND truly contributed to the survival. The aim of this study was to clarify an optimal lymph node dissection area for GC with ELM. **Methods:** Therapeutic value index (TI) was investigated using an integrated dataset of JCOG0001, 0405, and 1002. Irinotecan and cisplatin, S-1 and cisplatin, or docetaxel, cisplatin and S-1 were used as neoadjuvant chemotherapy in JCOG0001, 0405, or 1002, respectively. Patients were classified into Bulky N group (only bulky N without PAN) and PAN group (PAN regardless of bulky N). TI was calculated by the following formula: proportion of metastasis (%) $\times$ 5-year relapse-free survival (5y-RFS, %) of patients with metastasis in the respective lymph node area / 100. Subgroup analysis by pathological response was also examined using the Japanese Classification of Gastric Carcinoma (JCGC) criteria. Grade $\geq$ 2a in JCGC criteria was defined as major pathological response (MPR). **Results:** A total of 122 patients were analyzed (Bulky N group: 68, PAN group: 54). Proportion of metastasis in peri-gastric (PG), supra-pancreatic (SP), and PAN areas were 80.9%, 54.4%, and 15.2% in Bulky N group, and 79.6%, 51.9%, and 44.4% in PAN group, respectively. 5y-RFS of patients with metastasis in PG, SP, and PAN areas were 38.2%, 29.7%, and 30% in Bulky N group and 23.3%, 7.1%, and 4.2% in PAN group, respectively. 30.9% of Bulky N group and 27.8% of PAN group showed MPR.TIs in each lymph node area are shown in the table. **Conclusions:** TI of PAN was extremely low in PAN group, especially in recent 2 trials (JCOG0405 and 1002) and when neoadjuvant chemotherapy was effective. These results suggest that advances in chemotherapy may make PAND unnecessary for GC with PAN metastasis in the future. Research Sponsor: The National Cancer Center Research and Development Funds (2023-J-03).

Therapeutic value index in each lymph node area.

		PG area		SP area		PAN area	
Population	**N**	**Bulky N group**	**PAN group**	**Bulky N group**	**PAN group**	**Bulky N group**	**PAN group**
Overall	122	30.9	18.5	15.0	3.7	5.6	1.9
JCOG0001	36	25.0	12.5	18.2	12.5	4.5	6.3
JCOG0405	42	40.9	25.0	15.4	0.0	3.8	0.0
JCOG1002	44	26.9	16.7	16.2	0.0	4.5	0.0
MPR	36	19.0	33.3	4.8	0.0	4.8	0.0

PG, peri-gastric; SP, supra-pancreatic; PAN, para-aortic node; MPR, major pathological response.

First-line (1L) tislelizumab (TIS) plus chemotherapy (CT) vs placebo (PBO) plus CT in advanced/metastatic esophageal squamous cell carcinoma (ESCC): RATIONALE-306 Japanese subgroup analysis with longer follow-up.

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Background: In the global phase 3 RATIONALE-306 study (NCT03783442), TIS + CT showed a significant overall survival (OS) benefit vs PBO + CT as 1L therapy for advanced/metastatic ESCC. After a minimum 3-year follow-up, the hazard ratio (HR) for OS was 0.70 for all randomized patients (pts) and for pts with programmed death-ligand 1 (PD-L1) Tumor Area Positivity (TAP) score ≥10%.We report results for the Japanese subgroup. **Methods:** Eligible pts enrolled in Japan were randomized (1:1) to receive intravenous TIS 200 mg or PBO every 3 weeks + investigator-chosen CT (platinum + fluoropyrimidine/paclitaxel) until disease progression or intolerable toxicity. The primary endpoint was OS in the intent-to-treat (ITT) population. Secondary endpoints included progression-free survival (PFS), objective response rate (ORR), safety, and OS in pts with PD-L1 TAP score ≥10%. **Results:** Of 649 randomized pts, 66 (10.2%) were Japanese, median age was 67 years, 89.4% were male, and 28.8% had PD-L1 TAP score ≥10%. At study entry, Eastern Cooperative Oncology Group performance status was 0 for 77.3% pts and 97.0% had metastatic disease. As of Nov 24, 2023, 78.8% vs 84.8% Japanese pts on TIS + CT vs PBO + CT received post-systemic therapy (ITT: 51.5% vs 57.9%). After a minimum follow-up of 37.9 months, TIS + CT showed improvements vs PBO + CT in median OS (24.5 vs 15.1 months [mo]; HR: 0.75) in all pts and in pts with PD-L1 TAP score ≥10% (HR: 0.79), median PFS (HR: 0.77) and ORR (63.6% vs 45.5%) (Table). Treatment-related adverse events (TRAEs) with TIS + CT vs PBO + CT in Japanese pts were 45.5% vs 36.4% for any grade (ITT: 69.8% vs 60.7%); 27.3% vs 6.1% for grade ≥3 (ITT: 32.1% vs 20.2%); 24.2% vs 3.0% for serious TRAEs (ITT: 19.8% vs 8.4%); TRAEs led to treatment discontinuation in 3.0% vs 6.1% (ITT: 13.3% vs 6.5%). No TRAEs leading to death were reported in Japanese pts (ITT: 1.5% vs 0.6%). **Conclusions:** After 3 years, TIS + CT continued to demonstrate robust efficacy and a tolerable safety profile in Japanese pts as 1L therapy for advanced/metastatic ESCC in the RATIONALE-306 study, consistent with the overall population. Clinical trial information: NCT03783442. Research Sponsor: BieGene LTD.

Efficacy outcomes.

	Japan TIS + CT (n=33)	Japan PBO + CT (n=33)	Overall TIS + CT (n=326)	Overall PBO + CT (n=323)
Median OS, mo	24.5	15.1	17.2	10.6
(95% CI)	(17.6, 26.9)	(8.0, 22.5)	(15.8, 20.1)	(9.3, 12.1)
HR (95% CI)	0.75 (0.43, 1.30)	-	0.70 (0.59, 0.83)[b]	-
PD-L1 TAP score ≥10%, n (%)	12 (36.4)	7 (21.2)	116 (35.6)	107 (33.1)
Median OS by PD-L1 TAP score ≥10%, mo	25.5	16.8	16.6	10.0
(95% CI)	(10.9, NE)	(0.9, NE)	(15.3, 23.4)	(8.6, 13.3)
HR (95% CI)	0.79 (0.26, 2.36)	-	0.70 (0.52, 0.95)[b]	-
Median PFS[a], mo	6.8	4.5	7.3	5.6
(95% CI)	(4.4, 8.5)	(4.1, 6.7)	(6.9, 8.3)	(4.9, 6.0)
HR (95% CI)	0.77 (0.45, 1.32)	-	0.60 (0.50, 0.72)[b]	-
ORR[a], n (%)	21 (63.6)	15 (45.5)	207 (63.5)	137 (42.4)

[a]Investigator assessed.
[b]Stratified.
CI, confidence interval; NE, not estimable.

Perioperative adebrelimab combined with chemotherapy for locally advanced resectable esophageal squamous cell carcinoma: A single-arm, open-label, multicenter study.

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Background: Neoadjuvant programmed cell death 1 (PD-1) inhibitors combined with chemotherapy have shown promising results in treating locally advanced esophageal squamous cell carcinoma (LA-ESCC). However, relatively little attention has been focused on programmed cell death ligand 1 (PD-L1) blockades. This study aims to evaluate the efficacy and safety of neoadjuvant adebrelimab (anti-PD-L1 antibody) plus chemotherapy followed by adjuvant adebrelimab for resectable LA-ESCC. **Methods:** In this single-arm, multicenter study, patients (pts) with newly diagnosed, locally advanced resectable thoracic ESCC (cT1b-2, N+ or cT3-4a, N0/+) received 2 cycles of neoadjuvant chemoimmunotherapy with adebrelimab (1200 mg), paclitaxel (175 mg/m^2) and cisplatin (75 mg/m^2) every 3 weeks, followed by esophagectomy and 16 doses of adebrelimab. The primary endpoints were safety and pathological complete response (pCR) rate. Secondary endpoints included surgery completion rate, R0 resection rate, major pathological response (MPR) rate, event-free survival (EFS) and disease-free survival (DFS). The trial was registered at Chinese Clinical Trial Registry, identifier: ChiCTR2300069179. **Results:** Between May 2023 and January 2024, 36 pts met inclusion criteria, with a median age of 65 years (range: 42-73). Of these, 66.7% (24/36) were classified as stage III-IVa. As of August 1, 2024, the median number of treatment cycles for all pts was 7 (interquartile range, 2-12). The neoadjuvant therapy completion rate was 91.7% (33/36). Thirty-two pts underwent minimally invasive esophagectomy, resulting in an overall surgical resection rate of 88.9% and an R0 resection rate of 90.6%. The pCR rate was 15.6% (5/32) and the MPR rate was 31.2% (10/32). The EFS and DFS outcomes were not mature. During the preoperative treatment period, 35 pts (97.2%) experienced any grade of treatment-related adverse events (TRAEs). Grade 3 or 4 TRAEs occurred in 7 pts (19.4%), and there were no grade 5 TRAEs. The most common grade 3/4 adverse events included neutropenia (11.1%), leukopenia (5.6%), diarrhea (5.6%) and hyponatremia (5.6%). Surgical complications of grade 3/4 were reported in 7 pts (21.9%), with anastomotic fistula (6.3%) being the most common. No perioperative deaths occurred. **Conclusions:** Neoadjuvant adebrelimab combined with chemotherapy and adjuvant adebrelimab demonstrated encouraging clinical efficacy and manageable safety in pts with LA-ESCC. Further study is warranted. Clinical trial information: ChiCTR2300069179. Research Sponsor: None.

A multicenter retrospective study of salvage surgery for patients with residual or recurrent esophageal squamous cell carcinoma after definitive chemoradiotherapy: SURGES study.

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Background: Salvage surgery is defined as an operation for patients with residual or recurrent tumors after definitive chemoradiotherapy (dCRT) with > 50 Gy. Although salvage surgery has been considered to improve prognosis, it is associated with a high incidence of postoperative morbidity and mortality. A previous meta-analysis by Faiz et al. reported anastomotic leakage, pulmonary disorder rates, and 90 day-mortality rates of 18.6%, 30.2%, and 8.8%, respectively. The aim of this study was to assess the efficacy and safety of salvage surgery in patients with residual or recurrent esophageal squamous cell carcinoma (ESCC) after dCRT and to explore the risk factors for postoperative complications and poor prognosis. **Methods:** This multi-institutional retrospective study enrolled patients treated with salvage surgery after dCRT between January 2017 and December 2021. The preoperative backgrounds of the patients were analyzed to assess their impact on survival or toxicity. **Results:** In total, 148 patients (59 with recurrent disease and 89 with residual disease) underwent salvage surgery at 10 leading hospitals in Japan. Post-dCRT ycT stages were T0/T1/T2/T3/T4: 12/25/30/78/3, and ycN stages were N0/N1/N2: 87/53/8. Among 136 patients who underwent esophagectomy, 108 underwent minimally invasive surgery and 28 underwent open surgery. Twelve patients had metastatic lymph nodes without primary disease. Severe postoperative complications (≧Clavien-Dindo Grade III) occurred in 39 (26.3%) patients and overall postoperative complications (≧Grade II) rate was 46.6%, including 19 (12.8%) anastomotic leakage and 25 (16.9%) pneumonia. Thirty- and 90-day mortality rates were observed in 0 and 5 (3.3%) patients, respectively. Multivariate logistic regression analysis showed that a total radiation dose ≥60 Gy (odds ratio [OR] 2.65, 95% confidence interval [CI] 1.07-6.55) and residual disease (OR 3.04, 95%CI 1.32-7.00) were associated with severe postoperative complications. The 3-year overall survival (OS) and progression-free survival (PFS) rates were 48.9% and 40.0%, respectively (median follow-up time: 42.1 months). Multivariate analysis revealed that patients with a long interval between dCRT and salvage surgery (≥180 days) had significantly longer OS (hazard ratio [HR] 0.60, 95% CI 0.38-0.95, p=0.032), and ycN0 was associated with longer PFS (HR 0.56, 95%CI 0.36-0.87, p = 0.011). The total radiation dose was not an independent prognostic factor for either OS or PFS. **Conclusions:** Salvage surgery is safe and provides substantial long-term outcomes. In cases with a total radiation dose of ≥60 Gy and residual disease, special caution is required because of the higher incidence of postoperative morbidity. A longer interval between dCRT and salvage surgery and ycN0 were prognostic factors. Research Sponsor: None.

Updated results from the phase Ib/II study of fruquintinib combined with SOX and toripalimab in patients with advanced metastatic gastric/gastroesophageal junction adenocarcinoma (GC/GEJC).

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Background: The efficacy of first-line treatment in advanced GC/GEJC patients (pts) with negative or low PD-L1 expression still needs to be improved. This phase Ib/II, open-label study (NCT05024812) evaluating fruquintinib (a highly selective VEGFR-1, -2, -3 inhibitor) plus toripalimab (anti-PD-1), and SOX has shown preliminary antitumor activity as first-line therapy in GC/GEJC. Here we update the results with longer follow-up duration and a specific focus on PD-L1 CPS features. **Methods:** The study of phase Ib employed a 3+3 dose escalation design, pts were treated with fruquintinib 3mg/d (dose level; DL1), 4mg/d (DL2), or 5mg/d (DL3) po, d1-14, in combination with fixed dose of toripalimab (240mg, iv, d1), oxaliplatin (130 mg/m^2, iv, d1) and S-1 (40-60mg based on BSA, po, d1-14) every 3 weeks. It had been reported in phase Ib that fruquintinib 5mg/d was defined as the RP2D. In phase II, a further 64 pts would be treated with the same regimen. Primary endpoint of phase II was PFS per RECIST 1.1. Secondary endpoints included ORR, DCR, OS, DOR and safety. **Results:** As of August 15, 2024, 32 pts (9 in phase Ib; 23 in phase II) had been enrolled. Pts characteristics included: median age 62 (range 38–73); 66% male; 88% with ECOG PS 1, and 31% with liver metastasis. 31 pts had PD-L1 CPS available. 40.6% were CPS $<$ 1 and 68.8% were CPS $<$ 5. Of the 30 pts evaluable for tumor response, the ORR was 63.5% (95% CI 43.9–80.1) with 4 pts achieving complete responses and DCR was 96.7% (95% CI 82.8–99.9). After a median follow-up of 10.94 months, the median PFS was 9.33 (95% CI: 5.68–NA) months and OS result was not reached. Pts with CPS $<$ 5 were more likely to achieve higher response rate (65.0 vs 55.6%) and longer PFS than CPS $\geq$5 (12.68 vs 8.11 months). Similar trends were observed in pts with CPS $<$ 1 and CPS $\geq$1 (ORR: 75.0 vs 52.9%; PFS: 12.68 vs 8.11 months). Treatment-related adverse events (TRAEs) were mainly grade 1-2 and the most common ones were hypoalbuminemia (50%), neutrophil count decreased (34%), anemia (41%), platelet count decreased (34%) and white blood cell decreased (34%). Grade 4 TRAEs occurred in 2 pts (impaired liver function, hypertriglyceridemia). There were no treatment related deaths in the trial. **Conclusions:** Fruquintinib combined with SOX and toripalimab provided favorable efficacy and manageable toxicity profile as first-line therapy for pts with advanced metastatic GC/GEJC, especially in pts with negative or low PD-L1 expression. More data including the potential predictive response biomarkers would be further analyzed and reported. Clinical trial information: NCT05024812. Research Sponsor: None.

An organ-preserving strategy in patients with esophageal squamous cell carcinoma treated with immunochemotherapy.

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Background: This study evaluated the strategy of dose-dense chemotherapy plus a PD-1 inhibitor with selective organ-preservation in patients with advanced esophageal squamous cell carcinoma (ESCC). **Methods:** Patients with locally advanced or metastatic (confined to the supraclavicular lymph nodes) ESCC were enrolled to receive treatment with a PD-1 inhibitor plus albumin bound-paclitaxel, cisplatin and 5-FU every two weeks. After six cycles of treatment, patients with clinical complete response (cCR) confirmed by computed tomography (CT), endoscopy and biopsy underwent an organ-preserving strategy with continuation of immunochemotherapy. The primary endpoint was cCR rate in the intention-to-treat population. A post hoc analysis was also conducted in patients with advanced ESCC who had a best response of CR after immunochemotherapy from three prospective studies we previously reported. Consecutive patients who had not received esophagectomy prior to immunochemotherapy, and had durable CR as confirmed by imaging, endoscopy with or without biopsy were identified to provide supporting evidence for the organ-preserving strategy. **Results:** A total of twenty patients were enrolled. At data cut-off (September 12, 2024), eight patients fulfilled the criteria of cCR (40%), while six patients actually adopted the organ-preserving strategy. 75% of the patients experienced ≥ grade 3 treatment-related adverse events (TRAEs), most commonly neutropenia and leukopenia. All TRAEs were managed with appropriate medical care. In the post hoc analysis, out of 71 patients who had a best response of CR to immunochemotherapy in three previous trials, sixteen consecutive patients who had not received esophagectomy before immunochemotherapy, and remained CR assessed with CT at data cut-off were identified. Although endoscopy was not mandatory during follow-up, twelve patients underwent endoscopic exams, and none of them had suspected tumor with endoscopic visualization. Biopsies were taken for eight patients, and were all negative for tumor. The median follow-up for these twelve patients reached 42.7 months, suggesting a long-term disease-free survival after immunochemotherapy. **Conclusions:** Dose-dense nab-paclitaxel, cisplatin and 5-FU combined with a PD-1 inhibitor showed encouraging cCR rate and manageable safety, making organ-preservation a possible subsequent approach in patients with advanced ESCC. Clinical trial information: ChiCTR2300072992. Research Sponsor: None.

The safety and short-term efficacy of induction DCF plus nivolumab therapy in patients with unresectable locally advanced esophageal squamous cell carcinoma.

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Background: Definitive chemoradiotherapy (dCRT) is the standard treatment for unresectable locally advanced esophageal squamous cell carcinoma (LA-ESCC) based on the results of the JCOG 0303 study. However, the prognosis in this population remains poor with a median overall survival (OS) of 13.1 months, and fistula formation was reported in about 20% of patients who received dCRT. In such a circumstance, a more effective and safer treatment strategy, such as induction chemotherapy followed by conversion surgery or dCRT, was developed. A phase II study of induction chemotherapy consisting of docetaxel (DTX), cisplatin (CDDP), and 5-fluorouracil (5-FU) (DCF) therapy (IC-DCF) showed a conversion rate of 37.5%, promising efficacy with a median OS of 33.8 months and a fistula formation rate of 4.2%. Therefore, we evaluated whether adding nivolumab to IC-DCF (IC-DCF+Nivo) could improve clinical outcomes. **Methods:** We retrospectively reviewed the medical records of patients with unresectable LA-ESCC who received IC-DCF+Nivo between Nov 2023 and Aug 2024. IC-DCF+Nivo (DTX 70mg/m2 day1, CDDP 70mg/m2 day1, 5-FU 750mg/m2 day1-5, Nivo 360 mg/body day1, every 3 weeks) was administered for 3 courses. We administered prophylactic levofloxacin from day 5 to day 15 during each cycle, and used granulocyte colony-stimulating factor (G-CSF) as a treatment. We evaluated adverse events according to the CTCAE ver5.0 during induction chemotherapy as safety, and objective response rate (ORR) of IC-DCF+Nivo according to the RECIST version 1.1., conversion surgery rate, complete resection (R0) rate, pathological complete response (pCR) rate and clinical CR (cCR) rate after dCRT as efficacy. **Results:** The median follow-up period was 5.5 months. We identified 23 eligible LA-ESCC patients, median age (range): 65 (44-76), 83% male, PS 0/1: 57%/44%, T status T3/T4b 44%/57%, clinical stage III/IVA/IVB 22%/57%/22%. Seven patients (30%) had supraclavicular lymph node metastases, and the most common unresectable factors were tracheal invasion (52%), bronchial invasion (22%) and aortic invasion (13%) of primary tumor or metastatic lymph nodes. Eighteen patients (78%) completed the 3 courses of IC-DCF+Nivo. The frequent grade 3-4 hematological adverse events were neutropenia (13/23, 57%) and anemia (3/23, 13%), and 1 patient (4.3%) had grade 3 diarrhea as non-hematological toxicity. Febrile neutropenia occurred in 3 patients (13%), and G-CSF was used in 7 patients (30%). One patient (4.3%) experienced primary tumor perforation. The ORR after 3 cycles of induction chemotherapy was 53%. Fourteen patients (61%) achieved resectability, and 10 patients (44%) underwent surgery. R0 resection rate was 90% (9/10), and the pCR rate was 20% (1/10). No one achieved cCR in dCRT. **Conclusions:** Induction DCF plus nivolumab therapy showed well-tolerated and well conversion rate. Research Sponsor: None.

A phase 2 study to assess the safety and efficacy of FLX475 (tivumecirnon) combined with pembrolizumab in patients with advanced or metastatic gastric cancer.

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Background: Regulatory T-cells (T_{regs}) suppress anti-tumor immunity by migrating into tumors via CCL17/CCL22 chemokines binding to CCR4. FLX475 (tivumecirnon), a selective CCR4 antagonist, inhibits T_{reg} recruitment, enhancing the effector T-cell (T_{eff}) response in the tumor microenvironment. Preclinical studies demonstrate that FLX475 increases the T_{eff}/T_{reg} ratio and boosts anti-tumor efficacy, both as monotherapy and in combination with checkpoint inhibitors (CPI). Here, we present findings on the safety and efficacy of FLX475 with pembrolizumab in a Phase 2 trial, aimed at improving therapeutic outcomes in advanced or metastatic gastric cancer (GC) patients. **Methods:** This Phase 2, open-label study (NCT04768686) assesses the safety and efficacy of FLX475 combined with pembrolizumab in advanced or metastatic GC patients. Patients received FLX475 100 mg PO QD and pembrolizumab 200 mg IV Q3W. Cohort 1 included CPI-naïve Epstein-Barr Virus (EBV)-negative GC patients who had progressed on at least two prior treatments. Cohort 2 included CPI-naïve EBV-positive GC patients who had received at least one prior treatment. Both cohorts were selected based on high CCR4 ligand expression and T_{reg} enrichment, expected to enhance treatment response. The primary objective of the study was to assess the anti-tumor efficacy of the study treatment, as assessed by the investigator using RECIST 1.1. **Results:** A total of 20 patients were enrolled in the study, with 10 patients in each cohort. The combination of FLX475 and pembrolizumab was well-tolerated across both cohorts. The most common treatment-related adverse events occurred in more than 10% of patients across cohorts were QTc prolongation (35.0%), and pruritus (15.0%), which were manageable and consistent with the known profiles of the individual agents. In Cohort 2, the confirmed objective response rate (ORR) was 60.0% (95% CI: 26.2 to 87.8). The median time to response (mTTR) was 2.7 months, with a median duration of response (mDOR) of 17.3 months. The median progression-free survival (mPFS) for this cohort was 10.4 months, while the median overall survival was not reached. In Cohort 1, no objective responses were observed, and the best response achieved was stable disease. The relationship between treatment outcomes and biomarkers including CCR4 expression levels and the T_{eff}:T_{reg} ratio was confirmed in exploratory analyses. **Conclusions:** The combination of FLX475 with pembrolizumab was well-tolerated and showed promising anti-tumor activity and durable responses in patients with advanced or metastatic EBV-positive GC. Clinical trial information: NCT04768686. Research Sponsor: None.

		Cohort 1 (EBV (-) GC) N=10	Cohort 2 (EBV (+) GC) N=10
ORR	n (%)	0 (0.0)	6 (60.0)
	(95% CI)	(-)	(26.2, 87.8)
mTTR	Months (95% CI)	-	2.7 (1.2, -)
mDOR	Months (95% CI)	-	17.3 (2.7, -)
mPFS	Months (95% CI)	1.4 (1.1, 1.5)	10.4 (1.0, -)

Single agent olaparib in advanced oesophago-gastric cancer: Primary results from the SOlar clinical phase II single arm trial.

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Background: A subset of pts with advanced oesophago-gastric adenocarcinoma (OGA) are characterised by deficient DNA damage repair (DDR) providing a rationale for PARP inhibition (PARPi) in these tumours. Potential activity was observed in the phase 3 gastric cancer trial with Olaparib + paclitaxel, however the efficacy of Olaparib monotherapy and response biomarkers have not been established. Herein we report the primary clinical results from the SOlar trial. **Methods:** SOlar is a prospective, open label, single arm phase II trial using a Simon's two stage optimal design, evaluating the efficacy and safety of Olaparib 300mg twice daily in pts with locally advanced/metastatic OGA who had previously progressed on ≥1 line of therapy in the advanced setting. Pts with measurable disease by RECIST v1.1 amenable to mandatory baseline biopsy were included. The primary end point was disease control rate (DCR) at 8 weeks from Olaparib initiation by RECIST v1.1. In this two-stage optimal design, if ≥5/27 pts had disease control in stage 1, a further 27 pts were to be recruited in stage 2. If ≥12/54 evaluable pts achieved disease control, Olaparib would warrant further investigation. Other key endpoints include; overall response rate, progression free survival and overall survival, and translational biomarker analyses (NCT03829345). **Results:** Between July 2019 - February 2024, 55 pts with locally advanced/metastatic OGA were enrolled. 51 pts received Olaparib and 50 were evaluable for the primary endpoint (5 withdrew). The median age was 61yrs (range: 54-67), 7 (14%) were *HER*2-positive and 2 had a known germline *BRCA* mutation. 19 (38%) pts had received ≥3 lines of therapy and 22 (44%) pts had received prior immunotherapy for OGA. 30 (60%) pts had objective responses to prior platinum chemotherapy (CR + PR). The DCR at 8-weeks was 28% (91% one-sided confidence interval lower bound: 19.4) with 14/50 evaluable pts having stable disease. No objective response was observed. 36/51 (71%) treated pts experienced a treatment-related adverse event (TRAE) of any grade, and 9 (18%) experienced a grade ≥3 TRAE (most common being anaemia, fatigue, vomiting). Two serious adverse events were reported as treatment-related: definitely related grade 1 pneumonitis, and possibly related grade 4 lung infection. **Conclusions:** In SOlar, Olaparib met its primary endpoint with 28% DCR in this heavily pre-treated group. No new safety concerns were observed. Ongoing translational analyses include; homologous repair deficiency scoring, whole exome sequencing, RNA expression, serial ctDNA, and multiplex immunofluorescent assessment of the tumour microenvironment, with a view to identify a subgroup who might benefit from PARPi to support ongoing PARPi combinatorial studies in advanced OGA. Clinical trial information: NCT03829345. Research Sponsor: AstraZeneca (supplied Olaparib).

Pembrolizumab plus chemotherapy versus placebo plus chemotherapy in patients with advanced HER2-negative gastric or gastroesophageal junction (G/GEJ) adenocarcinoma enrolled in the Republic of Korea: KEYNOTE-859.

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Background: After a median follow-up of 41.6 mo, data from the global phase 3 KEYNOTE-859 study (NCT03675737; N = 1579) continued to show that use of pembrolizumab (pembro) plus chemotherapy (chemo) provided a significant and clinically meaningful improvement in overall survival (OS), with manageable safety, versus placebo (pbo) plus chemo for patients (pts) with locally advanced or metastatic HER2-negative G/GEJ adenocarcinoma. This post hoc analysis evaluated the safety and efficacy of pembro plus chemo in pts enrolled in KEYNOTE-859 in the Republic of Korea. **Methods:** Eligible pts aged ≥18 years with locally advanced unresectable or metastatic HER2-negative G/GEJ adenocarcinoma, measurable disease per RECIST v1.1, and an ECOG performance status (PS) of 0 or 1 were randomly assigned 1:1 to receive pembro 200 mg or pbo IV Q3W for ≤35 cycles. All pts received investigator's choice of chemo (FP or CAPOX). The primary end point was OS. Secondary end points included progression-free survival (PFS), objective response rate (ORR), and duration of response (DOR) per RECIST v1.1 by blinded independent central review, and safety. Efficacy end points were evaluated in all randomly assigned pts (intention to treat). **Results:** A total of 150 pts (66, pembro plus chemo; 84, pbo plus chemo) from the Republic of Korea were enrolled. The median time from randomization to database cutoff (August 22, 2023) was 48.1 mo (range, 32.7-56.8). Baseline characteristics were generally well balanced between treatment groups. Most pts were male (71.3%) with an ECOG PS of 1 (66.7%); 43.3% of pts had ≥3 metastases. The median OS was 19.5 mo (95% CI, 14.2-26.2) for pembro plus chemo versus 15.2 mo (95% CI, 11.7-19.8) for pbo plus chemo (HR, 0.80; 95% CI, 0.55-1.15). Additional efficacy data are reported in the Table. Treatment-related adverse events occurred in 63 pts (95.5%) in the pembro plus chemo group and 78 pts (92.9%) in the pbo plus chemo group; 39 (59.1%) and 31 (36.9%) pts, respectively, were grade 3 or 4. No treatment-related deaths occurred. **Conclusions:** Consistent with the global results, clinical outcomes favored pembro plus chemo versus pbo plus chemo with manageable safety in pts from the Republic of Korea with HER2-negative G/GEJ adenocarcinoma enrolled in KEYNOTE-859. Clinical trial information: NCT03675737. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

	Pembro + chemo n = 66	Pbo + chemo n = 84
Median PFS (95% CI), mo[a]	11.0 (6.9-17.6)	7.2 (5.6-8.9)
PFS HR (95% CI)[b]	0.74 (0.48-1.13)	
ORR, % (95% CI)[c]	53.0 (40.3-65.4)	46.4 (35.5-57.6)
Median DOR (range), mo	15.0 (1.4+ to 50.8+)	6.9 (1.4+ to 44.3+)
Responses lasting ≥24 mo,[a] %	42.6	23.6

[a]Based on Kaplan-Meier estimates.
[b]Based on the unstratified Cox regression model with the Efron method of tie handling with treatment as a covariate.
[c]Based on the stratified Miettinen and Nurminen method with strata weighting by sample size.

A phase I/II clinical trial of preoperative docetaxel/oxaliplatin/S-1 (DOS) combination therapy for esophagogastric junction adenocarcinoma.

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Background: Esophagogastric junction (EGJ) cancer is difficult to cure with surgery alone, necessitating adjuvant chemotherapy. However, administering aggressive adjuvant chemotherapy after invasive surgery is difficult. Neoadjuvant chemotherapy, on the other hand, is expected to enhance compliance, increase the rate of curative resection, and reduce recurrence. **Methods:** We conducted a single center Phase I/II study to evaluate the safety and efficacy of neoadjuvant chemotherapy using docetaxel, oxaliplatin, and S-1 (DOS), followed by adjuvant chemotherapy with S-1 in patients with cT3-4aN0-3M0 adenocarcinoma of the EGJ. In Phase I, we estimated the maximum tolerated dose and dose-limiting toxicity of neoadjuvant DOS therapy and determined the recommended dose (RD). In Phase II, we assessed the safety and efficacy of the RD established in Phase I. The primary endpoint was the curative resection rate, with secondary endpoints including the 3-year overall survival rate, 3-year recurrence-free survival rate, treatment completion rate, response rate to neoadjuvant chemotherapy, pathological response, and adverse events. Neoadjuvant DOS chemotherapy was administered in three 3-week cycles. The starting dose of docetaxel was 50 mg/m^2, with plans to escalate it to 60 mg/m^2. Docetaxel and oxaliplatin (100 mg/m^2) were administered on day 1, while S-1 (120 mg/body) was given orally on days 1-14. Postoperative adjuvant chemotherapy consisted of S-1 (120 mg/body), administered for 4 weeks followed by a 2-week rest period, repeated for 8 cycles. **Results:** Based on the 6 patients enrolled in Phase I, the RD of docetaxel was determined to be 60 mg/m^2. A total of 40 eligible patients, including 3 from Phase I, were enrolled in the study. Two patients discontinued protocol treatment during neoadjuvant chemotherapy due to adverse events and 1 patient discontinued due to positive peritoneal cytology. The R0 resection rate was 87.5% (35/40). Surgical procedures included esophagectomy in 18 patients, total gastrectomy in 15, and proximal gastrectomy in 4. Postoperative complications of Grade III or higher, as per the Clavien-Dindo classification, occurred in 15.0% of patients, with no surgery-related deaths. The response rates to neoadjuvant chemotherapy were 12.5% for PR, 2.5% for PD, and 85.0% for non-CR/non-PD. Pathological response rates were: Grade III in 8.1%, Grade II in 35.1% and Grade I in 56.8%. During neoadjuvant chemotherapy, Grade 3 or higher adverse events were observed, including neutropenia (82.5%), nausea (5%) and diarrhea (5%). The neoadjuvant chemotherapy completion rate was 92.5%, although 59.5% of patients required dose reductions. **Conclusions:** Neoadjuvant DOS therapy for esophagogastric junction cancer shows promise, achieving a high R0 resection rate and substantial pathological efficacy, with acceptable levels of adverse events. Clinical trial information: UMIN000022210. Research Sponsor: None.

Perioperative chemoimmunotherapy for patients with gastric or gastroesophageal junction cancer: A systematic review and meta-analysis.

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Background: Immune checkpoint inhibitors (ICIs) targeting the PD-1/PD-L1 axis, in combination with chemotherapy, are known to improve survival in advanced/metastatic gastric (GC) or gastroesophageal junction cancer (GEJC). Therefore, several trials incorporated ICIs in the neoadjuvant/peri-operative setting with promising preliminary results. The purpose of this meta-analysis was to determine the safety and efficacy of neoadjuvant chemoimmunotherapy for resectable GC or GEJC. **Methods:** PubMed/MEDLINE and Embase were systematically searched for randomized control trials (RCTs) investigating the efficacy of neoadjuvant anti PD-1/PD-L1 ICIs for locally advanced GC or GEJC up to 07/27/2024. Outcomes of interest were surgical outcomes such as pathological complete response (pCR) rate and major pathological response (MPR) rate. Risk ratio (RR) with 95% confidence intervals (CI) were pooled using a fixed model meta-analysis. Statistical analysis was performed with Review Manager 5.4.1. I^2 statistics was used to assess heterogeneity. The Cochrane risk of bias assessment tool was used to assess the risk of bias. All analyses were conducted at a significance level of 0.05. **Results:** Seven RCTs comprising 2,911 GC or GEJC patients were included. Among these, 1,454 patients received neoadjuvant chemoimmunotherapy and 1,457 patients received chemotherapy, respectively. Neoadjuvant chemoimmunotherapy significantly improved pCR rate (RR 2.94; 95% CI 2.34-3.70; P<0.00001; I^2=33%), and MPR rate (RR 1.47; 95% CI 1.25-1.72; P<0.00001; I^2=30%) compared to chemotherapy alone. Chemoimmunotherapy was associated with slightly higher incidence of grade 3 or worse treatment-related adverse events than chemotherapy (RR 1.07; 95% CI 1.01-1.14; p=0.03; I^2=0%). **Conclusions:** This meta-analysis showed that adding ICIs to chemotherapy significantly improved the pCR and MPR rates and was feasible for patients with locally advanced GC or GEJC. Results of survival outcomes in the included trials and ongoing trials are awaited to further determine the efficacy and utility of chemoimmunotherapy and identify patient populations that benefit from this strategy. Research Sponsor: None.

Included studies.

Author	Year	Trial#	Design	Cancer; Stage	Name of IO	Target	Chemo regimen	Extracted data		
pCR	**MPR**	**TRAE**								
Ding X	2023	NCT04982939	P2	G/GEJ; Stage2-3	Sintilimab	PD-1	SOX	○	○	
Janjigian YY	2023	NCT04592913	P3	G/GEJ; ≥T2Nany/T0-4 N+	Durvalumab	PD-L1	FLOT	○		○
Li C	2023	NCT04208347	P3	G/GEJ; T3-4aN+	Camrelizumab	PD-1	SOX(+apatinib)	○	○	
Lin JX	2024	NCT04195828	P2	G; T2-4N+	Camrelizumab	PD-1	S-1+nabPTX	○	○	○
Lorenzen S	2024	NCT03421288	P2	G/GEJ; ≥T2 and/or N1	Atezolizumab	PD-L1	FLOT	○	○	
Peng Z	2024	NCT04661150	P2	G/GEJ HER2+; resectable	Atezolizumab	PD-L1	CAPOX+Tmab	○		
Shitara K	2024	NCT03221426	P3	G/GEJ; ≥T3N+	Pembrolizumab	PD-1	FLOT or XP/FP	○		○
Yuan SQ	2024	NCT04250948	P2	G/GEJ; T3-4N+	Toripalimab	PD-1	SOX/CAPOX	○	○	○

FG-M108 plus capecitabine and oxaliplatin (CAPOX) for first-line treatment of CLDN18.2+/HER2- advanced gastric/gastroesophageal junction adenocarcinoma (GC): Update results of a phase I/II trial to determine RP2D.

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Background: There exists unmet need for previously untreated pts with CLDN18.2+/HER2- advanced GC. FG-M108, an mAb specifically targeting CLDN18.2 and exhibiting enhanced ADCC activity, has the potential to benefit these pts. **Methods:** This phase I/II trial evaluated safety, pharmacokinetics and preliminary antitumor activity of FG-M108 plus CAPOX in previously untreated pts with CLDN18.2+/HER2- advanced GC, where FG-M108 was administered intravenously at 300 mg/m^2 or 600 mg/m^2 Q3W. Here we compare results of FG-M108 at 300mg/m^2(Cohort A) with 600mg/m^2(Cohort B) to establish RP2D. **Results:** As of August 28, 2024, 63 pts with CLDN18.2+ (IHC 1/2/3+≥10%) GC were treated, 70% of whom had CLDN18.2 Medium-High expression (CMH, IHC 2/3+≥40%). Baseline characteristics were comparable between Cohorts A & B. Increasing the FG-M108 dosage did not result in a higher incidence or severity of treatment-related adverse events (TRAE) overall. However, nausea and vomiting, identified as CLDN18.2 on-target toxicities, were significantly more frequent and severe in Cohort B compared to Cohort A.When 300 or 600 mg/m^2 of FG-M108 was administered intravenously Q3W, serum trough concentration exceeded the target drug concentration predicted by in vitro ADCC activity. The 300 mg/m^2 dose is expected to provide sufficient drug exposure as the effective dose.Moreover, in Cohort A, pts with CMH expression had a confirmed ORR of 78% (unconfirmed ORR=81%) and a median PFS of 11.0 months (95%CI 6.9-12.7) , significantly better than those in Cohort B. Based on safety, PK and efficacy trends, the RP2D was determined to be FG-M108 at 300mg/m^2 plus CAPOX. **Conclusions:** FG-M108 plus CAPOX had a manageable safety profile and promising antitumor activity in GC, with RP2D at 300mg/m^2. A phase III trial is ongoing to confirm the efficacy of FG-M108 at 300mg/m^2 plus CAPOX as first-line treatment (NCT06177041). Clinical trial information: NCT04894825. Research Sponsor: None.

	Cohort A n=52	Cohort B n=11
CMH %	71	64
Primary lesion-Stomach %	92	100
Metastatic organs ≥3 %	27	9
ORR/DCR with CMH %	81/97	57/86
PFS/DOR with CMH months	11.0/9.9	5.5/4.2
≥G3 TRAE %	37	18
TRAE-Nausea % / ≥G3 Nausea %	38/2	64/9
TRAE-Vomiting % / ≥G3 Vomiting %	23/2	73/9

The safety and short-term efficacy of neoadjuvant chemotherapy in elderly patients with resectable locally advanced esophageal squamous cell carcinoma.

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Background: The standard treatment for resectable locally advanced esophageal squamous cell carcinoma (LA-ESCC) is neoadjuvant chemotherapy followed by surgery. However, the efficacy and safety of neoadjuvant chemotherapy for patients aged 76 or older have not been established because the JCOG1109 trial, which established the current standard of care, was limited to patients younger than 76 years. In our hospital, even patients aged 76 or older are treated with docetaxel, cisplatin, and 5-FU (DCF) or 5-FU / leucovorin, oxaliplatin, and docetaxel (FLOT), if the patient's general condition and organ function are maintained, but there have been few reports on the efficacy and safety of these therapies for elderly population. **Methods:** The subjects were 76 years old or older resectable LA-ESCC patients diagnosed as cT1N1-3M0, cT2-3N0-3M0, and cT1-3N0-3M1 (UICC-TNM 8th), who received at least one cycle of neoadjuvant DCF or FLOT in our institution from 2014 to 2023. Adverse events (CTCAE ver. 5.0) and histopathological responses were evaluated. **Results:** Patients who received neoadjuvant DCF (D group) and FLOT (F group) were 19 and 25, respectively. Patients background in the D / F group was as follows; median age 76 / 78 (years), male 84% / 64%, PS1 37% / 64%, clinical, T3 74% / 84%, clinical N1 32% / 52%, N2 47% / 32%, clinical stage III 53% / 64%, stage IVB 21% / 28%. Of the 19 patients in the D Group, 17 completed 3 cycles of treatment (completion rate 89%), 18 underwent surgery, including 2 patients who discontinued during the treatment (1 with disease progression and 1 with adverse event), and 1 underwent CRT. Of the 25 patients in the F group, 16 completed 4 cycles of treatment (completion rate 64%), 22 underwent surgery, including 8 patients who discontinued during the treatment (1 with disease progression, 6 with adverse events, and 1 with other), 2 underwent CRT, and 1 was placed on best supportive care. In the D and F groups, hematological toxicities were more common in the F group, especially grade 3 or more neutropenia was 26% / 96%, but febrile neutropenia was similar at 5% and 12%, respectively. Nonhematological toxicities were more common in the F group for all grades (nausea 42% / 64%, anorexia 63% / 80%, and fatigue 52% / 88%, while grade 3 or more toxicity was not significantly different from the D group (nausea 0% / 4%, anorexia 0% / 4%, and fatigue 5% / 4%). In the 18 patients of the D group and 22 patients of the F group who underwent surgery, R0 resection was performed in 94% / 95%, and pathological complete response (pCR) was 11% / 23%, respectively. **Conclusions:** Neoadjuvant triplet chemotherapy was well tolerated and resulted in high R0 resection rates and pathological responses even in patients aged 76 or older. Further investigation with long-term follow up is warranted to elucidate the clinical benefit of neoadjuvant chemotherapy for elderly LA-ESCC patients. Research Sponsor: None.

Non-cancer-related deaths in patients with resectable locally advanced esophageal squamous cell carcinoma: A supplementary analysis of JCOG1109.

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Background: Neoadjuvant docetaxel, cisplatin, and 5-FU (DCF) has become the standard treatment for resectable locally advanced esophageal squamous cell carcinoma (LA-ESCC) in Japan, based on the result of JCOG1109. In this study, more non-cancer-related deaths were observed in the cisplatin, and 5-FU (CF) plus radiotherapy (CF-RT) arm compared to the CF and DCF arm. However, the details of these non-cancer-related deaths remain unclear. We aimed to evaluate the influence of neoadjuvant therapies on non-cancer-related deaths using the data from JCOG1109. **Methods:** We analyzed patients' characteristics, classification, time of onset, risk factors associated with onset, and cumulative incidences in patients with non-cancer-related deaths. Cumulative incidences of non-cancer-related deaths in each arm were estimated, considering cancer-related deaths as a competing risk. Risk factors were explored using multivariable Fine-Gray model with Firth penalization. **Results:** Among 601 enrolled patients, 164 in the CF arm, 169 in the DCF arm, and 173 in the CF-RT arm were completed the surgery. Of those, non-cancer-related deaths were observed in 19 (12%), 12 (7%), and 32 (18%) patients, respectively. Patient characteristics of non-cancer-related deaths in the CF/DCF/CF-RT arm were as follows: male, 89/92/88%; median age, 69/69/69 years old; ECOG PS 0, 89/100/97%; lower thoracic esophagus, 21/25/19%; previous history of ischemic disease of heart, 5/0/0%; brain, 0/0/0%; other malignancies, 5/8/0%; comorbidity of hypertension, 26/25/47%; COPD, 0/8/3%; thoracoscopic approach, 37/42/38%. The median follow-up time was 5.7 years (range, 0.3-10.4 years). The causes of non-cancer-related deaths in the CF/DCF/CF-RT arm were pulmonary, 4 (21%)/1 (8%)/10 (31%); cardiovascular, 1 (6%)/0/2 (6%); infectious, 2 (11%)/1 (8%)/3 (9%); other malignancy, 2 (11%)/3 (25%)/5 (16%); treatment-related, 0/1 (8%)/1 (3%), and the median time from surgery to onset of non-cancer-related death was 3.3/3.8/4.3 years, respectively. Cumulative incidences of non-cancer-related deaths after 3, 6, 9 years in the CF/DCF/CF-RT arm were 4.9/3.0/7.0%, 10.7/6.1/15.6%, 12.4/8.8/26.1%, respectively. In multivariable analyses, ≥ 65 years old (hazard ratio [HR]: 3.243, 95% confidential interval [CI]: 1.797–5.852, vs. < 65 years old), ECOG PS 0 (HR: 3.579, 95% CI: 1.148–11.162, vs. PS 1), and serum CRP ≥ 1.0 mg/dl (HR: 2.238, 95% CI: 1.003–4.995, vs. <1.0 mg/dl) were significantly associated with the onset of non-cancer-related deaths. Whereas the CF-RT arm was not significantly associated (HR: 1.600, 95% CI: 0.885–2.892, P = 0.12, vs. CF). **Conclusions:** The incidence of non-cancer-related deaths was relatively higher in the CF-RT arm, and pulmonary-related complications tended to be a particularly common cause of death. Research Sponsor: None.

NRG-GI007: Secondary endpoints of safety assessment and clinical complete response (cCR) of chemoradiation with endoscopically injected oncolytic virus OBP-301 in medically inoperable esophageal cancer.

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Background: Definitive chemoradiation (CRT) is a standard-of-care for patients (Pts) with medically inoperable esophageal cancer (EC). The NRG/RTOG 0436 study of cisplatin/paclitaxel and RT (+/- cetuximab) as definitive therapy showed that 58% of control arm Pts have locally persistent disease. OBP-301 is a conditionally-restricted, replication-competent adenovirus derived from human adenovirus type 5 that adds a human Telomerase Reverse Transcriptase gene promoter, replicating only in tumor cells to cause lysis.It may cause immunogenic cell death, enhance RT increasing radiosensitization by blocking DNA repair and improve local control. **Methods:** In NRG-GI007, a phase 1 study (NCT04391049), OBP-301 was added to weekly carboplatin/paclitaxel and RT (50.4 Gy/28 fxs) for medically inoperable EC Pts with pathologically proven adenocarcinoma or squamous cell carcinoma (SCC) of the esophagus or Siewert Type I/II gastroesophageal junction. Pts receive 1-2mL of intratumoral OBP-301 1×10^{12} virus particles/ml via endoscopy 3 days prior to and then at days 12 and 26 of CRT. The primary endpoint was protocol-defined dose-limiting toxicity, already reported. Secondary endpoints reported here are treatment-related (definitely or probably related to any protocol treatment) AEs - all and grade 4+ non-hematologic, and cCR in the primary tumor, defined as negative EGD/biopsy 6-8 weeks post-CRT. Per the protocol design, no statistical testing is done for these endpoints. **Results:** Initially, 6 evaluable (eligible and started protocol treatment) Pts were enrolled from June 2020 to April 2023. As initial dose level was deemed safe, an additional 9 Pts were enrolled from August 2023 to July 2024, totaling 15 evaluable Pts for secondary endpoints. Median age was 74 years, 9 Pts (60%) with ECOG PS 1. 11 Pts (73%) had adenocarcinoma and 4 had SCC; 11 Pts had N+ disease. 14 Pts (93%) received all planned OBP-301 injections and received 50.4 Gy RT. The most common treatment-related grade 3/4 AEs were decreased neutrophil (6 Pts; 40%) and lymphocyte counts (5 Pts; 33%), expected with CRT. No grade ≥3 OBP-301-related non-hematologic AEs were reported. 2 Pts died prior to restaging neitherdefinitely or probably related to OBP-301: 1 Pt developed grade 5 respiratory failure 6 weeks after CRT (suspected RT pneumonitis), while a 2nd Pt had an unrecognized tumor invasion of the bronchus at baseline, which worsened during week 3 of CRT, resulting in a grade 5 tracheo-esophageal fistula. 2 Pts have not yet reached time for restaging. All 11 Pts who have been restaged had a cCR, for a preliminary cCR rate of 100% (95% CI: 74.1%, 100%). **Conclusions:** OBP-301 + CRT is safe and the preliminary cCR rate compares favorably to the historical control from NRG/RTOG 0436. Updated safety and cCR data will be presented. A randomized study is being planned. Clinical trial information: NCT04391049. Research Sponsor: NRG Oncology SDMC; U10CA180822; NRG Oncology Operations; U10CA180868; IROC; U24CA180803; Oncolys BioPharma.

Transarterial infusion chemotherapy and embolization (TAICE) as a pre-operative therapy for patients with locally advanced gastric cancer (LAGC): A prospective, single-arm, pilot study.

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Background: Interventional therapy is a local intensive therapy for tumor control, often utilized in palliative treatment for solid organ tumors. Yet, its application in tumors of hollow viscera still lacks evidence. Our previous retrospective study revealed transarterial infusion chemotherapy and embolization (TAICE) not only effectively managed tumor-related hemorrhage but also exhibited a notable anti-tumor efficacy in gastric cancer. This prospective study aims to further explore the feasibility and safety of TAICE as a pre-operative therapy for locally advanced gastric cancer (LAGC). **Methods:** Patients diagnosed with locally advanced adenocarcinoma of stomach and gastrointestinal junction (GEJ) with no distant metastasis (cT3-4N+M0) were enrolled. They were given 4 cycles of pre-operative treatments: 2 cycles of TAICE-SOX (Oxaliplatin [85mg/m^2 transarterial infusion on Day 1] and S-1 [40/50/60mg po bid on Day 1-14]) and 2 cycles of SOX (Oxaliplatin [130mg/m^2 IV on Day 1] and S-1 [same as above]), alternately. Then, they received D2 radical gastrectomy, and 4 cycles of SOX post-operatively. In brief, the TAICE procedure was summarized into 4 steps: 1) Celiac arteriography, 2) Super-selection of tumor feeding artery (TFA), 3) Infusion chemotherapy and embolization of TFA, 4) Celiac re-arteriography to ensure the success of embolization. The primary endpoint was pathological complete response (pCR) rate. Other endpoints included major pathological response (MPR) rate, overall survival (OS), progression-free survival (PFS), treatment-related adverse events (TRAEs). **Results:** Between December 2020 and June 2023, twenty patients were enrolled and all of them received TAICE. Among them, 18 (90.0%) patients received 2 cycles of TAICE. The pCR and MPR rate were 55.0% and 70.0%, respectively. The 1-year and 3-year OS rate were 100.0% and 82.6%. The 1-year and 3-year PFS rate were 90.0% and 67.3%. There was no difference in OS between the groups of high and low tumor residual rate with the threshold of 50%. Patients achieving low tumor residual rate had significantly longer PFS (NR vs. 12.17 months, p=0.018) than those with high rate after TAICE. Moreover, we divided TFA into two subtypes: dispersive and branching, depending on the vascular pattern presented in the angiography at the first cycle TAICE. Patients with dispersive TFA had higher risk of recurrence after TAICE (NR vs. 15.17 months, p=0.0264). All patients had TRAEs (vomiting, nausea, abdominal pain and fever) of grade 1 or 2 after the first cycle of TAICE. Only one patient experienced nausea of grade 3 after the second cycle of TAICE. The most common TRAEs were vomiting and nausea. **Conclusions:** TAICE is a promising pre-operative therapy for patients with LAGC for it displaying the satisfactory oncological effectiveness and safety profile. Clinical trial information: NCT05396326. Research Sponsor: Natural Science Foundation of Shanghai (23ZR1410700); Natural Science Foundation of Shanghai (23Y11902100); Natural Science Foundation of Fujian Province (2023J011692).

Chemoimmunotherapy plus radiotherapy in advanced esophageal squamous-cell carcinoma.

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Background: Chemotherapy plus Immune checkpoint inhibitors (ICIs) have improved survival outcomes of patients with advanced or metastatic esophageal squamous-cell carcinoma in both first- and second-line settings. However, the benefit of additional radiotherapy for the residual esophageal lesion after chemo-immunotherapy remains unclear. **Methods:** We conducted a one-arm, three-center, open label study involving patients who were treatment naïve, radiological and histological confirmed advanced or metastatic squamous-cell esophageal carcinoma. Enrolled patients were expected the survival time was three months or longer; and ECOG performance status score was 0 or 1. In induction phase, patients received four cycles of TP regimen chemotherapy (docetaxel, 75/mg^2, iv on day 1 and cisplatin, 75mg/m^2, iv on day 1) combined with PD-1 inhibitor (sintilimab, 200mg intravenous infusion on day 1) every 21 days. Patients who finished four cycles of chemo-immunotherapy with stable disease (SD) or partial response (PR) were initiated 50Gy/25f irradiation for residual tumors. 3-4weeks after radiotherapy, maintenance sintilimab therapy was administered every 21 days for another 31 cycles or until disease progression or intolerable toxicity. **Results:** Total of 39 patients were enrolled in this study, 30 of them could be evaluated in efficiency and toxicities. All of them were male and were squamous-cell carcinoma in histology. 12/30 (40.0%) patients were in stage III, and 18/30 (60.0%) were in stage IV. 29/30 (96.7%) patients completed four cycles chemo-immunotherapy, the complete response (CR) rate was 6.7% (2/30), the partial response (PR) rate was 53.3% (16/30) and disease control rate (DCR) was 86.7% (26/30). The median depth of response (DpR) was 34.5% for chemoimmunotherapy and was 64.0% after radiotherapy. The progression-free survival (PFS) was 16.4 months and the overall survival (OS) was not reached. The most common grade 3 or worse adverse event was neutropenia and occurred in 3 (10.0%) patients. **Conclusions:** Four cycles sintilimab plus cisplatin and docetaxel followed by radiotherapy for residual tumors and maintained sintilimab had high response rate, prolonged PFS and tolerable toxicities as first line treatment in patients with advanced or metastatic esophageal squamous-cell carcinoma. Longer OS is promising and more patients enrolled in this study is needed in future. Clinical trial information: NCT06138028. Research Sponsor: None.

Clinicopathological features and patient factors.

Factors	No. of patients (n=30) (%)
Gender	
Male	30 (100.0)
Female	0 (0)
Age (year)	
< 65	27 (43.3)
≥ 65	17 (56.7)
PD-L1 (CPS)	
< 5	9 (30.0)
≥ 5	21 (70.0)
Clinical Stage	
III	12 (40.0)
IV	18 (60.0)
Metastatic Pattern	
Regional lymph node	12 (40.0)
Cervical & Supraclavicular	4 (13.3)
Abdomen	5 (16.7)
Lung	6 (20.0)

Efficacy of pembrolizumab in advanced esophageal carcinoma: A systematic review and meta-analysis.

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Background: Pembrolizumab, a monoclonal immunoglobulin G4 antibody is a programmed cell death protein 1 inhibitor, approved for treatment of advanced (metastatic or recurrent) esophageal squamous cell carcinoma (ESCC) and esophageal adenocarcinoma (EAC). We performed a pooled analysis to assess the efficacy of pembrolizumab in esophageal carcinoma (EC) and to compare it with chemotherapy alone. **Methods:** We conducted a systemic literature search, using PubMed, EMBASE, and ClinicalTrials.gov, till July 30, 2024. Initial search yielded 412 articles. We excluded duplicates and irrelevant studies and included 4 clinical trials (CT) that reported pembrolizumab's efficacy in EC. We estimated pooled objective response rate (ORR), partial response (PR), and hazard ratio (HR) for overall survival (OS) and progression-free survival (PFS), along with 95% confidence interval (CI) and p-value in RevMan v.5.4. Subgroup analysis, based on combined positive score (CPS) for programmed cell death ligand-1 (PDL-1) expression and tumor histology type was also performed. **Results:** We included four CTs in our study; their characteristics are shown in the table. The estimated pooled ORR and PR of pembrolizumab in all patients was 21% (CI: 0.10-0.41) ($p < 0.001$) and 19% (CI: 0.08-0.37) ($p < 0.001$), respectively. Pooled median PFS was 3.05 mo (CI: 0.92-5.17) ($p = 0.004$), and pooled median OS was 8.05 mo (CI: 5.21-10.88) ($p < 0.001$). Pembrolizumab use in EC showed a reduced risk of disease progression vs chemotherapy alone [HR: 0.85 (CI: 0.75-0.96) ($p < 0.05$)] and a 21% reduction in mortality [HR: 0.79 (CI: 0.71-0.88) ($p < 0.001$)]. Safety analysis revealed less grade ≥3 adverse events [OR: 0.56 (CI: 0.35-0.90), $p < 0.05$)] with pembrolizumab compared with chemotherapy alone. On subgroup analysis between ESSC and EAC, pembrolizumab exhibited statistically nonsignificant responses with ORR (OR: 1.29, CI: 0.89-1.88, $p = 0.18$), OS (pooled HR: 0.94, CI: 0.76-1.15, $p = 0.53$). Pembrolizumab in patients with CPS ≥10 vs CPS <10 also showed statistically non-significant responses with ORR [OR: 1.70, CI: 0.99-2.91, $p = 0.06$), OS (pooled HR: 0.63 vs 1.00) ($p = 0.05$) and PFS (pooled HR: 0.62 vs 0.83, $p = 0.08$). **Conclusions:** Pembrolizumab alone or combined with chemotherapy compared with chemotherapy alone exhibited favorable responses in terms of ORR, OS and PFS with a manageable toxicity profile, setting a benchmark for future studies. Research Sponsor: None.

Characteristics of included studies.

Trial ID, phase	Histology	Line	Number of patients	Regimen	Median follow-up (mo)	ORR (%)	Median PFS (mo)	Median OS (mo)
KEYNOTE-590, III	ESCC, EAC	1st	749	Pembro + chemo vs chemo	58.8	45 vs 29.3	6.3 vs 5.8	12.3 vs 9.8
KEYNOTE-181, III	ESCC, EAC	2nd	628	Pembro vs chemo	7.1	13.1 vs 6.7	2.1 vs 3.4	9.3 vs 6.9
KEYNOTE-180, II	ESCC, EAC	3rd or later	121	Pembro	5.8	9.9	2.0	5.8
KEYNOTE-028, 1b	ESCC, EAC	3rd or later	21	Pembro	7	30	1.8	7

Pembro: pembrolizumab, mo: month.

Correlation between immune-related adverse events and clinical outcomes in patients treated with first-line nivolumab plus chemotherapy for advanced gastric or gastroesophageal junction cancer: A multi-center retrospective study in Japan.

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Background: The CheckMate 649 trial and ATTRACTION-4 trial showed the efficacy of first-line nivolumab plus chemotherapy (Nivo-CT) for advanced gastric or gastroesophageal junction (GEJ) cancer. Recent studies have shown that immune-related adverse events (irAEs) caused by immune checkpoint inhibitors were associated with survival benefit in patients with solid tumors. However, there were few data on correlation between irAEs and efficacy of first-line Nivo-CT for advanced gastric or GEJ cancer. **Methods:** This multi-center retrospective study included patients with histologically confirmed advanced gastric or GEJ adenocarcinoma who were started on first-line Nivo-CT between December 2021 and December 2023. We divided the patients into two groups according to occurrence of irAEs; those with irAEs (irAE group) or those without (non-irAE group) and assessed the efficacy in both groups. Efficacy was evaluated by overall survival (OS), progression-free survival (PFS) and objective response rate (ORR). Landmark analysis at 2 months after initiating Nivo-CT was performed to adjust effects of early progression or death, in which patients who had events up to 2 months were excluded. **Results:** A total of 271 patients were included (median age, 70 [range 18–85] years; male/female, 64/36%; performance status (PS) 0/1/2, 47/49/4%; primary tumor site gastric/GEJ, 90/10%; status unresectable/recurrent, 81/19%; histology intestinal/diffuse, 30/70%; PD-L1 CPS <1/1~5/>5, 15/33/53%). Median follow-up time was 11.3 (range 1-30) months, and 80 (30%) patients developed irAE. In the landmark analysis, the median PFS was 11.1 months in the irAE group and 7.2 months in non-irAE group (HR 0.70 [95%CI 0.50-0.97], p = 0.034), and the median OS was 22.5 months in the irAE group and 17.9 months in non-irAE group (HR 0.77 [95%CI, 0.50-1.17], p = 0.219). The ORR were 67.4% in the irAE group and 66.7% in non-irAE group (p = 0.934). Multivariate analysis indicated that PS ≥1 (HR 1.66 [95%CI, 1.20-2.30], p = 0.02), elevated ALP (HR 1.61 [95%CI, 1.13-2.27], p = 0.008), and absence of irAEs (HR 1.55 [95% CI, 1.08-2.22], p = 0.018) were associated with poor prognosis. The most frequent irAEs were pneumonitis (n = 16), and grade ≥3 irAEs were observed in 26 (9.6%) patients: pneumonitis (n = 6), hepatitis/hypophysitis (n = 5). **Conclusions:** Development of irAEs was associated with improved outcomes in advanced gastric or GEJ cancer patients receiving first-line Nivo-CT.
Research Sponsor: None.

HLX22 plus trastuzumab and XELOX for first-line treatment of HER2-positive locally advanced or metastatic gastric/gastroesophageal junction cancer (G/GEJC): Updated results with additional patients.

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Background: Approximately 12–23% G/GEJC patients have human epidermal growth factor receptor 2 (HER2)-positive disease. Despite survival benefit from combination therapy with trastuzumab and chemotherapy, the prognosis remains unsatisfactory; more effective treatments are needed. This phase 2 study is evaluating the combination of HLX22 (an anti-HER2 monoclonal antibody targeting a different epitope than trastuzumab), trastuzumab, and XELOX chemotherapy as first-line treatment for patients with advanced/metastatic G/GEJC. Following the most recent report of 17 to 18 patients in each group at ASCO 2024 Annual Meeting, herein we present the updated efficacy and safety results with 31 patients in each group. **Methods:** Patients with locally advanced or metastatic HER2-positive G/GEJC and no prior systemic antitumor therapy were enrolled. Herein reported are results from Stage 2 of the study. Eligible patients were randomized to receive either HLX22 + trastuzumab + XELOX or placebo + trastuzumab + XELOX in 3-week cycles. Primary endpoints were independent radiology review committee (IRRC)-assessed progression-free survival (PFS) and objective response rate (ORR) per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1. Secondary endpoints included other efficacy and safety endpoints. **Results:** As of June 30, 2024, 62 patients were randomized to the respective groups (31 vs 31), of whom 51 (82.3%) patients were male. Median follow-up duration was 20.3 and 24.0 months for the respective groups. The efficacy results are shown in the table. Treatment-emergent adverse events (TEAEs) were reported in 30 (96.8%) and 31 (100%) patients, and HLX22- or placebo-related TEAEs of grade 3 or higher were reported in 9 (29.0%) and 6 (19.4%) patients in the respective groups. One patient (3.2%) in the placebo + trastuzumab + XELOX group had a grade 5 HLX22-/placebo-related TEAE. **Conclusions:** With a manageable safety profile, the addition of HLX22 to first-line treatment with trastuzumab plus XELOX conferred survival benefit for HER2-positive G/GEJC patients. Clinical trial information: NCT04908813. Research Sponsor: Shanghai Henlius Biotech, Inc.

Updated efficacy, IRRC-assessed.

	HLX22 + trastuzumab + XELOX (n=31)	placebo + trastuzumab + XELOX (n=31)
Median PFS, months (95% CI)	NR (23.49, NE)	8.3 (5.7, 12.7)
HR (95% CI)	0.2 (0.06, 0.45)	-
12-month PFS rate (95% CI)	73.8 (50.3, 87.4)	34.2 (12.0, 58.1)
24-month PFS rate (95% CI)	61.5 (30.4, 82.0)	24.0 (10.3, 27.3)
Confirmed ORR, % (95% CI)	87.1 (70.2, 96.4)	80.6 (62.5, 92.5)
Week 48 ORR, % (95% CI)	38.7 (21.8, 57.8)	9.7 (2.0, 25.8)
Median OS, months (95% CI)	NR (17.6, NE)	22.0 (10.6, NE)
HR (95% CI)	0.5 (0.20, 1.21)	-
Median DOR, months (95% CI)	NR (22.1, NE)	9.7 (4.6, 20.0)
HR (95% CI)	0.1 (0.04, 0.41)	-

CI, confidence interval. NE, not evaluable. NR, not reached.

Phase II preoperative pembrolizumab for MSI high or MSS/PD-L1 gastric cancer followed by surgery and adjuvant therapy with pembrolizumab (NCT03257163): Results of a multicenter study.

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Background: Data suggest that patients with tumors with microsatellite instability (MSI-H), EBV expression, or positive PD-L1 expression can benefit from immunotherapy. This trial evaluated PD-1 immunotherapy in this subset of patients with operable gastric cancer. **Methods:** We present results of a phase 2 multi-institutional clinical trial (NCT03257163). Patients with cT2-T4, N0-N3, M0 gastric adenocarcinoma with MSI-H, PDL1 CPS > 1%, or EBV+ were included. Patients received 2 cycles preop pembrolizumab 200 mg IV q3 weeks followed by surgery. Pembrolizumab was given for 16 additional cycles postoperatively, concurrently with adjuvant chemoradiation (45 Gy) and 5 cycles of capecitabine (825 mg/m2 days 1-14 during cycles 3, 6 and 7; 625 mg/m2 BID days 1-14 during cycles 4 and 5 with radiation). Patients were defined as evaluable if they received > one cycle of postop pembrolizumab. Primary endpoint is DFS, powered to detect 3-yr DFS of 70% with 89% power. **Results:** All enrolled patients received two cycles of pembrolizumab preoperatively (n=45). Eight patients did not undergo surgery (1 due to frailty, 2 with distant progression on preoperative imaging, 2 with peritoneal disease on surgical exploration and 3 with locally advanced, unresectable tumors). Six patients did not receive adjuvant therapy (2 due to surgical complications, 2 for failure to thrive, 1 refusal and 1 pending adjuvant therapy). 31 patients received adjuvant therapy and are included in the predetermined, evaluable analytic cohort. These 31 patients had a median age of 67 years (range 44 – 85) with 22.6% Hispanic, 29.0% Asian, 25.8% Black and 25.8% White. Median follow-up was 31.1 months (range 3.45-63.8). In these patients, 14 (45.2%) had MSI-H, 2 (6.5%) EBV+, and 29 (93.5%) had PDL1 expression >1% tumors. Most tumors were advanced stage with 81% ≥ cT3 and 58% cN+. In the 31 evaluable patients, the 3-year DFS was 79.4%; 3-year OS was 78.9%. For MSI-H patients, 3-year DFS was 81.8%. For patients with MSS/PD-L1+ tumors, 3-year DFS was 77.8%. For the 45-patient cohort, 3-year DFS was 70%; 3-year OS was 65%. Three patients had pCR, two with MSI-H tumors and one with PD-L1 CPS of 4%. Downstaging occurred in 62.2% of patients, and 40% of surgical specimens had pT0/T1 tumors consistent with significant pathologic responses. **Conclusions:** A biomarker-driven preoperative treatment strategy for MSI-H, EBV+, and MSS/PDL1+ operable gastric adenocarcinoma patients is promising and warrants additional evaluation. This treatment strategy is novel and effective, and it incorporates immune checkpoint immunotherapy, low dose chemotherapy, and radiation thus avoiding more intensive combination chemotherapy regimens for patients with these biomarkers. Clinical trial information: NCT03257163. Research Sponsor: Merck.

Paclitaxel oral solution versus paclitaxel injection as a second-line therapy in advanced gastric cancer: A randomized, open-label, non-inferiority phase 3 trial.

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Background: Paclitaxel injection (paclitaxel IV) is category 1 preferred regimen in the second-line therapy of gastric cancer, while it has shortcomings including vehicle-led safety risk, long time injection, premedication or frequent hospital visit. Paclitaxel oral solution (Liporaxel), the world's first successfully developed oral formulation, is an alternative. This study aimed to establish non-inferiority in efficacy and comparable safety profile of paclitaxel oral solution versus paclitaxel IV, as a second-line monotherapy in gastric cancer. **Methods:** This is a randomized, open-label, non-inferiority phase 3 trial conducted at 53 centers in China. Patients with unresectable or recurrent or metastatic gastric cancer progressed after fluoropyrimidine- or fluoropyrimidine plus platinum-based first-line therapy were randomly assigned at 1:1 ratio (stratified by gastrectomy, ECOG PS and prior chemotherapy) to receive paclitaxel oral solution (200mg/m^2 twice daily on days 1, 8, 15 of a 28-day cycle) or paclitaxel IV (175mg/m^2 on day 1 of a 21-day cycle). Co-primary endpoints were progression-free survival (PFS) assessed by blind independent review committee (BIRC) and overall survival (OS), with non-inferiority margin of hazard ratio (HR) of 1.18 for PFS and 1.16 for OS. **Results:** From April 22, 2019, to January 31, 2022 (data cut-off), 536 patients were randomized to receive either paclitaxel oral solution (n=268) or paclitaxel IV (n=268), with a median follow-up of 13.4 vs. 12.6 months. PFS met non-inferiority criteria, with BIRC-assessed median PFS of 3.02 months (95% confidence interval [CI]: 2.69, 3.71) in the oral group vs. 2.89 months (95% CI: 2.53, 3.48) in the IV group (HR=0.894, 95% CI: 0.719, 1.112, p=0.311). OS (cut-off on February 15, 2023, for OS primary analysis) demonstrated superiority for paclitaxel oral solution, with median OS of 9.13 months (95% CI: 7.72, 10.97) vs. 6.54 months (95% CI: 5.75, 7.26), showing 2.59-month improvement (HR=0.770, 95.5% CI: 0.635, 0.934, p=0.006). For the treatment-related adverse events (TRAEs), the oral group showed lower incidences of neuropathy (22.3% vs. 38.7% all grade), alopecia, fatigue, musculoskeletal and connective tissue disorders, and no hypersensitivity occurred without premedication. The most common ≥Grade 3 TRAEs were neutrophil count decreased (47.9% in oral vs. 54.5% in IV), white blood cell count decreased (41.5% vs. 35.3%) and anemia (16.6% vs. 10.9%). Grade 5 TRAEs were rare with comparable rates across two groups (four [1.5%] vs. three [1.1%]). **Conclusions:** Paclitaxel oral solution demonstrated non-inferiority in PFS and superiority in OS compared to paclitaxel IV, with clinically manageable and favorable safety profile, supporting paclitaxel oral solution as a second-line treatment option for gastric cancer. Clinical trial information: CTR20190050. Research Sponsor: None.

Neoadjuvant adebrelimab plus chemotherapy for resectable locally advanced esophageal squamous cell carcinoma: A phase 2 trial.

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Background: Several clinical trials have demonstrated the enhanced pathological response with the addition of immune checkpoint inhibitor to neoadjuvant chemotherapy in patients with resectable esophageal squamous cell carcinoma (ESCC), including the phase 3 ESCORT-NEO trial. Adebrelimab, an anti-programmed cell death-ligand 1 antibody, has shown potential as neoadjuvant monotherapy in the phase 1b NATION-1907 trial of resectable ESCC, with a 2-year overall survival rate of 92% in 30 patients. This study evaluated the efficacy and safety of adebrelimab plus chemotherapy as neoadjuvant therapy in patients with resectable locally advanced ESCC. **Methods:** In this single-center phase 2 trial (NCT06178211), patients aged 18-75 years with stage II-III (cT2N1-2M0 or cT3N0-2M0) ESCC received adebrelimab (1200 mg, day 1) plus albumin-bound paclitaxel (100 mg/m^2, days 1, 8, and 15) and carboplatin (area under the curve of 5 mg/mL/min, day 1) every 3 weeks for 2 cycles. Esophagectomy was performed 4-6 weeks after neoadjuvant therapy. The primary endpoint was pathological complete response (pCR; ypT0N0) rate. Secondary endpoints were clinical complete response (cCR) rate (defined as no residual tumor after neoadjuvant therapy, based on computed tomography, positron emission tomography-computed tomography [PET-CT], endoscopic biopsy, and endoscopic ultrasound-guided fine-needle aspiration [EUS-FNA]), margin-free (R0) resection rate, major pathological response (MPR; ≤10% residual tumor cells) rate, event-free survival, overall survival, and safety. **Results:** Between January 2024 and May 2024, a total of 36 patients were enrolled. Most patients (75.0%) had clinical stage III ESCC. Thirty-five patients completed 2 cycles of neoadjuvant therapy, while one patient refused the last dose of albumin-bound paclitaxel. Two patients refused surgery while 34 underwent esophagectomy. The pCR rate was 38.9% (95% confidence interval [CI], 23.1-56.5) in all patients and 41.2% (95% CI, 24.6-59.3) in the surgery set; the R0 resection rate was 94.4% (95% CI, 81.3-99.3) and 100% (95% CI, 89.7-100), and the MPR rate was 63.9% (95% CI, 46.2-79.2) and 67.6% (95% CI, 49.5-82.6), respectively. The cCR rate was 47.2% (95% CI, 30.4-64.5) in all patients. Of 36 patients, grade ≥3 treatment-emergent adverse events during neoadjuvant therapy included decreased neutrophil count (2 [5.6%]), decreased white blood cell (2 [5.6%]), decreased platelet count (1 [2.8%]), impaired hearing (1 [2.8%]), hyponatremia (1 [2.8%]), immune-mediated hepatitis (1 [2.8%]), intestinal obstruction (1 [2.8%]), and esophageal ulcer (1 [2.8%]). Surgical complications occurred in 17 (50%) of 34 patients. **Conclusions:** Neoadjuvant adebrelimab plus chemotherapy shows promising pathological response and favorable safety profile in patients with resectable locally advanced ESCC. Survival follow-up is ongoing. Clinical trial information: NCT06178211. Research Sponsor: Jiangsu Hengrui Pharmaceuticals Co., Ltd.

Preliminary results of benmelstobart plus anlotinib combined with SOX in the first-line treatment of advanced gastric or gastroesophageal junction (G/GEJ) adenocarcinoma with low PD-L1 expression: A single-arm, multicenter phase II clinical trial.

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Background: First-line chemotherapy for advanced HER2-negative gastric/gastroesophageal (G/GEJ) adenocarcinoma has limited efficacy. PD-1 inhibitors plus chemotherapy show promise but need improvement, especially for patients (pts) with low PD-L1 expression. Anlotinib, a multi-targeted TKI for tumor angiogenesis/proliferation, is approved in China. This study aims to evaluate the efficacy and safety of benmelstobart a PD-L1 blockade) in combination with anlotinib and SOX (S-1 plus oxaliplatin) as a first-line regimen for advanced G/GEJ adenocarcinoma in patients with low PD-L1 expression, Preliminary findings will be reported promptly. **Methods:** Pts with HER2-negative, unresectable, locally advanced, or metastatic G/GEJ adenocarcinomas and a PD-L1 Combined Positive Score (CPS) < 5, who had not received prior systemic therapy were included. They received benmelstobart (1200mg, iv, d1, q3w) combined with anlotinib (10mg, po, d1~14, q3w), oxaliplatin (130mg/m^2, d1, iv, q3w) and S-1 (40mg, po, bid, d1~14, q3w) for 6 cycles as initial therapy. Maintenance therapy with benmelstobart (1200mg, iv, d1, q3w) plus anlotinib (10mg, po, d1~14, q3w) followed for non-progressive diseaseuntil PD or unacceptable toxicity occurred. Tumor responses were evaluated by RECIST 1.1 criteria. The target sample size was 37, with ORR as the primary endpoint, and safety, DCR, DoR, PFS, and 1-year OS rate as secondary endpoints. **Results:** From Jun 2023 to Sep 2024, 28 pts were enrolled and 27 were available for efficacy and safety evaluation. The patients' characteristics were as follows; median age (range): 59 (38-77) years, Male/Female: 19 (70%)/8 (30%), ECOG PS 0/1: 2 (7%)/25 (93%), Surgical history yes/no: 6 (22%)/21 (78%), CPS: $<1/\geq1$: 4 (15%)/23 (85%), respectively. In the response assessment, 19 PR (70.4%), 6 SD (22.2%), and 2 NE (7.4%) were observed, yielding an ORR of 70.4% (95% CI: 49.8-86.2) and DCR of 92.6% (95% CI: 75.7-99.1). As of September 17, 2024, 6 of 27 patients discontinued (3 due to PD, 3 voluntary), leaving 21 ongoing. The longest DoT was 15.08 months, and the median PFS was not reached. Common TEAEs ≥10% included thrombocytopenia (37%), anemia (33%), leukopenia (22%), HFS (19%), fatigue (15%), liver function abnormalities (15%), and appetite loss (11%). Grade ≥3 TEAEs were thrombocytopenia (4%) and anemia (4%). **Conclusions:** Preliminary findings indicate that the combination of benmelstobart and anlotinib with the SOX regimen as a first-line treatment for advanced G/GEJ adenocarcinoma with low PD-L1 expression demonstrates promising therapeutic efficacy and tolerable adverse events. Further validation of these results in a larger, consecutive patient population is warranted. Clinical trial information: ChiCTR2400085396. Research Sponsor: None.

Fruquintinib plus camrelizumab combined with paclitaxel liposome and nedaplatin as first-line treatment for advanced esophageal squamous cell carcinoma (ESCC): A single-arm, phase II clinical trial.

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Background: Camrelizumab in combination with chemotherapy has become a standard of care for the first-line treatment of advanced ESCC. Previous studies have indicated synergistic effect of fruquintinib in combination with chemotherapy or immunotherapy. Hence, we conducted a phase II trial to evaluate the efficacy and safety of fruquintinib plus camrelizumab, paclitaxel liposome, and nedaplatin as a first-line therapy for advanced ESCC. **Methods:** This is a single-arm, phase II study consisting of a dose-finding and a dose-expansion phase. A total of 33~36 patients aged 18~80 years with previously untreated advanced ESCC and an ECOG PS of 0-2 were planned to be enrolled. The dose-finding phase followed a 3+3 approach in order to determine the recommended phase II dose (RP2D) of fruquintinib for dose expansion. Patients received fruquintinib (3 mg, 4 mg, and 5 mg once daily on days 1-14, starting at 4 mg) orally, plus intravenous camrelizumab 200 mg d1, paclitaxel liposome 135 mg/m^2 d1, and nedaplatin 70 mg/m^2 d1, 21 days as one cycle. A maximum of 6 cycles was conducted, followed by maintenance therapy with fruquintinib in combination with camrelizumab. The primary endpoint was objective response rate (ORR, RECIST 1.1). Secondary endpoints included RP2D, disease control rate (DCR), progression-free survival (PFS), and adverse events (AEs). **Results:** As of August 24, 2024, 11 patients (9 males) were enrolled with a median age of 66 years (range 50-76). All patients had distant metastases, and the median number of sites of metastasis was 2 (range 1-5). In the dose-finding phase, 3 patients were treated with 4 mg of fruquintinib, and 6 patients were treated with 5 mg of fruquintinib. No dose-limiting toxicities (DLTs) were observed. Thus, the RP2D of fruquintinib was established as 5 mg once daily on days 1-14, 21 days as one cycle. Of the 7 evaluable patients, 6 achieved partial response, and 1 had stable disease. The ORR was 85.7% (95% CI 42.1%-99.6%), and the DCR was 100.0% (95% CI: 59.0%-100.0%). The median PFS was not reached. The most common treatment-related adverse events (TRAEs) were anemia (7/11), hypercholesterolemia (6/11), hypoalbuminemia (5/11), oral mucositis (5/11), hematuria (4/11), hypertension (4/11), fatigue (4/11), constipation (3/11), and neutropenia (3/11). Grade 3 TRAEs were identified in 2 patients, including oral mucositis (2/11), and anemia (1/11). No treatment-related serious adverse events (SAEs) or deaths occurred. **Conclusions:** The combination of fruquintinib, camrelizumab, paclitaxel liposome, and nedaplatin demonstrated significant efficacy and manageable toxicity profiles as a first-line treatment for advanced ESCC, suggesting a potential new treatment strategy. Clinical trial information: NCT06010212.
Research Sponsor: None.

Prediction of relapse-free survival in stage III gastric cancer patients treated with postoperative adjuvant chemotherapy: Random survival forest analysis from the START-2 trial.

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Background: The START-2 trial demonstrated that docetaxel plus S-1 (DS) was superior to S-1 alone (S-1) in terms of recurrence-free survival (RFS) and overall survival (OS) as the adjuvant chemotherapy for stage III gastric cancer (GC) patients (pts) (J Clin Oncol 2019, Gastric Cancer 2022). We conducted the START-2 AR study as a retrospective analysis using data from the START-2 trial to develop a prognostic model for stage III GC pts treated with postoperative chemotherapy. **Methods:** Among 912 pts in the START-2 trial, 599 pts signed the consent form to participate in the START-2 AR study. There were 542 pts (265 pts in DS group and 277 pts in S-1 group) without missing values of prognostic candidates. Pts were randomly divided into a training set and a validation set with a ratio of 7:3. Prognostic model for RFS was developed using random survival forest (RSF) with the training set. The RSF model was applied to the validation set to assess its performance, evaluated using Harrell's C-index. We calculated tertiles of risk scores from the RSF model to divide the training set into low-, middle-, and high-risk subgroups. These cut-off values were used to classify the validation set. A Cox proportional hazard model was applied to compare the prognosis among the risk groups. **Results:** There were no statistically significant differences in patient characteristicsbetween the 599 pts in the START-2 AR study and the 912 pts in the STRAT-2 trial. RFS was comparable between the two groups. The RSF model for the training set (n = 380) yielded a C-index of 0.873, identifying the number of metastatic lymph nodes, serum albumin, CEA, tumor diameter, the primary tumor location, platelet count, age as prognostic factors in order of importance. In the validation set (n = 162), the C-index was 0.648. Both the low-risk group (n = 35) (hazard ratio [HR]: 0.34, 95% confidence interval [CI]: 0.17 – 0.68, $p < 0.01$) and the middle-risk group (n = 74) (HR: 0.55, 95% CI: 0.33 – 0.89, p = 0.02) in the validation set had significantly better RFS compared to the high-risk group (n = 53). Among the 76 pts in the low-risk group treated with DS, 44 were categorized into the low-risk group when assuming they had been treated with S-1 alone. The analysis of 599 pts after multiple imputation yielded results consistent with those from the complete data. **Conclusions:** We developed the RSF-based predictive model for stage III GC pts, which demonstrated good performance. The number of metastatic lymph modes was identified as the most important prognostic factor. Risk stratification based on this model effectively differentiated prognostic outcomes. Future external validation of the model is required. Clinical trial information: UMIN000011438. Research Sponsor: Taiho Pharmaceutical Co., LTD.; Japan Clinical Cancer Research Organization.

Results of a phase 2 trial of ramucirumab plus docetaxel as second-line treatment for patients with advanced gastric cancer (HGCSG 1903).

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Background: Ramucirumab (RAM) has shown efficacy in combination with paclitaxel (PTX) or irinotecan in the second-line treatment of advanced gastric cancer (AGC). However, the efficacy and safety regarding the combination therapy of RAM and docetaxel (DTX) have not been reported. We hypothesized that RAM plus DTX for patients with AGC could be as effective as RAM plus PTX and could reduce the incidence of neuropathy. **Methods:** HGCSG 1903 was a non-randomized, single arm, phase 2 trial carried out at 20 institutes in Japan. AGC patients with refractory or intolerance to primary chemotherapy were eligible. RAM (8 mg/kg on day1 and 15) plus DTX (60 mg/m^2 on day1) combination therapy administered in 28-day cycle were continued until disease progression or emergence of adverse events requiring discontinuation. The primary endpoint was response rate (RR) with threshold value of 10.7% and an expected value of 27.9%. A total of 35 cases are planned for registration. The secondary endpoints were overall survival (OS), progression-free survival (PFS), and safety. This trial was registered with Japan Registry of Clinical Trials, number jRCTs011200010. **Results:** Between Nov 2020 and Dec 2023, 36 patients were enrolled. Median age was 70 (range, 36–82); female/male were 7/29; ECOG PS 0/1, 16/20, respectively. One patient did not initiate study treatment at the onset of bowel obstruction after enrollment. In the efficacy and safety analysis set of 35 patients, RR was 25.7% (95%C.I., 12.5–43.3%) and disease control rate was 74.3% (95%C.I., 56.7–87.5%). Median OS and PFS were 11.9 months (95%C.I., 3.0–20.7) and 3.1 months (95%C.I., 2.1–4.2), respectively. Grade 3 or higher adverse events that occurred in more than 5% of patients were neutropenia (68.6%), leucopenia (57.1%), febrile neutropenia (17.1%), hypertension (14.3%), anorexia (11.4%), anemia (8.6%), and fatigue (5.7%). Grade 4 neutropenia occurred in 60.0% of patients. The protocol was amended to allow primary G-CSF prophylaxis during study period. The number of patients who received primary G-CSF prophylaxis after the amendment was 7/13 (53.8%). Neuropathy occurred in 65.7% of all grades, with no incidence of grade 3 or higher. No death and new safety signals with a causal relation to study treatment were observed. **Conclusions:** The primary endpoint of response rate was met in HGCSG 1903. Although neutropenia and febrile neutropenia may be more frequent, they were manageable. RAM plus DTX might be considered as an option for second-line treatment of patients with AGC. Clinical trial information: jRCTs011200010. Research Sponsor: Nipro Corporation.

Updated results from the trastuzumab deruxtecan (T-DXd) 5.4 mg/kg triplet combination of DESTINY-Gastric03 (DG-03): First-line (1L) T-DXd with fluoropyrimidine (FP) and pembrolizumab in advanced/metastatic HER2-positive (HER2+) esophageal adenocarcinoma, gastric cancer (GC), or gastroesophageal junction adenocarcinoma (GEJA).

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Background: T-DXd is a HER2-directed antibody-drug conjugate; T-DXd 6.4 mg/kg monotherapy is approved for patients with metastatic HER2+ GC/GEJA who have received a prior trastuzumab-based regimen. Preliminary results from DG-03 Part 2 showed feasibility and promising antitumor activity with 1L T-DXd 6.4 mg/kg (arm D) or 5.4 mg/kg (arm F) with FP and pembrolizumab in patients with esophageal adenocarcinoma/GC/GEJA. T-DXd 6.4 mg/kg and FP with pembrolizumab was associated with higher than expected toxicity; alternatively, early safety data from T-DXd 5.4 mg/kg triplet combination showed a manageable safety profile. We report updated results for arm F from DG-03 Part 2 with an approximate time-matched analysis with arm D. **Methods:** DG-03 (NCT04379596) is a Phase 1b/2 multicenter, open-label, dose-escalation (Part 1) and -expansion (Parts 2, 3, and 4) study. In Part 2, patients with HER2+ (immunohistochemistry [IHC] 3+ or IHC 2+/in situ hybridization–positive by local testing) esophageal adenocarcinoma/GC/GEJA, irrespective of programmed cell death ligand 1 status, and no prior treatment for metastatic disease were enrolled. In arm D, patients received T-DXd 6.4 mg/kg intravenous (IV) infusion every 3 weeks (Q3W) and FP (5-fluorouracil [5-FU] 600 mg/m^2 continuous IV or capecitabine [cape] 1000 mg/m^2 twice daily [BID]) with pembrolizumab 200 mg IV Q3W; in arm F, patients received T-DXd 5.4 mg/kg IV Q3W and FP (5-FU 600 mg/m^2 continuous IV or cape 750 mg/m^2 BID) with pembrolizumab 200 mg IV Q3W.Primary endpoint was confirmed objective response rate by investigator assessment (INV) per RECIST 1.1. Secondary endpoints included duration of response by INV and progression-free survival by INV. Safety and tolerability were also assessed. **Results:** A time matched analysis was performed to compare the efficacy of arm D (n=43), data cut-off (DCO) October 27, 2022, and arm F (n=32), DCO May 6, 2024, given the limited follow up of the 5.4mg/kg. The median duration of follow up for the two cohorts was 4.1 months and 4.6 months for arm D and arm F, respectively. Objective response rate for arm D was 41.9% and 59.4% for arm F. Median PFS was 6.4 months (95% CI: 5.0, NC) for arm D and 5.8 months (95% CI: 5.6, NE) for arm F. At DCO, the median overall survival in both arms was not reached. Updated time matched safety and efficacy data will be provided at the time of presentation. **Conclusion:** The time matched analysis shows that lowering the dose of T-DXd to 5.4 mg/kg from 6.4 mg/kg and lowering the starting dose of capecitabine, did not result in decrease in efficacy of the combination of T-DXd + FP + pembrolizumab. Funding: This study is sponsored by AstraZeneca in collaboration with Daiichi Sankyo. In March 2019, AstraZeneca entered into a global development and commercialization collaboration agreement with Daiichi Sankyo for trastuzumab deruxtecan (T-DXd; DS-8201). Editorial acknowledgment: Under guidance of the authors and in accordance with Good Publication Practice, medical writing and editorial support was provided by Carmen Grimaldos, PhD, of Helios Medical Communications, part of Helios Global Group, and was funded by AstraZeneca. Clinical trial information: NCT04379596. Research Sponsor: AstraZeneca in collaboration with Daiichi Sankyo.

449 **Poster Session**

Clinical results of a phase 1 trial evaluating a HER2 directed CAR-T product selective for the tumor microenvironment (TME) in patients with GI and other solid tumor malignancies.

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Background: A major limitation of CAR-T in solid tumors has been the inability to safely target antigens shared between healthy and malignant tissue. One approach to improving selectivity is to regulate CAR affinity through the unique metabolism of the tumor microenvironment (TME). This first in human trial evaluated an autologous (41BB) CAR-T product encoding a tumor metabolism regulated HER2 CAR. **Methods:** This Phase 1, dose escalation, basket trial evaluated the safety, tolerability, pharmacokinetics and efficacy in 12 patients (pts) with HER2 IHC 3+ / FISH + Stage IV r/r solid tumor malignancies (gastric, esophageal, colorectal, breast). They had failed multiple lines of therapy (66.7% patients had failed 4~10 lines of therapies previously). Dose cohorts of 3E5 (n=3), 1E6 (n=3), or 1E7 (n=6) CAR-T cells/kg were evaluated with a 28-day in-patient DLT observation period. Cell product was manufactured from whole blood, with a 12-day closed process and release on day 17. Lymphodepletion regimen was 500 mg/m^2 Cy and 30 mg/kg^2 Flu X 3 prior to infusion of CAR-T. **Results:** Data cutoff 09/16/24. All 12 patients were evaluable for safety and efficacy. No DLTs, ICANS or SAEs were observed. For HER2 positive normal tissues of special concern (e.g. cardiomyocytes), longitudinal functional, biochemical and clinical cardiovascular tests showed no signs of OTOT. One patient experienced Grade 2 CRS and recovered. No AE's lead to study discontinuation. In one patient, there were two Grade 3 AEs attributed to product (neutropenia and hypoxia); both which resolved. With a median follow-up of 302 days for cohort 3, 12-month survival rate in this cohort is 83.3%, compared to the 12-month survival rate of 33.3% for cohorts 1 and 2 combined. Encouraging signals of efficacy were observed with an ongoing (at 24 wks) deep PR (100% reduction SLD) in a GC patient. The median peak copy number of CAR-T cells in peripheral blood (n=12) was 1,818 copies/μg DNA (range 403 – 10,466). **Conclusions:** The tumor metabolism regulated HER2 CAR-T product was generally well tolerated, with no DLTs, and evidence of radiologic objective response in patients with gastroesophageal cancer. Combined, the safety and activity observed in this study suggests that a previously challenging solid tumor antigen may be safely targeted by a CAR with affinity regulated by the metabolism of the TME. Enrollment for the study was complete as of 28-Feb-2024 (NCT04511871). Clinical trial information: NCT04511871. Research Sponsor: Shanghai Perhum Therapeutics, LTD; EXUMA Biotech, Hong Kong.

Tislelizumab (Tisle) combined with POFI (irinotecan, paclitaxel, oxaliplatin and 5-FU/levoleucovorin) as first-line treatment of advanced gastric/gastroesophageal junction adenocarcinoma (AGC): OS analysis results of the SYLT-023.

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Background: Tisle is an anti-PD-1 antibody. The combination of Tisle + XELOX/FP is a first-line treatment of AGC in China. Both irinotecan and paclitaxel have also shown antitumor activity in AGC. In this phase I/II study, we explore the safety, tolerability, and efficacy of Tisle + POFI as first-line treatment of HER-2 negative, pMMR AGC. **Methods:** In a phase I dose finding study using a standard 3+3 design, subjects received four escalating dose levels (dl) of irinotecan/paclitaxel (mg/m^2): 135/45 (dl #1), 150/45 (dl #2), 135/67.5 (dl #3), and 135/90 (dl #4) in combination with tisle 200 mg plus oxaliplatin 85 mg/m^2, levoleucovorin 200 mg/m^2, and 5-FU 2400 mg/m^2 for 46 hours every 2 weeks. Primary endpoints were safety, tolerability, and RP2D. **Results:** Fifteen subjects with treatment naïve AGC were enrolled (3 each in dl #1, dl #2, and dl #3 and 6 in dl #4). The median age was 65 years (range 36-72); 80% were male. Two subjects (13.3%) were diagnosed with GE junction cancer, 11 (73.3%) had undifferentiated disease, and 5 (33.3%) had liver metastasis. PD-L1 CPS scores were: 5 (n=5); 1 (n=1); and 0 (n=9). All subjects were evaluated for DLT. One DLT (grade 4 neutropenia) occurred within 28 days in dl #4. No maximum tolerated dose was reached; the RP2D was dl #4. As of 18th Sep 2024, confirmed objective response rate in 13 subjects with measurable disease was 100% (1 CR, 12 PR) per RECIST 1.1. Of the 2 subjects with non-measurable disease, one was a CR and the other was non-CR/non-PD. The median PFS was 10.51 (95% CI: 7.44, 13.58) months (versus 6.9 [5.7-7.2] months for ITT population for RATIONALE 305, ESMO 2023), the median DOR was 7.39 (95% CI: 5.34, 9.44) months, and the median OS was 14.75 (95% CI: 5.48, 24.02) months. The OS showed no significant differences between CPS ≥ 1 and CPS < 1. All subjects (100%) had AEs. Seven (n=7, 46.67%) of the 15 subjects reported grade $\geq$3 AEs, including neutropenia (n=7; 46.67%), leukopenia (n=3; 20%), anemia (n=2; 13.33%). **Conclusions:** Tisle+POFI was well tolerated and showed preliminary antitumor activity in AGC. A phase II study is ongoing. Clinical trial information: NCT05319639. Research Sponsor: None.

Neoadjuvant immunotherapy and nonoperative management of dMMR gastroesophageal adenocarcinoma (GEA): A retrospective case series.

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Background: Perioperative chemotherapy with surgery is the standard of care for patients (pts) with resectable gastric or gastroesophageal junction adenocarcinoma (GEA). However, recent studies have shown more favorable responses after neoadjuvant immunotherapy (NIO) in patients with mismatch repair deficient (dMMR) cancers. Herein, we present the outcomes of 10 cases with dMMR GEA. **Methods:** We retrospectively reviewed patients with GEA at Mayo Clinic Arizona who met the following criteria: have dMMR disease by IHC, nonmetastatic operable disease, and received NIO +/- chemotherapy. Patient demographics, disease characteristics, and clinical outcomes were extracted. The primary endpoint was clinical complete response (cCR) in patients treated with non-operative management (NOM) defined as complete radiological response and no residual disease on endoscopic ultrasound (EUS) and biopsy. **Results:** 10 pts with dMMR GEA with a median age of 70 (43-85) were identified. 6 pts had T2N0, 1 had T4N0, 3 had N+ disease. 8 pts received NIO monotherapy upfront, 1 received NIO after progressing on chemotherapy, and 1 concurrently with chemotherapy. No grade 3/4 toxicities were reported, however, 2 pts discontinued therapy due to arthralgia and colitis. After NIO, 9/10 pts had significant or complete resolution of FGD avidity on PET/CT, while 1/10 had stable disease. 3/10 pts underwent surgical resection, with 2 achieving pathological complete response (pCR) and the third showing T1N0 disease with no recurrence 1 year after surgery. Notably, one of the pts with pCR passed away due to surgical complications. 7/10 patients had NOM, of whom, 4 had cCR and 3 had radiological response only, as no EUS data was available. 4/7 NOM pts had follow up data at 6 months with no evidence of progression, of whom one pt remained progression-free at 33 months. ctDNA testing was performed in 6/10 pts: 2 had undetectable ctDNA at baseline, while 3 became MRD-negative after NIO, consistent with radiological and/or pathological response, and 1 had persistent ctDNA both before and after NIO, despite cCR. **Conclusions:** This study demonstrates exceptional and prolonged response, along with good tolerability to NIO in pts with dMMR GEA, supporting the feasibility of non-operative management in such patients. Research Sponsor: None.

Phase II study of neoadjuvant NALIRIFOX followed by chemoradiation with paclitaxel and carboplatin in locally advanced gastroesophageal cancer.

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Background: We performed a single-arm phase II study of neoadjuvant NALIRIFOX and chemoradiation (CRT) with concurrent carboplatin/taxol (C/T) followed by surgery in patients with locally advanced gastroesophageal (GE) cancer. **Methods:** Patients were enrolled on an NCI sponsored, prospective, single arm study (NCT04656041). Key eligibility criteria included: histologically confirmed T3/4 or lymph node (LN) positive GE junction or esophageal cancer, ECOG PS ≤1, age 18+, and life expectancy > 3 months. Exclusion criteria included: metastatic disease, prior chemotherapy or RT, or prior targeted therapy for diagnosis. Extensive LN disease beyond the surgical field (supraclavicular or para-aortic) was permitted if deemed feasible to be encompassed within a RT field. Laparoscopy was not required. Pts were treated with neoadjuvant NALIRIFOX x 4-8 cycles pending patient tolerance and physician discretion, restaging, CRT (50.4 Gy in 28 fractions) with concurrent C/T, restaging, followed by surgical resection. Dose reductions were at discretion of the treating physician. The primary study endpoint was the pathologic complete response rate following neoadjuvant treatment with NALIRIFOX and CRT with concurrent C/T. Major pathologic response (mPR) was defined as Tumor Regression Grade 0 and 1. Secondary endpoints included: 1) acute toxicity; 2) clinical response per RECIST criteria; 3) PFS and 4) OS. **Results:** From June 2021 to October 2023, 40 patients were enrolled. Median age was 64 (range: 47-82), and 31 patients were male (78%). All patients started NALIRIFOX, and the median number of NALIRIFOX received was 5 (range: 1-8). Reasons for discontinuation of NALIRIFOX were: physician discretion (n=29), subject withdrew from treatment (n=2), progressive disease (n=1), death (n=1). Rates of grade 3+ toxicity for overall, gastrointestinal, and hematologic attributed to NALIRIFOX were 55%, 35%, and 13% respectively. 35 patients started chemoRT (88%) and 33 patients completed chemoRT (83%). Rates of grade 3+ toxicity for overall, gastrointestinal, and hematologic attributed to chemoRT were 86%, 11%, and 9% respectively. Of 33 who completed chemoRT, two patients declined surgery (including 1 with clinical CR), and 31 (78%) patients went for surgical exploration. No patients were found with intraoperative metastases. Therefore, 31 (78%) patients underwent surgical resection. In total, 9 patients had pCR (29% in resected cohort, 23% in ITT cohort) and 15 had a major PR (48% in resected cohort, 38% in ITT cohort). In total, 10 (25% in ITT) patients had pCR or clinically CR at 1 year. **Conclusions:** Neoadjuvant NALIRIFOX followed by CRT is feasible with acceptable rates of treatment completion and grade 3+ toxicity. In our phase II trial, the rate of pCR is promising and further studies incorporating this regimen should be considered. Clinical trial information: NCT04656041. Research Sponsor: None.

Non-metastatic gastric cancer (GC) in patients aged 80 years or older: Immediate and long-term outcomes of combined treatment.

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Background: Comorbidities and increased risk of chemotherapy-induced toxicity in elderly patients presents significant challenges in management of GC. Here we presents a single-center experience in treatment patients aged 80 years or older with non-metastatic GS, reporting both immediate and long-term outcomes of surgical treatment (ST). **Methods:** This retrospective analysis includes 61 patients (pts) diagnosed with non-metastatic GC or cardio-esophageal junction (CEJ) cancer. The primary endpoint was relapse-free survival (RFS); secondary endpoints included overall survival (OS) and the incidence of surgical complications (SC). **Results:** Among the 61 pts (median age 82.7 years, range 80.0-89.2) 39.3% were men. ECOG PS was 0-1 in 49 pts (80.6%) and 2 – in 12 pts (19.4%). The primary tumor site was the stomach in 57 pts (93.4%) and the CEJ in 4 pts (6.6%). Tumor stage had been defined as cT_{1-2} in 33.3% pts, cT_{3-4} in 57.1%; cN_0 – in 36.1%, stage remained unknown in 6.3%. Radical surgery was performed in pts 78.7% (n=48), 29 pts (47.5%) of them underwent ST only without peri-operative chemotherapy (CT). Neoadjuvant CT (NACT) received 23 pts (37.7%): platinum-based doublet regimen – 82.6% (n=19), while 17.4% (n=4) received capecitabine only. ST was performed in 47.8% (n=11) pts received NACT, other 12 pts didn't undergo ST due to progression of disease (8.3%), comorbidities (33.3%), or patient preference (16.7%) in 1, 4 and 2 cases, respectively. Adjuvant CT was administered to 9 pts (14.3%). Surgical interventions included gastrectomy in 16.6% pts (n=9), proximal or distal resection – in 8.3% (n=4) and 74.1% (n=35), respectively. R0-resection was achieved in 42 pts (88.6%). SC has 20.8% (n=10) pts: classified as Clavien-Dindo grade (G) II – 4.4%, G IIIa – 8.9%, G IIIb - 8.9%; these included anastomotic leakage (6.7%), abscess formation (2.2%), cardiac events and thrombosis (4.4%), and eventration (2.2%). SC G IV or V not reported. The median duration of postoperative hospitalization was 8 days. In the overall cohort, the 3-y RFS was 53% (95%CI 28.9-53.9), 3-y OS – 67% (95% CI 27.9-75.2). Among the pts with stage IA (n=15), the 3-y RFS was 72% (95%CI 49.1-85.1), with a maximum follow-up period of 80 months and no cancer-related deaths. In patients with $cT_{2-4}N_{any}$ 3-y RFS was 70% in the overall cohort and 74% in the surgery-only group (HR 1.09; 95%CI 0.45-2.65, p=1). **Conclusions:** Our findings demonstrate that over 65% of patients aged 80 years or older survive for at least 3 years following surgery for GC, with 53% living without evidence of metastases. Additionally, the rate of SC in this elderly cohort is comparable to that observed in younger populations. The role of perioperative or neoadjuvant CT in this age group remains a topic of ongoing research and discussion. It is critical to manage such population at referral oncological centers with expertise in treating elderly cancer pts. Research Sponsor: None.

Updated efficacy results of ASKB589 combined with CAPOX and PD-1 inhibitor as first-line treatment for metastatic gastric/gastro-esophageal junction (G/GEJ) adenocarcinoma from a phase Ib/II study.

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Background: ASKB589 is a non-fucosylated fully human IgG1 monoclonal antibody that binds CLDN18.2 with high affinity, specificity, and improved ADCC and CDC activities. Previous findings indicated that ASKB589 exhibited a well-tolerated safety profile at doses up to 20 mg/kg and demonstrated promising antitumor effects. A dose of 6 mg/kg has been selected for further evaluation. Here, we present the updated data from dose expansion part in NCT05632939 study. **Methods:** This phase 1b/2 study was consisted of dose-escalation part and dose-expansion part. The expansion part aimed to assess the safety and efficacy of ASKB589 combined with CAPOX plus a PD-1 inhibitor as the initial treatment for advanced G/GEJ cancer with CLDN18.2 positive, regardless of PD-L1 expression. CLDN18.2 and PD-L1 expression were evaluated through IHC analysis using clone DS-3 and clone E1L3N at a central lab. All pts were given ASKB589 intravenously at a dosage of 6 mg/kg. Responses were assessed by RECIST 1.1 every 6 weeks. Adverse events were graded using CTCAE v5.0. **Results:** As of July 29, 2024, 49 evaluable pts with CLDN18.2 M/H expressions (≥2+ membrane staining intensity in ≥40% of tumor cells) regardless of PD-L1 CPS score. Forty pts (81.6%) achieved a partial or complete response, with a confirmed objective response rate (cORR) of 73.5% (95%CI: 58.9, 85.1). The median duration of response was 11.14 months (6.93, NE). Three subjects underwent surgical treatment as a result of tumor reduction following treatment. All 49 pts achieved SD or better with a disease control rate of 100%. cORR was 76.2% and 74.1% in the CPS<1 and CPS<5 groups, respectively. Of all treated pts (N=50), PFS rate at 9 months was 58.1% (95% CI: 42.2, 71.0) and OS rate at 12 months was 77.1% (95% CI: 61.2, 87.2). Mature PFS and OS data will be reported at the time of the conference. The safety profile of the triplet is generally consistent with the safety data of ASB589 plus CAPOX combination in first-line G/GEJ cancer pts presented previously, which was mainly characterized by manageable on-target-off-tumor effects, including hypoalbuminemia, nausea, and vomiting. **Conclusions:** The combination of ASKB589 with CAPOX and a PD-1 inhibitor as the initial treatment for pts with G/GEJ cancer was well tolerated with no unexpected safety signals. It has shown a high confirmed ORR, 100% disease control, deep and durable antitumor activity, regardless of PD-L1 expression levels. A phase 3 study of the triple regimen for the 1L G/GEJ cancer is actively enrolling pts in China. Clinical trial information: NCT05632939. Research Sponsor: None.

cORR based on CLDN18.2 and PD-L1 expression.

n/N (%)	PD-L1CPS<1	PD-L1CPS 1-5	PD-L1 CPS≥5	Total
CLDN18.2 M&HcORR	16/21(76.2%)	7/10 (70.0%)	13/18 (72.2%)	36/49 (73.5%)
CLDN18.2 Moderate cORR	2/3(66.7%)	0/1(0.0%)	6/9(66.7%)	8/13(61.5%)
CLDN18.2 High cORR	14/18(77.8%)	7/9(77.8%)	7/9(77.8%)	28/36(77.8%)

Individual patient data (IPD) pooled analysis on the optimal therapeutic management of patients with microsatellite instability-high (MSI) resectable gastro-esophageal adenocarcinoma (GEA).

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Background: Patients with resectable MSI/dMMR GEA showed improved survival and modest if any benefit from chemotherapy. Preoperative treatment with immune checkpoint inhibition (ICI) showed high rate of major-complete pathologic response in single arm trials possibly allowing the design of chemotherapy/surgery-free approaches. **Methods:** This was a multinational IPD analysis including patients with resectable GEA with MSI/dMMR status enrolled in INFINITY and NEONIPIGA phase II trials, with dual CTLA-4/PD-(L)1 ICI followed by surgery +/- adjuvant ICI; PROSECCO retrospective study, with perioperative FLOT chemotherapy and surgery, and the dataset of our previous IPD analysis on MAGIC, CLASSIC, ARTIST and ITACA-S randomized trials of patients treated with surgery alone or plus older perioperative/adjuvant chemo(radio)therapy regimen. Primary endpoint was the evaluation of rates of pathologic complete response (pCR) defined as TRG1a Becker and major-complete pathologic response (pCR/MPR) defined as TRG1a/b Becker according to preoperative treatment schedule in patients who underwent surgery. Univariable and multivariable analyses were conducted using a random effects logistic model adjusted with propensity score. Secondary endpoints were event-free survival (EFS) and overall survival (OS) according to the therapeutic strategy in the overall study population. Multivariable mixed-effects Cox models weighted with propensity score were performed. **Results:** The IPD included 197 patients. Of these, 49 received ICI +/- surgery, 27 FLOT chemotherapy plus surgery, 33 surgery alone and 88 older chemo(-radio)therapy regimens plus surgery. In the 69 patients resected after neoadjuvant ICI or FLOT standard of care treatment, ICI demonstrated a higher rate of pathologic response compared to chemotherapy (pCR 61.9% vs 3.7%, OR 54.8 p=0.002; pCR/MPR 78.6% vs 10%, OR 39.3 $p<0.001$). In ITT population, no significant difference in OS and EFS was shown in patients treated with ICI, FLOT plus surgery, old chemo(radio)therapy plus surgery or surgery alone. **Conclusions:** In resectable MSI/dMMR GEA, upfront ICI showed comparable survival outcomes to surgery alone, with limitations of study design and sample size. The impact on survival of ICI versus surgery alone should be investigated prospectively to avoid overtreatment or identify specific risk categories with benefit. The high rate of major-complete pathologic response may allow to study or perform organ sparing surgery procedures or non-operative management to reduce surgical morbidity/mortality and improve quality of life. Research Sponsor: None.

	OS			EFS		
	HR	2.5-97.5% CI	p	HR	2.5-97.5% CI	p
Ref: Surgery only	-	-	-	-	-	-
Chemo+surgery	1.16	0.27-4.98	0.84	1.10	0.39-3.07	0.85
FLOT+surgery	1.10	0.19-6.22	0.92	2.38	0.90-6.27	0.08
ICI +/- surgery	1.97	0.43-8.96	0.38	1.39	0.51-3.77	0.52

Safety and preliminary efficacy of ATG-022 in patients with advanced/metastatic gastric cancer (CLINCH).

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Background: ATG-022 is an antibody-drug conjugate (ADC) targeting Claudin 18.2 (CLDN 18.2) tumor antigen broadly expressed in gastric and other solid tumors. ATG-022 consists of a humanized monoclonal antibody that binds to CLDN 18.2 with high affinity of sub-nM grade and a conjugated linker-payload VC-MMAE, demonstrating activity across a wide range of CLDN18.2 expression levels, including both high and low/ultra-low expression levels. **Methods:** The CLINCH study included dose escalation phase and dose expansion phase. In the dose escalation, patients(pts) with advanced solid tumors regardless of CLDN 18.2 expression received ATG-022 once every three weeks (0.3-3.0 mg/kg Q3W) to evaluate the safety, tolerability, and pharmacokinetics. CLDN 18.2-positive patients were enrolled in dose expansion to receive ATG-022 at recommended phase 2 doses (RP2D) to evaluate the efficacy and safety. The primary endpoints included dose-limiting toxicity (DLT) and adverse events (AEs). The efficacy endpoints included objective response rate (ORR) and disease control rate (DCR), evaluated per RECIST1.1 criteria. **Results:** As of Aug 21, 2024, 16 pts with advanced solid tumors, including 7 gastric cancer (GC), were treated with ATG-022 from 0.3 to 3.0 mg/kg in dose escalation, with 1 DLT of grade 3 nausea found at 3 mg/kg among 6 pts, and 21 GC pts were enrolled in dose expansion to receive RP2D of 2.4 mg/kg Q3W. Among all 37 pts, median age was 61 years. Baseline ECOG were 0 (9 pts) and 1 (28 pts); 33 (89.2%) pts had stage IV disease; 26 (70.3%) pts had received ≥ 2 prior lines of systemic therapy and 14 (37.8%) pts had prior anti-PD-1/L1 therapy. At 2.4 mg/kg in expansion, 3 (14.3%) pts had ≥ 1 Serious TRAEs; 8 (38.1%) pts had grade ≥ 3 TRAEs; the most common TRAEs included neutrophil count decreased (47.6%), nausea (38.1%) and white blood cell count decreased (33.3 %). In dose escalation, 7 GC pts were enrolled, including 3 CLDN 18.2-positive pts. One PR and 1 CR was observed in pts with GC at the dosage of 1.8 mg/kg and 2.4 mg/kg, respectively. Among 12 GC pts who had at least 1 tumor evaluation in dose expansion, 5 PR and 7 SD were observed, resulting in an ORR of 41.7%, and a DCR of 100%. One CR and 1 PR were observed among 3 GC pts receiving 2.4 mg/kg and with CLDN 18.2 expression < 2+, 5%. The Cmax of both total antibody and ADC drug is increased as dose increased from dose 0.3 – 3.0 mg/kg. No pronounced exposure accumulation was found after multiple dose treatment. Preliminary data showed Dose-Normalized AUC of MMAE for ATG-022 is comparable with other vcMMAE ADC drugs. **Conclusions:** ATG-022 demonstrated a manageable safety profile, comparable PK properties in current dosing levels, and encouraging preliminary antitumor effects in GC pts from high CLDN 18.2 expression to low CLDN 18.2 expression, suggesting further clinical investigation in pts with variable CLDN 18.2 expression. The enrollment of GC and other solid tumors are ongoing. Clinical trial information: NCT05718895. Research Sponsor: Antengene.

Robot-assisted esophagectomy with extended mediastinal lymph node dissection: The learning curve and surgical outcomes.

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Background: We have performed thoracoscopic surgery (TMIE) and robot-assisted surgery (RAMIE) for esophageal cancer with extensive mediastinal lymph node dissection, including resection of the sympathetic nerve layers and the thoracic duct. We studied the learning curve, short-term results, and mid-to-long-term outcomes of RAMIE for esophageal cancer at our institution. **Methods:** We compared surgical times at different stages of the procedure. For mid to long-term outcomes, we analyzed 360 cases of TMIE/RAMIE performed between January 2009 and June 2022, comparing the incidence of postoperative complications, recurrence-free survival, and patterns of recurrence between RAMIE and TMIE. Recurrent laryngeal nerve palsy (C-D I) was evaluated using postoperative endoscopy or contrast swallowing tests to assess even mild paralysis. **Results:** The average console time in the thoracic procedure was 234.4 minutes, with Group A (294.5 minutes, range: 211-397 minutes) showing significantly longer times compared to Group B (190.5 minutes, range: 130-257 minutes). Blood loss was almost 0ml in 10 cases (50%) in Group A and 25 cases (62.5%) in Group B. Postoperative complications included pneumonia in 5 cases (8.3%) and recurrent laryngeal nerve palsy (C-D III) in 5 cases (8.3%), with no significant difference between the groups. However, surgical time for lymph node dissection around the recurrent laryngeal nerve was significantly shorter in Group B compared to Group A. Long-term analysis included 326 TMIE and 34 RAMIE cases. Mild recurrent laryngeal nerve palsy (>C-D I) was observed in 92 cases (28%) of TMIE and 7 cases (21%) of RAMIE, showing a lower tendency in the RAMIE group (p=0.338). Pneumonia (>C-D II) occurred in 57 cases (18%) in the TMIE group and 3 cases (9%) in the RAMIE group, with a relatively lower incidence in the RAMIE group (p=0.178). For patients with cT1N0M0 who did not receive preoperative chemotherapy, the 3-year recurrence-free survival rates were 95% for TMIE and 93% for RAMIE. **Conclusions:** RAMIE demonstrated a significant reduction in operative time due to the standardization of the procedure, particularly in the dissection of recurrent laryngeal nerve lymph nodes and other upper mediastinal maneuvers. RAMIE also showed a potential reduction in the incidence of recurrent laryngeal nerve palsy and pneumonia compared to TMIE. In terms of long-term outcomes, RAMIE maintained local control similar to TMIE, with further improvements in safety expected. Research Sponsor: None.

458 **Poster Session**

Impact of time to surgery on outcomes in patients with advanced esophageal squamous cell carcinoma following neoadjuvant therapy: An exploratory analysis of phase III trial JCOG1109.

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Background: JCOG1109, a multicenter three-arm phase III trial, evaluated three neoadjuvant treatment regimens for advanced esophageal squamous cell carcinoma (ESCC), cisplatin and 5-fluorouracil (CF), docetaxel, cisplatin, and 5-fluorouracil (DCF), and cisplatin, 5-fluorouracil, and radiation therapy (CF-RT). The trial demonstrated the superiority of neoadjuvant DCF therapy in improving overall survival (OS). This exploratory analysis investigates the impact of the interval time from completion of neoadjuvant therapy to surgery (time to surgery, TTS) on both short- and long-term outcomes across different neoadjuvant therapy regimens. **Methods:** This analysis included patients from JCOG1109 who underwent surgery after neoadjuvant therapy. Patients were categorized into four TTS subgroups within each arm, based on quartiles of the overall cohort. Short-term outcomes such as perioperative complications, as well as long-term outcomes including OS and progression-free survival (PFS) were evaluated across these TTS subgroups. **Results:** Of the 601 patients enrolled in JCOG1109, 546 patients proceeded to surgery following neoadjuvant therapy. The median TTS was 35 days (range: 16-81) for the CF arm, 38 days (range: 17-109) for the DCF arm, and 41 days (range: 14-98) for the CF-RT arm. Baseline characteristics were well-balanced across all TTS subgroups in each treatment arm. Short-term outcomes revealed that operative time and overall proportions of postoperative complications were comparable. However, in the CF-RT arm, a longer TTS was associated with increased blood loss (200 ml, 210 ml, 300 ml, 370 ml; p-value for trend test = 0.010) and tended to increase the risk of anastomotic leakage (6%, 7%, 14%, 18%; p-value for trend test = 0.073). No significant differences in OS or PFS were observed across TTS subgroups in any treatment arm, with hazard ratios indicating no clear trend toward improved or worsened long-term outcomes. **Conclusions:** The timing of surgery following neoadjuvant therapy does not appear to affect long-term prognosis in patients with advanced ESCC, irrespective of the neoadjuvant therapy regimen. However, for patients undergoing neoadjuvant CF-RT, earlier surgery may be advantageous in reducing the risk of increased surgical complexity and anastomotic leakage.
Research Sponsor: None.

Anlotinib plus benmelstobart as adjuvant therapy in patients with esophageal squamous cell carcinoma: A phase II clinical trial (ALTER-E005).

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Background: Surgery represents a primary radical treatment for esophageal cancer, However, the risk of recurrence remains noteworthy after surgical resection, underscoring the need for more effective and tolerable postoperative regimens for esophageal squamous cell carcinoma (ESCC). Benmelstobart, a novel PD-L1 blockade, demonstrated promising antitumor activity when combined with anlotinib, an antiangiogenic agent, in the treatment of various tumors including advanced ESCC (NCT05038813) and biliary tract cancer (NCT03996408). Therefore, ALTER-E005 study aimed to evaluate the efficacy and safety of combining anlotinib with benmelstobart as adjuvant therapy in patients (pts) with ESCC. **Methods:** This was a single-arm, multi-center phase II clinical trial. Thirty pts (≥18 years) with histologically confirmed T1-2N1-3M0 or T3-4NanyM0 ESCC who had undergone radical (R0) resection with no recurrence 6 to 12 weeks after surgery and an ECOG PS ≤ 1 were eligible for enrollment. Enrolled pts received oral anlotinib (12 mg, days 1-14) and intravenous benmelstobart (1200 mg, day 1) every 3 weeks for up to 16 cycles or until disease recurrence. The primary endpoint was disease recurrence free (DFS). while secondary endpoints included safety, 1-year DFS rate, 3-year DFS rate, 1- year overall survival (OS) rate, and 3-year OS rate. **Results:** As of the data cut-off date (September 20, 2024), a total of 30 patients were enrolled with a median age of 67. Among the 30 surgically resected patients with ESCC, 26 (86.7%) had an ECOG score of 1, and 25 (83.3%) were male. According to the eighth edition of the AJCC Cancer Staging Manual, 9 (30.0%) were classified as stage II, and 21 (70.0%) were classified as stage III. As of the data cut-off date, the median follow-up time was 8.9 months (95% CI: 7.9-12.3). Four pts experienced disease recurrence, and the median duration of DFS has not yet been reached, with 6-month and 1-year DFS rates of 95.7% (95% CI, 72.9%-99.4%), 84.0% (95% CI, 57.7%-94.6%) respectively. Currently, 13 pts are still undergoing therapy. Out of the total 30 patients, 9 (30%) experienced grade 3 treatment-emergent adverse events (TEAEs). No grade 4 or higher TEAEs were observed. The grade 3 TEAEs included hand-foot syndrome (10.0%), hypertension (3.3%), abnormal liver function (3.3%), immune-related hepatitis (3.3%), pneumonia (3.3%), hyperglycemia (3.3%) Lymphocyte count decreased (3.3%), acute cholecystitis (3.3%) and syncope (3.3%). **Conclusions:** This study highlighted the promising efficacy and safety preliminarily of combining anlotinib with benmelstobart as adjuvant therapy in patients with ESCC who underwent radical (R0) resection and showed no recurrence 6 to 12 weeks after surgery. Clinical trial information: NCT05252078. Research Sponsor: None.

Clinical study of radiotherapy combined with tegafur bolus synchronization in the treatment of inoperable elderly middle and advanced esophageal cancer.

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Background: For patients with inoperable oesophageal cancer, simultaneous radiotherapy is the standard treatment option. However, elderly patients are often unable to tolerate CCRT, so a safe, effective and tolerable treatment option is needed. Tegafur, a derivative of 5-FU, has shown efficacy and safety in the treatment of oesophageal cancer. This clinical study evaluated the efficacy and toxicity of radiotherapy combined with a Tegafur bolus in elderly patients with oesophageal cancer. **Methods:** In this study, 62 patients with unresectable intermediate and advanced ESCC (stage IIA to IVA) between March 2022 and February 2024 were screened and divided into two groups: 32 in the experimental group (simultaneous radiotherapy and chemotherapy) and 30 in the control group (radiotherapy alone). The total radiotherapy dose was 50.4-64 Gy/5.6-6.4 wGy (1.8-2.0 Gy per fraction) in all cases, and the chemotherapy regimen: tegafur bolus (0.5 g, qd/bid, d1-14,q3w) was given from the first day of radiotherapy. The primary endpoint was near-term efficacy, and secondary endpoints included objective remission rate (ORR), disease control rate (DCR), T-cell subpopulation indices, and safety. **Results:** The CR, PR, SD and PD of radiotherapy combined with tegafur bolus synchronized treatment were 1 (3.1%), 17 (53.2%), 13 (40.6%) and 1 case (3.1%), respectively. The incidence of CR, PR, SD, and PD in the radiotherapy group was 0 (zero), 8 (26.7%), 19 (63.3%), and 3 (10%), respectively. The ORR of the simultaneous radiotherapy group and the control group were 56.3% and 26.7%(P=0.018), and the DCR of the two groups was 96.9% and 90% of the disease control rate of the patients in both groups (P=0.346). The results of univariate analysis showed that the influencing factor of ORR was the treatment regimen (P=0.018), and multivariate logistic regression analysis of variables with $P<0.2$ showed that the recent efficacy was correlated with the treatment regimen (P=0.039), and that the patients with radiotherapy synchronized with tigafluoroembolization chemotherapy had a better recent efficacy. After treatment, the levels of $CD3^+$, $CD4^+$, $CD8^+$, $CD4^+/CD8^+$were higher in the synchronized radiotherapy group than before treatment,$CD3^+$ and $CD8^+$ were higher in the control group than before treatment, and $CD4^+$ and $CD4^+/CD8^+$ were lower than before treatment. In terms of adverse reactions, the incidence of acute hematological toxicity (P=1), nausea and vomiting (P=1), radiation esophagitis (P=0.197), radiation pneumonitis (P=0.238), and esophageal fistula (P=0.613) were slightly increased in the simultaneous radiotherapy group compared with the control group. **Conclusions:** Radiotherapy synchronized with Tegafur suppositories is effective, with relatively mild side effects, convenient and safe for the treatment of inoperable advanced elderly patients with oesophageal cancer. Clinical trial information: ChiCTR2300073845. Research Sponsor: None.

Fruquintinib combined with PD-1 inhibitors and chemotherapy in the first-line treatment of HER2-negative advanced gastric or gastroesophageal junction adenocarcinoma (FDZL-FIX): A single-arm, open-label phase 2 study.

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Background: For unresectable locally advanced or metastatic HER2-negative gastric cancer (GC), the combination of chemotherapy with immunotherapy, specifically PD-1 inhibitors, has emerged as a new standard of care in first-line treatment of advanced GC, but the patients' outcomes remain poor. Early clinical studies investigating the combination of immunotherapy with anti-angiogenic drugs have demonstrated favorable efficacy. Therefore, it is worth exploring whether the addition of fruquintinib, a potent VEGFR-1,2,3 inhibitor, to the combination of PD-1 inhibitor and chemotherapy can further enhance the survival benefits for patients with unresectable locally advanced or metastatic HER2-negative GC. **Methods:** This is an open-label, single-center clinical study. The enrolled patients were all no less than 18 years old with HER2 negative locally advanced unresectable or metastatic gastric or gastrooesophageal junction adenocarcinoma, without any systemic anticancer treatment for advanced disease. Eligible patients received 6 cycles of combined first line treatment with fuquinitinib (4mg daily, d1-14, q3w, oral) combined with PD-1 inhibitor (sintilimab 200mg or nivolumab 360mg intravenously q3w) and chemotherapy (XELOX or SOX) regimen. The following maintenance treatment was fuquinitinib combined with PD-1 inhibitor and S-1/capecitabine until disease progression or intolerable toxicity. The primary endpoint was progress free survival (PFS). Secondary endpoints included objective response rate (ORR), overall survival (OS), duration of response (DoR), disease control rate (DCR) and safety. **Results:** At the cutoff date on Sep 18, 2024, a total of 27 patients were enrolled and received at least one dose of treatment. 23 patients had received the response evaluation, with an ORR of 60.9% (14/23; 95%CI 38.5-80.3) and DCR of 100% (23/23 95%CI 85.2-100.0). In the intention to treatment (ITT) population, the median PFS was 9.79 months (95% CI 5.29-NA). Safety profile indicated that 22.2% (6/27) experienced grade≥3 adverse events. The most common grade ≥ 3 adverse reactions mainly include neutropenia (7.4%, 2/27) and thrombocytopenia (7.4%%, 2/27). **Conclusions:** The combination of fruquintinib and chemotherapy with PD-1 blockade in the first-line treatment for unresectable locally advanced or metastatic HER2-negative GC had shown promising efficacy and acceptable safety profile. Enrollment in the trials is still ongoing. Clinical trial information: NCT06158919. Research Sponsor: None.

Surgical treatment results for oligo-recurrence of gastric cancer.

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Background: Despite advancements in diverse treatments, approximately 40-50% of advance gastric cancer patients experience recurrence even after curative resection surgery. The condition where there is a limited recurrence after the control of the primary tumor site is referred to as oligo-recurrence. However, there is scarce reporting on oligo-recurrence after gastric cancer surgery. This study compiled cases of patients who underwent surgery for recurrent gastric cancer after R0 surgery. Since there are no specific treatment guidelines for recurrence after gastric cancer surgery, this research could be beneficial in providing insights and assistance. **Methods:** This retrospective study includes data from 25 patients who underwent surgery for suspected oligo-recurrence of gastric cancer among those who had undergone surgery from June 2008 to May 2018. If a solitary mass suspected of recurrence is identified during surveillance, a PET-CT is performed. Patients with suspicion of peritoneal dissemination, carefully excluded through a multidisciplinary approach, are selectively omitted. All patients undergo excision surgery for both treatment and tissue examination purposes. **Results:** Out of 25 cases, there were 6 cases of paraaortic lymph node enlargement, 9 cases liver nodules, 8 case lung nodules, and 1 case tumor bed mass. In the postoperative tissue analysis, 4 cases of paraaortic lymph node enlargement, 5 cases of the liver nodules, and 1case tumor bed mass were confirmed to be recurrences of gastric cancer. All of lung nodule were diagnosed with lung cancer. All 10 cases underwent chemotherapy after metastatectomy. Among them, 3 cases experienced recurrence and died an average of 12.3 months after surgery. Seven cases did not experience recurrence and have survived with an average survival time of 102 months to date. **Conclusions:** The metastatectomy is safe and effective for patients with oligo-recurrence from GC and can improve their prognosis. For an accurate diagnosis, oligo-metastectomy is essential. However, our results need to be confirmed by more randomized controlled clinical studies. Research Sponsor: None.

Treatment result after metastatectomy.

Case	Diagnosis	Chemotherapy aftermetastatectomy	Recurence after metastatectomy	DFS after metastatectomy	OS after metastatectomy	OS after gastrectomy	Status
1	LN metastasis 1/1	15cycles FOLFOX		147	147	156.4	ALIVE
2	LN metastasis 1/1	37cycles TAS-118	liver metastasis	22	40.6	74.9	DEATH
3	LN metastasis 1/3	12cycles FOLFOX		98.1	98.1	125.5	ALIVE
4	LN metastasis 1/1	35cycles FOLFOX		50.9	50.9	71.3	ALIVE
5	Liver metastasis	ONO tiral	Liver, bone mets.	3.73	27.9	64.6	DEATH
6	Liver metastasis	12cycles FOLFOX		68.3	68.3	122.2	ALIVE
7	Liver metastasis	12cycles FOLFOX		96	96	105.8	ALIVE
8	Liver metastasis	82cycles FOLFOX		55	55	77.4	ALIVE
9	Liver metastasis	12cycles FOLFOX		94.4	94.4	103.9	ALIVE
10	Tumor bed	12cycles FOLFOX		9	25.5	24.4	DEATH

Efficacy, safety and DNA methylation analysis of cadonilimab combined with taxane and cisplatin as first-line treatment for advanced esophageal squamous cell carcinoma (ESCC): Updated results from an open-label, multicenter phase II trial (AK104-IIT-014).

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Background: Immune checkpoint inhibitors combined with chemotherapy had become the first-line standard treatment for advanced ESCC and cadonilimab, as a bispecific antibody simultaneously targeting PD-1 and CTLA-4, may further boost anti-tumor activity with a satisfied safety profile. Here, we present the updated data for the safety and efficacy of cadonilimab combination therapy as the first-line treatment in advanced ESCC. The correlation between DNA methylation and clinical response was also investigated. **Methods:** Treatment-naïve patients (pts) with unresectable locally advanced or metastatic ESCC were enrolled. Cadonilimab (10mg/kg, iv, d1, q3w) combined with paclitaxel or nab-paclitaxel (175 mg/m^2, iv, d1, q3w) and cisplatin (65 to 75 mg/m^2, iv, d1, q3w) were administrated for up to 6 cycles, then Cadonilimab (10mg/kg, iv, d1, q3w) monotherapy continued as maintenance until progressive disease or unacceptable toxicity, with a maximum of 24 months. The primary endpoint was objective response rate (ORR). Secondary endpoints were progression-free survival (PFS), overall survival (OS), disease control rate (DCR) and safety. Plasma cell-free DNA samples were collected before and after treatment and prepared for methylation level sequencing. **Results:** As of September 20, 2024, 43 pts were enrolled with a median age of 61 years (range 44-75), 81.4% were male, 39.5% had PD-L1 CPS≥10 and 95.3% had metastatic diseases. All pts were evaluable for safety and efficacy. The ORR was 81.4% (95%Cl: 66.1%-91.1%) and DCR was 97.7% (95%Cl: 86.2%-99.9%). The median PFS (mPFS) was 7.05 months (mo) (95%Cl: 5.86-8.24) and OS analysis was immature. In the PD-L1 CPS≥10 pts, the ORR was 100.0% (95%Cl: 77.1%-100.0%) and mPFS was 7.05 mo (95%Cl: 5.19-8.91). In the PD-L1 CPS < 10 pts, the ORR was 77.3% (95% Cl: 54.2%-91.3%) and mPFS was 7.05 mo (95%Cl: 5.87-8.24). Six hyper-methylated CpG sites, EPTIN9、PKNOX2、DLEU7、SOX7、CNRIP1 and LINC00554 might be the candidate biomarkers as the mean pretreatment methylation levels were significantly higher in the PR pts than those in the non-PR pts ($p=7.8\times10^{-7}$). Grade 3-4 treatment-related adverse events (TRAEs) were reported in 44.2% (19/43) pts, mainly including neutropenia (25.6%), leukopenia (9.3%) and hyponatremia (7.0%). The infusion-related reactions (IRR) occurred in 14.0% (6/43) and grade ≥3 IRR occurred in 4.7% (2/43) pts. 7 pts discontinued cadonilimab administration due to TRAEs. **Conclusions:** The updated results suggested that bispecific antibody cadonilimab combined with taxane and cisplatin as first-line treatment continued to show encouraging anti-tumor activity and manageable safety in pts with advanced ESCC. DNA methylation level might be a potential biomarker for guiding patient outcomes. Clinical trial information: NCT05522894. Research Sponsor: None.

Pembrolizumab or placebo plus chemotherapy for advanced HER2-negative gastric or gastroesophageal junction (G/GEJ) adenocarcinoma: An updated analysis of KEYNOTE-859 for patients enrolled in Asia.

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Background: After a median follow-up of 41.6 months, data from the global phase 3 KEYNOTE-859 study (NCT03675737; N = 1579) continued to show that use of pembrolizumab (pembro) plus chemotherapy (chemo) improved overall survival (OS), progression-free survival (PFS), and objective response rate (ORR), with manageable safety, versus placebo (pbo) plus chemo for patients (pts) with advanced HER2-negative G/GEJ adenocarcinoma. Here, we report updated results from the subgroup analysis of pts enrolled in Asia after an additional 11 months of follow-up from the first interim analysis. **Methods:** Eligible pts aged ≥18 years with locally advanced unresectable or metastatic HER2-negative G/GEJ adenocarcinoma, an ECOG performance status (PS) of 0 or 1, and measurable disease per RECIST v1.1 were randomly assigned 1:1 to receive pembro 200 mg or pbo IV Q3W for ≤35 cycles; all pts received investigator's choice of chemo (FP or CAPOX). The primary end point was OS. Secondary end points included PFS, ORR, and DOR per RECIST v1.1 by blinded independent central review, and safety. Efficacy end points were evaluated in all randomly assigned pts (intention to treat). **Results:** A total of 525 pts (263, pembro plus chemo; 262, pbo plus chemo) were enrolled in KEYNOTE-859 in Asia. The median time from randomization to database cutoff (August 22, 2023) was 39.2 months (range, 26.0-56.8). The median OS was 17.3 months (95% CI, 14.8-19.5) for pembro plus chemo versus 13.0 months (95% CI, 11.8-14.4) for pbo plus chemo (HR, 0.75; 95% CI, 0.62-0.91). The median PFS was 8.4 months (95% CI, 7.1-9.6) for pembro plus chemo versus 5.8 months (95% CI, 5.6-6.9) for pbo plus chemo (HR, 0.72; 95% CI, 0.58-0.89). The ORR was 61.2% (95% CI, 55.0-67.1; 36, complete response [CR]; 125, partial response [PR]) for pembro plus chemo and 48.5% (95% CI, 42.3-54.7; 24, CR; 103, PR;) for pbo plus chemo. The median DOR was 10.0 months (range, 1.2+ to 50.8+) for pembro plus chemo and 5.8 months (range, 1.3+ to 44.3+) for pbo plus chemo; 28.9% and 23.3% of pts, respectively, had a response lasting ≥24 months. Grade 3-5 treatment-related adverse events (AEs) occurred in 155 pts (59.2%) in the pembro plus chemo group and 119 pts (45.4%) in the pbo plus chemo group. Treatment-related AEs led to death in 1 pt (0.4%) in the pembro plus chemo group (unknown cause) and 2 pts (0.8%) in the pbo plus chemo group (cerebral hemorrhage and abnormal liver function). Immune-mediated AEs and infusion reactions occurred in 85 pts (32.4%) in the pembro plus chemo group and 35 pts (13.4%) in the pbo plus chemo group. **Conclusions:** The addition of pembro to chemo improved OS, PFS, and ORR in pts with advanced HER2-negative G/GEJ adenocarcinoma from Asia enrolled in KEYNOTE-859, with no new safety signals. These results further support first-line pembro plus chemo as a treatment option for this population. Clinical trial information: NCT03675737. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

Exploring the role of bispecific and CAR-T cell therapies in gastric cancer: A systematic review and meta-analysis.

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Background: Gastric cancer (GC) is associated with significant mortality due to the limited efficacy of current conventional treatments, necessitating the need for novel therapies. In this systematic review and meta-analysis, we aim to explore the potential role and outcomes of bispecific antibodies and chimeric antigen receptor T-cell (CAR-T) immunotherapy in treating GC. **Methods:** Using PRISMA guidelines, searches were conducted on PubMed, Cochrane, and Clinicaltrial.gov for 'Gastric cancers', 'bispecific antibodies', and 'CAR-T therapy' as of March 30, 2024. Out of 35 studies, 9 were selected for pooled analysis in R (v4.3.3) using the Der Simonian-Laird Estimator, calculating inter-study variance and extracting data with 95% CI. **Results:** 138 gastric cancer (GC) patients from 2 phase I (22.22%), 3 phase Ib (33.33%), 3 phase II (33.33%), and 1 case report (11.11%). Median age was 57 (28-77) years, and 76% (31/41) were male. Bispecific antibodies were MCLA-128 (25/138), catumaxomab (15/138), AK104 PD-1/CTLA-4 (16/138), ABL 001 DLL4VEGFA (19/138), and KN026 (25/138) while CART therapy was CT041 (38/138). The pooled overall response (OR), partial response (PR), and complete response (CR) for bispecific antibodies was 37% (95% CI, 0.12-0.71, I2=76%, n=100, p <0.01), 21% (95% CI,0.05-0.55, I2=73%, n=85, p<0.01), and 11%(95% CI, 0.02-0.38, I2=63%, n=85, p=0.03) respectively, while the pooled OR, PR, and CR for CART was 56% (95% CI, 0.40-0.71, I2=0%, n=38, p=0.41), 53%(95% CI, 0.36-0.69, I2=0%, n=38, p=0.38,), and 12% (95% CI, 0.02-0.42, I2=38%, n=38 p=0.2), respectively. The pooled disease control rate was 75% (95%CI,0.46-0.92, I2=79%, n=80, p < 0.01) for bispecific antibodies and 27% (95% CI,0.16-0.40, I2=0%, n=65, p=0.37) had the stable disease while pooled incidence of progressive disease was 18% (95% CI,0.09-0.31, I2=0%, n=55, p=0.53). The median progressive free survival was 6.74 months, with a median overall survival of 14.75 months. Common adverse effects included anemia, diarrhea, fatigue, and cytokine release syndrome grades 1 or 2. **Conclusions:** This analysis shows promising results in GC patients using bi-specific antibodies and CAR-T cell therapy. However, further clinical trials are needed to fully explore their potential. Research Sponsor: None.

466 **Poster Session**

Phase II study of neoadjuvant FLOT and chemoradiation for trimodality therapy of esophageal/GEJ adenocarcinoma.

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Background: Recent data support use of perioperative FLOT (5-FU/leucovorin/oxaliplatin/docetaxel) chemotherapy over pre-operative chemoradiation (CRT) for patients with resectable esophageal/GEJ adenocarcinoma with limited studies evaluating combined FLOT/CRT. This prospective phase II study (NCT04028167) evaluated a neoadjuvant approach of FLOT followed by CRT for resectable esophageal/GEJ adenocarcinoma (EAC). **Methods:** Operable patients with cT1-2 N1-2 or cT3-4N any4 Nany non-metastatic EAC were eligible. Neoadjuvant treatment consisted of FLOT x 3 cycles followed by restaging PET and chemoradiation (41.4 Gy / 23 fractions to the primary site and regional lymphatics with weekly carboplatin/paclitaxel). Adjuvant nivolumab for non-pathologic complete response (pCR) was permitted. The primary endpoint was pCR rate among patients receiving the study regimen and undergoing resection, with secondary endpoints including disease free survival (DFS), overall survival (OS), toxicity, and correlative studies. DFS/OS was measured using the Kaplan Meier (KM) method. Interim results are presented after a planned safety analysis. **Results:** Since 4/2020, 23 patients with median age of 62 (range 40-73) have enrolled. cT3-4 and cN+ disease was present in 20/23 (87%) and 16/23 (70%) of patients, respectively. FLOT was completed by 21/23 patients (reasons for discontinuation included allergic reaction and G5 sepsis/endocarditis). All 21 patients initiating chemoradiation completed planned treatment and 17/21 underwent resection (2 refused surgery after achieving cCR, 1 died of cardiac arrest, 1 patient is awaiting surgery). A pCR was observed in 5/17 resected patients (29%), with pCR or near pCR observed in 9/17 (53%). The complete response rate (pCR + cCR for patients refusing surgery) was 7/19 (37%). Adjuvant nivolumab was administered in 9 patients. The median follow-up among all patients was 20 months with 2-year DFS and OS estimates of 61% and 73%. The regimen was tolerated with G3+ hematologic, G3+ gastrointestinal, and G3+ other toxicity observed in 39%, 13%, and 22% of patients, respectively. **Conclusions:** In a locoregionally advanced population, neoadjuvant sequential FLOT followed by CRT was well tolerated with encouraging DFS/OS and a high rate of complete response compared to prior studies. This hybrid platform may enhance both locoregional and distant control and is of interest for expanded evaluation. Clinical trial information: NCT04028167. Research Sponsor: American Cancer Society; Cancer League of Colorado.

Identification of candidate biomarkers for gastric cancer by bioinformatics analysis of pooled microarray gene expression datasets in Gene Expression Omnibus.

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Background: Gastric cancer (GC) remains one of the most prevalent malignant diseases worldwide. The high mortality rate is largely due to the limited treatment options and the challenges in diagnosing the cancer at an early stage. Understanding the genetic differences driving the proliferation of GC is crucial for advancing diagnostic methods and developing targeted treatments. This study aims to identify biomarkers involved in the pathogenesis of GC through integrated bioinformatics analysis. **Methods:** Extracted from the Geo Expression Omnibus, three DataSets (GSE118916, GSE79973, and GSE13861) were analyzed for differentially expressed genes (DEGs) using the R statistical language. The three data sets were then compared for common DEGs, which would serve as the focus of this study. These DEGs were functionally analyzed for pathways by Gene Ontology (GO) and the Kyoto Encyclopedia of Genes and Genomes (KEGG). A protein–protein interaction (PPI) network was constructed to screen out candidate genes. **Results:** The results revealed that 123 DEGs of the three datasets containing 26 upregulated genes and 97 downregulated genes were ascertained in our study. The GO and KEGG enrichment analysis results showed that the functions of DEGs were mainly involved in the retinol metabolic process, xenobiotic metabolic process, collagen type I trimer, potassium: proton exchanging ATPase complex, benzaldehyde dehydrogenase (NAD+) activity, alcohol dehydrogenase (NADP+) activity, Collecting duct acid secretion, and Tyrosine metabolism. Through the PPI analysis network, the core genes with the highest degree of 7 nodes were selected: COL1A1, COL1A2, COL11A1, THBS2, ATP4A, COL10A1, and CXCL8. **Conclusions:** The 7 genes identified in this study may play a vital regulatory role in the occurrence and development of GC and may also be potential therapeutic targets. Further mechanistic studies of the pathogenesis and treatment of GC may be able to identify new targets using these shared pathways. These findings could contribute to a better understanding of the origins of gastric cancer and facilitate the development of early interventions. The bioinformatics analysis conducted in this study offers new insights and directions for further research on gastric cancer. Research Sponsor: None.

Relationship in gene amplification between *ERBB2* and other oncogenes: Implications for the therapeutic efficacy of trastuzumab (Tmab)-based chemotherapy (CTx) in HER2 positive gastric cancer.

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Background: This study aimed to elucidate the molecular characteristics, focusing on the co-amplification of *ERBB2* and other oncogenes, and to investigate their impact on treatment efficacy in HER2 positive GC. **Methods:** We conducted a phase 2 clinical trial (UMIN 00017602) evaluating the efficacy of S-1, oxaliplatin, and Tmab. Next-generation sequencing was performed using the TS-170 (Illumina). **Results:** 32 patients were enrolled, of whom 6 (19%) were IHC2+/FISH+. The median progression-free survival (PFS) and overall survival (OS) were 11.8 and 28.8 months. *ERBB2* amplification with a cut-off value of 2.84 was present in 84.4% with a median copy number (CN) of 9.3 (range 2.4-76.6). Amplification of oncogenes other than *ERBB2*, such as *KRAS, FGFR2, MET,* and *CCNE1,* was observed in 87.5%, and the median CN of the highest CNs among oncogenes other than *ERBB2* was 7.8 (range 2.4-22.9). Eventually co-amplification was observed in 75%, and CNs of *ERBB2* and other oncogenes showed an inverse correlation (r=0.345, p=0.058). In all tumors having *ERBB2* CN $\geq$ 7.25 obtained by the ROC analysis, CN of *ERBB2* was the highest among oncogenes (*ERBB2* dominant). Conversely, in 92% of the tumors having *ERBB2* CNs $<$ 7.25, CNs of oncogenes other than *ERBB2* was higher than ERBB2 (*ERBB2* non-dominant) ($p<0.001$). Moreover, the highest CN of the oncogenes other than *ERBB2* was higher in tumors with *ERBB2* CNs $<$ 7.25 than that with *ERBB2* CNs $\geq$ 7.25 (median 12.6 vs 6.4 p=0.018). Patients with *ERBB2* non-dominant tumors showed significantly shorter PFS (median 6.9 vs 17.0 months, HR 6.40, p=0.001) and OS (median 14.8 vs 35.5 months, HR 2.72, p=0.016) and numerically lower response rate (63.6 vs 90.0%, p=0.151) compared with those with *ERBB2* dominant tumors. **Conclusions:** Most patients had co-amplification of *ERBB2* and other oncogenes, and the dominance of oncogene amplification in the tumor was associated with treatment efficacy of Tmab-based CTs in HER2 positive GC. These results suggest that the dominance of oncogene CNs within the tumor may determine tumor drivenness and influence treatment efficacy. Research Sponsor: None.

469 **Poster Session**

A liquid biopsy analysis of exosomal miRNA to detect lymph node metastasis in early gastric cancer.

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Background: Additional surgical resection is required to achieve a complete cure in patients with early gastric cancer (EGC) due to the potential risk for lymph node metastasis (LNM) after pathological analysis. However, LNM is estimated to occur in approximately 10% of patients with high-risk EGC. In this study, we investigated a blood-based liquid biopsy assay of exosomal miRNA for the noninvasive detection of LNM in patients with high-risk EGC. **Methods:** Two genome-wide miRNA expression profiling datasets (GSE164174 and TCGA) were analyzed to prioritize biomarkers in pretreatment plasma samples from clinical training and validation cohorts of GC patients. An integrated exosomal miRNA panel was developed and a risk stratification model combining the miRNA panel with clinical risk factors was established. **Results:** Using comprehensive expression profiling of public datasets, we identified a transcriptomic panel of four miRNAs (miR-34b, miR-130a, miR-375, and miR-627) that robustly identified patients with LNM [area under the curve (AUC) = 0.86, 95% confidence interval (CI) = 0.77-0.92]. We assessed panel performance in a training cohort (AUC=0.86, 95% CI=0.67-0.96) and validated it in an independent validation cohort (AUC=0.84, 95% CI=0.64-0.95). Our risk stratification model was more accurate than the panel and was an independent predictor of LNM identification (AUC=0.97). **Conclusions:** A novel noninvasive, liquid biopsy-based method for patients with EGC may predict those traditionally classified as high-risk patients with LNM who are unlikely to benefit from surgical resection. Research Sponsor: None.

Quantitative amplification ratio for established and emerging biomarkers in gastroesophageal (GE) cancer.

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Background: Gene copy number (CN) amplifications (amps) and/or protein overexpression in *ERBB2*, *EGFR*, *FGFR2*, and *MET* represent established and emerging clinical targets in GE. Various methodologies are used to report gene amp or overexpression with some data suggesting FISH ratio may predict therapy response. Quantitative methods for NGS CN and relationship to ISH/IHC remains largely unresolved. We studied the landscape of targetable CN amps in GE to understand correlations of quantitative CN amp ratio and outcomes. **Methods:** GE tissue (N = 20,468) and liquid (N = 1,760) samples underwent NGS-based comprehensive genomic profiling (CGP). A genome-wide CN model for each sample was generated, segmented, and combined with variant allele frequencies of heterozygous SNPs to estimate the sample purity, ploidy, and CN for each segment. Genes were considered amplified at CN ≥ ploidy + 3 for *ERBB2* and + 4 for other genes. Sub-analysis using amp ratio (ratio of the gene CN to the sample ploidy) ≥ 3 was performed. ctDNA tumor fraction (TF) was quantified using a composite algorithm (PMID: 38990098). The Flatiron Health-Foundation Medicine clinico genomic database, a nationwide (US-based) de-identified EHR-derived database linked to CGP data, was queried to assess external HER2 FISH or IHC and describe real-world (rw) progression-free survival (rwPFS) and overall survival (rwOS) of GE patients (pts) with *ERBB2* amp on trastuzumab-based regimens. **Results:** ≥1 gene amps were detected in 15,341 (75%) GE tissue cases, and in 67% of cases using an amp ratio ≥3 threshold. The rates of amps in current clinical targets using each threshold were *ERBB2*: 17%/12%, *KRAS*: 13%/12%, *EGFR:* 8.7%/7.2%, *MET*: 5.2%/4.0%, *FGFR2:* 3.1%/2.7% of tissue samples. The median amp ratio and percent of amps with ratio ≥ 3 were *ERBB2*: 7.1/73%, *KRAS*: 14/92%, *EGFR*: 13/82%, *MET*: 6.3/76%, and *FGFR2*: 29/88%. Amp ratio ≥ 3 was associated with focal amplicons vs amp ratio < 3 (median amplicon size 0.95 vs. 3.8 Mbp, $p < 0.001$). Of 1,382 pts with HER2 NGS and IHC results, concordance with IHC3+ was 68% and increasing *ERBB2* amp ratio was associated with IHC 3+ status. Of 604 pts with HER2 NGS and FISH, concordance was 80% (118 positive, 366 negative). Higher *ERBB2* amp ratio was also associated with longer rwOS and rwPFS as a continuous variable on first-line trastuzumab-based regimens. ≥1 gene amps were detected in 597 (34%) liquid samples (median TF 18%) and 32% with the amp ratio ≥3 threshold, increasing to 79% and 73% with TF ≥ 20% (n=355). **Conclusions:** Gene amps are common established or emerging GE targets and *ERBB2* amp ratio is associated with outcomes for pts on trastuzumab-based regimens. Amp detection in liquid correlates with higher TF and amp ratio. As the number of therapies targeting gene amps and overexpression grows, quantification of the degree of amplification and harmonization across methodologies may be an important tool to select pts for optimal outcomes. Research Sponsor: Foundation Medicine.

Role of intraperitoneal paclitaxel in eosinophil activation and recruitment in the peritoneal cavity and anti-tumor effects against peritoneal metastasis from gastric cancer.

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Background: Recent studies indicate that the efficacy of chemotherapy is significantly influenced by the tumor immune microenvironment. Peritoneal cavity contains many immune cells, however, the relationship between immune response within the peritoneal cavity and tumor response remains poorly understood. **Methods:** Single cell suspensions were obtained from ascites or peritoneal lavages from 41 patients with PM from GC. In 28 patients, cells were collected both before and after 1-3 courses of IP chemotherapy. These samples were immunestained with mAbs targeting to specific lymphoid or myeloid subsets, and their proportions in CD45(+) leukocytes were analyzed using flowcytometry. Eosinophils were purified by magnetic cell sorting for subsequent RNA-Seq analysis. **Results:** Among 41 patients with PM, tumor leukocyte ratio (TLR) calculated as CD326(+) tumor cells divided by CD45(+) leukocytes varied from 0.002% to 58.1%. TLR was not correlated with any ratios of immune cells, however, the ratios of CD8(+) T cells, NK cells and CD16(-) CD193(+) eosinophils tended to decrease with the increase in TLR. Conversely, the ratios of CD19(+) B cell, CD14(+) macrophages and CD16(+) neutrophils increased with elevated TLR. Mean survival time of these patients was 17.7 months. None of the leukocyte subsets showed significant association with patient outcome. In 28 patients, the ratios of lymphocytes mostly decreased with significant difference in CD4(+) T cells and CD19(+) B cells after IP treatment. In contrast, the ratio of CD11b(+) myeloid cells increased in peritoneal cavity after IP treatment. Among the myeloid cells, the ratio of CD16(-) CD193(+) eosinophils significantly increased. The change was particularly prominent in in post-treatment negative peritoneal lavage cytology (CY0) patients (M=0.47% vs M=10.0%, $p<0.0001$), while the difference was not significant in post-treatment positive peritoneal lavage cytology (CY1) patients (M=0.93% vs M=0.97%, p=0.84). Patients with eosinophil ratio of ≥2% after IP chemotherapy in peritoneal cavity had significantly longer overall survival compared to those with lower eosinophil ratios (17.3 mo vs 26.7 mo, p=0.034; HR =0.23[0.06-0.89]). The peritoneal eosinophils after IP treatment exhibited elevated levels of CD11b and CD63 expression compared to circulating eosinophils ($p<0.01$), while SSC-A levels tended to be lower. Gene ontology pathways revealed enrichment of cytokine-mediated signaling pathway, extracellular matrix organization and response to interferon-gamma in peritoneal eosinophils ($p<0.0001$). **Conclusions:** IP-PTX induces eosinophil activation and recruitment in the peritoneal cavity, while also affecting other types of immune cells. These effects may potentially contribute to the anti-tumor response against peritoneal metastases.
Research Sponsor: None.

Unraveling the role of interferon-stimulated neutrophils: A promising biomarker for immunotherapy efficacy and prognosis in gastric adenocarcinoma.

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Background: Despite significant advancements in immunotherapy, the identification of effective predictive biomarkers for treatment response in gastric adenocarcinoma (GAC) remains a critical challenge. Discovering such biomarkers could enhance patient stratification and improve therapeutic outcomes. **Methods:** We conducted a comprehensive analysis of 18 specimens obtained from 10 GAC patients, both prior to and following immunotherapy. Utilizing single-cell RNA sequencing, we compared the cellular composition and functional phenotypes of these specimens, with particular emphasis on tumor-infiltrating immune cells and the identification of previously uncharacterized neutrophil subclusters. **Results:** Our analysis identified eight distinct neutrophil subclusters, including Neu01_S100A12, Neu02_VEGF, Neu03_CCL3_CCL4, Neu04_IL1β, Neu05_CXCR4, Neu06_IFN-Stimulated, Neu07_Proliferative, and Neu08_B-cell_related. Notably, the presence of IFN-Stimulated neutrophils was positively correlated with the efficacy of immunotherapy in GAC, a finding that was further validated through analyses of public datasets and in-house cohort studies employing multiplex immunohistochemistry. Additionally, IFN-Stimulated neutrophils served as prognostic indicators across multiple datasets. Investigations revealed that these cells engage in significant cellular communication with other immune components within the tumor microenvironment, including CD8+ T cells, macrophages, and B cells, thereby promoting a pro-inflammatory and anti-tumor phenotype. Importantly, pan-cancer analyses demonstrated the consistent prognostic and immunotherapy efficacy-predicting potential of IFN-Stimulated neutrophils across various cancer types. **Conclusions:** IFN-Stimulated neutrophils represent a promising biomarker for predicting immunotherapy response and patient prognosis in GAC. Our findings offer novel insights into the underlying mechanisms of immunotherapy and highlight potential avenues for personalized treatment strategies in the management of GAC. Research Sponsor: National Nature Science Foundation of China.

The prognostic value of HER2 gene dosage and heterogeneity at a single-cell resolution in gastroesophageal adenocarcinoma.

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Background: HER2 gene dosage (i.e. level of amplification) and heterogeneity are key prognostic variables for HER2+ gastroesophageal adenocarcinoma (GEA), yet no standardized approach to test these in routine clinical use exists. Here we modify the analytical pipeline for HER2 FISH to quantitatively estimate these prognostic variables and correlate them with clinical outcomes. **Methods:** Patients with metastatic HER2+ GEA who received trastuzumab-based therapy between 2011-2024 at Dana-Farber Cancer Institute and had available pre-treatment tissue for FISH analysis were included. HER2 FISH was conducted in a CLIA approved clinical pathology laboratory. Standard HER2/CEP17 ratio (an average of 50 cells) and individual cell HER2/CEP17 ratio were calculated. Kaplan-Meier method was used to estimate progression free (PFS) on trastuzumab-based therapy and overall survival (OS). Survival outcomes were adjusted for age, gender, tumor site, grade, prior resection, number of metastatic sites, and HER2 IHC using cox proportional hazard regression models. **Results:** A total of 77 patients were included. Median age was 60 years, 88% were male, and 92% had HER2 3+ tumors. Individual cells with HER2/CEP17 ratio $\geq$ 2.0 were defined as amplified (same cutoff as current bulk standard) and those with HER2/CEP17 ratio $\geq$ 4.95 (population median) were labeled highly HER2 amplified providing objective metrics to assess heterogeneity and gene dosage. Using these metrics we defined three subgroups: 1. Heterogenous (Het, n = 15): <75% cells HER2+; 2. Homogenous, low gene dosage (HL, n = 30): >75% cells HER2+, <50% cells with high HER2 amplification; 3. Homogenous, high gene dosage (HH, n = 32): >75% cells HER2+, >50% cells with high HER2 amplification. Cutoffs for both metrics were determined using data distribution patterns. The subgroups were prognostically relevant and PFS and OS were significantly lower for heterogenous tumors and numerically lower for homogenous tumors with low gene dosage after multivariable adjustment (Table). **Conclusions:** HER2 gene dosage and heterogeneity significantly impact outcomes in HER2+ GEA. More importantly by deconvoluting clinical HER2 FISH data at single cell level, we demonstrate an objective metric with prognostic value that could be adopted for routine clinical use and for patient stratification. Further insights into differing biology of these molecular subtypes are needed. Research Sponsor: Dana Farber Cancer Institute.

Outcomes of patients by HER2 heterogeneity and gene dosage. Hazards ratio (HR) and p values are adjusted using multivariable cox proportional hazard regression model.

		PFS			OS		
	Patients, No.	Median, mo	HR (95%CI)	*P* Value	Median, mo	HR (95%CI)	*P* Value
HH	32	16.2	1		16.4	1	
HL	30	12.1	1.70 (0.79-3.66)	0.173	15.7	1.71 (0.93-3.14)	0.085
Het	15	6.7	3.12 (1.35-7.2)	0.008	12.4	2.72 (1.28-5.75)	0.009

Blood TCTP as a biomarker associated with immunosuppressive features and resistance to immunotherapy in metastatic gastric cancer.

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Background: Cancer biomarkers are essential for determining the prognosis of the disease and/or predicting its response to specific treatments. However, established biomarkers representing specific myeloid cell populations and representative biomarkers in gastric cancer remain unavailable. Therefore, this study aimed to explore the prognostic and immunological relevance of plasma translationally controlled tumor protein (TCTP) in patients with advanced gastric cancer treated with immune checkpoint inhibitor (ICI) and cytotoxic chemotherapy. **Methods:** Plasma samples were prospectively collected from the cohorts of patients with gastric cancer who were treated with 1st line fluoropyrimidine plus platinum chemotherapy (n = 143, cohort 1) and 3rd line nivolumab (n = 165, cohort 2). Plasma TCTP levels were quantified using ELISA, and multiplex proteomic analysis (Olink) was conducted to assess expression levels of immune-related proteins. An external single-cell RNA sequencing (scRNA-seq) dataset was employed to validate the findings. **Results:** Patients with high plasma TCTP levels (TCTP-high group) exhibited poor survival outcomes with 1st line chemotherapy compared to those with low levels (TCTP-low group) in cohort 1. In the TCTP-high group, proteins associated with immunosuppressive myeloid cells, angiogenesis, and immune exclusion of T/NK cell function were upregulated, whereas proteins involved in T-cell activation/exhaustion were significantly upregulated in the TCTP-low group. The scRNA-seq analyses of myeloid cells revealed that TCTP pathway-associated genes (i.e., *TLR2*, *CXCL1*, and *CXCL2*) were specifically expressed in immunosuppressive *APOE*$^{+}$ macrophages, *THBS1*$^{+}$macrophages, and *CD1C*$^{+}$conventional type 2 dendritic cells. In the TCTP-high group, patients treated with nivolumab (cohort 2) also experienced poor survival outcomes. **Conclusions:** Plasma TCTP is a readily measurable prognostic biomarker, reflecting immunosuppressive signals of myeloid cells in patients with gastric cancer treated with ICI and chemotherapy. Research Sponsor: This research was supported by the Korea Drug Development Fund, funded by the Ministry of Science and ICT, Ministry of Trade, Industry, and Energy, and Ministry of Health and Welfare, Republic of Korea).

Effect of tRF-28-ZJ47D112Z4DZ on the pathogenesis of gastric cancer through targeting cyclin D1/CDK4 and its early diagnostic value.

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Background: Gastric cancer exhibits high incidence and mortality rates worldwide. Current research on gastric cancer aims to discover stable and effective early diagnostic markers, as well as uncover the disease's pathogenesis. tRNA-derived fragment (tRF) is a newly discovered regulatory non-coding RNA, which plays an important role in tumor diagnosis and treatment. **Methods:** This study, through transcriptome sequencing and MINTBase v2.0 database, screened out a new molecule tRF-28-ZJ47D112Z4DZ (tRF-28) related to gastric cancer. An absolute quantitative method was established to quantify the level of tRF-28 in plasma. ROC curve, survival curve, Cox regression model were constructed, and clinicopathological factors were collected to analyze the diagnostic value of tRF-28. "Overexpression" and "knockdown" tRF-28 gastric cancer cell models and xenograft tumor models were established to study the effects of tRF-28 on cell proliferation. Using bioinformatics, qRT-PCR, Western blot, dual luciferase reporter gene, fluorescence in situ hybridization, RNA binding protein immunoprecipitation assay, rescue experiment and immunohistochemistry to study the regulatory role of tRF-28 by binding to the 3'UTR of CCND1. **Results:** This study clarified the tRF-28 levels in the plasma of healthy individuals, and gastric benign and malignant lesions, providing a new candidate biomarker for the early diagnosis of gastric cancer. Through upregulating and downregulating the level of tRF-28 in gastric cancer cells, it was revealed that tRF-28 inhibits the proliferation of gastric cancer cells in vivo and in vitro by regulating key proteins such as Cyclin D1, CDK4, Rb, and p-Rb. In studying specific molecular mechanisms, tRF-28 and CCND1 were co-located in the cytoplasm. tRF-28 combines with Ago2 protein to form a complex and plays a role in silencing the target gene CCND1 in the cytoplasm. The rescue experiments revealed that silencing the target gene CCND1 can reverse the carcinogenesis effect of tRF-28 inhibitors, clarifying the biological functions of tRF-28 by regulating CCND1. **Conclusions:** Based on a series of research findings, we elucidated that tRF-28 form a complex with Ago2 protein, binding to the 3' UTR of CCND1 to downregulate Cyclin D1, CDK4, and other critical proteins, thereby inhibiting the proliferation of gastric cancer cells. Our study aims to offer new insights into identifying potential early diagnostic markers and therapeutic targets for gastric cancer. Research Sponsor: National Natural Science Foundation of China; 82302611.

Predictive role of circulating tumor DNA for locally advanced esophageal squamous cell carcinoma before surgery.

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Background: A precise preoperative tumor monitoring method that reflects tumor burden during neoadjuvant treatment is required to guide individualized perioperative treatment strategies for esophageal squamous cell carcinoma (ESCC). This study examined the clinical significance of preoperative circulating tumor DNA (ctDNA) in the plasma of patients undergoing neoadjuvant chemotherapy (NAC) followed by esophagectomy. **Methods:** Plasma samples were collected longitudinally for ctDNA analysis as well as genomic DNA from primary lesions from patients with histologically confirmed ESCC who received neoadjuvant chemotherapy (NAC) followed by subtotal esophagectomy. Next-generation sequencing was used to identify mutations in both the plasma and primary tumors. We evaluated the relation ship between ctDNA alterations and recurrence in patients with locally advanced ESCC. **Results:** Pretreatment samples from 25 patients (100%) showed the same mutations in both ctDNA and primary tumors; therefore, they were classified as ctDNA-positive before treatment. In the cohort of 25 patients analyzed, those who tested positive for ctDNA after NAC had a significantly higher risk of recurrence; the 36-month recurrence-free survival rates were 92% for ctDNA-negative patients and 8% for ctDNA-positive patients ($p < 0.001$). **Conclusions:** Preoperative ctDNA status may be a promising prognostic biomarker that can be assessed before surgery in patients with ESCC who received NAC. Expanded cohort validation will allow for more personalized multidisciplinary treatment approaches for ESCC tailored to ctDNA analysis. Research Sponsor: None.

Phase 1b study of futibatinib plus pembrolizumab in patients with esophageal carcinoma: Evaluating the immunological effects.

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Background: Combination therapy using FGFR inhibitors and Immune checkpoint inhibitors (ICIs) are being explored as a novel strategy for esophageal cancer (EC) patients (pts). We reported the tolerability and encouraging antitumor activity of futibatinib (futi) plus pembrolizumab (pem) in cohort (CO) A (≥2nd line, ICIs naïve) and COB (≥2nd line, ICIs refractory) and futi plus pem with chemotherapy in COD (1st line, ICIs naïve) with EC pts at Ooki et al ASCO 2024. The purpose of this Cohort E (COE) was to evaluate the effects of futi as a single agent or futi in combination with pem on tumor immune microenvironment by detecting immune cells and immune-related gene expression. **Methods:** In COE of the study, advanced or metastatic EC pts with at least one prior therapy with a fluorouracil and platinum-based drug received a single agent of futi the first 7 days, followed by futi 20 mg once daily (continuous dosing) plus pem 200 mg/every 3 weeks (up to a maximum of 35 cycles). Regardless of FGFR overexpression level and ICIs naïve or refractory pts were include on the study. Tumor samples were collected by biopsy during the screening period, 7 days after the administration of futi, and 21 days after the combination administration of futi and pem. Expression of 18 biomarkers including CD3, CD4, CD8 were analyzed by multiplex IHC (NeoGenomics). Gene expression related to immunity in tumor microenvironment were evaluated by nCounter system (NanoString). **Results:** As of March 10, 2024, 18 pts (ICIs naïve, 3 pts; ICIs refractory, 15 pts) were enrolled. As a result of multiplex IHC, an increase of $CD3^+CD8^+$ T cells was observed in the samples 7 days after a single treatment of futi comparing with baseline samples and a trend towards greater increased $CD3^+CD8^+$ cells was observed 21 days after combination treatment. An increasing CD8A gene expression was observed by nCounter analysis after a single treatment of futi comparing with baseline and a trend towards greater increase in CD8A gene expression was observed after combination treatment. Confirmed partial responses were observed in 3 pts (2 pts were ICIs naïve and 2 pts were FGFR positive). ORR was 16.7% (3/18; 1 pt had no target lesion, 95% CI: 3.6, 41.4). Most common TRAEs were hyperphosphatemia (77.8%), diarrhea (33.3%), ALT increased, and AST increased (22.2%). **Conclusions:** The preliminary multiplex IHC and nCounter results suggest that futi promoted $CD8^+$ T cells infiltration into EC tumor microenvironments by single treatment and following combination treatment with pem at tolerable dosage. Research Sponsor: Taiho pharmaceutical co., LTD. Merck & Co., Inc.

Summary of $CD3^+CD8^+$ T cell and CD8A gene expression after futibatinib and combination treatment (% change relative to the baseline).

	N	Mean (S.D)	95%CI
Cell:$CD3^+CD8^+$ T cell			
C1D7 (futi)	15	176.4 (88.8)	[127.2, 225.5]
C2D21 (Combination)	12	306.3 (229.8)	[160.3, 452.3]
Gene:CD8A			
C1D7 (futi)	16	139.4 (54.0)	[110.6, 168.2]
C2D21 (Combination)	14	245.9 (190.8)	[135.8, 356.1]

Concordance between human epidermal growth factor receptor 2 (HER2) status by immunohistochemistry (IHC) and next-generation sequencing (NGS) in tissue and blood samples from gastric and gastroesophageal junction cancers (GC/GEJC).

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Background: An estimated 18% of patients (pts) with GC/GEJC have HER2-positive (HER2+; IHC 3+ or 2+/in situ hybridization–positive [ISH+]) tumors. Although IHC/ISH are the standard testing methods for determining HER2 positivity, NGS can identify *HER2* (*ERBB2*) amplification alongside other biomarkers. This study assessed the concordance of detecting *HER2* amplification by tissue/plasma NGS with IHC/ISH HER2 status in pts with GC/GEJC. **Methods:** Pts were retrospectively identified in the Tempus database. The primary analysis assessed tissue and plasma NGS *HER2* copy number (CN) concordance with a HER2+ (IHC 3+ or 2+/ISH+) or HER2− (IHC 0, 1+, or 2+/ISH−) status; NGS samples were collected in ≤30 days of the IHC sample collection date. *HER2* amplification was defined as tissue CN ≥8 or plasma log2(CN) ≥0.5; additional amplifications were captured by manual inspection. A secondary analysis assessed the correlation between paired tissue and plasma *HER2* log2(CN) in pts with plasma and tissue NGS samples collected ≤30 days apart. The secondary analysis included pts irrespective of IHC sample collection date. **Results:** Of 1578 pts with GC/GEJC and a curated IHC score evaluated in the primary analysis, 1507 and 305 pts underwent tissue and plasma NGS testing, respectively. Among pts with tissue and plasma NGS results, 902 (59%) and 201 (66%) had primary tumors of the stomach, respectively; all remaining pts had primary tumors of the cardia.According to IHC/ISH, 229 (15%) and 40 (13%) pts were HER2+ among those with tissue and plasma NGS testing, respectively. Of the pts identified as *HER2* amplified by tissue and plasma NGS,95% and 92% were HER2+ by IHC/ISH, respectively. Positive percent agreement (PPA) of *HER2* amplification by tissue and plasma NGS with IHC/ISH HER2+ status was 70% and 58%, respectively. Concordance rates by PPA, negative PA (NPA), and overall PA (OPA) are summarized in Table. In the secondary analysis, a positive correlation in NGS *HER2* log2(CN) was observed in pts with paired tissue and plasma NGS (mutant variant allele frequency >5%; R=0.85; P<2.2e−16; n=307). **Conclusions:** Based on PPA, we observed moderate concordance between NGS *HER2* amplification detection (plasma or tissue) and standard HER2 IHC assessment. NGS is not sufficient to detect all pts with HER2+ GC/GEJC. Further testing is warranted to determine if NGS testing could complement IHC/ISH in identifying pts who may benefit from HER2-directed therapies. Research Sponsor: AstraZeneca in collaboration with Daiichi Sankyo.

	IHC 3+ or 2+/ISH+ (HER2+), n	IHC 0, 1+, or 2+/ISH− (HER2−), n	Total
Total	**229**	**1120**	
Tissue *HER2* amplified, n	160	8	**168**
Tissue *HER2* not amplified, n	69	1112	**1181**
	PPA: 69.9%	**NPA: 99.3%**	**OPA: 94.3%**
	IHC 3+ or 2+/ISH+ (HER2+), n	**IHC 0, 1+, or 2+/ISH− (HER2−), n**	**Total**
Total	**40**	**233**	
Plasma *HER2* amplified, n	23	2	**25**
Plasma *HER2* not amplified, n	17	231	**248**
	PPA: 57.5%	**NPA: 99.1%**	**OPA: 93.0%**

Concordance between PD-L1 assays 28-8 and SP263 and their respective scoring algorithms in procured upper gastrointestinal adenocarcinoma samples.

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Background: PD-L1 is being studied as a predictive immunohistochemical (IHC) biomarker for gastroesophageal junction (GEJ), gastric (GC), and esophageal (EAC) adenocarcinomas, and anti-PD-(L)1 therapies are used to treat them. Here, we compare the VENTANA PD-L1 (SP263) IHC assay and its tumor area positivity (TAP) algorithm with the clinically validated Agilent PD-L1 IHC 28-8 pharmDX combined positive score (CPS) algorithm in GEJ, GC, and EAC samples. The degree of concordance between SP263 TAP $\geq$ 5% and 28-8 CPS $\geq$ 5 is important for interpreting real-world settings and emerging clinical trial data. **Methods:** GEJ, GC, and EAC tumor samples procured from company-sponsored clinical trials (NCT02545504, NCT02862535, NCT02864381) and external vendors with informed consent were analyzed 4 ways: SP263 TAP, SP263 CPS, 28-8 TAP, and 28-8 CPS. Readouts were evaluated by a single pathologist and were analyzed with Pearson correlation coefficients. **Results:** 286 patient samples were analyzed (GEJ, n = 106; GC, n = 88; EAC, n = 92). The agreement among the SP263 TAP $\geq$ 5% and 28-8 CPS $\geq$ 5 prevalence is shown in the Table. The overall correlation coefficient between 28-8 CPS and SP263 TAP was high (R = 0.95; $P < 2.2 \times 10^{-16}$). In addition, when applying the same scoring algorithm to both assays (SP263 TAP vs 28-8 TAP or SP263 CPS vs 28-8 CPS), the correlation coefficients remained high (R = 0.97; $P < 2.2 \times 10^{-16}$). When both algorithms were applied to the same IHC assay (SP263 TAP vs SP263 CPS or 28-8 TAP vs 28-8 CPS), the correlation coefficients were also high (R = 0.98; $P < 2.2 \times 10^{-16}$). **Conclusions:** In this controlled experiment the PD-L1 assays 28-8 and SP263 concordance was high. The scoring algorithms for GEJ, GC, and EAC samples were observed as highly correlated; minor differences were likely driven by the distinct PD-L1 monoclonal antibody clones utilized in the IHC assays. These data suggest that the 2 assays are comparable for evaluating PD-L1 expression at the TAP $\geq$ 5% and CPS $\geq$ 5 cutoffs. Clinical trial information: NCT02545504, NCT02862535, NCT02864381. Research Sponsor: Gilead Sciences, Inc.

Concordance of SP263 TAP ≥ 5% with 28-8 CPS ≥ 5.

Positive percentage agreement: 0.96 Negative percentage agreement: 0.95 Overall percentage agreement: 0.95	28-8 CPS ≥ 5: Positive	28-8 CPS ≥ 5: Negative
SP263 TAP ≥ 5%: Positive	116	9
SP263 TAP ≥ 5%: Negative	5	156

480 **Poster Session**

Genomic landscape study of esophago-gastric adenocarcinoma (EGAC) in young patients under the age of 30 years.

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Background: Despite recent advances in treatment of EGAC, prognosis continues to be poor with reported 5-year survival of less than 30%. EGAC is more frequently diagnosed at an old age with the median age at diagnosis around 65 years. However, approximately 10% of EGAC are diagnosed at a younger age of less than 50 years and those patients present with more advanced stages resulting in worse prognosis. Our study investigated the genomic landscape of EGAC diagnosed under the age of 30 years to better understand the characteristics of EGAC occurring in very young population. **Methods:** 8,001 cases of clinically advanced EGAC who underwent comprehensive genomic profiling (CGP) from 2012 to 2024 were included in the study. Patients aged less than 30 years were classified as 'EGAC-young' and patients aged 50 and older were classified as 'EGAC-old'. All classes of genomic alterations (GA), microsatellite instability high (MSI-high) status, tumor mutation burden (TMB), genomic ancestry, genomic signature, and homologous recombination defect signature (HRDSig) were determined from the sequencing data. PD-L1 was measured by IHC using the Dako 22C3 tumor proportional score (TPS) system. Comparisons utilized the Fisher Exact method with the Benjamini-Hochberg adjustment to reduce the false discovery rate. **Results:** Out of 8,001 cases, 7389 (92.4%) were classified as 'EGAC-old' and 26 (0.3%) as 'EGAC-young'. Both groups had a higher frequency of male patients but there was no significant difference between the two groups (86.4% vs 76.9%; NS). Median GA per tumor was 6 in both groups. The 'EGAC-old' featured a significantly higher frequency of EUR ancestry (92.0% vs 73.1%; P=0.026). There were no significant differences in clinically important GA between both groups, including similar ERBB2 GA (21.2% vs 30.8%; NS) and FGFR3 GA (1.2% vs 7.7%; NS). There was a trend of more frequent GA of AKT2, CCNE1, and KEAP1 noted in 'EGAC-young' (0.9% vs 11.5%, 8.5% vs 26.9%, 1.3% vs 11.5%, respectively; P=0.071). MSI-high status was 3.1% in the 'EGAC-old' but not detected in the 'EGAC-young' (NS). The median TMB was higher in the 'EGAC-old' vs 'EGAC young' (3.8 mutations/Mb vs 1.9 mutations/Mb; P=.003). The frequencies of a positive HRDSig were similar in both groups (5.6% vs 8.0%; NS). There were no significant differences in COSMIC trinucleotide signatures. PD-L1 expression was identified in greater than 20% of the 'EGAC-old' but was not measured in the 'EGAC-young'. **Conclusions:** Clinically advanced EGAC in young patients under the age of 30 years features a genomic landscape that differs from EGAC identified in older patients, including lower TMB and a trend of increased GA frequency of genes related to poor prognosis in other types of cancer. Further studies are warranted to utilize CGP findings in identifying potential implications for treatment and prognostication in these rare cases of EGAC occurring in very young patients. Research Sponsor: None.

Relationship between PD-L1 expression and the number of biopsy specimens in advanced gastric cancer.

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Background: Programmed cell death ligand 1 (PD-L1) expression shows spatial heterogeneity in gastric cancer. Compared with single biopsies, multiple biopsies may provide reliable PD-L1 expression results. However, the optimal number of biopsy specimens remains unclear. This study aimed to assess the relationship between PD-L1 expression and the number of biopsy specimens in advanced gastric cancer. **Methods:** We retrospectively analyzed patients with advanced gastric cancer who received first-line chemotherapy at a single institution between December 2021 and June 2024. Those in whom PD-L1 expression was evaluated using tumor-containing biopsy specimens were included. PD-L1 expression was measured using the Dako PD-L1 IHC 28-8 pharmDx assay. PD-L1 positivity was defined as a combined positive score of ≥5. The association between PD-L1 positivity and the number of biopsy specimens was tested using the chi-square or Fisher's exact test. **Results:** Of the 183 patients screened, 110 were included. The patient characteristics were as follows: median age, 71 (range: 34–87) years; sex (male/female), 75/35; primary site (gastric/gastroesophageal junction), 102/8; and histological type (diffuse/intestinal/other), 59/34/17. The PD-L1 positivity prevalence was 71.8%. In 97 patients, human epidermal growth factor receptor 2 (HER2) expression was detected in the same specimens as PD-L1, with a HER2-positive rate of 14.4%. The mean numbers of biopsy and tumor-containing biopsy specimens were 5.14 (range: 2–10) and 4.25 (range: 1–10), respectively. The PD-L1 positivity prevalence was significantly higher when the number of biopsy specimens was ≥5 compared with ≤4 (77.5% vs 56.7%, P=0.03; Table). The same trend was observed according to the histological type (diffuse type, 75.0% vs. 46.7%, P=0.04; intestinal type, 79.2% vs. 60.0%, P=0.39). In HER2-negative cases, PD-L1 positivity was significantly more prevalent when the number of biopsy specimens was ≥5 compared with ≤4 (83.6% vs 54.5%, P=0.006; Table). However, there was no such difference in HER2-positive cases (54.4% vs 66.7%, P=1). **Conclusions:** The data suggest that collecting at least five biopsy specimens increases the likelihood of identifying PD-L1 positivity in advanced gastric cancer, particularly in HER2-negative cases. This approach may improve PD-L1 evaluation accuracy and lead to better treatment decisions. Research Sponsor: None.

Relationship between the number of biopsy specimens and PD-L1 positivity.

Number of biopsy specimens	PD-L1 positivity (%)	PD-L1 negativity (%)	*P* value
Overall			0.03
≥5	77.5	22.5	
≤4	56.7	43.3	
HER2-negative			0.006
≥5	83.6	16.4	
≤4	54.5	45.5	

The expand indication of HER2 targeted therapies by monitoring of circulating tumor cells.

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Background: Despite success in the HER2 targeted therapies for metastatic gastric cancer patients, the conventional assessment of HER2 status in tumor tissue specimens is still under debate. **Methods:** On-chip sorting system was used for the isolation of circulating tumor cells (CTCs). The fluorescence signal intensity > 50 is set to be a threshold value for distinguishing between positive and negative HER2 on CTCs. Epithelial mesenchymal transition (EMT) was assessed by the vimentin and cytokeratin. DNA was extracted from CTCs and applied to the cancer gene panel. **Results:** In 27 patients with metastatic gastric cancer, 13 patients showed HER2 positive in tumor tissues (Group A) who were treated with cytotoxic agents and HER2-targeted therapy while 14 patients exhibited negative and treated with cytotoxic agents. In these 14 patients, 8 patients showed HER2 positive (Group B) and 6 patients displayed HER2 negative (Group C) on CTCs. HER2 expression on CTCs was associated with EMT ($p<0.001$). Increased expression of EMT was seen in Group A and B, indicating they are likely to have metastasis. Group B exhibited worse PFS than others (13.8 M in A, 7.0 M in B and not reached in C, $p=0.012$). In addition, genomic analysis identified cancer gene mutations, some of which were involved in mechanisms underlying EMT and were shared between Group A and B. Considering HER2 expression on CTCs in Group B, an additional HER2-targeted therapy may be beneficial. **Conclusions:** Our study suggests that monitoring of circulating tumor cells leads to the expand indication of the HER2 targeted therapies in patients with metastatic gastric cancer. Research Sponsor: None.

Use of nucleotyping and tumor stroma ratio to predict prognosis in patients with resected intestinal-type gastric cancer.

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Background: Heterogeneity exists in recurrence and survival among patients with intestinal-type gastric cancer (IGC). Nucleotyping is the alterations in chromatin organization. We sought to identify the prognostic utility of nucleotyping, tumor stroma ratio (TSR) and DNA ploidy in these patients. **Methods:** Radical resection of gastric carcinoma (> 15 LNs removed) was performed in 283 patients (stage I, n=73; stage II, n=102; stage III, n=108) at our university hospital between 2010 and 2018. Automated image cytometry tools were used to quantitatively measure nucleotyping, TSR and DNA ploidy. Cutpoints of nucleotyping and TSR were 0.044 and 50%, respectively. Associations of these features with clinicopathological characteristics, time-to-recurrence (TTR) and cancer specific survival (CSS) were analyzed by univariate and multivariate analysis. **Results:** Among 283 patients, chromatin heterogeneity, high TSR and non-diploid were significantly associated with larger tumor size, higher T and N stage, and poorer TTR and CSS at 5 years. In a Cox multivariable analysis, nucleotyping and TSR were independently associated with TTR and CSS ($P < 0.05$) (Table). The top three contributors to TTR were N stage (33.9%), nucleotying (29.7%) and TSR (25.5%). The top three contributors to CSS were N stage (35.6%), nucleotying (29.7%) and TSR (18.0%). As a combined variable, the percent relative contribution of nucleotyping and TSR ranked first for TTR and CSS (56.0%, 50.0%), ahead of N stage (33.4%, 34.6%) and TILs (8.4%, 10.6%). **Conclusions:** Nucleotyping and TSR were each significantly and independently associated with TTR and CSS, and their combined variable was more strongly prognostic than either alone. Research Sponsor: None.

Cox multivariable analysis of TTR and CSS.

Features	TTR HR (95%CI)	TTR P value	CSS HR (95%CI)	CSS P value
Tumor size (>3cm vs ≤3cm)	1.10 (0.48-2.54)	0.83	0.77 (0.35-1.70)	0.56
T stage (T3-T4 vs T1-T2)	1.07 (0.46-2.49)	0.87	1.41 (0.53-3.76)	0.49
N stage (N2N3 vs N0N1)	2.84 (1.39-5.78)	<0.01	7.69 (3.37-17.55)	<0.01
Adjuvant Therapy (with vs without)	1.32 (0.72-2.40)	0.37	0.85 (0.48-1.52)	0.59
TB (high vs low)	1.29 (0.62-2.69)	0.50	1.61 (0.85-3.07)	0.15
TILs (low vs high)	1.73 (0.96-3.13)	0.07	1.90 (1.06-3.40)	0.03
Nucleotyping (Chromatin heterogeneous vs Chromatin homogeneous)	2.19 (1.21-3.99)	0.01	3.44 (1.96-6.03)	<0.01
TSR (high vs low)	2.12 (1.16-3.82)	0.01	2.51 (1.43-4.41)	<0.01
DNA ploidy (Non-diploid vs Diploid)	1.20 (0.55-2.63)	0.65	0.98 (0.44-2.15)	0.95

1 TTR, time-to-recurrence; 2 CSS, cancer specific survival.

Histopathological growth patterns of gastric cancer liver metastasis and their association with patient prognosis.

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Background: The histopathological growth patterns (HGPs) of liver metastasis reflect the complicated and varied interactions between tumor cells and the host microenvironment. We studied HGPs of gastric cancer liver metastasis (GCLM) and sought to predict them using radiomic and clinical variables. **Methods:** Hepatectomy was performed in 62 patients with GCLM at our university hospital between 2011 and 2021. HGPs were evaluated according to international consensus guidelines, and were classified as either desmoplastic HGP (dHGP) or non-dHGP. CD4, CD8, VEGF and CD31 were analyzed by immunohistochemistry. Eight hundred and fifty-one radiomics features were extracted from CT portal venous phase images and the optimal radiomics signature was determined by univariate analysis and LASSO regression. Multivariable regression analyses and ROC curves were used to assess the predictive performance of HGPs with a radiomic and clinical combined model. **Results:** Among 62 patients, 14 were categorized as dHGP, and 48 as non-dHGP. Non-dHGP vs. d-HGP was associated with higher tumor N stage and CEA level, and with decreased $CD8^+$ T cells, and lower VEGF and CD31 expression. HGP status was independently associated with patient cancer-specific survival (CSS) ($P < 0.05$) multivariately.Analysis of percent relative contribution revealed that HGP ranked first for CSS (35.9%), ahead of T (28.9%) and N (28.9%) stage. A model with combined radiomic and clinical features demonstrated robust performance with area under the curve (AUC) values of 0.993 and 0.933 in the training and validation sets, respectively. **Conclusions:** Non-invasive determination of radiomic and clinical features was shown to predict HGPs types. Compared to dHGP, GCLM with non-dHGP exhibited significantly lower immune cell infiltration and decreased angiogenesis. Among features examined in GCLM, HGP was the most robust for prognostication. Research Sponsor: None.

Univariate and multivariate analysis of prognostic factors for cancer-specific survival.

	Univariate analysis		Multivariate rergression analysis	
Characteristics	**HR (95%CI)**	**P**	**HR (95%CI)**	**P**
Gender (Women vs Men)	1.13(0.61-2.09)	0.71		
Age (>60years vs ≤60years)	1.08(0.60-1.70)	0.98		
HGP (non-desmoplastic HGP vs desmoplastic HGP)	3.63(1.78-7.41)	0.00	3.48(1.37-8.88)	0.01
T stage (T3-T4 vs T1-T2)	2.03(1.08-3.80)	0.03	2.29(1.15-4.58)	0.02
N stage (N1-N3 vs N0)	3.46(1.66-7.23)	0.00	2.54(1.17-5.56)	0.02
CEA (>5ng/ml vs ≤5ng/ml)	2.37(1.25-4.48)	0.01	1.56(0.70-3.48)	0.27

A study of the homologous recombination deficiency signature (HRDSig) status of advanced esophageal squamous cell carcinoma (AESCC) by utilizing comprehensive genomic profiling (CGP).

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Background: The role of poly-ADP-ribose polymerase (PARP) inhibitors is currently under investigation as a potential therapeutic option for AESCC. Homologous recombination deficiency signature status (HRDSig) positivity has been shown to predict biallelic loss of BRCA1/2 which may simultaneously portend a worse prognosis and a sensitivity to poly-ADP-ribose polymerase (PARP) inhibitors. In this study, we attempt to characterize the genomic alterations present in AESCC based on HRDSig status utilizing comprehensive genomic profiling (CGP) techniques. **Methods:** 2,029 cases of AESCC underwent comprehensive genomic profiling with an examination of all genomic alterations (GA). MSI high status, tumor mutation burden (TMB) levels, genomic ancestry and genomic trinucleotide signatures were determined from the sequencing data. HRDSig status was calculated using a broad set of genome-wide copy number features (PMID 37769224). Results were compared using the Fisher exact system with the Benjamini-Hochberg adjustment to correct for false discovery. **Results:** 154 (7.6%) of the 2029 AESCC cases featured a positive HRDSig status (HRDSig+). The age (66-67) and gender (58%-63% male) distribution were similar in the HRDSig+ and HRDSig- AECSS cases as was the frequency of GA/tumor (9 for both). The median TMB was higher in the HRDSig+ (6.3 vs 3.8; P<.0001) as was frequency of TMB > 10 mutations/Mb (19.5% vs 10.0%; P=.004). An APOBEC trinucleotide signature was also more frequent in the HRDSig+ AESCC (11.0% vs 5.4%; P=.05). As anticipated, GA in genes associated with HRD including *BRCA1* (10.4% vs 1.4%; P<.0001) and *BRCA2* (14.3% vs 2.0%; P<.0001) was present. MTOR pathway activating mutations were also more frequent in the HRDSig+ AESCC group including *PTEN* (13.0% vs 7.3%; P=.06). **Conclusions:** With a 7.6% frequency, HRDSig+ status is a relatively rare event in AESCC. CGP with determination of HRDSig status in AESCC may prove useful in tailoring PARP inhibitor-based treatment regimens and may potentially uncover other genomic alterations that can aid in designing targeted therapy combinations in future. Research Sponsor: None.

Characteristics of comprehensive genomic profiling of clinically advanced esophageal squamous cell carcinoma by HRDSig status.

	AESCC HRDSig- (n=1875)	AESCC HRDSig+ (n=154)	P-value
Pathogenic genomic alterations			
BRCA1	1.4% (26)	10.4% (16)	0.00
BRCA2	2.0% (36)	14.3% (22)	0.00
PTEN	7.3% (137)	13.0% (20)	0.06
COSMIC trinucleotide signature			
APOBEC	5.4% (101)	11.0% (17)	0.05
Tumor mutational burden (TMB)			
Median TMB (range) (IQR)	3.8 (0-85) (2.5-6.3)	6.3 (0-61) (3.8-8.8)	0.00
TMB≥10 mut/Mb	10.0% (188)	19.5% (30)	0.00

Prognostic value of tumor regression grade (TRG) in patients with gastric/gastroesophageal junction cancer treated by peri-operative chemotherapy and surgery: A single-center experience (Institut Mutualiste Montsouris, Paris, France).

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Background: Management of locally advanced gastric and gastroesophageal junction (GOJ) cancer relies on peri-operative chemotherapy and radical surgery. Response to neoadjuvant chemotherapy can be evaluated by Mandard Tumor Regression Grade (TRG) performed on the removed specimens. The aim of this study was to investigate the prognostic value of TRG, in terms of post-surgical disease-free survival (DFS) and overall survival (OS). **Methods:** Retrospective analysis of all consecutive patients (Jan 2010 to Dec 2021) who underwent oncological gastrectomy for gastric or GOJ adenocarcinoma after neoadjuvant chemotherapy was performed. TRG was evaluated according to Mandard classification, ranging from 1 (complete pathological response) to 5 (absence of response), TRG 1-2 being considered as "good responders" and TRG 3-5 as "bad responders". Univariate and multivariate analysis were performed for DFS and OS. **Results:** One hundred and ninety-nine patients were included, 157 treated with FOLFOX and 42 with FOLFOX + Taxanes (either FLOT or FOLFOX + Abraxane). TRG 1-2 was observed in 30% in patients (pts) treated with FOLFOX and 24% in pts treated with FOLFOX + Taxanes. With a median follow-up over 5 years (yrs), median DFS and OS were not reached (DFS rates at 3, 5 and 10yrs of 66.5%, 64.1% and 55.7%, respectively; OS rates at 3, 5 and 10yrs of 75.6%, 66.8% and 60.4%, respectively). In univariate analysis, TRG 1-2 was significantly associated with favorable outcome compared to TRG 3-5 in terms of DFS (HR 2.4, p-value 0.0056) and OS (HR 2.6, p-value 0.0061), as well as ECOG at diagnosis (0 vs. 1), ypT stage (yp T0-2 vs. yp T3-4), ypN stage (ypN- vs. ypN+), tumor localization (antrum + body vs. cardia + esophagus), and lymphatic, vascular and peri nervous embolisms (negative vs. positive) for both DFS and OS. However, in multivariate analysis (selection backward Cox-model), only ypT stage and vascular embolisms for DFS, and ypT and ypN stages for OS were found to be independent prognostic factors. **Conclusions:** Although TRG is a prognostic factor in patients with gastric or GOJ cancer, it does not appear as an independent factor for DFS and OS in this series of 199 pts. These results raise the issue of "good responder" definition, especially for TRG 3. Prognostic impact of TRG needs to be investigated for patients treated with combination of immune checkpoint inhibitors and chemotherapy in neoadjuvant regimens.
Research Sponsor: None.

SOX/FOLFOX+nivolumab followed by conversion surgery for HER2 negative primary gastric or gastroesophageal junction cancer with oligo-metastasis or borderline resectable tumors: Early results of a consecutive 41 cases.

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Background: Gastric or gastroesophageal junction cancer with oligo-metastasis or borderline resectable tumors could be cured by conversion surgery after primary chemotherapy. Because SOX/FOLFOX plus Nivolumab showed high clinical response regardless of CPS score, this nivolumab-based chemotherapy followed by surgery may be promising strategy for these tumors. **Methods:** The study examined the patients who had HER2 negative primary gastric or gastroesophageal junction cancer with synchronous oligo-metastasis or borderline resectable tumors. Oligo-metastasis was defined as peritoneal dissemination localized to the upper abdomen, para-aortic lymph node metastasis, liver metastasis limited up to 3 segments, mediastinal lymph node metastasis, ovarian metastasis, or esophageal intramural metastasis. Borderline resectable tumors were defined as regional lymph node metastasis forming bulky mass or primary tumors invading adjacent organs. The patients with these tumors were treated with primary chemotherapy with SOX/FOLFOX plus Nivolumab for 3 to 22 months, then were proceeded to conversion surgery in case the tumors are responded to chemotherapy and are considered to achieve R0 resection based on the MDT conference. **Results:** During November 2021 and February 2024, 41 consecutive patients were included in the study. Disease status was peritoneal dissemination in 1, para-aortic lymph node metastasis in 26, liver metastasis in 5, mediastinal lymph node metastasis in 1, ovarian metastasis in 1, esophageal intramural metastasis in 2, bulky nodal metastasis in 6, and primary tumors invading pancreas in 1 (distant oligo-metastasis in 34 patients and borderline resectable in 7). CPS score was >5 in 13, 5> and >1 in 23, and >1 in 5 patients. Chemotherapy regimen was SOX+Nivo in 38 and FOLFOX+Nivo in 3. Of 41 patients, 33 (80.5%) (SOX+Nivo: 30, FOLFOX+Nivo: 3) were proceeded to the conversion surgery, and 32 patients achieved R0 resection. 6 patients showed disease progression and 2 patients were judged as unresectable. Major pathological response (disappearance more than 80% of the primary tumor) rate was 54.5% (18/33) and pCR rate was 33.3% (11/33). **Conclusions:** SOX/FOLFOX plus Nivolumab for HER2 negative gastric or gastroesophageal junction cancer with oligo-metastasis or borderline resectable tumors achieved a high R0-conversion surgery rate and pathological response rate. Research Sponsor: None.

Use of immune repertoire sequencing analysis to detect differences in treatment responses for advanced esophageal squamous cell carcinoma treated with camrelizumab and platinum-based chemotherapy.

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Background: The study evaluated the efficacy and safety of camrelizumab combined with platinum-based chemotherapy (platinum plus paclitaxel [TP] or fluorouracil [FP] agents) as first-line treatment in patients with advanced esophageal squamous cell carcinoma (ESCC), and explored potential biomarkers for treatment response using immune repertoire (IR) sequencing. **Methods:** In this multi-center, prospective cohort study, advanced ESCC patients were treated with camrelizumab and TP or FP chemotherapy. Primary outcome was 1-year progression free survival (PFS). Secondary outcomes included 1-year overall survival (OS), objective response rate (ORR), disease control rate (DCR) and safety. Patients were categorized based on their treatment responses into the non-ORR and ORR groups. IR sequencing was exploratorily performed on the peripheral blood mononuclear cells from 15 patients in each group. **Results:** From June 2020 to April 2023, 88 out of 98 screened patients were enrolled, with a median follow-up of 15.9 months. The 1-year PFS was 56.8%, OS was 68.2%, ORR was 64.8%, and DCR was 91.1%. Most treatment-related adverse events were grade 1-2, though 12.5% experienced grade ≥3 toxicities. The FP regimen's efficacy was comparable to the TP regimen. The analysis of the complementarity-determining region 3 polypeptide sequences revealed significant differences in amino acid composition between the non-ORR and ORR groups in T-cell receptor beta-chain (TRB) and immunoglobulin heavy chain (IGH) repertoire. Furthermore, the ORR group exhibited a more concentrated distribution of clones within TRB. Two V genes (TRBV29-1 and TRBV4-1) and one J gene (TRBJ1-3) in the TRB region, along with six V genes (IGHV1-45, IGHV3-20, IGHV3-48, IGHV3-49, IGHV4-4, and IGHV5-51) in the IGH region, were differentially expressed between the two groups. **Conclusions:** Camrelizumab with platinum-based chemotherapy is effective and well-tolerated as first-line treatment in advanced ESCC. IR sequencing may provide insights into the underlying mechanisms influencing treatment response. Clinical trial information: ChiCTR2000037942. Research Sponsor: None.

Objective response and disease response.

Efficacy evaluation	Total (N=88)	Camrelizumab plus T (n=69)	Camrelizumab plus FP (n=19)	p
1-year PFS rate	50/88 (56.8%)	39/69 (56.5%)	11/19 (57.9%)	1.000
1-year OS rate	60/88 (68.2%)	47/69 (68.1%)	13/19 (68.4%)	1.000
ORR	57/88 (64.8%)	43/69 (62.3%)	11/19 (57.9%)	0.793
DCR	81/88 (91.1%)	63/69 (91.3%)	15/19 (79.0%)	0.213
ORR group, n (%)	54/88 (61.4%)	43/69 (62.3%)	11/19 (57.89%)	0.793
Non-ORR Group, n (%)	34/88 (38.6%)	26/69 (37.7%)	8/19 (42.1%)	1.000
2-year PFS rate	38/88 (43.2%)	30/69 (43.5%)	8/19 (42.1%)	1.000
2-year OS rate	42/88 (47.7%)	33/69 (47.8%)	9/19 (47.4%)	1.000

*$p<0.05$, a statistically difference.

Transcriptomic features of Epstein-Barr virus-associated gastric cancer based on PD-L1 status.

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Background: Epstein-Barr virus (EBV)-associated gastric cancer (EBVaGC) accounts for 5-10% of all gastric cancers and has a relatively favorable prognosis compared to other subtypes. While the genomic profile of EBVaGC has been previously reported to exhibit extensive CpG island methylation, elevated levels of PD-L1/2, and an absence of TP53 mutations, transcriptomic features remain less explored. The response to immunotherapy of EBVaGC has been various. This study comprehensively investigates the transcriptomic characteristics of EBVaGC in relation to PD-L1 status. **Methods:** RNA sequencing was conducted on formalin-fixed, paraffin-embedded samples from 27 EBV-positive and 12 EBV-negative patients with HER2-negative gastric cancer. Gene expression was quantified using the RSEM package. Predict-IO, a machine learning-based predictive algorithm developed at Auristone Pte Ltd based on initial work done at the Genome Institute of Singapore (GIS), assessed the likelihood of a patient responding to immunotherapy based on transcriptome expression data. PD-L1 immunohistochemistry was performed using the 22C3 pharmDx assay, and status was classified by CPS. **Results:** A total of 200 differentially expressed genes (DEGs; log2FC > 2, FDR < 0.05) were identified in EBV-positive versus negative samples, with 157 genes down-regulated and 43 genes up-regulated. Upregulated genes included immune response related genes such as DMBT1, IDO1, PLA2G2A, and IL13RA2, while downregulated genes were primarily associated with cell adhesion, including LAMA1 and CCN6. In the EBV positive group, immune response pathways were enriched; however, within the PD-L1-Low group (CPS < 1), activated signaling pathways such as MYC, E2F, and MTORC1 were observed. Conversely, the PD-L1-High group (CPS ≥ 5) exhibited upregulated snoRNAs. In the EBV negative group, the PD-L1-Low cohort displayed suppressed immune response and ECM remodeling factors, whereas the PD-L1-High cohort showed enrichment of ECM remodeling factors. The Predict-IO Score, which predicts immunotherapy response, was significantly higher in EBV-positive compared to EBV-negative samples ($p = 0.003$), and scores were elevated in the PD-L1-High group compared to the PD-L1-Low group. **Conclusions:** In summary, EBVaGC includes subsets of different PD-L1 expression patterns with distinct activated signaling pathways, differences in ECM remodeling factors, and varying Predict-IO Scores. With these information, we could properly select the responsive patients to immunotherapy among EBVaGC. Research Sponsor: None.

Biomarkers of response in patients with gastric/gastroesophageal junction adenocarcinoma (GEA) treated with telisotuzumab adizutecan.

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Background: c-Met protein (MET protein) overexpression (OE) and *MET* amplification (amp) occur in several cancer types, including GEA, and are associated with poor prognosis. The c-Met–directed antibody-drug conjugate telisotuzumab adizutecan (ABBV-400) is being assessed in advanced solid tumors in an ongoing, first-in-human study (NCT05029882). Results showed promising antitumor activity and a tolerable safety profile in GEA (Strickler et al. ESMO 2024, 1439P). Here, we present our analysis on associations between c-Met OE, *MET* amp, and ctDNA-derived molecular response with clinical outcomes. **Methods:** c-Met expression was evaluated retrospectively in FFPE tissue (archival/fresh) collected at study screening using IHC (SP44 Ab clone) or whole transcriptomic RNA sequencing (Caris). Plasma ctDNA collected at baseline (BL) and cycle 3 day 1 (C3) was analyzed with the GuardantINFINITY assay. Molecular response (MR) was assessed using circulating tumor fraction (cTF), estimated on the basis of ctDNA variant allele frequencies in a 74-gene panel (Guardant360 CDx) and methylation signals (GuardantINFINITY methylome platform). MR was defined as a 50% decrease in cTF from BL. *MET* amp status was determined retrospectively by the Guardant Health algorithm. Radiographic response was assessed per RECIST v1.1. **Results:** Clinical data from 42 patients (pts) with advanced GEA were included in this biomarker analysis. There was a negative correlation between c-Met expression and tumor size change from BL (R=-0.59). Of 42 pts, 12 (29%) had tumors with *MET* amp per ctDNA analysis/local site reports, which was higher than the bioinformatic analysis of real-world data (Caris-ConcertAI linked RWD360 and PT360; ~3%). Concordance between c-Met expression per IHC and RNA was observed (R=0.75, $p<10^{-6}$). There were 12 responders; 7 (58%) had *MET* amp, and 5 (42%) did not. Responders had lower ctDNA percentage change from BL vs nonresponders (1.5-fold and 1.6-fold difference based on 74-gene and methylation panel, respectively). Pts with MR had a greater decrease in tumor size from BL vs those without MR. A positive correlation between change in ctDNA levels (BL to C3) and change in tumor size was observed using the 74-gene panel and methylation panel (R=0.73 and 0.69, respectively). Additional biomarker analyses focused on the association between *MET* and other key biomarkers relevant in GEA will be discussed. **Conclusions:** In pts with advanced GEA treated with telisotuzumab adizutecan, c-Met OE/*MET* amp and ctDNA-based MR were associated with enhanced antitumor activity. Compared with real-world data, *MET* amp was enriched in this study population. Phase 2 studies will continue to explore c-Met OE and *MET* amp prevalence and their potential predictive value with other biomarkers in a clinically representative population. Clinical trial information: NCT05029882. Research Sponsor: AbbVie; n/a.

491 **Poster Session**

The Gastric Cancer Registry: A multi-omic cellular and molecular resource for cancer biomarker and therapeutic discovery.

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Background: The Gastric Cancer Registry (GCR) is a comprehensive clinical and genomic data resource focused on gastric cancer (GC) patients and individuals with a family history of GC or a germline CDH1 mutation. As part of the GCR's ongoing enrollment, patients contribute cancer history information, archival tumor samples and other biospecimens. The cancer samples undergo extensive molecular and cellular analysis that includes genomic sequencing and spatial analysis. We have generated a robust dataset, publicly accessible online through the GCR Genome Explorer (GCR-GE). This genomic, molecular and cellular resource provides a rich data set that accelerates multidisciplinary research aimed at improving detection and treatment strategies for GC. GC disproportionately affects Latin American populations - we have included new cohorts from Latin America. The GCR enables researchers, clinicians and patients to address specific questions about the molecular and cellular basis of GC. **Methods:** GC samples underwent whole genome, whole exome, and bulk RNA sequencing were conducted. The genomic features identified include copy number variation, gene mutations, gene expression, microbiome composition, estimation of tumor-infiltrating immune cells, tumor neoantigens, MSI/MSS status, and HLA subtypes. In addition to genomic data, clinical data was compiled from survey responses and pathology reports and integrated into the GCR-GE. We have incorporated single cell and spatial analysis that includes diverse cell types, single cell gene expression and cellular architecture in native tumor tissues. This data was released to the GCR Genome Explorer website. **Results:** A total of 817 subjects have enrolled in the GCR including 598 diagnosed with GC and 219 at high-risk for GC through family history of GC and/or a CDH1 mutation. Of the 598 with GC, some met multiple criteria including at least 40 with a family history of GC, 10 with a CDH1 mutation, and 18 with both a family history of GC and a CDH1 mutation. The GCR-GE includes data from 257 gastric tumors in addition to external datasets from TCGA. In the latest data release, clinical and genomic data from 78 gastric tumors from patients in Latin America were integrated into the GCR-GE. **Conclusions:** The GCR is a comprehensive database of crucial clinical and genomic data necessary for driving forward GC research. Through global expansion, the GCR has generated a more representative dataset by increasing racial diversity and including underrepresented populations that are at higher risk for GC. Research Sponsor: Gastric Cancer Foundation.

Claudin 18.2 determination in locally advanced gastric and gastroesophageal adenocarcinoma treated with neoadjuvant chemotherapy.

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Background: Gastric and esophago-gastric adenocarcinoma (GAC) represent a major unmet need. For tumors amenable with curative intent, prognosis remains poor with 5-year survival rates below 40%. Several strategies have been added to surgery including peri-operative chemotherapy which represents the standard of care for locally advanced GAC. Claudins is a family of transmembrane junctional proteins which acts in paracellular permeability, transport and communication. Claudin 18.2 (CLDN), a tissue-specific variant, is physiologically expressed in both gastric mucosa and gastric cancer. In metastatic GAC, CLDN positivity settles around 40% of cases and the addition of target therapy zolbetuximab to chemotherapy resulted in a significant improvement in overall survival. However, limited data are present about CLDN in early-stage disease. The aim of our study is to assess rates of expression of CDLN in locally advanced GAC treated with neoadjuvant chemotherapy and correlate it with clinical outcomes. **Methods:** We conducted a multi-omics prospective trial at IEO enrolling patients with locally advanced GAC treated with peri-operative chemotherapy and surgery. Pre-chemotherapy endoscopic samples were selected for CDLN determination. Positive cases were defined as moderate-to-strong membranous staining in ≥ 75% of tumor cells by IHC using the VENTANA CLDN18 (43-14A) Assay. CLDN expression was analyzed for correlation with pathologic complete response (pCR), tumor regression grade (TRG), event-free survival (EFS), and overall survival (OS). Survival was calculated using Kaplan–Meier method and log-rank tests. Statistical analyses were performed using SPSS software (version 29.0.2.0). **Results:** We included 48 patients, 67% male, median age 62 years (35-78). 31 tumors originated from gastric body, 14 from the junctional region and 4 from distal esophagus. Most frequent histotypes were diffuse (43%) and tubular (39%). Five cases (11%) presented microsatellite instability. All patients received neoadjuvant chemotherapy, mainly FLOT (92%) and underwent surgery, with 93% of R0 resection. Three patients (6%) achieved pCR and 18 had a TRG 1A or B. Of the total, 20 cases were defined as positive (42%). After a median follow-up of 20 months, 12 patients relapsed and 3 died. No significant association was observed for CDLN positivity and pCR or TRG1 rate, as well as for primary site or histotype. PFS and OS were not statistically different according to claudin status, even though numerically shorter in CDLN positive cases (PFS: 31 vs 33 months, p=.43; OS: 36 vs 39 months, p =.10). **Conclusions:** In our study of locally advanced GAC, rate of CDLN positivity was 40%, similarly to advanced stage. However, no differences in pCR rate, regression grade and survival outcomes have been observed. Larger studies and longer follow-up are required to assess the impact of CDLN in early-stage GAC. Research Sponsor: None.

The immunogenomic landscape of primary and metastatic gastroesophageal adenocarcinoma.

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Background: Metastatic gastroesophageal adenocarcinoma (GEA) is a heterogeneous disease with an overall poor prognosis. The tumour microenvironment (TME) between the primary and metastatic compartment is not well characterized. This study aims to describe the immunogenomic features of matched primary and metastatic GEA and their correlation with clinical outcomes. **Methods:** We performed whole exome sequencing (WES, n=36) and total RNA sequencing (RNA-seq, n=36) on a prospectively collected cohort of metastatic GEA, of which 14 had matched primary and metastatic tissues. We used TME deconvolution (Bagaev et al., 2021) comparing between the primary and metastatic compartments. Using an orthogonal technology, we further integrated immune checkpoint-directed multiplex immunohistochemistry (mIHC) for 22 patients with matched primary and metastatic samples. **Results:** The median age of our cohort was 61 (range 27-89), primarily male (n=28, 77%), and non-Asian (n=32, 89%). All had metastatic GEA at time of diagnosis. Differential expression showed activation of immune pathways such as tumor necrosis factor signalling and interferon-gamma response in the metastatic compartment. Deconvolution of the TME demonstrated 29% of primary tissues having an immune-inflamed TME compared to 61% of metastatic tissues (p=0.032). Exploring immune cell types, metastatic compartment is characterized by higher abundance of naïve and mature B cell populations ($p<0.001$), while primary compartment is enriched with CD4+ T cells ($p<0.01$). In matched cases, only 2 of 14 (14%) had concordant TME subtypes between primary and metastatic samples suggesting high degree of immune divergence. Furthermore, expression of common immune checkpoints aligned more with TME subtypes than within patient compartments suggesting common pathways of immune evasion. M-IHC showed both T-cell and immune checkpoint markers are enriched at the tumor margins compared to the tumor center in both primary and metastatic compartments (CD4; $p<0.0001$, CD8; $p<0.001$, CD68; $p<0.001$). There was a significant increase in the number of CD68+/CD163+/PDL1+ M2-like macrophages in the tumor margins ($p<0.0001$). Consistent with our deconvolution analysis, there are higher CD4+ T cells in the matched primary compared to metastatic samples. Long-term survivors (>16 months) had decreased number of CD68+ macrophages in the tumor center (p=0.019). **Conclusions:** Primary and metastatic GEA have divergent TME. This may partially explain varying responses to chemo-immunotherapy approaches. Therapies aimed at modifying TME may provide personalized treatment options. Research Sponsor: Physician Services Incorporated Foundation.

Molecular profiling of a gastroesophageal adenocarcinoma (GEA) multicohort using next-generation sequencing (NGS).

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Background: The mutational landscape of GEA is rapidly evolving with the identification of promising molecular targets. However, histological subtypes are insufficient to capture the molecular heterogeneity in these patients (pts). We aim to assess the histopathological and molecular features in a GEA cohort using NGS techniques, and to evaluate the mutational frequency and its prognostic correlation. **Methods:** We conducted a retrospective study on GEA from 2019 to 2024. Immunohistochemistry (IHC) was used to evaluate HER2 expression, microsatellite instability (MSI), Epstein-Barr virus (EBV) and PD-L1 status. In-house and commercial NGS platforms were used to detect molecular alterations from tumor samples. Cox regression model was used to analyze overall survival (OS) based on the clinicopathological and molecular subgroups. **Results:** Using NGS techniques, we identified the following alterations with potential clinical relevance in the 115-pts included: pathogenic mutations in *PIK3CA* (5.2%), *BRCA1/2* (3.5%) and *POLD1* (0.9%), along with copy number alterations (CNA) in *ERBB2* (10.4%), *EGFR* (8.7%), *MET* (5.2%) and *FGFR2* (4.3%) genes, among others. Additionally, a high tumor mutational burden (≥ 13 Mut/Mb) was observed in 6 pts (5.2%). Clinical characteristics are detailed in the table. We detected a higher rate of alterations in *CDH1* (10.42% vs 4.48%; p=0.22) and *PIK3CA* (8.33% vs 2.99%; p=0.20) genes in early (<50y) vs late-onset (≥50y) GEA. Additionally, we identified a significantly higher mutation frequency in diffuse vs intestinal tumors; *ARID1A* (17.6% vs 2.2%; p=0.01) and *CDH1* (11.8% vs 0%; p=0.02) genes. Also, pts with peritoneal involvement showed a greater prevalence of mutations in *CDH1* (12% vs 0%; p=0.04) and *RHOA* (10% vs 0%; p=0.06) compared to those with liver disease. We observed a significantly worse OS in pts with diffuse vs intestinal tumors (15.7m vs 19.1m; p=0.04) and a trend toward shorter survival in pts with *CDKN2A* mutations (11.97m vs 17.83m; p=0.42). Although not significantly, CNA in *ERBB2* were associated with a better prognosis (22.2m vs 16.2m; p=0.09). **Conclusions:** Our results support the utility of NGS platforms in detecting novel molecular targets with prognostic and clinical impact, beyond *ERBB2*. Additionally, we observed poorer prognosis in diffuse subtype tumors, which exhibited a higher frequency of mutations in *ARID1A* and *CDH1* genes. Research Sponsor: None.

Clinicopathological characteristics of the 115-pts with GEA.

		N=115 (%)
Age	54 [18-83]	
Median years [range]		
Sex	Male	69 (60.0)
Tumor site	Gastric	72 (62.6)
	Gastroesophageal junction	37 (32.2)
	Esophagus	6 (5.2)
Histological subtype	Diffuse or pooly cohesive	51 (44.3)
	Intestinal or tubular	46 (40.0)
	Others	18 (15.7)
Stage at diagnosis	Advanced	74 (66.9)
	Localized	41 (33.1)
Metastatic site	Peritoneum	53 (46.1)
	Liver	34 (29.6)
Molecular profile (IHC)	HER2 (+)	20 (17.4)
	MSI	5 (4.3)
	EBV (+)	2 (1.8)
	PDL1 (CPS) ≥1	35 (30.4)

Exploration of novel biomarkers and novel therapies for immune checkpoint inhibitors (ICIs) in genetic variants of esophageal squamous cell carcinoma (ESCC): A study of 1,059 Japanese cases.

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Background: In Japan, cancer genome profiling (CGP) tests will be covered by insurance in 2019, and more than 1,000 ESCC cases have been accumulated in the national database. ICIs are currently the key drugs for esophageal cancer, and MSI and TMB status can work as biomarkers for ICIs in solid tumors. Therefore, to improve the prognosis of esophageal squamous cell carcinoma (ESCC), we aim to search for gene variants that may predict the efficacy of ICIs and explore the possibility of new treatment from genetic data accumulated in the Japanese nationwide C-CAT database. **Methods:** We retrospectively analyzed 1,059 ESCC cases from C-CAT data accumulated from June 2019 to February 2024. We used the registered clinical and genetic information to examine gene variants and carcinogenic pathways that could serve as biomarkers for ICIs. **Results:** Of the 1059, 152 (14.4%) had TMB-H and 3 (0.3%) had MSI-H, with a median TMB of 5.0 Muts/Mb (range: 0-268). *XPO1* and *TP53* were found to be positively and negatively correlated as genes affecting TTF in nivolumab monotherapy (FDR P value < 0.01). TMB has the potential to be a biomarker for ICI, but TMB was significantly higher in the group with variants in the *ARID* family (*ARID2*, *ARID1A* plus *ARID5B*) (median TMB 5.0 vs 8.0 Muts/Mb, $P<0.001$). In addition, 709 patients (66.9%) had any variants in three pathways (TP53, RTK and cell cycle), which are thought to be involved in the carcinogenesis of ESCC. PIK/Akt/mTOR and epigenetic modulation pathways were also found to be abnormal in many cases, and abnormalities in the epigenetic modification pathway were found to contribute significantly to TMB-H (FDR P value<0.01, Logarithmic value=19.2). **Conclusions:** This is the first case-specific report on the relationship between mutations in the *ARID* family and TMB in ESCC and the signalling pathways leading to carcinogenesis. In the present study, the association between the *ARID* family and TMB, and between abnormalities in the epigenetic modification pathway and TMB, which has been previously reported in other cancer types but not in esophageal cancer, suggests that multiple pathological mutations including these may be new biomarkers for ICI in 1059 Japanese patients. Furthermore, multiple pathways are often involved in ESCC carcinogenesis at the same time, and a separate, combined approach that takes these into account may be necessary in the development of new drugs. Research Sponsor: None.

Molecular characteristics of early-onset versus average-onset gastroesophageal adenocarcinoma.

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Background: Incidence of early-onset gastroesophageal adenocarcinoma (eoGEA, ages <50 years) has been increasing in the United States since the 1990s, the etiology of which is still unclear, and limited data exists on molecular drivers. This study evaluates somatic and germline profiles in eoGEA compared to average-onset GEA (aoGEA, ages≥ 50 years). **Methods:** This is a retrospective, cross-sectional study utilizing data from de-identified records of GEA (esophageal, gastroesophageal junction and gastric adenocarcinomas) patients who underwent somatic NGS testing via the Tempus xT assay (595-648 gene DNA panel) from 12/2017 to 07/2024. This assay assesses somatic tumor mutations (including SNVs, Indels, CNVs, and select SVs), MSI, TMB, and incidentally detected germline SNVs and small indels in matched normal tissue. Clinical characteristics and immunotherapy markers were compared between eoGEA and aoGEA by Wilcoxon Rank Sum or Chi-squared test. Somatic and germline mutations were compared between two age groups with false discovery rate adjustments. **Results:** We compared the demographic, clinical, and molecular features of a total of 5863 patients with eoGEA (n=785, median age 43) and aoGEA (n=5078, median age 68). Over 80% of patients were diagnosed with stage IV disease. The eoGEA group had a higher proportion of females, non-white, and Hispanic or Latino patients compared to the aoGEA group (p<0.001). Prevalence of MSI-H/ dMMR, and TMB-H was lower in eoGEA vs aoGEA (p<0.001), while PD-L1 expression was similar. Prevalence of somatic *CDH1*, *CDKN2A*, and *TP53* SNVs, and *FGFR2* and *KRAS* CNAs were higher in eoGEA vs aoGEA (q<0.001). In a subset of 3286 patients with tumor/normal match testing (eoGEA=464 and aoGEA=2822), eoGEA also had a higher prevalence of germline CDH1 mutations (q<0.001). **Conclusions:** eoGEA has a unique somatic and germline mutation profile compared to aoGEA; further study evaluating the molecular landscape including epigenetic changes could shed light on underlying mechanisms responsible for the rising incidence of eoGEA. Research Sponsor: None.

	eoGEA (n=785)	aoGEA (n=5078)	p-value/q-value*
Baseline characteristics			
Median age at diagnosis (IQR)	43 (38, 47)	68 (61,74)	<0.001
Gender: male	509 (65%)	3798 (75%)	<0.001
Race (white)	280 (69%)	2347 (80%)	<0.001
Ethnicity (not hispanic or latino)	206 (60%)	1768 (86%	<0.001
Stage IV	495 (85%)	2883 (81%)	0.073
Immunologic markers			
TMB ≥ 10	25 (3.2%)	519(10%)	<0.001
MSI-H	13 (1.7%)	261 (5.1%)	<0.001
dMMR	7 (2.2%)	120 (5.9%)	0.007
Somatic mutations profiles *			
CDH1	126 (16%)	334 (6.6%)	<0.001
TP53	515 (66%)	3764 (74%)	<0.001
CDKN2A	101 (13%)	1011 (20%)	<0.001
KRAS	92 (12%)	931 (18%)	<0.001
NOTCH1	12 (1.5%)	228 (4.5%)	0.002
Germline mutation profiles *			
	eoGEA (N=464)	aoGEA (N= 2,822)	
Overall prevalence	46 (9.9%)	207 (7.3%)	
CDH1	10 (2.2%)	8 (0.3%)	0.001
TP53	3 (0.6%)	0 (0%)	0.039
BRCA2	2 (0.4%)	38 (1.3%)	0.6
BRIP1	4 (0.9%)	10 (0.4%)	0.6

Impact of Lauren's subtype in locally-advanced gastric cancer (LAGC) prognosis: A call for change.

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Background: Patients with LAGC should undergo perioperative chemotherapy and surgery. Lauren's diffuse subtype and signet-ring cell carcinoma have been linked to worse outcomes and chemoresistance in this setting. This study aimed to assess the impact of Lauren's classification on response to neoadjuvant chemotherapy and prognosis in LAGC. **Methods:** This analysis involved adult patients from a prospectively collected gastric cancer database who had histologically confirmed LAGC and received curative-intent perioperative chemotherapy and surgery. Patients were classified into two groups according to Lauren's classification: intestinal and non-intestinal subtypes (including diffuse, signet-ring cell carcinoma and mixed histologies). Clinicopathological and treatment characteristics (age, gender, primary site, HER-2 status, staging laparoscopy, neoadjuvant chemotherapy, surgery and pathological complete response) of the two groups were compared using chi-square tests. Survival curves were obtained using Kaplan-Meier, and log-rank and Cox proportional models used to compare survival between groups. **Results:** Between 2015 and 2022, 174 patients met the inclusion criteria, with a median follow-up of 62.2 months (95% CI: 57.9-66.9). The intestinal subtype comprised 72 patients (41.4%) and the non-intestinal subtype included 102 (58.6%). The median age of diagnosis was significantly higher for the intestinal subtype (63 years) compared to the non-intestinal subtype (56 years; $p<0.001$). Female patients represented 26.4% of the intestinal subtype and 46.1% of the non-intestinal subtype ($p=0.013$). The primary gastric site was more prevalent in the non-intestinal subtype (78.4%) compared to the intestinal subtype (51.4%; $p<0.001$). HER-2 positivity was observed in 12.5% of intestinal and 4.5% of non-intestinal subtypes ($p=0.14$). Staging laparoscopy was conducted in 67 out of 72 (93%) patients with the intestinal subtype and in 96 out of 102 (94.1%) patients with the non-intestinal subtype, $p=1$. FLOT and FOLFOX/CAPOX protocols were administered in 50% and 40.2% of intestinal subtype cases, and in 53.9% and 36.2% of non-intestinal subtype cases, respectively ($p=0.14$). Pathological complete response rates were 8.5% for intestinal and 11.5% for non-intestinal subtypes ($p=0.46$). Three-year recurrence-free survival rates were 67.1% for intestinal and 64.6% for non-intestinal subtypes (HR=0.99, 95% CI: 0.61-1.61, $p=0.96$), while five-year overall survival rates were 73.3% and 67.2%, respectively (HR=1.12, 95% CI: 0.63-1.97, $p=0.70$). **Conclusions:** Our findings demonstrate that the non-intestinal subtype is more common in younger, female patients with primary gastric sites. However, Lauren's subtype did not significantly correlate with pathological staging post-neoadjuvant chemotherapy or prognosis in LAGC. Lauren's subtype should not influence treatment decisions in LAGC. Research Sponsor: None.

Surrogacy between pathological complete response and overall survival: An individual patient data analysis of eight RCTs on neoadjuvant treatment plus surgery for esophageal cancer.

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Background: Overall survival (OS) is the gold standard endpoint of treatment efficacy but requires an extended follow-up period. This study aimed to determine the validity of pathological complete response (pCR) as a surrogate endpoint for OS. **Methods:** The individual-level surrogacy between categorical outcome and OS was assessed using Kendall correlation coefficient, τ. Because no method has been reported to estimate τ between categorical variables and OS, we proposed the new method using an inverse probability of censoring weighting (IPCW) estimator adjusted for tied data that occur when applied to categorical outcomes. To evaluate the validity of this new method, we conducted simulations to compare its estimation accuracy with existing methods. A systematic review for all phase III RCTs comparing therapies in perioperative settings for resectable advanced esophageal and gastroesophageal junction cancer was performed and individual patient data (IPD) were requested from all included trials. Finally, surrogacy between pCR and OS was assessed using our proposed method. **Results:** Across all scenarios with varying τ values, sample sizes, and censoring proportions, the proposed method showed the lowest bias compared to the existing statistical methods. In total, 23 eligible trials were identified and IPD were available from eight RCTs (JCOG1109, JCOG9907, JCOG9204, FFCD9901, FFCD9102, SAKK75/08, CROSS, and KOK), including 1688 patients who received neoadjuvant therapy. For trial-level analysis, pCR showed a strong correlation with a determination of correlation of 0.65 when limited to neoadjuvant chemotherapy (NAC) trials. For individual-level analysis, in patients who received NAC, the hazard ratio (HR) for OS of pCR was 0.25 (95% CI: 0.13 - 0.47), and τ between OS and pCR was 0.225. While in patients who received neoadjuvant chemoradiotherapy (NACRT), the HR was 0.54 (95% CI: 0.45 - 0.66), and τ was 0.189. Additionally, τ between OS and pathological grade (pGrade), OS and tumor regression grade (TRG) was 0.196 and 0.143, respectively. As a reference, hypothetical data showed that in order to achieve a τ of 0.8 between pCR and OS, it is necessary to be able to clearly identify the prognosis to the extent that an HR of 0.09 (95% CI: 0.08 - 0.11) can be obtained. **Conclusions:** While pCR is sensitive enough to stratify prognosis and exhibit strong trial-level surrogacy, no individual-level surrogacy was demonstrated, making them insufficient to replace OS. It should be noted that there are substantial differences in prognostic stratification and potential use of pCR as primary endpoints in clinical trials. Research Sponsor: None.

499 **Poster Session**

Development and validation of a prognostic model for advanced gastric and oesophageal carcinoma (AGOC) using individual patient data from two clinical trials.

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Background: Estimating survival in advanced gastric and oesophageal carcinoma (AGOC) remains a significant challenge. We developed a prognostic model incorporating readily accessible patient and clinical data and specific patient-reported outcomes (PROs), utilizing individual participant data from two randomized trials. **Methods:** We used data from 2 trials comparing regorafenib to placebo; AGITG INTEGRATE IIa (n=251) for model development and AGITG INTEGRATE (n=152) for validation. Both trials enrolled metastatic or AGOC following failure of 1 or more prior lines of systemic therapy. Candidate variables were chosen from systematic literature review and expert consultation. Significant prognostic factors for inclusion in the multivariable model were identified using univariable Cox proportional hazards models with p-value threshold of 0.1. Multivariable models were developed via Lasso regression. An initial model was created using clinical variables alone, followed by the integration of PROs. C statistics were used to discriminate the model's efficiency. **Results:** Univariable analysis identified 9 clinical variables and 4 PRO domains as significant, including body mass index (BMI)(p=0.08), ECOG PS (0.02), extent of cancer ($p<0.001$), liver involvement (p=0.04), treatment with regorafenib (p=0.005), neutrophil-lymphocyte ratio (NLR) ($p<0.001$), LDH ($p<0.001$), albumin ($p<0.001$), and CA 19-9 (p=0.007). Significant baseline PROs included appetite loss ($p<0.001$), constipation ($p<0.001$), fatigue ($p<0.001$), and pain ($p<0.001$). Gastrectomy violated the Cox model proportional hazards assumption necessitating that the analysis be stratification by gastrectomy. The primary model (M1) incorporated region (Asia vs non-Asia, p=0.25, was included because it was significant in previous analysis (INTEGRATE), ECOG PS status, cancer extent, treatment with regorafenib, NLR, BMI, LDH, CA 19-9, and albumin. The final model (M2), which included PROs, demonstrated superior discriminative ability with the highest c-statistic values in both the gastrectomy and non-gastrectomy strata. The c-statistic increased from 0.695 for M1 to 0.723 for M2, indicating the added prognostic value of including PROs. **Conclusions:** The prognostic model that integrated both clinical data and PROs, proved robust for predicting survival in AGOC. The addition of PROs significantly enhanced prognostic accuracy, highlighting their importance in survival models. Further validation and refinement of the model is ongoing. Research Sponsor: NHMRC CTC University of Sydney.

Assessment of clinical value and barriers to use of new investigational targeted agents in the community setting: A study of claudin 18.2 and zolbetuximab.

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Background: Rapid integration of novel agents in clinical practice can be lifesaving. However, community-based practices may experience extended adoption times, due to practical barriers. REFLECT was developed to assess new drug implementation in the community setting. **Methods:** REFLECT consists of a 20-minute prerecorded presentation, followed by a data-collection event for community oncologists. It was developed to evaluate and assess challenges of implementing novel targeted agents in the community practice setting. In this pilot study, awareness of zolbetuximab (zolbe), a potential first-in-class claudin 18.2 (CLDN18.2) monoclonal antibody and companion diagnostic, was assessed. Between April and August 2024, 53 data-collection events were held representing 26 states across the US. Data from 492 oncologists were recorded and analyzed. **Results:** Nearly half the community oncologists (47%) were not familiar with CLDN18.2 as a therapeutic target in gastric/gastroesophageal junction (G/GEJ) cancer. Awareness of zolbe data was low among community oncologists; 79% indicated they had not seen the results from the phase 3 studies. However, 96% of oncologists reported the data will at least moderately impact their treatment of CLDN18.2+ patients with metastatic (m) G/GEJ cancer. Only 40% of oncologists reported having access to CLDN18.2 testing. Uncertainty regarding availability of CLDN18.2 testing was high among oncologists in the Central US (52%), representing a significant awareness gap for CLDN18.2 testing. Opinions were divided regarding the treatment approach for CLDN18.2+, immunosensitive patients (Table). In a HER2−, CLDN18.2+ patient with mG/GEJ cancer and a combined positive score (CPS) of 10; 58.1% would recommend first-line chemotherapy (chemo) plus immunotherapy (IO) and 40.4% would recommend chemo plus zolbe. **Conclusions:** First-in-class targeted agents often encounter additional challenges and knowledge gaps regarding adoption in the community practice setting, particularly associated with molecular testing. Timely education on patient identification and correct utilization may accelerate adoption of new regimens, and potentially save lives. Data-sharing and -collection platforms including REFLECT can be instrumental in the identification of challenges associated with widespread and rapid implementation of novel agents in the community oncology setting. Research Sponsor: None.

What treatment approach would you recommend in a patient with the following characteristics, if all treatments were available? mGEJ adenocarcinoma, HER2− disease, CLDN18.2+ staining in >75% of tumor cells, CPS 10.

Treatment		Total: N = 492 % (n)
Chemo	**mFOLFOX6**	**1.0 (5)**
	CAPOX	**0 (0)**
Chemo + IO	**mFOLFOX6 + IO**	**52.8 (260)**
	CAPOX + IO	**5.3 (26)**
Chemo + zolbe	**mFOLFOX6 + *zolbe***	***33.9 (167)***
	CAPOX + *zolbe*	***6.5 (32)***
	Other	**0.4 (2)**

501 **Poster Session**

Influence of data review on community physicians' treatment preferences in frontline esophageal cancer (EC).

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Background: EC remains a global health challenge with a rising incidence and poor prognosis despite advancements in treatment modalities. Standard therapies include surgery, chemotherapy (CT), radiation therapy, and recently, targeted and immunotherapy. The PD-L1 inhibitors pembrolizumab (pembro) and nivolumab (nivo) received FDA approval for frontline treatment in EC in March and May 2021, respectively, based in part on the KEYNOTE-590 (KN590) and CheckMate 648 trials. In this study, we aimed to uncover rationale in physician preference for PD-L1 inhibitors in EC. **Methods:** US-based oncologists convened at two in-person meetings in April 2024 to discuss clinical and real-world data presented at the 2024 ASCO GI Cancers Symposium. Among data discussed included the KN590 5-year update. Survey questions were fielded at the meeting to capture participants' impressions. Demographics were captured via an online survey ahead of the meeting. Responses from participants who manage EC and responded to all survey questions were aggregated. **Results:** 86 participants qualified; and, collectively, are 76% community physicians with 18 years average in clinical practice, see 18 patients median on clinic days, and spend 83% of their time in direct patient care on average. ECOG PS status (50%), tumor histology (52%), and location/stage (48%) were reported as critical factors in treatment decision-making; only 14% said PD-L1 status was a major factor. Participants were queried on a hypothetical patient case of esophageal squamous cell and adenocarcinoma before and after reviewing the KN590 data. Prior to reviewing KN590, physician preferences in the squamous cell 1L setting were split between nivo+CT and pembro+CT; following data review, pembro+CT was favored nearly 3:1. In the adenocarcinoma 1L setting, preference shifted from 60/40 nivo+CT to 60/40 pembro+CT, see Table. The cohort were most impressed with the overall survival (87%) and objective response rates (40%), and 55% would consider pembro+CT for both squamous cell and adenocarcinoma disease while 37% would consider it for PD-L1 CPS>10 squamous cell disease only. **Conclusions:** Our study showed PD-L1 status was not a major factor in clinical decisions. The shift in treatment preference toward pembro+CT for both patient cases following KN590 data review suggests an impact for data review to influence physician behavior. This highlights the importance of educational initiatives to disseminate clinical updates to providers. It is still to be determined the optimal platform and length for these initiatives, and to quantify their effectiveness. Research Sponsor: None.

65-year-old male patient with advanced EC; PD-L1 CPS ≥10; ECOG PS of 1	Squamous cell disease		Adenocarcinoma	
	Before	After	Before	After
Fluoropyrimidine (FP) + platinum (P)-based CT	1%	1%	1%	0%
FP +P-based CT + nivo	55%	27%	60%	36%
FP + P-based CT + pembro	43%	71%	38%	63%
Nivo + ipilimumab	1%	1%	0%	1%
None of the above	0%	0%	0%	0%

Impact of inpatient palliative care consultation on end-of-life care among patients with esophageal cancer.

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Background: Esophageal cancer is an aggressive malignancy with a high global prevalence. Patients often suffer from various debilitating symptoms like dysphagia and cachexia, particularly with more advanced stage disease. Most patients continue to receive aggressive interventions until terminal stages of disease without timely initiation of formal palliative care (PC). We sought to investigate the impact of formal inpatient palliative care consultation on end-of-life (EOL) care among patients with esophageal cancer. **Methods:** The National Inpatient Sample (NIS) was queried to identify all hospitalizations with esophageal cancer utilizing ICD-10 codes C15.x from 2016 to 2020. Hospitalizations with documented inpatient mortality events were extracted and stratified based on reception of PC consultation (ICD-10 code Z51.5). Demographic and clinical data were analyzed using chi-squared tests and independent sample t-tests. A threshold of p-value less than 0.05 was set to determine statistical significance. **Results:** A total of 17,745 patients with esophageal cancer were included, of which 10,370 (58.4%) received PC consultation. Patients that received PC consultation at EOL were more likely to have higher comorbidities per CCI. No significant difference between age and sex was observed between the two cohorts. Patients who received PC consultations at EOL had higher rates of Do Not Resuscitate (DNR) code status (78.1% vs 43.2%, $p<0.001$) and lower rates of aggressive interventions such as chemotherapy (0.9% vs 1.6%, $p<.001$), blood transfusions (12.3% vs 18%, $p<.001$), and mechanical ventilation (28.5% vs 41%, $p<0.001$). No significant difference was observed in the rates of vasopressor use between the two groups ($p=0.83$). Hospitalizations with PC consultations were associated with significantly lower total hospitalization cost ($97,879 vs $146,128, $p<.001$). **Conclusions:** Inpatient PC consultations among patients with esophageal cancer were associated with lower rates of aggressive interventions and higher rates of DNR code status in addition to significantly lower total hospitalization costs. These findings underscore the importance of inpatient PC consultation at the EOL in helping minimize aggressive interpretation and facilitate a greater focus on comfort and quality of life.
Research Sponsor: None.

	No Palliative N=7375	Palliative N=10370	p-value
Age (in years)	67.58 ± 10.9	67.58 ± 10.9	.98
CCI	9.1 ± 3.5	9.4 ± 3.3	**<.001**
Mean length of stay (days)	8.9 ± 14.9	7.5 ± 11.3	**<.001**
Total Charges	146128 ± 321830	97879 ± 195868	**<.001**
DNR	3170 (43.2%)	8100 (78.1%)	**<.001**
Blood transfusion	1320 (18.0%)	1280 (12.3%)	**<.001**
Mechanical ventilation	3015 (41.0%)	2960 (28.5%)	**<.001**
Vasopressor	585 (8.0%)	835 (8.1%)	0.83
Chemotherapy	115 (1.6%)	95 (0.9%)	**<.001**

503 Poster Session

The impact of protein-energy malnutrition on clinical outcomes in hospitalized gastric cancer patients: A population-based analysis.

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Background: Protein-energy malnutrition (PEM) is considered an adverse prognostic factor in various clinical conditions. Patients with gastric cancer are at an increased risk of malnutrition because of factors such as loss of appetite due to the tumor, difficulty swallowing, malabsorption, and metabolic changes caused by the cancer. However, the implications of protein-energy malnutrition (PEM) in gastric cancer patients are not well studied. Through this study, we wanted to examine how PEM affects clinical outcomes in hospitalized patients with gastric cancer. **Methods:** We used the National Inpatient Sample (NIS) from 2018 to 2020 to detect patients with a primary diagnosis of gastric cancer and a secondary diagnosis of protein energy malnutrition (PEM). We used the chi-square and student T-test to analyze categorical and continuous variables. Confounders were adjusted using multivariable regression analysis, and the adjusted odds ratio (aOR) was calculated. **Results:** We detected a total of 1673915 patients with the primary diagnosis of gastric cancer using ICD-10 codes. Of them, 401,739 (24%) of the patients had a secondary diagnosis of PEM. The mean age of the patients was 66 years (p-value <0.01). Of these, 56% of the patients were male, while the remaining 44% were female (p-value <0.001). Based on race, 68% of the patients belonged to the White Race, 12% were African Americans, and 11% were Hispanics (p-value <0.001). It was observed that the presence of PEM was associated with higher mortality (aOR 1.6, p-value <0.001). There was also higher hospitalization cost ($109522 vs. 75725, p-value <0.001) and mean length of stay (LOS) (9.1 days vs. 5.6 days, p-value <0.001). PEM in gastric cancer patients was also associated with higher odds of sepsis (aOR 1.6, p-value <0.001), gastritis (aOR aOR 1.2, p-value <0.001), gastrointestinal (GI) bleeding (aOR 1.1, p-value <0.001) and acute kidney injury (AKI) (aOR 1.7, p-value <0.001). **Conclusions:** Our study demonstrates that protein-energy malnutrition (PEM) is associated with increased mortality, higher hospitalization costs, and longer stays in hospitalized gastric cancer patients. These findings emphasize the need for regular nutritional assessment and management to improve clinical outcomes in this population. Research Sponsor: None.

Comparison of outcomes in hospitalized gastric cancer patients with and without protein-energy malnutrition (PEM).

Hospital Outcomes	Gastric Cancer Patients with PEM	Gastric Cancer Patients without PEM	Adjusted Odds ratio (aOR)	95% Confidence Interval	P-value
Mortality	9.1% (n=36999)	5.3% (n=68100)	1.6	1.5-1.7	<0.001
Length of Stay	9.1	5.6	3.0	2.7-3.3	<0.001
Sepsis	10% (n=41915)	6.4% (n= 82035)	1.6	1.5-1.6	<0.001
Gastritis	3.9% (n=16174)	3.3% (n=41879)	1.2	1.1-1.31	<0.001
GI Bleeding	7.1% (n= 28949)	6% (n= 28949)	1.1	1.0-1.2	<0.001
AKI	28% (n=117439)	19% (n=1268134)	1.7	1.7-1.8	<0.001

TPS504 **Trials in Progress Poster Session**

A randomized phase II clinical trial evaluating the safety and efficacy of initial dose adjustment of zolbetuximab plus chemotherapy in patients with HER2-negative and CLDN18.2-positive unresectable advanced or recurrent gastric, gastroesophageal junction cancer or esophageal adenocarcinoma (GENTLE-Z).

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Background: Zolbetuximab, a monoclonal antibody targeting CLDN18.2, was recently approved in Japan for patients with HER2-negative/CLDN18.2-positive metastatic gastric or gastro-esophageal junction cancer (mGC/GEJC). On-target GI toxicities such as nausea and vomiting are commonly observed during first infusion, occasionally leading to early treatment discontinuation. Effective management of these toxicities is essential for the successful use of zolbetuximab in clinical practice. This study aims to assess whether omitting the loading dose in the first cycle can reduce zolbetuximab-related GI toxicities without compromise efficacy. **Methods:** GENTLE-Z is a prospective, open-label, multi-intuitional, randomized phase II trial designed to compare the safety and efficacy of fixed-dose of zolbetuximab without loading dose vs. the standard zolbetuximab dose in combination with first-line mFOLFOX6 or CAPOX for patients with HER2-negative/CLDN18.2-positive mGC/GEJC. Patients are randomly assigned (1:1) to receive either a fixed dose of zolbetuximab (400 mg/m^2 q2w or 600 mg/m^2 q3w) or standard zolbetuximab regimen (800 mg/m^2 loading dose followed by 400 mg/m^2 q2w or 600 mg/m^2 q3w) with chemotherapy. The primary endpoint is the vomit-free rate during cycle 1, and secondary endpoints include efficacy outcomes such as objective response rate, progression-free survival and overall survival. The sample size is set based on the hypothesis that omitting loading dose will improve the vomit-free rate from 50% to 75%. The planned sample size is 86 patients, with a one-sided alpha error of 5% and power of 75%, over an estimated enrollment period of 1.5 years. Enrollment has been ongoing since September 2024 across 55 sites in Japan. This trial is registered in Japan Registry of Clinical Trials (jRCTs031240347). Clinical trial information: jRCTs031240347. Research Sponsor: None.

TPS505 **Trials in Progress Poster Session**

Phase 2 trial of defactinib in combination with avutometinib in patients with advanced diffuse-type gastric cancer.

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Background: Diffuse-type gastric cancer (DGC) accounts for approximately one third of gastric cancer diagnoses. DGC is characterized by poor differentiation, discohesive growth, frequent peritoneal spread, chemoresistance, and inferior survival compared to intestinal-type gastric cancer. Moreover, DGC is commonly genomically stable and often lacks clinically actionable targets such as *ERBB2* amplification. Alterations enriched in DGC including loss-of-function *CDH1* mutations and gain-of-function *RHOA* mutations can facilitate increased focal adhesion kinase (FAK) signaling that drives tumor proliferation, invasiveness, and metastasis. Recent preclinical studies demonstrated that the FAK inhibitor defactinib can inhibitor DGC growth *in vitro* and *in vivo*. FAK inhibitor monotherapy leads to compensatory mitogen-activated protein kinase (MAPK) pathway upregulation, which can be overcome with combined FAK and MEK inhibition. In preclinical DGC xenograft models, defactinib plus the RAF/MEK clamp avutometinib led to deep tumor regressions, suggesting synergistic activity of the combination. The goal of this study is to investigate the efficacy and tolerability of avutometinib plus defactinib in patients with advanced DGC refractory to standard therapy. **Methods:** This investigator-initiated, multicenter phase 2 trial will enroll 27 patients with metastatic/unresectable gastric or gastroesophageal junction carcinoma classified as diffuse, poorly cohesive, signet ring cell, or mixed type with progression on at least one line of therapy including platinum/fluorouracil chemotherapy. Patients with known pathogenic *CDH1* and/or *RHOA* alterations regardless of histology are also eligible. Patients must have ECOG PS 0/1 and can have either measurable or evaluable non-measurable disease per RECIST 1.1 criteria. Avutometinib is given 3.2 mg orally twice weekly and defactinib is given 200 mg orally twice a day, both for 3 weeks on/1 week off for each 28-day cycle. The primary endpoint is 6-month PFS rate, with secondary endpoints including ORR, OS, DOR, and tolerability. Exploratory studies will interrogate FAK/MAPK pathway activation, ctDNA dynamics, and the impact of the combination of avutometinib/defactinib on the tumor microenvironment from pre- and on-treatment biopsies. The trial was opened for enrollment in August 2024. Clinical trial information: NCT06487221. Research Sponsor: Gateway for Cancer Research.

TPS506 **Trials in Progress Poster Session**

Neoadjuvant cadonilimab plus anlotinib followed by surgery for locally advanced esophageal squamous cell carcinoma: A single arm, phase 2 trial.

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Background: Esophageal cancer (EC) is the seventh most common cancer in China, where most of patients have locally advanced esophageal squamous cell carcinoma (LA-ESCC) at the time of diagnosis. Neoadjuvant chemotherapy (nCT) or chemoradiotherapy (nCRT) followed by surgery is standard treatment strategy for LA-ESCC in China. Immune checkpoint inhibitors (ICIs) have revolutionized the treatment of ESCC. Programmed death 1 (PD-1) inhibitors are the most commonly used ICIs in treatment of ESCC. Lately, the phase 3 ESCORT-NEO/NCCES01 trial demonstrates that neoadjuvant PD-1 inhibitor plus chemotherapy obtains superior pCR rates compared to chemotherapy alone for LA-ESCC. Except for PD-1 inhibitors, cytotoxic T lymphocyte-associated antigen-4 (CTLA-4) inhibitor is another effective ICI for ESCC, especially combined with PD-1 inhibitor according to the CHECKMATE-648 study. Although neoadjuvant immunotherapy plus chemotherapy has brought earth-shaking changes to the treatment of LA-ESCC, the chemo-based treatment modality still results in a high incidence of adverse events, reduces the quality of life and compliance of patients. Cadonilimab is a dual immune antibody drug comprised of PD-1 antibody and CTLA-4 antibody. Anlotinib is an oral, small molecule anti-angiogenic targeted agent that has been approved for use in advanced or metastatic ESCC in China. Therefore, we launched this trial to assess the efficacy and safety of cadonilimab plus anlotinib, a novel "chemo-free" strategy, followed by surgery for LA-ESCC. **Methods:** Study design: This is a single-arm, phase 2 study (NCT06426797). The primary endpoint of this study is the pCR rate. The secondary endpoints include safety, major pathological response (MPR) rate, objective response rate (ORR), R0 resection rate, and event-free survival (EFS). Study procedures: Cadonilimab will be intravenously administered at day 1 with a dose of 10 mg/Kg every three weeks for three cycles. Anlotinib will be orally administered daily at day 1 to day 14 with a dose of 12mg every three weeks for three cycles. In the end, the subjects will receive subtotal esophagectomy 4 to 6 weeks later after the last dose of cadonilimab. Major inclusion criteria: 1. Age $>$ 18 years. 2. Histopathologically confirmed ESCC: c-stage II–IVA (8th ed. AJCC). 3. Enough tissue for PD-L1 IHC test. 4. Eligible for anlotinib treatment. 5. ECOG performance status $\leq$ 2 points. Statistics: We hypothesized that the expected pCR rate would be 50%. 24 subjects are needed at an α-level of 0.05 (one-sided) and power of 80%. So, 25 subjects in total should be enrolled. The descriptive statistical method will be applied to the presentation of baseline characteristics, toxicities and measures of effectiveness. The Kaplan–Meier method will be used to calculate survival rates. Current enrollment: The study started on August 1st, 2024, and 3 of planned 25 patients have been enrolled. Clinical trial information: NCT06426797. Research Sponsor: Akeso Pharmaceuticals, Inc.

TPS507 **Trials in Progress Poster Session**

CLARITY-Gastric 01: A randomized phase 3 study of AZD0901, a Claudin18.2 (CLDN18.2)-targeted antibody-drug conjugate, in second- or later-line (2L+) advanced gastric or gastroesophageal junction cancer (GC/GEJC).

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Background: GC/GEJC is the fifth most commonly diagnosed cancer and the fourth leading cause of cancer-related death worldwide (Sung H, et al. *CA Cancer J Clin* 2021). Prognosis is poor in GC/GEJC; median overall survival (OS) is less than 10 months in the 2L setting and less than 6 months in the 3L+ setting (Wilke H, et al. *Lancet Oncol* 2014; Shitara K, et al. *Lancet Oncol* 2018). New treatment options are needed. CLDN18.2 has emerged as a new therapeutic target in GC/GEJC. AZD0901, a potential first-in-class CLDN18.2-targeted antibody conjugated to monomethyl auristatin E via a protease cleavable linker, has demonstrated clinical activity in a Phase 1 study in advanced GC/GEJC (Xu R-H, et al, *J Clin Oncol* 2023). CLARITY-Gastric 01 (NCT06346392) is a randomized, open-label, sponsor-blinded, global Phase 3 study that will assess the efficacy and safety of AZD0901 versus investigator's (INV) choice of therapy after ≥1 prior therapy for advanced GC/GEJC expressing CLDN18.2. **Methods:** Participants (pts) will be randomized 1:1:1 to AZD0901 dose level 1 intravenous (IV) every three weeks (Q3W; Arm 1), AZD0901 dose level 2 IV Q3W (Arm 2), or INV choice of therapy (2L: ramucirumab [ram] + paclitaxel [PTX], PTX, or docetaxel [for pts with contraindication to ram only]; 3L+: irinotecan, TAS-102 [excluding China], or apatinib [China only]; Arm 3). One AZD0901 arm (Arm 1 or Arm 2) will stop enrollment after a pre-planned dose selection decision. Randomization will continue in a 1:1 ratio to the two remaining arms (selected AZD0901 dose arm and Arm 3). Eligible pts must have histologically confirmed, unresectable, locally advanced or metastatic GC/GEJC (non-HER2+, CLDN18.2 expression). Pts must have disease progression on/after ≥1 prior regimen (including platinum-fluoropyrimidine) and have an ECOG performance status of 0 or 1. Pts previously treated with CLDN18.2-targeting therapy (other than naked monoclonal antibody) are excluded. Recruitment has begun and pts are planned to be enrolled across 16 countries in Asia, Europe, and North America. Dual primary endpoints are progression-free survival (PFS; intent-to-treat [ITT] pts) and OS (3L+ pts). The secondary endpoints include OS (ITT pts), PFS (3L+ pts), objective response rate and duration of response (ITT and 3L+ pts), and safety/tolerability. Efficacy data will be summarized and analyzed in the ITT population. All efficacy endpoints will assess the selected AZD0901 dose arm versus Arm 3. Safety and tolerability will be assessed in all pts who receive ≥1 dose of study treatment. Previously presented at European Society for Medical Oncology - Gastrointestinal Cancers Congress 2024, Final Publication Number: 495TiP, Kohei Shitara et al. Reused with permission. Clinical trial information: NCT06346392. Research Sponsor: AstraZeneca.

TPS508 **Trials in Progress Poster Session**

Valemetostat and trastuzumab deruxtecan (T-DXd) in previously treated advanced or metastatic human epidermal growth factor receptor 2 (HER2)-positive gastric cancer or gastro-esophageal junction (GEJ) adenocarcinoma.

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Background: Valemetostat tosylate (valemetostat), a novel and selective dual inhibitor of enhancer of zeste homolog (EZH)2 and EZH1, has demonstrated clinical efficacy and tolerability in multiple hematologic cancers (200 mg orally [PO] once daily [QD]). EZH2 inhibition by valemetostat is expected to upregulate DNA/RNA helicase Schlafen 11 expression and sensitize tumor cells to DNA damaging agents such as topoisomerase I inhibitor antibody–drug conjugates (ADCs). Topoisomerase inhibitors cause DNA strand breaks, which may synergize with valemetostat. HER2 gene amplification or protein expression has been associated with gastric cancer (GC) and GEJ adenocarcinoma. Patients with HER2+ locally advanced or metastatic gastric adenocarcinoma who progress after first line treatment have poor prognosis and limited therapeutic options. T-DXd is approved globally for HER2+ locally advanced or metastatic GC previously treated with a trastuzumab-based regimen. This study evaluates the clinical activity and safety of valemetostat in combination with T-DXd in patients with GC or GEJ adenocarcinoma as part of a Master Protocol trial evaluating valemetostat in combination with ADCs in solid tumors. **Methods:** This global, phase 1b, multicenter, open-label Master Protocol study (NCT06244485) will enroll ~ 70 patients per sub-protocol in multiple regions, including the US and Japan. The gastric cohort enrolls patients with previously treated advanced or metastatic HER2+ GC or GEJ adenocarcinoma. The dose-escalation (Part 1) will determine the recommended dose for expansion (RDE), combining valemetostat 50–200 mg PO QD with T-DXd 5.4 or 6.4 mg/kg intravenously every 3 weeks. The dose-expansion (Part 2) will assess the efficacy of valemetostat + T-DXd at the RDE. Primary endpoints include safety and tolerability in Part 1 and the objective response rate in Part 2. Secondary endpoints include response duration, progression-free and overall survival, pharmacokinetics, and HER2 expression by immunohistochemistry and in situ hybridization with clinical response. An interim futility analysis will be performed when 20 patients are enrolled with ≥ 6 months of follow-up. Clinical trial information: NCT06244485. Research Sponsor: Daiichi Sankyo Inc.

TPS509 **Trials in Progress Poster Session**

Beamion BCGC-1: A phase Ib/II trial of the HER2-selective tyrosine kinase inhibitor (TKI) zongertinib (BI 1810631) + trastuzumab deruxtecan (T-DXd) or trastuzumab emtansine (T-DM1) for patients with metastatic breast cancer (mBC) and metastatic gastric, gastroesophageal junction, or esophageal adenocarcinoma (mGEAC).

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Background: Approximately 15% of mGEAC tumors overexpress HER2. Currently, the HER2-targeted antibody-drug conjugate (ADC), T-DXd, is approved in mGEAC as a second-line or later-line treatment option following the failure of prior trastuzumab-containing therapy. Zongertinib (BI 1810631) is a novel, orally administered HER2-selective TKI that spares EGFR. In a Phase I trial (NCT04886804), zongertinib showed encouraging tolerability and efficacy in patients with HER2 aberration-positive advanced solid tumors, including mGEAC. Beamion BCGC-1 (NCT06324357) is a Phase Ib/II dose escalation/optimization trial investigating zongertinib plus T-DXd or T-DM1 in patients with HER2+ mGEAC or mBC. Here we focus on the mGEAC cohorts, in which patients will receive zongertinib in combination with T-DXd. **Methods:** Across the whole trial, approximately 240 patients will be enrolled from approximately 48 sites in 6 countries. In the mGEAC cohorts, approximately 80 patients with histologically/cytologically confirmed locally advanced/metastatic, unresectable, HER2-positive, pre-treated mGEAC will be enrolled. All patients must have investigator-assessed progression and documented HER2-positive disease (overexpressed and/or amplified; confirmed by immunohistochemistry and *in situ* hybridization). In Phase Ib, approximately 20 patients will receive escalating doses of zongertinib plus a fixed approved dose of T-DXd (6.4 mg/kg, once every 21 days). Dose escalation will be guided by a Bayesian Logistic Regression Model, with overdose control. In Phase Ib, the primary endpoint is the occurrence of dose-limiting toxicities (DLTs) during the maximum tolerated dose evaluation period (first 21-day cycle); secondary endpoints include objective response (RECIST v1.1), DLTs during the on-treatment period and pharmacokinetics. In Phase II, up to 60 patients with mGEAC will be randomized 1:1 to receive one of two zongertinib dose levels (dose levels informed by Phase Ib), plus a fixed approved dose of T-DXd. In Phase II, the primary endpoint is objective response; secondary endpoints include progression-free survival, disease control, treatment-emergent adverse events leading to dose reduction, pharmacokinetics and patient-reported outcomes. Patients will remain on treatment until disease progression, undue toxicity or any other protocol-defined stopping criterion. Enrollment is ongoing. Clinical trial information: NCT06324357. Research Sponsor: Boehringer Ingelheim.

TPS510 **Trials in Progress Poster Session**

A safety trial of antibiotic fecal microbiota transplantation for esophageal and gastric cancer patients treated with immune checkpoint inhibitors (Biorich2 trial).

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Background: The development of immune checkpoint inhibitors (ICIs) has improved the prognosis of unresectable advanced or recurrent esophageal and gastric cancer. However, the prognosis remains poor. The human intestinal microbiota plays various roles in digestion, immunomodulation, and metabolism and is known to influence tumor suppression. Previous studies reported that intestinal microbiota can impact immune responses and the effectiveness of ICIs. Therefore, this study investigates the safety and efficacy of antibiotic fecal microbiota transplantation (A-FMT) in patients with unresectable advanced or recurrent esophageal or gastric cancer. **Methods:** Eligible patients are diagnosed with unresectable advanced or recurrent esophageal squamous cell carcinoma or gastric adenocarcinoma. The primary endpoint is to examine the dose-limiting toxicity (DLT) of A-FMT. Secondary endpoints include response rate, disease control rate, progression-free survival, overall survival, and biomarkers of efficacy and adverse events. This trial consists of two parts: safety confirmation, which assesses DLT incidence, and an expansion part, which examines efficacy and safety. The safety confirmation part will enroll 3-6 patients in cohort 1 (combination therapy with ICIs and chemotherapy) and cohort 2 (combination of anti-PD-1 and anti-CTLA4 antibodies). The expansion part will enroll untreated patients: 10 for esophageal cancer (ICIs + chemotherapy), 10 for anti-PD-1 + anti-CTLA4 antibodies, and 11 for gastric cancer (ICIs + chemotherapy). After enrollment, patients receive antibacterial agents for 1 week before FMT. Patients then undergo total colonoscopy after bowel cleaning with polyethylene glycol plus ascorbic acid. A 200 mL fecal suspension from a healthy donor is transferred to the patient's cecum, and ICIs with/without chemotherapy are initiated thereafter. Biomarker analysis explores diagnostic or surrogate biomarkers, aiming to identify molecular targets, cell populations, microbial species, and functions involved in efficacy by analyzing the effects of A-FMT and ICIs on the tumor microenvironment and intestinal microbiota using methods such as proteome analysis. This trial is registered in the Japan Registry of Clinical Trials as jRCTs031240170 and was initiated in August 2024, with enrollment ongoing. Clinical trial information: jRCTs031240170. Research Sponsor: Metagen Therapeutics Inc.

TPS511 **Trials in Progress Poster Session**

KEYMAKER-U06: Phase 1/2 umbrella platform study of TROP2 antibody-drug conjugate sacituzumab tirumotecan plus pembrolizumab plus chemotherapy (substudy 06C) and sacituzumab tirumotecan plus paclitaxel (substudy 06D) in gastroesophageal adenocarcinoma.

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Background: Treatment options for patients with gastroesophageal adenocarcinoma (GEA) are limited and prognosis is poor. KEYMAKER-U06 is a multicenter, open-label, phase 1/2, umbrella platform study designed to evaluate investigational agents with or without pembrolizumab and/or chemotherapy for GEA. Substudy 06C (NCT06469944) will be conducted to evaluate sacituzumab tirumotecan(TROP2 antibody-drug conjugate; formerly MK-2870/SKB264) plus pembrolizumab plus investigator's choice of chemotherapy as first-line therapy for patients with advanced or metastatic GEA. Substudy 06D (NCT06445972) will be conducted to evaluate sacituzumab tirumotecan plus paclitaxel as second-line therapy for patients with advanced or metastatic GEA. **Methods:** In both substudies, patients aged ≥18 years with confirmed advanced or metastatic HER2-negative GEA (gastric, gastroesophageal junction, or esophageal adenocarcinoma), measurable disease per RECIST v1.1 by investigator review and verified by blinded independent central review (BICR), and an Eastern Cooperative Oncology Group performance status score of 0 or 1 are eligible; patients in substudy 06C should be treatment naive and patients in substudy 06D should have disease progression on or after 1 prior therapy. Both substudies will have a safety lead-in phase of ≤10 patients to determine the recommended phase 2 dose of sacituzumab tirumotecan when used in combination with pembrolizumab plus chemotherapy (substudy 06C) or paclitaxel (substudy 06D). In the safety lead-in phase, patients will receive sacituzumab tirumotecan 3 mg/kg or 4 mg/kg IV per a Bayesian optimal interval design. In the efficacy phase of substudy 06C, ≅120 patients (including 10 patients in safety lead-in) will be allocated or randomly assigned to receive pembrolizumab 400 mg IV every 6 weeks plus investigator's choice of chemotherapy (arm 1) or sacituzumab tirumotecan IV on days 1, 15, and 29 of each 6-week cycle plus pembrolizumab plus investigator's choice of chemotherapy (arm 2). In the efficacy phase of substudy 06D, ≅80 patients (including 10 patients in safety lead-in) will be allocated or randomly assigned to receive ramucirumab 8 mg/kg IV on days 1 and 15 of each 4-week cycle plus paclitaxel 80 mg/m^2 IV on days 1, 8, and 15 of each 4-week cycle (arm 1) or sacituzumab tirumotecan IV on days 1, 15, and 29 of each 6-week cycle plus paclitaxel (arm 2). Primary outcomes for both substudies are safety and tolerability for the safety lead-in and ORR per RECIST v1.1. by BICR for the efficacy phase (calculated based on pooled data from safety lead-in and efficacy phase); secondary outcomes include DOR and PFS per RECIST v1.1 by BICR, OS, safety, and tolerability. Enrollment is ongoing in both substudies. Clinical trial information: NCT06469944; NCT06445972. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

TPS512 **Trials in Progress Poster Session**

A single arm phase II trial of trastuzumab deruxtecan in patients with gastro-oesophageal adenocarcinoma cancer who are ctDNA and HER2 positive: DECIPHER.

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Background: Oesophagogastric adenocarcinoma (OGA) is globally prevalent and frequently advanced at presentation. Less than 50% of patients with operable OGA are cured when treated with multimodality therapy. Patients in the UK with operable OGA, are treated with FLOT prior to and following surgery. However, patients who have circulating tumour DNA (ctDNA) in their blood after surgery have worse survival than ctDNA negative patients. Therefore, use of novel treatment approaches aiming to cure this micrometastatic disease are considered a rational and appropriate treatment approach. Trastuzumab deruxtecan (T-DXd), a novel HER2-targeting antibody drug conjugate, is licensed to treat advanced, previously treated HER2 positive OGA. Therefore, there is a need to evaluate T-DXd in resectable HER2 and ctDNA positive OGA, where options are limited. **Methods:** Co-ordinated by the Cancer Research UK Southampton Clinical Trials Unit, DECIPHER is a single arm, multicentre, phase II trial testing the effect of T-DXd on reducing micrometastatic disease burden in HER2 positive OGA patients who are ctDNA positive after chemotherapy and surgery. To be eligible, participants must have been treated with chemotherapy before surgery for at least 6 weeks and must have recovered from surgery with no evidence of metastatic disease on post-surgical imaging. HER2 positive patients are recruited to the tissue testing stage of the trial prior to surgery. Patients who are ctDNA positive after surgery will be treated with T-DXd at a dose of 6.4 mg/kg intravenously every 21 days for a maximum of 8 cycles or until disease recurrence. If required, patients may dose reduce, but no dose re-escalation will be permitted. A clinically validated, personalised, tumour-informed ctDNA assay (signatera, Natera, Inc.) will be utilised to detect ctDNA in patient plasma samples. The primary endpoint is the percentage of patients who are ctDNA negative after 4 cycles of treatment. Secondary endpoints include ctDNA clearance after each cycle, disease free survival, overall survival, quality of life and frequency of adverse events. DECIPHER (NCT05965479) opened in April 2024 and will run in approximately 15 UK secondary care hospitals with the aim of recruiting 25 evaluable patients (recruitment to the tissue testing stage is currently 27 on 23-Sep-2024). Clinical trial information: NCT05965479. Research Sponsor: Astra Zeneca.

TPS513 **Trials in Progress Poster Session**

A phase I trial of intraperitoneal (IP) mesothelin (MSLN)-targeted CAR T-cell therapy in patients with MSLN-positive esophagogastric cancer (EGC) with peritoneal carcinomatosis.

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Background: Outcomes for patients with advanced EGC remain poor. The peritoneum is an immune-privileged niche and a common site of metastases occurring in up to 40% of patients with metastatic EGC; this is characterised by poor survival because of resistance to immuno-therapy and a lack of measurable disease, which excludes trial participation. MSLN was selected as a target for CAR T cell therapy as it is overexpressed in up to 60% of EGC, is associated with aggressive disease biology and has low levels of expression in normal tissues. In pre-clinical *in vivo* models of peritoneal carcinomatosis, improved efficacy was observed with regional administration of IP CAR T cells, compared with systemic CAR T cells. No significant toxicity was observed at clinically relevant doses of IP administered CAR T-cells. M28z1XXPD1DNR CAR T-cells are a next-generation MSLN-targeted CAR, equipped with a modified CD3z (1XX), and a PD-1 dominant negative receptor (PD1DNR) that provides T-cell intrinsic checkpoint blockade.
Methods: This is a single-arm phase I study assessing the safety, maximum tolerated dose (MTD) and preliminary efficacy of IP administered M28zXXPD1DNR CAR T cells in patients with MSLN-positive (IHC $\geq$ 25%) EGC with evidence of peritoneal carcinomatosis (imaging and/or cytology). Patients with concomitant extraperitoneal disease are eligible. Patients must have received at least one prior line of treatment for advanced or metastatic disease and have at least one measurable or evaluable lesion per RECIST 1.1. A minimum of 4 patients and maximum of 18 patients will be treated with 4 escalating doses (with 1 "fallback" dose) of CAR T cells to a maximum of 3 x 10^7 cells/kg administered via peritoneal catheter (Table). Lymphodepletion with fludarabine 30mg/m^2/day and cyclophosphamide 300mg/m^2/day is given for 3 days prior to the CAR T cell infusion. Adverse events, response rates (RECIST 1.1), survival outcomes and immuno-monitoring (immune cell phenotyping and serum/peritoneal fluid cytokine analysis) will be assessed. The trial is currently enrolling patients. Clinical trial information: Will be registered before ASCO GI. Research Sponsor: U.S. Department of Defense; Memorial Sloan Kettering Cancer Center.

Dose-escalation scheme based on the continuous reassessment method.

Dose Level	Dose (cells/kg)
1	1 x 10^6
2 (Starting dose)	3 x 10^6
3	6 x 10^6
4	1 x 10^7
5	3 x 10^7

TPS514 **Trials in Progress Poster Session**

A phase II study of neratinib (Nera) in combination with chemotherapy/ trastuzumab/pembrolizumab (C/T/P) in HER2 overexpressing gastroesophageal cancers (GECs).

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Background: Human epidermal growth factor receptor 2 (HER2) is overexpressed in about 15-20% of GECs and is tested using immunohistochemistry and/or in-situ hybridization. Per 3rd interim analysis of KN-811 study, addition of P to C/T improved overall response rate (ORR) and overall survival (OS) in patients with HER2 positive tumors. HER2 heterogeneity and activation of other signaling pathways ultimately leading to T resistance limits its potential to have a durable response. Nera is an anilinoqiunoline derivative intracellular oral kinase inhibitor that irreversibly binds to epidermal growth factor receptor (EGFR), HER2 and HER4. In this study, we hypothesize that targeting other oncogenic pathways with Nera may help derive additional therapeutic benefit in HER2 positive GECs when combined with C/T/P. **Methods:** This is an open-label, single arm, multi-institutional Phase II study of Nera in combination with C/T/P in HER2 overexpressing GECs as 1st line treatment. The first 6 pts as a lead-in phase will be evaluated for toxicity. If no dose limiting toxicity (DLT), a total of 30 patients (include the first 6pts in the lead-in phase) will be used to evaluate efficacy in the phase II portion. All pts will receive standard dose mFOLFOX/trastuzumab q2w along with pembrolizumab 400 mg q6w. Nera will be dosed at 240 mg daily. In case of DLTs observed with this dose, a dose of 160 mg/day for Nera will be considered. Estimated sample size is 30-36 pts using Simon 2 stage design. Primary endpoint in ORR. The null hypothesis will be rejected if 26 or more responses are observed in 30 patients. Key inclusion criteria include: HER2 positive GECs (as determined by local testing), treatment naïve for stage IV disease, measurable disease, ECOG 0-1, preserved cardiac function. Uncontrolled brain metastases or impaired organ function that poses risk with treatment are key exclusion criteria. Primary endpoint is ORR and secondary endpoints include safety, duration of response, clinical benefit rate and OS. The trial is registered on clinical-trials.gov under NCT06109467. Clinical trial information: NCT06109467. Research Sponsor: National Comprehensive Cancer Network (NCCN) through a grant provided by Puma Biotechnology.

TPS515 **Trials in Progress Poster Session**

A phase II study of agenT-797 (invariant natural killer T-cells), botensilimab (Fc-enhanced CTLA-4 inhibitor) and balstilimab (anti-PD-1) in patients with advanced, refractory gastroesophageal adenocarcinoma.

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Background: Esophagogastric (EG) cancer, the fifth most common cancer worldwide and second leading cause of cancer-related mortality, results in 1.3 million deaths annually. Despite recent advances in frontline management, nearly all patients with metastatic disease experience progression, highlighting the urgent need for novel therapies. Botensilimab (BOT), an Fc-enhanced, multifunctional anti-CTLA-4 antibody, has shown promise in combination with balstilimab (BAL, anti-PD-1) across multiple treatment-refractory cancers. Additionally, agenT-797, an allogeneic off-the-shelf invariant Natural Killer T (iNKT) cell therapy, enhances immune responses, including in settings refractory to immune checkpoint inhibitors (ICI), without requiring HLA matching or lymphodepletion. In a cohort of 82 previously treated patients, agenT-797 demonstrated a favorable safety profile with no severe Cytokine Release Syndrome, Graft-versus-Host Disease, or immune effector cell-associated neurotoxicity syndrome. Notably, in patients with relapsed/refractory gastric cancer, agenT-797 plus anti-PD-1 showed important clinical activity that warrants further exploration. Given their complementary mechanisms of action in modulating anti-tumor immunity, we aimed to evaluate the safety and efficacy of combining BOT, BAL, and agenT-797 with standard ramucirumab (anti-vascular endothelial growth factor receptor-2 antibody) and paclitaxel chemotherapy in patients with advanced EG cancer who have progressed on frontline therapy. **Methods:** This is an investigator-initiated, single-arm phase II trial in adult patients (≥18 years) with esophageal, gastric, or gastro-esophageal junction (GEJ) adenocarcinoma who have received one prior line of therapy, performed at Memorial Sloan Kettering Cancer Center. Eligible patients receive a single infusion of agenT-797 on cycle 1 day 1, BOT 75 mg on day 1 of cycles 1 through 3, BAL 240 mg on days 1 and 15, ramucirumab 8 mg/kg on days 1 and 15, and paclitaxel chemotherapy 80 mg/m^2 on days 1, 8, and 15 of a 28-day cycle. Treatment is continued until disease progression (defined by Response Evaluation Criteria in Solid Tumors [RECIST] version 1.1), unacceptable toxicity, or withdrawal of consent. The primary endpoint is overall response rate. The secondary endpoints are safety and tolerability, disease control rate, duration of response, progression-free survival, and overall survival. The regimen will be considered worthy of further investigation if at least 15 out of 37 patients achieve an objective response. To date, 12 patients have been enrolled and started on treatment. This trial is registered with ClinicalTrials.gov, NCT06251793. Clinical trial information: NCT06251793. Research Sponsor: None.

TPS516 **Trials in Progress Poster Session**

A phase II double window of opportunity trial evaluating the oral TGF-beta receptor I inhibitor vactosertib in patients undergoing standard of care preoperative therapy for locally advanced esophageal adenocarcinoma.

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Background: Esophageal adenocarcinoma (EAC) is the seventh most common cancer globally and ranks sixth in overall mortality. Despite improvements in pre-operative therapy, EAC has a high recurrence rate, especially among patients whose tumors do not achieve pathologic complete response (pCR), thus there is a critical need to enhance pCR rate and improve long-term outcomes. We recently discovered that a subset of EACs are driven by tumor-intrinsic oncogenic TGFβ signaling and have identified two potential biomarkers to predict response to anti-TGFβ therapy. One of these biomarkers is enriched in poorly differentiated EAC. We hypothesize that TGFβ signaling blockade will exert a tumor-inhibitory effect in patients with EAC and propose a double window-of-opportunity trial to test if TGFβ receptor I inhibitor vactosertib can improve pCR and induce tumor metabolic response in patients undergoing neoadjuvant therapy for localized EAC. **Methods:** This phase II double window of opportunity study assesses the efficacy and pharmacodynamics of the oral TGFβ Receptor I inhibitor vactosertib in patients with locally advanced esophageal adenocarcinoma (EAC). Patients will be treated with single-agent vactosertib in two windows of opportunity: the first will be prior to initiation of standard of care neoadjuvant therapy and the second after completion of neoadjuvant therapy prior to surgery. Patients will be eligible if they have poorly differentiated EAC and are appropriate for neoadjuvant therapy followed by resection as per standard of care. The primary objectives are improvement in pCR compared to historical controls and metabolic response (decrease in fluorodeoxyglucose uptake by PET CT) in the primary tumor after the first window. Secondary objectives will evaluate biomarkers of response identified in preclinical studies. Pre- and post-first window of opportunity treatment biopsies will be obtained for correlative and exploratory analyses. Clinical trial information: NCT06044311. Research Sponsor: None.

TPS517 **Trials in Progress Poster Session**

Preclinical and phase 1/2 data of the CHK1 inhibitor BBI-355 in development for esophageal and gastric cancers (EGC) with *EGFR* or *FGFR2* amplifications.

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Background: High-copy number amplifications of oncogenes (e.g., *EGFR*, *FGFR2*) frequently occur on extrachromosomal DNA (ecDNA), highly transcribed units of circular non-chromosomal DNA. While targeted therapies have improved survival for patients with oncogene mutations, they have limited activity in oncogene amplified cancers. In Barrett's esophagus-associated esophageal adenocarcinoma (EAC) ecDNA can be found early in the transition from high-grade dysplasia to EAC. In EAC and gastric cancer (EGC), ~7% and ~3% have *EGFR* and *FGFR2* amplifications, respectively, with possibly >50% occurring on ecDNA. Tumor cells with oncogene amplifications, particularly on ecDNA, have increased DNA replication stress and are sensitive to inactivation of checkpoint kinase 1 (CHK1). We developed BBI-355, an oral, potent, and selective small molecule inhibitor of CHK1, which is in Phase 1/2 clinical development. **Methods:** BBI-355-101 is a first-in-human Phase 1/2 study of BBI-355 alone or in combination with targeted therapies for advanced or metastatic solid tumors with oncogene amplification (NCT05827614). Part 1 is dose escalation of BBI-355 (PO Q2D). In Parts 2 and 3, patients with *EGFR* or *FGFR1-3* amplifications are treated with BBI-355 in combination with the EGFR inhibitor erlotinib (150 mg PO QD) or the FGFR inhibitor futibatinib (20 mg PO QD). While this study is enrolling all tumor types, EGC are some of the most likely to harbor ecDNA amplifications. Amplification-driven cancers rarely contain driver oncogene mutations or fusions, and so are excluded. **Results:** *Preclinical:* Tumors that harbor oncogene amplifications on ecDNA can evade targeted therapeutic pressure via ecDNA-based resistance mechanisms. BBI-355 in combination with agents targeting the amplified oncogenes (*FGFR2* or *EGFR*) prevented resistance to targeted therapies in multiple ecDNA+ gastric cancer CDX and PDX models. *Clinical:* Safety and PK data of BBI-355 alone and in combination with erlotinib or futibatinib will be presented. The most common drug-related TEAEs, SAEs, and DLTs were hematologic (i.e., leukopenia, neutropenia, lymphopenia and thrombocytopenia) which are on target for CHK1 inhibition and generally well managed. Dose escalation is ongoing and the RP2D/MTD has not been determined. PK showed dose-dependent increase in exposure of BBI-355. Target engagement as determined by pCHK1 IHC as a PD biomarker was observed at all dose levels of BBI-355. **Conclusions:** The first ecDNA directed therapy, BBI-355, demonstrated significant synergistic anti-tumor activity in combination with targeted therapies in multiple ecDNA+ oncogene amplified EGC models. BBI-355 alone and in combination with erlotinib/futibatinib was well tolerated. Clinical testing in patients with *EGFR* and *FGFR2* oncogene amplifications is ongoing with a strong rational for EGC. Clinical trial information: NCT05827614. Research Sponsor: None.